Investigations into Distributional Semantics for Cognate Detection and Phylogenetics

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2021

To

Late Smt. Saraswati Devi,

Late Mrs. Roseline Jose,

and

Kobe Bean Bryant.

"For their fond memories will linger in my eyes, forever."

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Abstract

Distributional semantics has become a key component of natural language processing (NLP) tasks as it allows computational models to understand natural languages. The sharing of vocabulary among different languages facilitates language learners to acquire a second language with much more ease, and it enables the computational models to perform cross-lingual learning for various NLP tasks. This thesis describes our work on two problems: cognate detection and false friends' detection. Our investigations in distributional semantics for both problems are based on the linguistic phenomena of shared vocabulary in related languages. Cognate words are variants of the same lexical form across different languages. For example, the words "fonema" in Spanish and "phoneme" in English, both mean "a unit of sound". The task of automatic detection of such word pairs, *i.e.,* automatic cognate detection, has implications on tasks such as cross-lingual information retrieval, machine translation, and computational phylogenetics. Similarly, false friends are word pairs from two different languages which pose a challenge to cross-lingual NLP tasks since they share similar spelling but differ in meaning (*e.g.*, the word "gift" in German, means "Poison" in English). Existing approaches rely heavily on phonetic and orthographic information but neglect cross-lingual semantic information. Our investigations use cross-lingual features to help detect cognates and false friends across languages. In the preliminary investigations, we describe our understanding of distributional semantics and present three studies: (a) efforts to create a dataset from a linked knowledge graph, *viz.*, Wordnets, (b) a semi-automated method to link these Wordnets, and (c) creation of monolingual and cross-lingual embedding models for NLP tasks.

Following this, we propose our approaches for the tasks of cognate and false friends' detection. We describe our efforts to create challenging datasets for the cognate and false friends' detection task and evaluate the state-of-the-art approaches on them. To detect cognates, we propose a weighted orthographic similarity-based measure in our initial

investigations and use various deep learning-based classification approaches to perform binary classification amongst cognate and non-cognate word-pairs. Our approaches use machine learning and deep learning techniques such as statistical classifiers and cross-lingual distributional word vectors to detect both cognates and false friends. Towards the former, we also propose the use of human cognition and collect gaze-data via eye-tracking experiments. With this collected gaze-data, we propose additional features for the task of cognate detection. Finally, we also perform the task of cognate detection for Indian and European languages to evaluate our approaches, on a different language family.

Additionally, we also explore the use of distributional semantics-based similarity for the task of computational phylogenetics and language typology. The use of cross-lingual features to create a typological tree for Indian languages demonstrates the efficacy of cross-lingual features. Computational phylogenetics is used to infer the evolutionary relationships between genes, species or taxonomic units. In our work, these taxonomic units are variant manuscripts of a written text, which has evolved over time. The goal of computational phylogenetics is to infer a tree via a grouping of these variants into families or "clades" with the use of computational methods. Phylogenetic analysis has shown to help philologists build the critical edition for a text, *i.e.,* an edition that attempts to construct a text of the work using all the available evidence. We propose a distributional semantics-based approach to generate better trees than the traditional phylogenetic approaches based on orthographic measures. We also propose the division of the text data into meaningful functional units which help the task of phylogenetic tree generation. Finally, we study the induction of various factors like available timeline, commentaries, sub-commentaries, quotations and printed editions of the text and present our insights into how these factors can help build the critical edition the said text. This thesis details our investigations related to the use of distributional semantics for cognate detection, false friends' detection and computational phylogenetics in terms of findings and insights drawn from our research.

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Chapter 1

Introduction

Natural Language Processing (NLP) lies under the purview of Artificial Intelligence (AI), which is a branch of computer science. Artificially intelligent systems aim to 'emulate' human behaviour and intelligence. A key aspect of human behaviour is to be able to communicate and understand natural languages. Humans have been trying to equip machines to understand natural languages since the early 1950s [Jones, 1994]. In this day and age, NLP is concerned with helping machines, not just understand natural languages, but also generate them. When NLP research started, rudimentary translation systems were deployed under the umbrella of Georgetown-IBM experiment [Hutchins, 2004] which dealt with translating sixty sentences from Russian into English.

As interest grew in being able to translate one natural language to another automatically, statistical models for machine translation were conceived. These advancements were only possible due to the significantly improved computational power and required resources, *i.e.,* machine-readable natural language data. With the advent of translation systems, a need for understanding human languages arose and the sub-area of "Semantics" came into the picture. Semantics primarily deals with helping machines understand human language and is a key component for natural language generation (NLG) tasks. In the past 30 years, various NLP tasks piqued the interest of theoretical linguistics community and the term "Computational Linguistics (CL)" was coined. Unlike early approaches which used rules and logic for each task, the use of machine learning and deep learning-based algorithms has helped computational linguistics advance towards fairly accurate and dependable results. NLP takes a more pragmatic stand with respect to the tasks involved, whereas computational linguistics deals with the tasks in a more granular sense.

CL studies human language to computationally understand how we as humans have the capacity to produce and understand language. With time, NLP tasks have become more statistical in nature, the credit for which, rightly goes to the advancements in Machine Learning (ML) algorithms. Machine learning algorithms like to focus on learning rules from the data (or examples), and then these learned models are able to predict the required output. In the past ten years or so, the use of simple linear classifiers like Naive Bayes and SVMs have now been replaced with complex models like Kernel SVMs and deep neural networks. These deep learning models, however, require large amounts of labelled data to estimate the parameters or the weights of the function reliably. The labelled data should contain sentences or vocabulary exhibiting all possible patterns so that the model can make appropriate predictions when those patterns are observed during the time of *testing*. Natural languages typically deal with huge vocabularies. Word distribution in natural languages follows *Zipf's law*, *i.e.*, a large portion of words appear only a few times in the corpus. Translation task deals with at least two languages and engages the learning of a machine via observing shared patterns in two languages. This brings us to the domain or multilingual or cross-lingual learning.

1.1 Multilingual or Cross-lingual Learning

Languages are related to each other due to various factors. The source of language relatedness may arise due to a genealogical relationship or via language contact. Related languages share phonological, orthographic and structural features among them. For example, Indian languages from the Indo-Aryan language family have high vocabulary overlap and similar word-order [V Subbãrão, 2012]. Some related languages share a significant amount of vocabulary amongst themselves.

Table 1.1 shows vocabulary overlap between the following Indian languages from Indo-Wordnet [Bhattacharyya, 2017]: Hindi (Hi), Bengali (Bn), Gujarati (Gu), Marathi (Mr), Punjabi (Pa), Malayalam (Ml), Tamil (Ta) and Telugu (Te). The Indo-Wordnet knowledge graph contains linked wordnets of several Indian languages. To estimate the percentage vocabulary overlap across Indian languages, we take a word from a language and look for an exact lexical match in the other language, for possible candidates. If the word has at least one matching sense in both the languages, we consider it a match.

	Bn	**Gu**	**Hi**	**Mr**	**Pa**	**Ml**	**Ta**	**Te**
Bn	-	19.51	29.45	11.39	2.45	1.05	0.34	0.78
Gu	13.90	-	51.75	20.14	4.46	1.06	0.30	1.22
Hi	12.76	31.47	-	15.22	4.43	0.78	0.21	0.95
Mr	11.81	29.31	36.42	-	3.40	0.62	0.27	0.92
Pa	4.26	10.88	17.79	5.71	-	0.22	0.16	0.40
Ml	1.19	1.70	2.04	0.67	0.14	-	0.72	2.48
Ta	0.43	0.54	0.62	0.33	0.11	0.80	-	0.25
Te	0.95	2.10	2.67	1.08	0.28	2.68	0.24	-

Table 1.1: Percentage of words in the source language (row) which also appear in the target language (column) and having at least one common synset obtained from IndoWordnet [Bhattacharyya, 2017].

For languages with different scripts, we use *Indic NLP library* [1] [Kunchukuttan et al., 2015] to transliterate it to a common script. We observe that *Bengali, Gujarati,* and *Marathi* share significant vocabulary with *Hindi* as compared to other languages. The values presented are an under-estimate of the actual vocabulary overlap across languages, as we do not factor into account for words that undergo slight phonological changes in the other language. For example, the Marathi word जीव (*jiva)* meaning *living animal* becomes জীব (*jiba*) in Bengali.

This experiment gives us a hint of relatedness among Indian languages. Vocabulary goes through orthographic changes when languages borrow words, and despite not taking such words into account; we report a significant number of identical words from the IndoWordnet knowledge graph. However, IndoWordnet is based on concepts (synsets) which contain synonymous words. Identical words from the same concepts are strong candidates for true cognate among Indian languages. We perform an additional experiment by increasing the number of languages and show the results in Table 1.2. The new language introduced for the results in this table are Nepali (Ne), Kannada (Kn), Oriya (Or), Assamese (As), Sanskrit (Sa), and Konkani (Ko). In this table, it can be seen that any two related languages from the Indian subcontinent share a significant amount of concepts which have at least one identical word. We have highlighted the language pair values which contain more than 25% of such shared concepts.

[1] Indic NLP Library

	Ne	Kn	Or	As	Bn	Te	Ml	Ta	Sa	Mr	Hi	Pa	Ko	Gu
Ne	-	13.07	23.61	16.38	**26.09**	2.83	1.90	0.20	17.20	**26.56**	**48.95**	10.11	17.10	**39.71**
Kn	6.95	-	7.41	4.90	8.27	5.02	3.01	0.75	4.11	12.47	17.44	3.34	7.49	16.53
Or	7.84	4.63	-	8.65	**32.43**	1.30	1.69	0.31	7.57	15.87	**30.92**	3.69	11.74	**25.82**
As	12.83	7.22	20.40	-	**27.87**	1.32	1.19	0.13	8.97	12.01	22.34	4.67	6.32	17.99
Bn	8.41	5.01	**31.49**	11.47	-	1.34	1.95	0.47	9.13	15.94	**34.77**	3.68	10.88	**26.39**
Te	1.57	5.24	2.17	0.93	2.31	-	5.14	0.53	0.97	2.75	6.04	0.59	1.84	5.06
Ml	0.74	2.20	1.97	0.59	2.36	3.60	-	1.27	0.77	1.45	3.99	0.21	1.02	3.26
Ta	0.09	0.65	0.44	0.07	0.68	0.44	1.51	-	0.27	0.51	1.00	0.18	0.26	0.86
Sa	5.22	2.35	6.92	3.48	8.61	0.53	0.60	0.18	-	9.36	18.76	1.45	4.60	13.36
Mr	9.48	8.37	17.06	5.47	17.65	1.77	1.33	0.40	11.00	-	**48.09**	5.24	**28.91**	**41.05**
Hi	14.20	9.52	**27.03**	8.28	**31.30**	3.15	2.98	0.63	17.93	**39.11**	-	10.95	**25.08**	**59.73**
Pa	3.66	2.27	4.02	2.16	4.13	0.38	0.19	0.15	1.72	5.31	13.65	-	3.48	9.25
Ko	6.19	5.10	12.80	2.92	12.22	1.20	0.95	0.21	5.49	**29.32**	**31.28**	3.48	-	**30.74**
Gu	13.06	10.23	**25.59**	7.56	**26.94**	3.00	2.76	0.62	14.47	**37.85**	**67.74**	8.41	**27.95**	-

Table 1.2: Percentage of concepts in the source language (row) with at least one identical word in the target language (column), in the same synset, *i.e.,* have the same meanings, obtained from Indo-Wordnet [Bhattacharyya, 2017].

Shared concepts with identical word-form help a learner with second language acquisition. As per the acquisition/learning hypothesis of the monitor theory [Gregg, 1984], second language (L2) learners perform subconscious learning based on their natural language (L1). This subconscious learning can be based on any meaningful conversation in the target language (L2). A shared set of words among L1-L2 which also mean the same definitely help learners acquire the second or the target language. Table 1.3, however, shows us the other side of vocabulary sharing among languages. When two languages belong to the same linguistic area, they may also drift apart. In such cases, the same word forms may also change their meaning with time; we perform another experiment to show this phenomenon based on the cross-lingual word-pairs in IndoWordnet. In table 1.3, we show the number of word-pairs across language pairs, which belong to different concepts but share an identical word-form (80% orthographic/lexical match).

Such word pairs can cause confusion among the second language learners and hence require us to perform more nuanced research in terms of computational linguistics, which not only helps computations models with the shared vocabulary but also avoids these perilous pitfalls. This leads us to the domain of shared vocabulary and how language change occurs, which is discussed in the next section.

	Bn	**Hi**	**Or**	**Mr**	**Sa**	**Kn**	**Te**	**Ko**	**Pa**	**Gu**	**Ml**	**As**	**Ne**	**Ta**
Bn	-	43586	43722	43338	43202	42997	42699	42524	41744	43111	41935	44425	44402	34716
Hi	96016	-	94830	97838	94987	95890	95748	96191	94619	96930	92674	96535	99772	76533
Or	50223	49718	-	49564	49252	48874	48362	48142	47811	49408	47900	50780	50909	38621
Mr	40397	41811	40904	-	40832	41555	41358	42439	40851	42438	40215	41178	42595	33565
Sa	46215	45398	45640	46132	-	46027	45965	45002	41321	45441	45731	46293	47697	38630
Kn	36502	39364	36762	40080	38751	-	42045	37998	32720	39633	39866	37168	41217	28807
Te	30274	31970	30065	32347	31646	33591	-	30725	26466	32068	32390	31062	33098	24274
Ko	29398	31371	29827	32112	30198	30745	30670	-	30162	31735	29531	30328	31598	23741
Pa	20987	22703	21807	22966	21137	21042	20133	21845	-	22513	18829	21356	23357	12856
Gu	57199	59513	58065	60966	57854	59507	59636	59480	57679	-	57669	58415	60941	46588
Ml	35005	35689	35022	35879	36710	37664	37789	34417	29240	35373	-	36073	38224	33841
As	20465	19427	20100	19341	19061	18367	18233	18088	16923	19188	17639	-	20831	11541
Ne	17919	17900	17660	17915	17795	17534	17501	17371	16471	17754	17313	17777	-	13127
Ta	27958	30389	28736	29303	31527	32435	31912	26829	23241	28827	33568	31072	32489	-

Table 1.3: The number of word-pairs in the source language and the target languages, which share a similar word-form (80% match) but in non-identical synsets (concepts), obtained from Indo-Wordnet [Bhattacharyya, 2017].

1.2 Understanding Shared Vocabulary in Languages

As discussed above, languages can exhibit similar linguistic features either through genealogical influence or through contact. In this section, we discuss various factors due to which languages are "close-to" or "far-apart" from each other. This relatedness among languages forms a key basis for our work on cognates and shared vocabulary in this thesis.

1.2.1 Etymological Relatedness

Languages exhibit genetic relatedness when they have descended from a common antecedent (language), *i.e.,* they have shared etymology. Any languages which descend from a common language form a *language family.* Apart from the Indo-Aryan language family, the most commonly cited example is how Latin and "Vulgar Latin" are considered to have been an antecedent for the set of "Romance Languages" [Rudder, 2012]. The term *family* here is used to reflect the hierarchical nature of relations typically observed in a biological family tree. A tree which depicts such a relation amongst various languages is known as phylogeny or a phylogenetic tree. Such trees depict the evolutionary informa-

tion about languages in the tree. Typological trees depict information about languages which can be used to measure the phylogenetic relatedness among languages. A typological classification groups languages into types according to their structural characteristics [Dunn et al., 2007].

Etymological relatedness is seen among related languages when a sub-community speaking a language migrates from the community to a different geographical location, and over the course of the time, the language spoken by the sub-community diverges from the original community[2]. Another reason for the linguistic change could be due to the influence of other languages. The language spoken by the sub-community now becomes a child language in the tree. *Ethnologue* estimates the number of living languages to be around $7,111$ and are distributed into 141 different languages families[3]. Classification of languages into language families is carried out by *comparative linguists*. *Comparative linguists* compare different characteristics of languages like phonology, morphology, lexicon, and, the syntax to determine the relationship between the languages. It is also possible that some languages do not have known related languages. Such languages are known as *language isolates.* These languages were part of a branch which, now, we have lost records of, and as a result, an entire branch is missing leaving just the leaf.

However, comparative linguistics is limited by the availability of historical records. Due to its over-dependence on historical records, we cannot fully trace the evolutionary tree of all the languages. As and when new evidence is obtained, comparative linguistics fills in the missing branches expanding our current knowledge of language relatedness. Languages within the same family form several *sub-families* and can exhibit significant differences between each sub-families. For example, *English* belongs to the *Germanic* sub-family and *Hindi* belongs to the *Indo-Aryan* sub-family. English exhibits canonical *SVO* (Subject-Verb-Object) word-order whereas Hindi exhibits canonical *SOV* (Subject-Object-Verb) word order.

1.2.2 Relatedness due to Language Contact

Languages can also be related to each other via contact. *Contact relatedness* is observed when two communities speaking different languages have significant social interaction,

[2]Migration and Language Contanct

[3]Ethnologue Statistics

and borrowing some features of the other language over time. The most common form of such contact is the borrowing of vocabulary from the other language. For example, English is known to contain borrowed words from Latin, French and the family of *Romance languages*. The influence of the other language could be substantial, leading to changes in morphology and grammar of the language.

Commonly spoken Indian languages like *Hindi*, *Marathi*, *Bengali*, *Gujarati*, *etc.* contain many words borrowed from the *Sanskrit* language. The Dravidian langauge family is also known to borrow words from *Sanskrit*. If there is no change in Sanskrit **word form** during adoption of these languages, the words are called *tatsama* words and if there is some change in the word form they are called as *tadbhava* words. *Tatsama* means "same as that" and *Tadbhava* means "arising from that". For *e.g.,* The Sanskrit word "*putra*" is borrowed "as-is" in Hindi and retians it's orthographic form in the word "*putra*", meaning "Son". In case of the Sanskrit word "*Satya*", the Hindi word takes an intermediary form first namely "*sacchh*", and later is borrowed in Hindi as "*sach*" meaning "Truth", in it's *Tadbhava* form.As a consequence, related languages share a lot of linguistic features and vocabulary among them.

Shared vocabulary among etymologically related and non-related languages is a known phenomenon where languages borrow words from each other. In case of related languages which belong to a single language family, for example, the Indo-Aryan language family, most of the shared vocabulary originates from a single historical antecedent. Indo-Aryan languages are known to borrow words from the Sanskrit language, and this borrowed vocabulary acts as a semantic link between any given language pair in the family. Such words share a formal or semantic affinity with each other. When such words are similar in form and meaning across related languages, they are known to be cognates.

1.3 From Cognate to Non-cognates: The Spectrum

Cognates are words with similarities in their spelling and meaning across two related languages L_1 and L_2. For example, the French and English word pair, *Liberté - Liberty*, reveals itself to be a true cognate through orthographic similarity. In some cases, similar words have a common meaning only in some contexts; such words are called **partial cognates**. For example, the word *"police"* in French can translate to *"police"*, *"policy"*

or *"font"*, depending on the context[4]. Manual detection of such cognate sets requires a human expert with a good linguistic background in multiple languages. Moreover, manual annotation of cognate sets is a costly task in terms of time and human effort. **False friends** are word pairs from two different languages which pose a challenge to downstream NLP applications such as Machine Translation, Cross-lingual Question Answering and Information Retrieval since they share similar spelling but differ in meaning (*e.g.*, the word "**gift**" in German, means "**Poison**" in English). **True cognates** share a similar orthographic form, but unlike False friends, mean the same across two different languages. However, then, there are confusing word-pairs which do not mean the same despite sharing a similar form. Such word-pairs are called False Friends, and these can severely affect the output of NLP models. False friends are often confused with being false cognates, but there is a certain distinction among both these types of word pairs.

In this thesis, we refer to the term *false friends* when we discuss word pairs, in related languages, which share a similar form, but do not mean the same. False cognates, on the other hand, are word pairs which can mean the same despite belonging to unrelated language pairs and are known to have different etymologies. We explain all these word-pair classes with the help of a diagram in Figure 1.1 below.

- True Cognates: class of word pairs which have similar origin, spelling, and meaning.
- False Friends: class of word pairs which have similar origin and spelling, but differ in meaning.
- False Cognates: class of word pairs which do not have a similar origin, but have similar spelling and meaning.
- Synonyms: having a similar meaning, but spelt very differently.
- Genetic Cognates: have similar origin and meaning, but differ a lot when it comes to spelling.
- Co-Etym: We coined this term for the class of word-pairs which have a similar origin, but do not belong to any class of cognates as they do not mean the same, and are spelt differently.

[4]Cognates can also exist in the same language. Such word pairs/sets are commonly referred to as *doublets*.

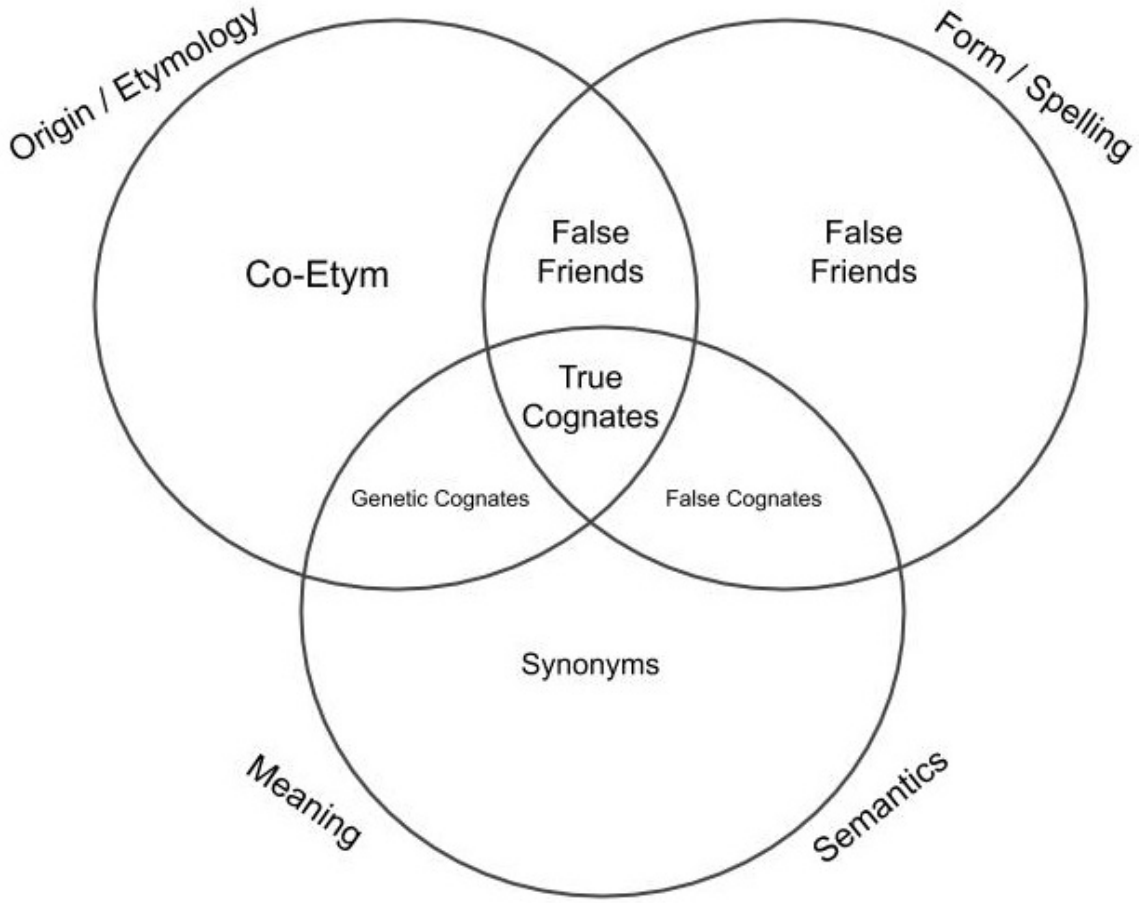


Figure 1.1: Classification of word-pairs across languages

In figure 1.2, we present a classification of word-pair classes we deal with, during our investigations. Here, we also present examples of cognates, false friends, false cognates to present a descriptive view of the word-pair classes in question.

1.4 The Cognate Detection Task

Automatic detection of cognates is a well-known task, which has been explored for a range of languages using different methods; and has shown to help Cross-lingual Information Retrieval [Meng et al., 2001], Machine Translation (MT) [Al-Onaizan et al., 1999], and Computational Phylogenetics [Rama et al., 2018]. If languages which belong to the same language family are concerned, we hypothesize that word with a high spelling and semantic similarity can be called 'cognates'. Obtaining a spelling similarity-based score for any candidate cognate pair can be performed via classical methods such as Normalized Edit Distance. However, what does it mean for two words to mean 'the same'? Semantic

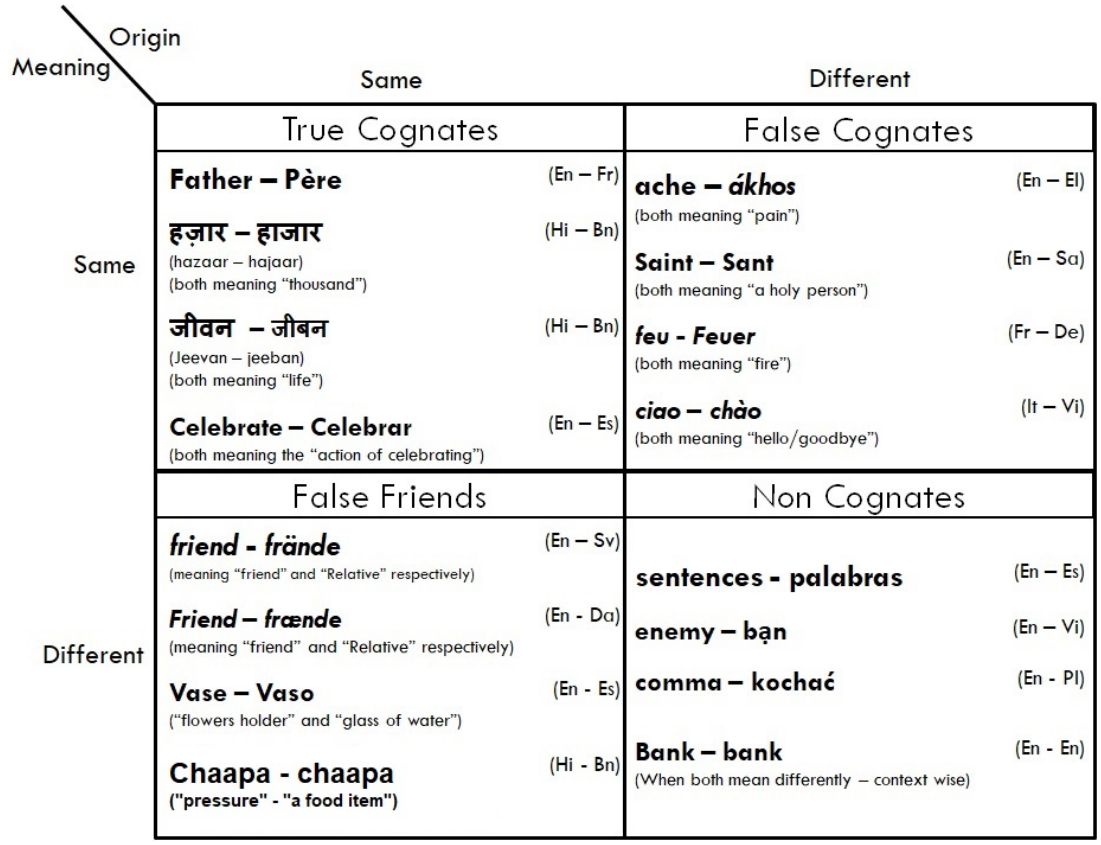


Figure 1.2: Examples from the shared vocabulary classes across two languages.

equivalence can be defined in functional terms: the meaning of a word w_1 in one language (L1) is aligned with a word w_2 in another language (L2) if the two words are used in the same contexts by L1 and L2 speakers. One reason why describing the semantic structure of natural languages is difficult is that word meanings, similar to other psychological constructs, are not directly observable. The most direct way to assess semantic equivalence would be to query L1–L2 speakers of multiple languages about the meanings of different words. For example, the English word 'impressed' has an unambiguously positive valence, whereas the valence of its Italian translation equivalent, 'impressionato', is relatively more negative. This difference in valence suggests that 'impressed' and 'impressionato' do not quite mean the same thing. However, this approach is challenging to implement at scale. For this reason, our attempts to quantify semantic similarity focus on using cross-lingual word embeddings, and then we use standard machine learning classifiers to generate models.

The two main classification approaches applied to the detection of cognates belong to the generative and discriminative paradigms. The first paradigm is based on the compu-

tation of a similarity score between potential candidate pairs. This score can be a simple Orthographic similarity [Jäger et al., 2017], phonetic similarity [Rama, 2016], or a distance measure with the scores learned from an existing parallel set [Mann and Yarowsky, 2001, Tiedemann, 1999]. The discriminative paradigm uses standard approaches to machine learning, which are based on (1) extracting features, e.g., character n-grams, and (2) learning to predict the transformations of the source word needed to [Jiampojamarn et al., 2010, Frunza and Inkpen, 2009]. Cognate identification approaches can confuse a false friend pair with being a cognate if orthographic similarity-based techniques are relied upon. For languages which do not share the same script, normalization techniques help compare the orthography among their words. Romanization of words is a common methodology used among such language pairs. False friends are especially problematic for language learners. We discuss then in the next section.

1.5 The False Friends' Detection Task

In our work, we also focus on the task of false friends' detection and learn to identify false friends' from among candidate pairs, by utilizing distributional semantics across languages. Applications of cognate detection in NLP include sentence alignment [Simard et al., 1993, Melamed, 1999], inducing translation lexicons [Mann and Yarowsky, 2001, Tufis, 2002], improving statistical machine translation (MT) [Al-Onaizan et al., 1999], improving information retrieval [Makin et al., 2007] and identification of confusable drug names [Kondrak and Dorr, 2004]. All these applications depend on an effective method of identifying cognates by computing a score, which reflects the likelihood that the two words are cognates among languages.[5]

Cognate identification approaches can confuse a false friend pair with being a cognate if orthographic similarity-based techniques are relied upon. False friends are especially problematic for language learners as learners tend to overgeneralize and assume that they know the meaning of these misleading words. Hence, we also focus on the task of false friends' detection across languages.

[5]Cognates in the same language are referred to as *doublets*.

1.6 Computational Phylogenetics

Phylogenetics is defined as the task of creating a Phylogenetic Tree which represents a hypothesis about the evolutionary relationship among a set of genes, species or any other taxa. It is the study of evolutionary history and relationships among various taxa. A 'Taxon' represents a group of one or more manuscripts written in Sanskrit in our case, where we analyze how the manuscripts are related to each other.

These relationships are discovered through phylogenetic methods that compute observed heritable traits in a manuscript, such as spelling errors, variations in text, text deletion, the morphology of the text, *etc.* under a model of the evolution of these traits. The result of these analyses is known as a phylogenetic tree, *i.e.,* a diagrammatic hypothesis about the history of the evolutionary relationships of a group of manuscripts (usually belonging to the same text).

The Computational purview of this part of our research problem deals with developing new methods for alignment estimation, phylogeny estimation, and species identification of metagenomic data. Computational historical linguistics, which involves the development of methods for estimating evolutionary histories of languages and of models of language evolution, is another research problem based on phylogenetics.

Phylogenetic methods are designed to recover the "true" evolutionary tree as often as possible. They do not guarantee to do so with high probability under reasonable conditions. Some methods which offer this guarantee vary considerably in their requirements [Warnow et al., 2001]. To rigorously establish the validity of such a phylogenetic approach, a fundamental question that must be addressed is whether the models in use are *identifiable*. From the theoretical distribution predicted by the model, is it possible to uniquely determine all parameters?

Using the currently available models, finding optimal phylogenetic trees using compatibility criteria is in its general case NP-Complete [Warnow, 1993]. Also, finding a Maximum Compatible Tree is NP-Hard [Roch, 2006]. Thus, it means that efficient algorithms to solve the problem, probably, cannot exist. By restricting the kinds of input to the problem, however, we may be able to solve it efficiently.

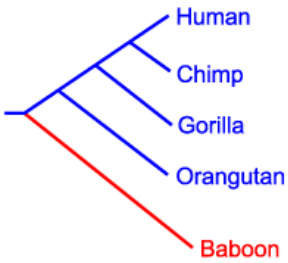


Figure 1.3: A Sample Rooted Tree

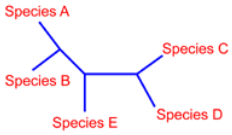


Figure 1.4: A Sample Unrooted Tree

1.6.1 Tree Generation

The trees generated by phylogenetic methods can be either *rooted* or *unrooted*, depending on **the input data** and **the algorithm used to build the tree**. A rooted tree is a directed graph that identifies a most recent common ancestor (MRCA). Unrooted trees plot the distances and relationships between input sequences without making assumptions regarding their descent. It can always be produced from a rooted tree, but root cannot usually be placed on an unrooted tree without additional data on divergence rates [Mount, 2004].

For identifying a root, it is usually required that that inclusion in the input data of at least one "outgroup" is known to be distantly related to the sequences of interest. A generic example of a phylogenetic tree is shown in figures 1.3 and 1.4 where the former represents a rooted phylogenetic tree and the latter depicts an unrooted phylogenetic tree.

An example of a phylogenetic tree, based on textual manuscripts, is shown below in figure 1.5 where a phylogenetic tree is drawn manually from Malayalam manuscripts for Kāśikāvṛtti.

Kulkarni [2006] show that M is the archetype source, and Ma, Mb and Mc are its hyperarche child nodes. M is decided as a source based on the analysis made on the variant readings. In this process, manuscripts with similar variants are grouped and named as M1, M2, M3 ..., M11.

1.6.2 Comparitive Approaches to Historical Linguistics

During the last two decades there has been an increasing interest in automatic approaches to historical linguistics, which is reflected in the large amount of literature on phylogenetic reconstruction (e.g. [Ringe et al., 2002]; [Gray and Atkinson, 2003]; [Brown et al.,

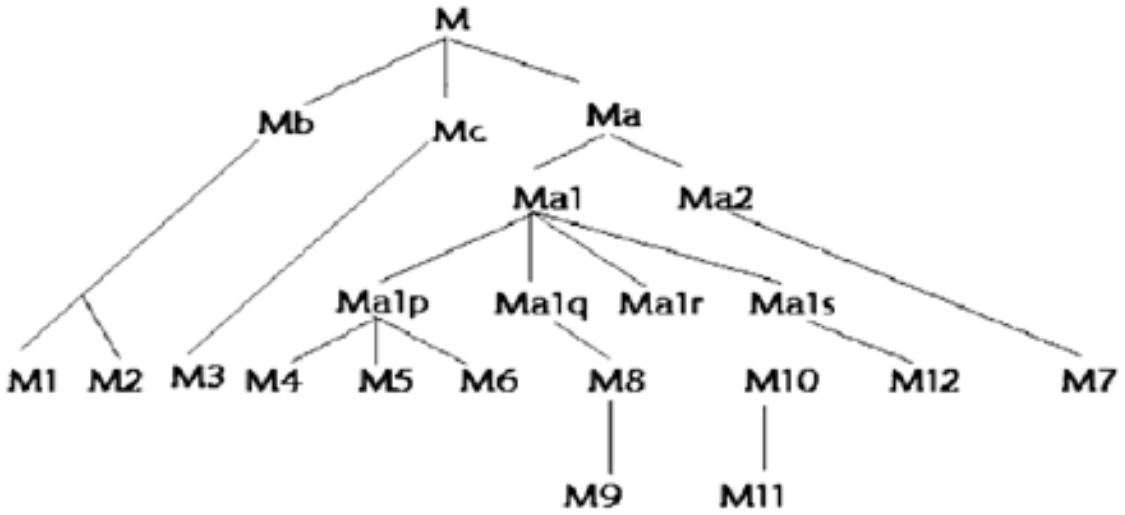


Figure 1.5: A Sample Stemma of Malayalam Manuscripts as shown by [Kulkarni, 2006]

2008]), statistical aspects of genetic relationship (e.g. [Baxter and Ramer, 2000]; [Kessler, 2001]), and phonetic alignment (e.g. [Kondrak and Hirst, 2002]; [Prokić et al., 2009]). While the supporters of these new automatic methods would certainly agree that their greatest advantage lies in the increase of repeatability and objectivity, it is interesting to note that the most crucial part of the analysis, namely the identification of cognates in lexicostatistical datasets, is still almost exclusively carried out manually.

Under the purview of 'diachronic linguistics', establishing relationships among languages which have been in contact for a long time has also been a topic of interest [Chevillet, 2000]. However, this topic has been much less explored in the computational linguistics community. Most of the previous work is focused on the reconstruction of phylogenetic trees for a particular language family using manually curated word lists [Gray and Atkinson, 2003, Atkinson et al., 2005, Atkinson and Gray, 2006, Nakhleh et al., 2005] or using synthetic data [Barbançon et al., 2013]. Phylogenetic reconstruction and analysis is based on a data matrix where the rows represent the languages to be studied, and the columns represent a linguistic feature or character [Nichols and Warnow, 2008]. Moreover, the methods inspired from glottochronology take a boolean matrix as input, which denotes the change in the state of the 'characters' (the 'characters' can be lexical, morphological or phonological) to infer the phylogenetic trees. A portion of our work also deals with establishing relationships among various Indian languages which are an important part of the Indo-European language family.

Phylogenetic analysis can be carried out by sequence comparison. It is one of the key tasks in historical linguistics. By comparing words or morphemes across languages, linguists can identify which words have sprung from a common source in genetically related languages, or which words have been borrowed from one language to another. By comparing words within a language, linguists can identify grammatical and lexical morphemes, cluster words into families, and shed light on the internal history of languages. So far, the majority of this work has been carried out manually. Linguists sift through dictionaries and fieldwork notes, trying to identify those words which reflect a shared history across languages. All etymological dictionaries available today have been based on manual word comparison, and their results fill thousands of pages. Even the largest databases which offer cognate judgments, such as the Austronesian Basic Vocabulary Database ([Greenhill et al., 2008]) or the Indo-European Lexical Cognacy Database ([Dunn, 2012]) are based on manual assessments of cognacy.

Historical linguists are sceptical about automating the methods for cognate identification ([Holman et al., 2011] and commentaries, as well as [List et al., 2017b]). First, the accuracy of automated methods is often low, failing to reproduce the analyses of linguistic experts. Especially, the use of the edit distance ([Levenshtein, 1965]) has been criticized for being linguistically too naive, conflating sound correspondences and lexical replacement, for being useful for subgrouping or cognate detection ([Holman et al., 2011]; [Greenhill, 2011]). Second, it is hard to verify many algorithms as they are seen as black-boxes which hide the crucial decisions leading to cognate judgments and subgroupings, making it difficult for scholars to determine whether similarities are due to inheritance or contact ([Jäger, 2015];[List et al., 2017a]). The non-transparency of automatic methods is highly problematic for computational historical linguistics: if we do not know what evidence decisions are based on, we cannot criticize and improve them.

Despite the scepticism of historical linguists, another important aspect of Historical linguistics (also popularly known as “Diachronic Linguistics”) when it comes to phylogenetic inference is, indeed, Cognate Detection. The study of cognates plays a crucial role in applying comparative approaches for historical linguistics, in particular, solving language relatedness and tracking the interaction and evolvement of multiple languages over time. A cognate instance in Indo-European languages is given as the word group: night (English), *nuit* (French), *noche* (Spanish) and *nacht* (German).

Information retrieval addresses the problem of scoring a document with a given query, which is used in every search engine. One can view the above problem as the construction of a suitable search engine, through which we want to find the cognate counterpart of a word (query) in a lexicon of another language (documents). Rama et al. [2018] establish that work on phylogenetic reconstruction can profit much from automatic cognate detection. Especially where scholars are merely interested in exploring the bigger picture of a language family's phylogeny, algorithms for automatic cognate detection are a useful complement for current research on language phylogenies. They test multiple cognate detection methods and describe the usefulness of automated cognate detection methods for phylogenetic inference. Although the cognate sets predicted by automated cognate detection methods yield phylogenetic trees that come close to expert trees, there is still room for improvement, and future research is needed to enhance automatic cognate detection methods further.

However, manually annotated cognate sets are also not free from errors, and it seems likewise useful to investigate how the consistency of cognate coding by experts could be further improved.

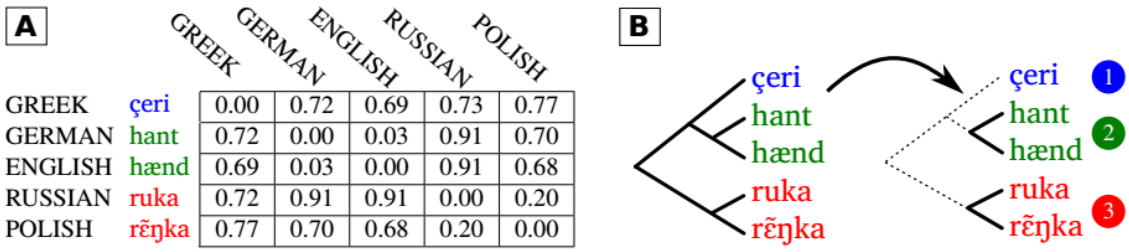


Figure 1.6: Rama et al. [2018] show how cognates can form phylogenetic trees; they use the English word 'hand' as a sample.

The existing studies on cognate detection involve experiments which distinguish between a pair of words whether they are cognates or non-cognates ([Ciobanu and Dinu, 2014]; [List, 2012]). These studies do not approach the problem of predicting the possible cognate of the target language if the cognate of the source language is given. For example, given the word 'nuit', could the algorithm predict the appropriate German cognate within the huge German word list?

Achieving good performance on automatic cognate identification can also benefit machine translation when dealing with two languages that share a certain quantity of

cognates, as cognates are usually translations and serve as anchors when aligning. Hence, we explore the task of Cognate detection, False Friends' detection and Computational Phylogenetics and show that Cognates help the downstream task of Machine Translation to a certain extent.

1.7 Thesis Contribution

Our investigations in distributional similarity for cognate detection and phylogenetics span three areas: (a) Understanding the phenomenon of shared vocabulary and cognates, (b) The task of true cognate and false friends' detection, and (c) Computational phylogenetics for variant manuscripts and language typology using distributional similarity. Specific contributions are as follows:

1. Approaches for cognate and false friends' detection based on the notion of semantic similarity. The notion of 'similarity' is explored using different measures which exist between words and their contexts: orthographic similarity (when word spellings are similar), phonetic similarity (where similarity among phonemes is computed), and semantic similarity (when cross-lingual distributional similarity among words is computed).

2. To the best of our knowledge, cross-lingual similarity-based approach to cognate and false friends' detection has not been explored in the past.

3. To the best of our knowledge, gaze tracking-based approach to cognate detection has not been explored in the past.

4. Introduction of novel research problems apart from cognate and false friends' detection, namely, computational phylogenetics for variant manuscripts. These result in the following contributions:

   (a) A first-of-its-kind approach to generate phylogenetic trees using word embeddings based scores.

   (b) A first-of-its-kind, open-source online tool which can help capture and display the historical evolution of a text.

5. Creation of challenge datasets for cognates and false friends: (a) cognate dataset across nine Indic languages, (b) false friend dataset, and (c) cross-lingual models for all the language pairs.

6. A study in language typology for Indic languages which accurately depicts the distance among Indian languages based on Wordnet data.

To the best of our knowledge, all of the above contributions are first-of-their-kind. Our work resulted in '*Textual History Tool*' which is a deployment of our phylogenetic tree generation and data accumulation approaches. The work on Textual History Tool was accepted as a full paper, and a demonstration was provided at ISCLS-19.

1.8 Thesis Organization

This thesis is organized as follows:

1. Chapter 2 describes the mathematical and computational concepts present in this thesis and presents a survey of the past work in the areas of cognate detection and computational phylogenetics.

2. Chapter 3 describes our investigations in creating relevant data for this thesis, and understanding distributed similarity, with respect to our work.

3. Following that, we discuss our preliminary investigations towards the task of Cognate detection in chapter 4.

4. In chapter 5, we discuss novel feature sets proposed towards the task of cognate detection.

5. Chapter 6 presents our approaches to false friends' detection.

6. Chapter 7 presents our approaches to computational phylogenetics and our work on Indian language typology.

7. We conclude the thesis in Chapter 8 and discuss future work along with it.

1.9 Publications

The publications resulting from the research described in this thesis are:

Related to our applications of Distributional Semantics to both the areas

1. Kanojia, Diptesh, Nikhil Wani, and Pushpak Bhattacharyya. "Is your Statement Purposeless? Predicting Computer Science Graduation Admission Acceptance based on Statement Of Purpose." In Proceedings of the 14th International Conference on Natural Language Processing (ICON 2017). 2017.

2. Kanojia, Diptesh, Kevin Patel, and Pushpak Bhattacharyya. "Indian Language Wordnets and their Linkages with Princeton WordNet." In Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018). 2018.

3. Patel, Kevin, Diptesh Kanojia, and Pushpak Bhattacharyya. "Semi-automatic WordNet Linking using Word Embeddings." In Proceedings of the 9th Global WordNet Conference (GWC 2018), p. 269. 2018.

4. Kanojia, Diptesh, Malhar Kulkarni, Pushpak Bhattacharyya, and Gholamreza Haffari. "Cognate Identification to Improve Phylogenetic Trees for Indian Languages." In Proceedings of the ACM India Joint International Conference on Data Science and Management of Data, pp. 297-300. 2019.

5. Kanojia, Diptesh, Kevin Patel, Pushpak Bhattacharyya, Malhar Kulkarni, and Gholamreza Haffari. "Utilizing Wordnets for Cognate Detection among Indian Languages." In the Global Wordnet Conference, p. 404. 2019.

6. Kanojia, Diptesh, Malhar Kulkarni, Pushpak Bhattacharyya, and Eivind Kahrs. "An Introduction to the Textual History Tool." In Proceedings of the 6th International Sanskrit Computational Linguistics Symposium, pp. 166-180. 2019.

7. Kanojia, Diptesh, Abhijeet Dubey, Malhar Kulkarni, Pushpak Bhattacharyya, and Gholamreza Haffari. "Utilizing Word Embeddings based Features for Phylogenetic Tree Generation of Sanskrit Texts." In Proceedings of the 6th International Sanskrit Computational Linguistics Symposium, pp. 152-165. 2019.

8. Kanojia, Diptesh, Sravan Munukutla, Sayali Ghodekar, Pushpak Bhattacharyya, and Malhar Kulkarni. "Keep Your Dimensions on a Leash: True Cognate Detection using Siamese Deep Neural Networks." In Proceedings of the 7th ACM IKDD CoDS and 25th COMAD, pp. 324-325. 2020.

9. Mantha, Yashasvi, Diptesh Kanojia, Abhijeet Dubey, Pushpak Bhattacharyya, and Malhar Kulkarni. "Harnessing Deep Cross-lingual Word Embeddings to Infer Accurate Phylogenetic Trees." In Proceedings of the 7th ACM IKDD CoDS and 25th COMAD, pp. 330-331. 2020.

10. Kanojia, Diptesh, Sayali Ghodekar, Malhar Kulkarni, Eivind Kahrs, and Pushpak Bhattacharyya, "Strategies of Effective Digitization of Commentaries and Sub commentaries Towards the Construction of Textual History". In Proceedings of the International Conference on Tradition of Commentaries and Flow of Knowledge, Shree Somnath Sanskrit University, Veraval, Gujrat, edited by D. N. Pandeya, D. V. Katira, J. Acharya, Chief Editor: Gopabandhu Mishra, as part of Shree Somnath Sanskrit University Grantha Series-22, ISBN: 978-93-83097-43-2, pp. 477-489. 2020.

11. Kanojia, Diptesh, Pushpak Bhattacharyya, Malhar Kulkarni, and Gholamreza Haffari. "Challenge Datasets of Cognate and False Friend Pairs for Indian Languages". In Proceedings of the Language Resources and Evaluation Conference (LREC). 2020.

12. Saurav, Kumar, Kumar Saunack, Diptesh Kanojia, and Pushpak Bhattacharyya. "A Passage to India": Pre-trained Word Embeddings for Indian Languages. In Proceedings of the SLTU-CCURL Workshop held jointly with LREC. 2020.

13. Kanojia, Diptesh, Raj Dabre, Shubham Dewangan, Pushpak Bhattacharyya, Gholamreza Haffari, and Malhar Kulkarni. "Harnessing Cross-lingual Features to Improve Cognate Detection for Low-resource Languages." In Proceedings of the 28th International Conference on Computational Linguistics, pp. 1384-1395. 2020.

14. Kanojia, Diptesh, Prashant Sharma, Sayali Ghodekar, Pushpak Bhattacharyya, Gholamreza Haffari, and Malhar Kulkarni. "Cognition-aware Cognate Detection."

In Proceedings of the 16th conference of the European Chapter of the Association for Computational Linguistics (EACL 2021). 2021.

Additional Publications (apart from what can be considered as contributions towards the thesis, but published during the PhD tenure):

1. Joshi, Aditya, Diptesh Kanojia, Pushpak Bhattacharyya, and Mark Carman. "Sarcasm Suite: a browser-based engine for sarcasm detection and generation." In Thirty-First AAAI Conference on Artificial Intelligence. 2017.

2. Mishra, Abhijit, Diptesh Kanojia, Seema Nagar, Kuntal Dey, and Pushpak Bhattacharyya. "Scanpath complexity: Modeling reading effort using gaze information." In Thirty-First AAAI Conference on Artificial Intelligence. 2017.

3. Mathias, Sandeep, Diptesh Kanojia, Kevin Patel, Samarth Agarwal, Abhijit Mishra, and Pushpak Bhattacharyya. "Eyes are the Windows to the Soul: Predicting the Rating of Text Quality Using Gaze Behaviour." In Proceedings of the 56th annual conference of the Association for Computational Linguistics. 2018.

4. Redkar, Hanumant, Rajita Shukla, Sandhya Singh, Jaya Saraswati, Laxmi Kashyap, Diptesh Kanojia, Preethi Jyothi, Malhar Kulkarni, and Pushpak Bhattacharyya. "Hindi Wordnet for Language Teaching: Experiences and Lessons Learnt." In Proceedings of the 9th Global WordNet Conference (GWC 2018), p. 317. 2018.

5. Gajjam, Jayashree, Diptesh Kanojia, and Malhar Kulkarni. "New Vistas to study Bhartrhari: Cognitive NLP." In Proceedings of the World Sanskrit Conference (WSC). 2018.

6. Kanojia, Diptesh, Preethi Jyothi, and Pushpak Bhattacharyya. "Synthesizing Audio for Hindi WordNet." In Proceedings of the 9th Global WordNet Conference (GWC 2018), page 392. 2018.

7. Panjwani, Ritesh, Diptesh Kanojia, and Pushpak Bhattacharyya. "pyiwn: A Python-based API to access Indian Language WordNets." In Proceedings of the 9th Global WordNet Conference (GWC 2018), page 382. 2018.

8. Salaskar, Swaraja, Diptesh Kanojia, and Malhar Kulkarni. “Some Strategies to Capture Karaka-Yogyata with Special Reference to apadana”. Sanskrit and other Indian Languages Technology (SoIL-Tech). 2019.

9. Sheoran, Akash, Diptesh Kanojia, Aditya Joshi, and Pushpak Bhattacharyya. “Recommendation Chart of Domains for Cross-Domain Sentiment Analysis: Findings of A 20 Domain Study”. In Proceedings of the Language Resources and Evaluation Conference (LREC). 2020.

Chapter 2

Research Background

In this chapter, we first describe the terminology and some foundational aspects related to this thesis. We discuss the mathematical terms and concepts, as well as terms related to machine learning, neural networks, and gaze behaviour that we use in the thesis. Later in this chapter, we discuss the relevant past literature related to our research. This chapter also provides a brief history of cognate detection and computational phylogenetics. It summarizes the related work on current automatic cognate detection approaches. We first cover approaches to cognate detection, followed by the past research on false friends' detection, and computational phylogenetics for text. We also describe some of the research done in natural language processing using eye-tracking.

Foundations and Terminology

2.1 Types of Evaluation Metrics

Cognate and False friends' detection methodologies use machine learning and deep learning-based classification approaches. We use the standard evaluation metrics used in the machine learning paradigm. In addition to the above, whenever we create a dataset, we also take into account measures of inter-annotator agreement.

2.1.1 Classification-based Measures

Classification-based models treat individual labels like 'true' or 'false' as individual ordinal classes and treat the overall cognate detection problem as a classification problem.

These types of cognate of false friends' detection methodologies use the following evaluation metrics:

1. Accuracy
2. Weighted F-Score

2.1.1.1 Accuracy

Accuracy is defined as the percentage of labels in the test set that have been classified correctly. Consider the following confusion matrix in Table 2.1. The accuracy in the example is **75%**.

Class	**a**	**b**
a	70	20
b	20	50

Table 2.1: Sample Table for the calculation of different evaluation metrics.

2.1.1.2 Weighted F-Score

Precision, recall and F-Score are defined per class. They are defined as follows:

- **Precision** is defined as the percentage of correct instances classified in a certain class among those that have been classified in that class.
- **Recall** is defined as the percentage of correct instances classified in a certain class among all the elements belonging to that class in the test set.
- **F-Score** is the harmonic mean between the precision and the recall.

Since precision, recall and F-Score are defined per class, in order to get the overall measures for the data, we either micro-average or macro-average the values. The micro-average method involves getting the **weighted mean** of the individual measures. This weighted mean is weighted on the number of members in each class. The macro-average method involves taking the **arithmetic mean** of the scores of all the classes.

Class	**a**	**b**
Precision	0.78	0.71
Recall	0.78	0.71
F-Score	0.78	0.71

Table 2.2: Class-wise precision (P), recall (R) and F-Score (F) values from Table 2.1.

Average	**P**	**R**	**F**
Micro-Average	0.750	0.750	0.750
Macro-Average	0.745	0.745	0.745

Table 2.3: Micro-average and macro-average precision (P), recall (R) and F-Score (F) values from Table 2.1.

2.1.2 Regression-based Evaluation Measures

In addition to the above system evaluation metrics, researchers also use correlation coefficients for regression-based systems. The most common of these is the Pearson Correlation Coefficient (Pearson [1895]) and the Spearman Rank Correlation Coefficient (Spearman [1907]).

2.1.2.1 Pearson Correlation Coefficient

The **Pearson Correlation Coefficient** is a measure of the linear correlation between two variables X and Y. It has a value ranging between $+1$ and -1, with $+1$ denoting perfect ***positive*** linear correlation 0 denoting ***no*** linear correlation, and -1 denoting perfect ***negative*** linear correlation.

2.1.2.2 Spearman's Rank Correlation Coefficient

Spearman's Rank Correlation Coefficient is a measure of the correlation of the ***ranking*** between the values of two variables, X and Y. While Pearson's Correlation measures linearity between the variables, Spearman's Correlation measures monotonicity between them.

2.1.3 Inter-Annotator Agreement Metrics

Inter-Annotator Agreement is a measure of the degree of agreement among annotators who label the data manually. This is especially important to see if the labels assigned by annotators is due to chance or not. There are a number of rating approaches that are used to measure Inter-Annotator Agreement, such as Cohen's Kappa (Cohen [1968]), Fleiss' Kappa (Fleiss and Cohen [1973]), and Gwet's Agreement Coefficient (Gwet [2014]).

2.1.3.1 Cohen's Kappa

Cohen's Kappa (κ) is a measure of agreement between 2 annotators. We consider each label as an individual class, that is decided on by our system. Cohen's kappa coefficient is defined and given by the following function:

$$k = \frac{p_0 - p_e}{1 - p_e} = 1 - \frac{1 - p_o}{1 - p_e} \tag{2.1}$$

where,

p_0 = relative observed agreement among raters.

p_e = the hypothetical probability of chance agreement.

p_0 and p_e are computed using the observed data to calculate the probabilities of each observer randomly saying each category. If the raters are in complete agreement then k = 1. If there is no agreement among the raters other than what would be expected by chance (as given by p_e), k 0.

2.1.3.2 Fleiss' Kappa

Fleiss' kappa assesses the reliability of agreement between a fixed number of annotators when categorical ratings are to be assigned to a number of items for classifying items. It can be used with binary or nominal-scale. This contrasts with Cohen's kappa, which only works when assessing the agreement between not more than two annotators. It is interpreted as expressing the extent to which the observed amount of agreement among annotators exceeds what would be expected if all annotators made their ratings entirely randomly. Cohen's kappa assumes the same two annotators have rated a set of items,

Fleiss' kappa specifically allows that although there are a fixed number of annotators (e.g., three), different items may be rated by different individuals.

2.1.4 Gaze Behaviour

Tracking a reader's eye movements is a way in which we see whenever a reader understands a text. For one of our cognate detection studies, we use the eye-tracking to collect gaze behaviour data from participants. This is based on the eye-mind hypothesis, which simply states that "whatever the eye sees, that is what the mind processes" (Just and Carpenter [1980]).

2.1.4.1 Interest Area

An **interest area** (IA) is the area of the screen that is of interest. Whenever a reader looks at the screen, their eyes can fixate at any point on the screen - either on a word or on the blank portion of the screen (where there are no words). Since we are interested only on the words that a reader is fixating on, ***only the area of the screen, corresponding to a word*** is of interest, and is called an interest area.

Interest areas are not just limited to *only* the word-level. We can also create sentence-level and paragraph-level interest areas, by combining the interest areas of adjacent words belonging to the same sentence/paragraph.

2.1.4.2 Fixation

A **fixation** is an event that takes place when the eye is focused at a point on the screen. That point could either be an interest area, or the screen's background.

In our work, we are only concerned with fixations that take place on interest areas. This is because, according to the eye-mind hypothesis mentioned earlier, whatever the eye sees, that is what the mind processes (Just and Carpenter [1980]).

2.1.4.3 Saccade

A **saccade** is the rapid movement of the eye from one fixation point to the next. There are two types of saccades - **regressions** and **progressions**. **Regressions** take place

when the eye moves from the current interest area to *an earlier* one. **Progressions** take place when the eye moves from the current interest area to *a later* one.

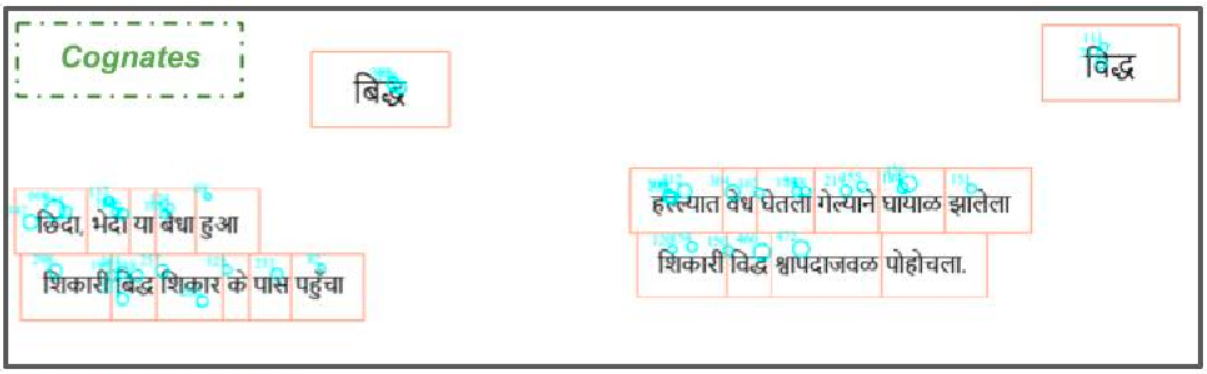


Figure 2.1: Sample Image from our eye tracking software showing a cognate word pair, along with fixations (turquoise circles), and interest areas (red rectangles).

Figure 2.1 shows the output from the eye-tracking software for a particular piece of text. The turquoise circles denote fixations, with the fixation point being the centre of the circle, and the size of the circle corresponding to the duration of the fixation. The red rectangles correspond to interest areas. Note that we consider the words alone as the interest areas in this example.

2.2 Types of Neural Networks

In this section, we describe the following types of neural networks:

1. Feed-Forward Neural Network
2. Convolutional Neural Network
3. Recurrent Neural Network
4. Long Short Term Memory Network
5. Transformer Networks

which we use in our thesis.

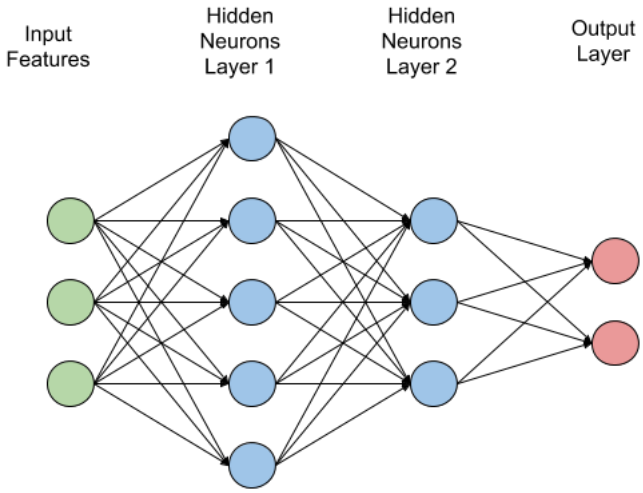


Figure 2.2: A Simple Feed-forward Neural Network

2.2.1 Feed-forward Neural Network (FFNN)

Perceptrons [Rosenblatt, 1958] are simple supervised linear classifiers which classify the input to one of the two classes. Perceptrons were invented to mimic the neurons in the brain. The input signals are sent to the perceptron where they are weighted according to the strength of the connection, and the resulting weighted inputs are summed up. If the sum is greater than a threshold the perceptron fires else it is said to be inhibited. However, perceptrons were criticized for being capable of learning only linear boundaries and unable to learn a simple XOR function.

Feed-forward Neural Networks contain multiple layers of perceptrons, *i.e.,* and perceptrons are organized in an array fashion comprising of a layer. The architecture of a simple feed-forward neural network is, as shown in Figure 2.2. These layers are connected both above and below with similar perceptron arrays. Each layer receives input from the previous layers through weighted connections. Unlike regular perceptrons, the weighted sum is passed through an additional layer of non-linearity, usually a sigmoid activation function. The output from the sigmoid activation function acts as inputs to the higher layers. The output from the last layer is then sent through the output layer. The non-linearity allows the network to not suffer from the same disadvantage as a perceptron. The network is trained through Back-propagation algorithm.

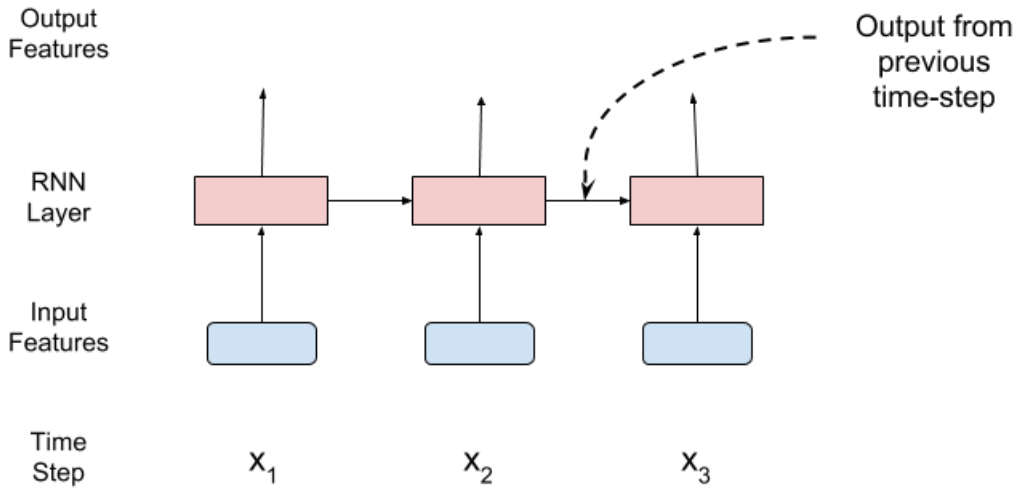


Figure 2.3: Architecture of a Vanilla RNN

2.2.2 Convolutional Neural Network (CNN)

Many NLP tasks involve working with variable length sequences. For example, a sequence of words forming a sentence, a sequence of characters forming a word, *etc.* Feed-forward neural networks have the disadvantage of requiring fixed-length input, thus rendering it infeasible for many NLP tasks. A convolutional neural network is a type of neural network which learns space-invariant features. For example, the representation for the phrase "Ram likes Shyam" is the same as that of "Shyam likes Ram". They are regularized versions of fully connected networks, that take advantage of patterns in data to assemble more complex patterns using more simple patterns.

2.2.3 Recurrent Neural Network (RNN)

Recurrent Neural Networks (RNN), also known as Vanilla RNN, provides an elegant solution to the problem of long-range dependencies usually present in the sequences. For instance, consider the task of language modelling where the goal is to predict the next word in a sentence given the context, *i.e.,* previous words. For a given time step t, the RNN looks at the current input x_t and the output for previous time-step h_{t-1} and then extracts a set of relevant features h_t. The mathematical formulation is described below,

$$h_t = \phi(Ux_t + Wh_{t-1}) \tag{2.2}$$

Here, ϕ is a sigmoid non-linearity function. The model remembers the history via h_{t-1}, which is again obtained from h_{t-2}. The architecture of a vanilla RNN is as shown in Figure 2.3. The training of the network happens via Backpropagation Through Time (BPTT) algorithm. The network is unfolded for every time-step and gradients at each time-step is accumulated till the first time-step, after which the gradient update proceeds.

Vanilla RNN using sigmoidal activation has been observed to suffer from the vanishing/exploding gradient issue [Bengio et al., 1994]. Vanishing gradient problem arises when the gradient of the loss function approaches zero resulting in the network not learning anything. Exploding gradient problem arises when the derivatives become very large, thus resulting in large updates to the network weights leading to instability in the network. Long Short Term Memory [Hochreiter and Schmidhuber, 1997a] and other variants of RNN are usually preferred over Vanilla RNNs. Recently, Le et al. [2015] demonstrated that carefully initializing RNNs and replacing sigmoid units with Rectified Linear Units (ReLU) can overcome the problem of vanishing/exploding gradients to a certain extent.

2.2.3.1 Long Short Term Memory (LSTM)

Long Short Term Memory [Hochreiter and Schmidhuber, 1997a] was proposed to overcome the vanishing/exploding gradients usually observed in Vanilla RNNs. LSTMs introduce the concept of a time-dependent memory cell state C_t. The information flow to and from the memory cell is regulated through various gate vectors.

Given the current input x_t and the output from LSTM at previous time-step h_{t-1}, the information flow through an LSTM proceeds as follows. The input flows through a *forget gate* which decides which components of the memory cell state to be forgotten before passing to the next stage.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{2.3}$$

Here, W_f and b_f are the *forget gate* weights and biases, f_t is the *forget gate vector*. In parallel, the input information passes through an *input gate* which decides which components needs to be passed to update the contents of the memory cell state.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{2.4}$$

Here, W_i and b_i are the *input gate* weights and biases, i_t is the *input gate vector*. The next step creates the new memory cell state vector $\tilde{C}_t$ based on which the memory cell state would be updated.

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \tag{2.5}$$

Here, W_C and b_C are weights and biases used to calculate the new memory cell state. The next step is to update the memory cell state which is done as follows,

$$C_t = f_t \circ C_{t-1} + i_t \circ \tilde{C}_t \tag{2.6}$$

We observe that the forget gate vector and input gate vector regulate how the current memory cell is updated based on the information passed from the previous cell state and the current input. The next step is to calculate the information to be passed to the higher layers. The *output gate* decides what information from the cell state is passed to the higher layers.

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{2.7}$$

$$h_t = o_t \circ \tanh(C_t) \tag{2.8}$$

Here, W_o and b_o are the *output gate* weights and biases, o_t is the *output gate vector*. h_t is the output vector from the LSTM component for the time-step t.

2.2.3.2 Gated Recurrent Units (GRU)

Gated recurrent Units (GRU) are a simplified version of the LSTM proposed by Cho et al. [2014]. Unlike LSTM, GRU do not have any memory cell state. A GRU has two gates, *reset gate* and *update gate* similar to the gates in LSTM. The reset gate vector r_t and update gate vector z_t are calculated as follows,

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r) \tag{2.9}$$

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z) \tag{2.10}$$

Here, W_r and b_r are the *reset gate* weights and biases, W_z and b_z are the *update gate* weights and biases. The information to be passed through is calculated as follows,

$$\tilde{C}_t = \tanh(W_h x_t + U_h(r_t \circ h_{t-1}) + b_h) \tag{2.11}$$

The output h_t from the GRU unit for the time-step t is now calculated as follows:

$$h_t = (1 - z_t) \circ h_{t-1} + z_t \circ \tilde{C}_t \tag{2.12}$$

2.2.4 Transformer Networks

Transformer networks [Vaswani et al., 2017] were proposed as a non-recurrent alternative to RNNs. Transformer networks take advantage of the *attention mechanism* and *position encoding* to handle the sequential nature of the data. As there is no recurrence in the model, the computation can happen for all time steps in parallel, leading to a significant speed-up.

Consider any task involving the processing of sequences as input. For example, let $X = \{x_1, \ldots, x_n\}$ be the sequence of words forming the sentence. The words are sent through a lookup table to obtain the corresponding word embedding sequence $\{h_1^0, \ldots, h_n^0\}$. This sequence is sent through an encoder layer. The encoder layer comprises of m identical layers and let us call the layers as *disambiguation* layers. The disambiguation layers consists of two sub-layers *multi-headed self-attention layer* and *position-wise feed-forward layer*.

2.2.4.1 Multi-headed Self-attention Layer

Given a sequence of inputs $\{h_1^{i-1}, \ldots, h_n^{i-1}\}$ from the lower-layer i-1, the input is sent through parallel *Scaled Dot-Product Attention* layers. Scaled Dot-Product Attention layer employs *queries* and *keys* to find parts of the sentence more relevant to the word at the current time-step. Every input is multiplied by three sets of matrices Query (Q), Key (K), and values (V). The query matrix is responsible for probing features which it is expecting from other words in the sequence. The Key matrix is responsible for broadcasting features exhibited by the words in the sequence. Given the word represented by input vector h_j^{i-1}

we first multiply the input vector with the query matrix to get the query vector. We now multiply the input vector of every word with the key matrix to get a set of key vectors. Multiplying query vector with the key vector gives the information if the word with the corresponding key vector holds any information useful to the word represented by the query vector or not. The query vector of jth word is multiplied with the key vector of every word which is later sent through a softmax function. The softmax layer gives a distribution over the words in the sequence indicating which words are relevant to the current j^{th} word. We now take a weighted average of the value vectors with the weights coming from the softmax output. The entire process can be concisely written as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V \tag{2.13}$$

here, d_k is the dimension of the input representation. The layer consists of l such parallel Scaled Dot-Product Attention modules, the output of which are concatenated and sent through a projection layer to project it back to d_k dimensions to get the sequence $h_1^{\text{pre}_i}, \ldots, h_n^{\text{pre}_i}$. The output is now sent through a residual connection followed by layer normalization. The output can now be written as LayerNorm $(h_j^{i-1} + h_j^{\text{pre}_i})$.

2.2.4.2 Position-wise Feed-Forward Networks

The output from the *multi-headed self-attention layer* is sent through a feed-forward neural network. The feed-forward neural network is applied to individual time-steps separately. The network consists of a linear layer followed by $ReLU$ activation and another linear layer. The output of the feed-forward neural network is sent through a residual connection and layer normalization step similar to the one described in *multi-headed self-attention layer*. The self-attention mechanism and the position-wise feed-forward network are responsible for obtaining relevant features for the word at that time-step.

2.3 Literature Survey: A Brief History

In this part of the chapter, we describe the relevant past literature starting with a brief history of related research work. Kay [1964] discuss the importance of cognates in historical linguistics. They theorize that construction of cognate words across two or more

languages from the same hypothetical source language should help reduce the number of phonemes required to construct new words.

A classic approach [Dolgopolsky, 1964] is based on the concept of sound classes. They cluster sounds into various classes to reduce the phonetic space for comparison across languages; classes which frequently occur in correspondence relation in genetically related languages. This work provides us with a somewhat rough but grounded sound class system and groups all consonants into ten classes which ignores the vowels. When converting all words in the data to their respective sound classes, different criteria can be used to assign words which resemble each other in their sound classes, to the same set of cognate words.

Haynes [1984] discuss the patterns among cognates and their role in acquiring a second language. De Groot and Nas [1991] identify that connected lexical representations among two languages have shared conceptual representations. They show that cognates, within a language, and across languages share conceptual representations; which is not the case for non-cognates. They compare within- and between-language repetition-priming and associative (semantic)-priming effects for performing their experiments.

Classical approaches to identify cognates between language pairs [Simard and Foster, 1992, Kondrak, 2001, Inkpen et al., 2005, Schulz et al., 2004] consist of methods to compute similarity among words based on orthographic and/or phonetic measures. Some of the measures in the first group are EDIT distance, LCSR (longest common subsequence ratio), Jaccard distance and measures based on the number of n-grams that are shared by words. Another frequently employed measure is the binary identity function. The phonetic approaches are Soundex and Editex, which try to take advantage of individual features to determine the similarity between words [Mackay and Kondrak, 2005].

In the area of computational phylogenetics, Sokal [1961] describe how distance can be used as a measure for computing similarity between various taxa. They describe the utilization of their work in the domain of zoology, but the same is applicable in linguistics when the taxa in consideration are various versions of the same text. Fitch and Margoliash [1967] present their work on the construction of phylogenetic trees which uses distance-based methods to generate phylogenies. Their work is generally applicable to all

the phylogenetic analyses whether it is on the domain of biology, zoology, or language processing. Tuldava [1980] discuss statistical fit and the possibilities of extrapolation beyond the limits of observable data are illustrated on the material of several languages belonging to different typological groups. Phylogenetic regression [Grafen, 1989] uses statistical regression analysis to perform phylogenetics among species using multiple techniques.

2.4 Automatic Cognate Detection

This section presents an overview of the work done prior to this thesis in the area of cognate detection. We organize it in different subsections based on methodologies used to detect cognates among languages.

2.4.1 Orthographic or String Similarity

Cognate detection has been explored vastly in terms of methodologies and for various languages. 'Orthographic' or 'String' similarity-based methods are often used as baseline methods for cognate detection. The most commonly used method amongst them is the similarity measure based on edit distance [Melamed, 1999].

Ciobanu and Dinu [2014] automatically detecting pairs of cognates employing an orthographic alignment method which aligned subsequences as features for support vector machine (SVM) algorithm, in order to infer rules for linguistic changes undergone by words when entering new languages and to discriminate between cognates and non-cognates. The authors further their research [Ciobanu and Dinu, 2015] and propose a computational approach for discriminating between cognates and borrowings from other languages. They show that orthographic features have discriminative power and analyze the underlying linguistic factors that prove relevant in the classification task.

Bhargava and Kondrak [2009] test the applicability of the Hidden Markov Model (HMM) algorithm for the tasks of multiple cognate alignments and cognate set matching. They test HMMs for two tasks, *viz.,* multiple cognate alignment and cognate set matching and report the HMMs can be further profiled for word-based NLP tasks.

While the LexStat algorithm by List [2014] uses a permutation method to compute individual segmental similarities between individual language pairs which are then fed to an alignment algorithm, the PMI similarity approach by Jäger [2014] infers general segmental similarities between sounds from an exhaustive parameter training procedure.

2.4.2 Phonetic Alignment and Phoneme Matching

Turchin et al. [2010] further formalized the classical approach proposed by Dolgopolsky [1964] and employed a modified sound class schema of nine vowel classes to test the Altaic hypothesis[1]. The Consonant Class Matching (CCM) approach, as they call it, reported relatively lesser false positives; but the number of false negatives was very high [List, 2014]. This was due to the lack of flexibility in the procedure, which hard-coded sounds to classes, ignoring the fact that a change in sound is usually based on fine-grained transitions.

Previous approaches to cognate detection using various aspects also involves the computation of similarity by decomposing phonetically transcribed words [Kondrak, 2000]. Another work which forms the basis of phonetic alignment for the task of cognate detection was proposed by Covington [1996]. However, Kondrak [2000] present a new algorithm that combines a number of techniques developed for sequence comparison with a scoring scheme for computing phonetic similarity on the basis of multivalued features.

LexStat [List, 2012] is a framework which combines various approaches to compare sequences in historical linguistics and evolutionary biology in a novel framework. This framework closely models the most important aspects of the comparative method. Their method was tested on a large gold standard of IPA encoded wordlists which showed highly consistent results across languages. Amengual [2012] investigate voice onset times (VOTs) to determine whether cognates enhance the cross-language phonetic influences in the speech production of Spanish–English bilingual speakers.

Rama and Borin [2011] explore the use of parallel corpus to estimate language relatedness with cognates. The use of gap-weighted subsequences for discriminating cognates from non-cognates was proposed by Rama [2015]. They propose a scheme to integrate pho-

[1]The Altaic Hypothesis

netic features into the feature vectors for cognate identification. To further their research, they also propose the use of, and evaluate two similarity-dependent Chinese Restaurant Process (sd-CRP) algorithms [Rama et al., 2015] at the task of automated cognate detection. Later, they also proposed the use of convolutional neural networks (CNN) for the purpose of cognate identification [Rama, 2016]. They come up with a siamese architecture that jointly learns phoneme level feature representations and language relatedness from raw words for cognate identification.

Rama et al. [2017] explore the use of unsupervised methods for detecting cognates in multilingual word lists. They use online expectation maximization (EM) to train sound segment similarity weights for computing similarity between two words. Rama and List [2019] also present an automated framework for the task of cognate detection. They also show that this framework performs the task of cognate detection comparatively faster than the previously proposed approaches while it achieves, more or less, the same accuracy when compared to other methods.

2.4.3 Semantic Matching based Approaches

Frunza and Inkpen [2006] discuss the detection of partial cognates by utilizing semi-supervised bilingual bootstrapping. Hauer and Kondrak [2011] incorporates a number of diverse word similarity measures and features that encode the degree of affinity between pairs of languages. The output of this classification algorithm is then used to generate cognate groups.

Dellert [2014] develop a method applied to an entire dictionary database, resulting in a polysemy network that spans around 30,000 German lexemes. They hypothesize that short paths in polysemy networks are expected to express possible semantic shifts. They investigate a number of true cognate pairs whose translations are connected in the polysemy network by shortest paths of different lengths and show that polysemy networks across languages can be used to identify cognates.

St Arnaud et al. [2017] explore various methods which use semantic similarity from corpus-based techniques to perform the task of cognate alignment and cognate matching.

Comesaña et al. [2012] explore how bilingual speakers represent words in two languages and which mechanisms are responsible for L2 (second language) acquisition. They answer these important questions in the bilingual and vocabulary acquisition literature. They do not, however, perform the task of automatic cognate detection.

Mulloni et al. [2007] purely orthographic approach to the cognate detection task by introducing semantic evidence in the form of monolingual thesauri and corpora to support the identification process. Their method on only four different language pairs suggested that the introduction of semantic evidence in cognate detection helps to increase the precision of cognate identification, which forms the basis of our work.

Kondrak [2001] introduce a procedure for estimating the semantic similarity of glosses that employs keyword selection and WordNet[Fellbaum, 2010]. They perform experiments on vocabularies of four Algonquian languages. However, their work indicates that their method is capable of discovering, on average, nearly 75% per cent of cognates with only 50% precision.

Labat and Lefever [2019] present develop a context-independent gold standard dataset by manually labelling word pairs of cognates and false friends in bilingual term lists. These annotated cognate pairs are used to train and evaluate a supervised binary classification model for the automatic detection of cognates. However, this approach is only tested for the English-Dutch language pair.

2.5 False Friends' Detection

False Friends are a particular and yet very frequent type of mainly inter-lingual error which can decrease the accuracy of downstream NLP tasks. The kind of error we are dealing with, usually, involves two different languages. Confusion arises because word A (which belongs to the foreign language being learned or used) looks or sounds exactly or nearly like word B, which belongs to the learner's mother tongue. The user then establishes an unwarranted inter-lingual equivalence on the basis of this total or partial similarity [Hayward and Moulin, 1984].

For our work, we consider Frunza and Inkpen [2009] the definition of cognates and false friends where cognates are pairs of words that are understandably similar and are translations candidates for each other. Pronunciation can be completely identical or not, for example, amor-amor and jefe (in Spanish) - chefe (in Portuguese). Frunza and Inkpen [2009] also consider as cognates, word pairs that are orthographically identical or have slightly different spellings. False Friends are pairs of words in two languages that are thought to be similar, but have different meanings (semantically different), depending on the context. for example, aula (class, in Portuguese) - aula (classroom, in Spanish).

Frunza and Inkpen [2009] propose an approach to identify cognates and false friends estimating various orthographic and phonetic similarity-based distance measures between word pairs. The authors propose to employ ML techniques to differentiate between cognates from false friends. Their work on the language pair French-Portuguese language pair considers orthographic and phonetic similarity measures as the classification feature set. The measures used by them are IDENT, PREFIX, DICE, TRIGRAM, XDICE, XXDICE, LCSR, NED, SOUNDEX, in addition to some other measures considered as generalizations of the metrics LCSR and NED, defined in [Kondrak and Dorr, 2004]. Frunza and Inkpen [2009] combine several techniques to disambiguate between cognates and false friends. According to them, there are several investigations dedicated to building cognate lists, but few studies centre on the identification of false friends.

Initially, Inkpen et al. [2005] focussed on French and English, but their methods are known to be applicable to other language pairs. They use several measures of orthographic similarity as features for classification among cognates and false friends. They study the impact of selecting different features, averaging them, and combining them through machine learning techniques for the task of false friends classification.

The work done by Mitkov et al. [2007] proposes novel methods for the automatic identification of both cognates and false friends from comparable para corpora. Their methods do not require parallel corpus and make use of monolingual corpora and a bilingual dictionary necessary for mapping the co-occurrence data across languages. Their methods do not require that the newly discovered cognates or false friends to be present in the dictionary and hence are capable of operating on out-of-vocabulary expressions.

Nakov et al. [2009] present several unsupervised algorithms for acquiring such pairs from a sentence-aligned bi-text. They try different ways of exploiting simple statistics about monolingual word occurrences and cross-lingual word co-occurrences in the bi-text. Using methods from statistical machine translation, they induce word alignments in an unsupervised way, from which they estimate lexical translation probabilities, which they, then, use to measure cross-lingual semantic similarity. They also experiment with a semantic similarity measure that uses the Web as a corpus to extract local contexts from text snippets returned by a search engine, and a bilingual glossary of known word translation pairs, used as "bridges". Finally, they combine all the methods to detect false friends across Russian-Bulgarian word pairs.

Torres and Aluísio [2011] evaluate the performance of (ML) algorithms in the identification of cognates and false friends, based on orthographic similarity measures, confirming that these measures are not sufficient for classification of false friends from cognates; however, when combined with other measures, they can achieve more satisfactory results.

Ljubešić and Fišer [2013] a corpus-based approach to automatic identification of false friends for Slovene and Croatian. They focus on measuring the difference in meaning between identically spelt words by using frequency and distributional information from the corpora.

For the detection of false friends and true cognates, Chen and Skiena [2016] present an unsupervised transliteration based approach but do not report F-scores for Indian languages. Their results for the detection of false friends are inconclusive not just for eight Indian languages they discuss, but for 50% of languages they report their work on, as stated in their work. For the task of False Friends' detection, they show only a high correlation of English-French and English-Spanish language pairs with manual validation.

The most recent work by Castro et al. [2018], which explores the problem of False Friends' detection, utilizes monolingual embeddings generated using word2vec [Mikolov et al., 2013c] for Spanish and Portuguese. They foray into the semantic layer and apply projection of two separate monolingual embeddings into a shared space, but using an inefficient method which our work shows to improve upon significantly.

2.6 Computational Phylogenetics

Automatic Cognate Detection (ACD) is an important task which can help phylogenetic reconstruction and complement current research on language phylogenies [Rama et al., 2018]. Kulkarni [2012] builds a phylogenetic tree for Malayalam manuscripts of the Kā ś ikāvrtti, with manual analysis, to show that M is the archetype source and Ma, Mb and Mc are its hyperarche child nodes. Kulkarni [2003] and Kulkarni [2008] build another similar tree for the Sharada manuscript variants of the KV. In recent years, computational phylogenetics has developed various methodologies [Felsenstein, 2004, Huelsenbeck et al., 2001, Saitou and Nei, 1987, Swofford et al., 1996] for bioinformatics.

The growth of phylogenetics as an area with significance to statistical methods is captured by Felsenstein [2001] in an article where he explains the developments of numerical methods for phylogenetic reconstruction. Computational linguistics has adopted many of these methods for the construction of phylogenetic trees. Computational linguistics, however, majorly depends on the lexical analysis of the textual variants and induces the dissimilarities present in phylogenetic approaches. A major disadvantage of using character-based or lexical distance-based methods is the need for manually curated word lists. Csernel and Patte [2007] discuss the LCS algorithm for preparing a critical edition of Sanskrit texts and provide a method for comparison of Sanskrit manuscripts. Among the many available methods [Huelsenbeck, 1995] to construct phylogenetic trees, UPGMA [Gronau and Moran, 2007] is widely used in historical linguistics. Their method assumes a constant rate of evolution and is not a well-regarded method for inferring interrelationships among the taxa unless this assumption has been justified for the dataset in use.

The UPGMA method constructs phylogenetic trees based on a distance matrix which can be computed in various ways. Saitou and Nei [1987] proposed a neighbor-joining method to construct phylogenies based on sequence analysis, which uses genetic distance as a clustering metric. Moret et al. [2002] study the sequence lengths required by neighbor-joining, greedy parsimony, and a phylogenetic reconstruction method based on disk-covering and the maximum parsimony criterion and show improvements in large scale phylogenetic reconstruction. However, Neighbor-joining is considered a more effi-

cient method compared to the more accurate Fitch-Margoliash [Fitch and Margoliash, 1967] method.

Symmetric cross-entropy is one of the methods which is a letter *n-gram* based measure similar to the one used by Singh [2006b] for language and encoding identification. Singh and Surana [2007] used corpus-based measures to show that corpus can be used for a comparative study of languages. They used both character n-gram distances and surface similarity [Singh, 2006a] to identify the potential cognates, which are used to estimate the inter-language distance. Rama and Singh [2009] also used measures based on cognate identification and feature n-grams to infer this matrix. Ellison and Kirby [2006] discussed establishing a probability distribution for every language through intra-lexical comparison using confusion probabilities and estimate distances using KL divergence and Rao's distance [Atkinson and Mitchell, 1981].

2.7 Cognitive Psycholinguistics for NLP tasks

It is intuitive that tasks involving subjective judgements can benefit from cognitive information of the readers. Using gaze behaviour information is one such way of collecting cognitive information. Eye-tracking is a means of tracking the eye movements of a reader as he/she reads a piece of text. Using eye-tracking, we are able to extract the gaze behaviour of the reader as he/she reads the text. The link between gaze behaviour and cognitive understanding was hypothesized by Just and Carpenter [1980].

Much work has been done to study the relationship between various linguistic features and the corresponding aspects of gaze behaviour. Word length has been shown to be positively correlated with fixation count ([Rayner, 1998]) and fixation duration ([Henderson and Ferreira, 1993]). Word predictability, (*i.e.,* how well the reader can predict the next word in a sentence) was also studied by Rayner [1998], where he found that unpredictable words are less likely to be skipped than predictable words. Devices like smartphones come with cameras that could be used to track rudimentary gaze movements. For example, Smart Scroll (Samsung smartphones and tablets) is an application that looks at a person's eye movements and automatically scrolls through a web page on the browser application. SR Research has also launched a portable eye tracker[2].

[2]SR Research Eyelink Portable Duo

Gaze behaviour has been used to aid in solving a number of NLP tasks ([Mishra and Bhattacharyya, 2018]). Collecting gaze behaviour information though is a costly task - in terms of time and money. Therefore, a lot of the research nowadays is being done to learn gaze behaviour and use that learnt gaze behaviour to solve various NLP tasks. Klerke et al. [2016], for instance, use a multi-task learning approach to shorten sentences. They bin the first fixation duration and regression duration features to normalize for varying reading speeds for different readers. González-Garduño and Søgaard [2017], Singh et al. [2016a] described a way to incorporate learning gaze behaviour to predict readability. Mishra et al. [2018] describes an approach to learn gaze behaviour in a multi-task learning framework for sentiment analysis on movie reviews.

2.8 Summary

In the first part of this chapter, we have described the different mathematical, computational, and gaze behaviour related terms that we would use later on in this thesis. We first looked at different types of system performance measures, correlation coefficients like Pearson's Correlation Coefficient (Pearson [1895]) and Spearman's Rank Correlation Coefficient (Spearman [1907]). After that, we defined basic gaze behaviour terms like interest areas, fixations and saccades. We described different types of neural network layers that we used in our thesis.

In the second part of this chapter, we have described the previous literature in the areas of automatic cognate detection, false friends' detection and computational phylogenetics. First, we described historical work done in these areas. We, then, explained a number of different methodologies that are used for automatic cognate detection. Next, we looked at a number of previously explored methodologies for the task of false friends' detection that are used in NLP tasks. After that, we discussed a few papers that establish the use of cognates in computational phylogenetics and dealt with the problem computational phylogenetics in general. We also described previous work which performs manual phylogenetic analysis for creating phylogenies of a single text. Finally, we looked at some of the research done in the area of utilizing cognitive psycholinguistics for natural language processing tasks.

Chapter 3

Distributional Semantics & Dataset Creation

In the thesis so far, we have presented past work related to cognates in linguistics and computational linguistics. We discussed various approaches to the cognate detection task, the false friends' detection task, and computational phylogenetics along with a brief use of cognitive psycholinguistics for NLP. This chapter presents our exploration in understanding the distributional hypothesis and its application to cognate detection, false friends' detection and computational phylogenetics. We also provide details of our foundational work done on the creation of linked datasets for the tasks investigated in this thesis.

3.1 Distributional Semantics

The distributional hypothesis in linguistics states that "words that occur in the same contexts tend to have similar meanings" [Harris]. This concept of "a word is known by the company it keeps" was proposed by Firth [1957]. This theory has formed a base for various tasks in statistical semantics which lead to improved performance in many NLP tasks [Collobert et al., 2011, Turian et al., 2010]. Deep learning techniques have been proposed as an alternative to traditional machine learning models for dealing with the semantics of a word. The deep learning techniques employ deep neural networks for modelling the NLP task. These deep neural networks are usually trained on the unlabeled data in an unsupervised way in the first stage. This stage is called *unsupervised pre-training*

phase and enables the model to learn better features from the available large unlabeled data. The model is later fine-tuned for the task in hand and takes advantage of the features learned during the unsupervised pre-training phase. *Word embeddings* [Mikolov et al., 2013c,d, Pennington et al., 2014, Dhillon et al., 2015, Bojanowski et al., 2017a] are an example of unsupervised pre-training phase in NLP which is based on distributional similarity among the vocabulary used for the pre-training phase. The objective, here, is to predict the middle word given the surrounding words (*CBOW variant*) using a simple feed-forward neural network. Such an objective takes advantage of all available unlabeled corpora in that language. A desirable side-effect of this objective is, the first layer of the neural network embeds the words in a d-dimensional real-valued vector space. The *word embeddings* are shown to form clusters containing other similar words in the vector space. These real-valued vectors are commonly referred to as word embeddings and have been shown to improve the end task performance when used in the fine-tuning phase as inputs or the first layer of the deep neural network [Turian et al., 2010]

Recently, deep learning has been extremely successful in a wide array of NLP applications. This is primarily due to the development of word embeddings, which have become a crucial component in modern NLP. Bengio et al. [2003] were the first to propose neural word embeddings. Many word embedding models have been proposed since then [Collobert and Weston, 2008, Huang et al., 2012, Mikolov et al., 2013b, Levy and Goldberg, 2014]. They have been efficiently utilized in many NLP applications: Part of Speech Tagging [Collobert and Weston, 2008], Named Entity Recognition [Collobert and Weston, 2008], Sentence Classification [Kim, 2014], Sentiment Analysis [Liu et al., 2015], Sarcasm Detection [Joshi et al., 2016]. Mikolov et al. [2013a] made a particularly interesting observation about the structure of the embedding space of different languages. They noted that there is a linear mapping between such spaces.

3.1.1 Monolingual Embeddings

In this section, we will look at algorithms for obtaining word embeddings in a particular language. These models are mostly trained on a large monolingual unlabelled corpus available for the language of interest.

3.1.1.1 word2vec

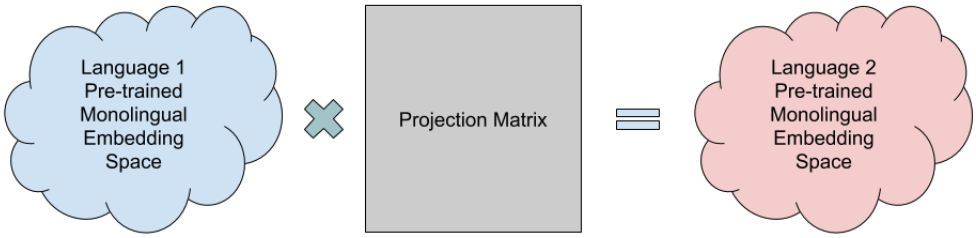
[Mikolov et al., 2013d] proposed a method to learn a continuous vectorial representation of words. In their method, every word is represented by a d dimensional real-valued vector. The objective is to maximize the likelihood of seeing the word given its context. This fits nicely into the distributional hypothesis that words with similar contexts learn similar representation, and hence they tend to be similar. Unlike other tasks, this method does not require any annotated corpus. It requires unannotated monolingual corpus, which is easy to obtain for many languages. It treats every word along with the context appearing in the corpus as training instances. The training is done using the stochastic gradient descent method.

The method has two variations: a) predict the middle word given its contextual (or neighbouring) words known as Continuous Bag-Of-Words (CBOW) approach and b) predict the contextual words given the middle word known as Skip-gram approach. Since the output is now of vocabulary size, which is large even for reasonably large corpus, Hierarchical Softmax or Negative Sampling is used.

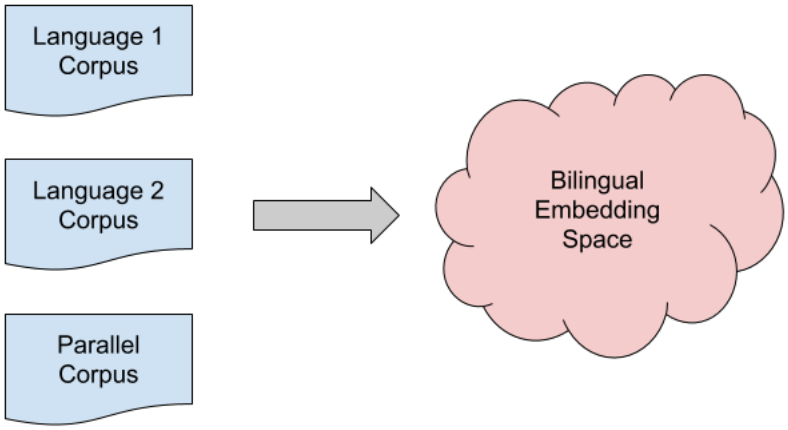
3.1.1.2 Fasttext Embeddings

A major setback with word2vec model is it neglects sub-word information. This might not be crucial for languages like English, but it is crucial for morphologically rich languages *i.e,* both agglutinative and inflectional languages. Every inflection of the same root form is considered as a different type. For example in Kannada, every inflectional form of the word ರಾಮ (raama) as in ರಾಮನು (raamanu), ರಾಮನಿಂದ (raamaninda) (from raama), ರಾಮನಿಗೆ (raamanige) (to raama), *etc.* is considered as a different type. As a result, there is no sharing of contextual information across these inflectional forms, which leads to some forms learning better representation compared to other forms. This is a major setback for downstream applications like Named Entity Recognition, Question Answering, Machine Translation, *etc.* which rely on word embedding features.

Recently, [Bojanowski et al., 2017a] proposed an extension to word2vec, which considers sub-word information while calculating word embedding. Given a dictionary, they find the most frequent character n-grams present in the character sequence forming the word,



(a) Offline Approach



(b) Online Approach

Figure 3.1: Different Approaches for Bilingual Embeddings

with n varying from say 3 to 6. The final embedding of the word is the sum of embeddings of all the character n-grams and the embedding of the word as a whole.

Given a word, all the frequent character n-grams present in it are listed. The word representation of the word is now the sum of its word embedding and the embedding of all the frequent character n-grams present in it. The algorithm is the same as word2vec except for this variation where the word representation contains information about its sub-words also.

3.1.2 Bilingual and Cross-lingual Embeddings

Word embeddings represent words of a language as d dimensional points in a d dimensional space. These word embeddings are usually trained from the monolingual corpus and are used as input features for various NLP tasks. The quality of such word embeddings depends on the availability of large monolingual corpus. For languages like English, we

have large amounts of monolingual corpora available which is not the case for low-resource languages.

However, to project features from one language to another, we need language-independent features to be fed into the system. Word embedding features, though valuable, are predominantly language-dependent making them inapplicable as cross-lingual features. However, several approaches have been proposed to obtain *Bilingual Embeddings* or *Multilingual Embeddings* where similar words across languages have similar embedding features.

Approaches to obtain bilingual or multilingual embeddings fall into broadly two categories (i) offline methods and (ii) online methods. Offline methods use existing pre-trained monolingual word embeddings and project them onto a common space using bilingual dictionaries and are demonstrated by Figure 3.1a. Online methods learn the bilingual embeddings directly from the corpus (Figure 3.1b.)

3.1.2.1 BilBOWA

BilBOWA (Bilingual Bag-of-Words Without Alignments) is an online approach for learning bilingual embeddings. The approach uses available monolingual corpora in both the languages and a sentence-aligned parallel corpus. The approach has a monolingual objective and a cross-lingual objective. The monolingual objective is the same as the Skip-gram objective using negative sampling as defined by [Mikolov et al., 2013d]. The cross-lingual objective minimizes the mean L_2 loss between the mean bag-of-words sentence vectors of the parallel corpus. Specifically, given a parallel sentence pair, each sentence on either side is represented as the mean of the word vectors of the words present in that sentence, also called as the *sentence vector*. As the two sentences are translations of each other, the obtained *sentence vectors* should be closer to each other. This is accounted for by the L_2 loss in the objective. This forces similar words (or translations) across the two languages to have similar word embeddings.

3.1.2.2 Bilingual Embedding Using Inverted Softmax

Smith et al. [2017] presented an offline approach to learn bilingual embeddings. The intuition is that similar words across languages should have similar contextual distribu-

tions. To better understand the approach, running word2vec ([Mikolov et al., 2013c,d]) on the same corpus twice will result in the same word having two different word embeddings. However, the relative similarity/dissimilarity between the words will be retained. A simple rotation of one of the word embedding space is sufficient to align the two embedding spaces. Similar reasoning holds valid for word embeddings trained on comparable corpus in the same language. [Smith et al., 2017] extended the above approach to the cross-lingual scenario, and the objective is to learn the rotation matrix. However, to learn the rotation matrix, we need some anchor points (or words) which are obtained from a bilingual dictionary. The approach maps words from both the languages to a common embedding space.

3.1.2.3 Multilingual Unsupervised and Supervised Embeddings (MUSE)

Lample et al. [2018] proposed an approach to learn bilingual embeddings in both supervised and unsupervised setting. Unlike the previous approach, they map the source language words to the target space, and the goal is to learn an orthogonal projection matrix. In the unsupervised setting, the model uses adversarial training to learn **cross-lingual embeddings** without using any bilingual dictionary. The quality of the projection matrix learned would be very poor. In the next stage, they build a synthetic parallel dictionary after the adversarial training step. The model then solves the supervised objective using the generated synthetic dictionary to obtain better projection matrix. They build the synthetic dictionary again using the new projection matrix obtained, and the same process is repeated again and again.

3.1.2.4 Cross-lingual Language Model (XLM) and XLM-RoBERTa (XLM-R)

Conneau and Lample [2019], Conneau et al. [2019] introduce a new unsupervised method for learning cross-lingual representations using cross-lingual language modelling and investigate two monolingual pre-training objectives. One unsupervised that only relies on monolingual data, and one supervised that leverages parallel data with a new cross-lingual language model objective. This work builds on top of Devlin et al. [2018], Howard and Ruder [2018], Radford et al. [2018] and provides a pre-trained cross-lingual model for many languages. Using the pre-trained models provided or by training XLM/XLM-R models on a custom corpus, word embeddings can be obtained via the lookup table of

cross-lingual language model (XLM) which can help obtain similarity scores for two words from different languages.

3.2 Dataset Creation

In this section, we shall describe the initial efforts towards creating a linked knowledge graph dataset for the task of cognate and false friends' detection tasks. We will describe the linked IndoWordnet knowledge graph, and our efforts to produce linkages with English Wordnet to expand our work of cognate detection among Indian languages to finding cognates among English and Indian languages. The two subsections below describe two research papers accepted at Global Wordnet Conference 2018 and Language Resources and Evaluation Conference 2018.

3.2.1 IndoWordnet & Linkages with Princeton WordNet

Wordnets [Fellbaum, 1998] have been useful in different Natural Language Processing applications such as Word Sense Disambiguation [TufiŞ et al., 2004, Sinha et al., 2006], Machine Translation [Knight and Luk, 1994], *etc.* Linked Wordnets are extensions of wordnets. In addition to language specific information captured in constituent wordnets, linked wordnets have a notion of an interlingual index, which connects similar concepts in different languages. Such linked wordnets have found their application in machine translation [Hovy, 1998], cross-lingual information retrieval [Gonzalo et al., 1998], *etc.*

Given the extensive application of wordnets in different NLP applications, maintenance of wordnets involves expert involvement. Such involvement is costly both in terms of time and resources. This is further amplified in case of linked wordnets, where experts need to have knowledge of multiple languages. Thus, techniques that can help reduce the effort needed by experts are desirable. With this work, we make the following contributions:

- We release the latest version of 18 wordnets under the IndoWordNet project as a single bundle.
- Using mappings between Princeton WordNet and Hindi wordnet, we create and release mappings between Princeton WordNet and these 18 languages wordnet.

Princeton WordNet or the English WordNet was the first wordnet and inspired the development of many other wordnets. EuroWordNet [Vossen et al., 1997] is linked wordnet comprising of wordnets for European languages, *viz*, Dutch, Italian, Spanish, German, French, Czech and Estonian. Each of these wordnets is structured in the same way as the Princeton WordNet for English [Miller et al., 1990] - synsets (sets of synonymous words) and semantic relations between them. Each wordnet separately captures language-specific information. In addition, the wordnets are linked to an Inter-Lingual-Index, which uses Princeton WordNet as a base. This index enables one to go from concepts in one language to similar concepts in any other language. Such features make this resource helpful in cross-lingual NLP applications like cognate detection and false friends' detection.

IndoWordNet [Bhattacharyya, 2010] is linked wordnet comprising of wordnets for major Indian languages, *viz*, Assamese, Bengali, Bodo, Gujarati, Hindi, Kannada, Kashmiri, Konkani, Malayalam, Manipuri, Marathi, Nepali, Oriya, Punjabi, Sanskrit, Tamil, Telugu, and Urdu. These wordnets have been created using the expansion approach with Hindi WordNet as a pivot, which is partially linked to English WordNet. We exploit these links to create mappings from English WordNet to wordnets of other languages.

3.2.1.1 Current Statistics: IndoWordnet

Table 3.1 shows the statistics of the released wordnets. These wordnets have, on an average, approximately 28,000 synsets, with Nepali and Hindi having the minimum and the maximum number of synsets respectively. The number of synsets in Hindi is maximum due to the fact that work on IndoWordNet started with the Hindi language. It should also be noted that the ratio of nouns, verbs, adjectives, and adverbs is also on an average 48:6:13:1; the trend being similar to Princeton WordNet.

3.2.1.2 Linkage between English and Indian Language WordNets

For linking Indian language wordnets with the Princeton WordNet, we link the Hindi Wordnet data with Princeton WordNet data manually with the help of lexicographers. This has been an ongoing work for many years, and a resource release was long-standing. We delve deep into the language-related issues in linking both the languages and ensure that only a valid relation is established between both the lexicons.

	Noun	Verb	Adjectives	Adverbs	Total
Assamese	9065	1676	3805	412	14958
Bengali	27281	2804	5815	445	36346
Bodo	8788	2296	4287	414	15785
Gujarati	26503	2805	5828	445	35599
Hindi	29807	3687	6336	541	40371
Kannada	12765	3119	5988	170	22042
Kashmiri	21041	2660	5365	400	29469
Konkani	23144	3000	5744	482	32370
Malayalam	20071	3311	6257	501	30140
Manipuri	10156	2021	3806	332	16351
Marathi	23271	3146	5269	539	32226
Nepali	6748	1477	3227	261	11713
Odiya	27216	2418	5273	377	35284
Punjabi	23255	2836	5830	443	32364
Sanskrit	32385	1246	4006	265	37907
Tamil	16312	2803	5827	477	25419
Telugu	12078	2795	5776	442	21091
Urdu	22990	2801	5786	443	34280

Table 3.1: Number of synsets in different wordnets

3.2.1.3 Current Statistics: Linkages for Language pairs

Table 3.2 shows the statistics of the released linkages. There are approximately 20,000 links for an English-Indian language pair on average, with Nepali and Hindi having the minimum and the maximum number of links. Again, the number of links in Hindi is maximum due to the fact that work on IndoWordnet started with the Hindi language, and we link Hindi directly with English. At times, the concept present in Hindi is not present in the other Indian languages, thus leading to the less number of linkages for the other languages, in some cases. Table 3.2 show the part-of-speech category-wise distribution of the linked synsets, and also indicated the number of directly linked synsets

	Nouns		Verbs		Adjectives		Adverbs		Total
	D	H	D	H	D	H	D	H	
Assamese	7019	679	1300	36	2744	0	294	0	12072
Bengali	11049	7680	1824	99	3356	3	312	0	24323
Bodo	6940	603	1594	64	2854	1	293	0	12349
Gujarati	10910	7533	1825	99	3356	3	312	0	24038
Hindi	11584	8221	1988	212	3542	4	344	0	25895
Kannada	7806	1973	1921	154	3453	3	133	0	15443
Kashmiri	9363	6261	1767	100	3240	2	294	0	21027
Konkani	10545	6952	1888	128	3391	2	328	0	23234
Malayalam	9146	4754	1970	206	3525	4	340	0	19945
Manipuri	7192	823	1324	43	2712	0	244	0	12338
Marathi	9874	6556	1839	144	3092	0	333	0	21838
Nepali	5217	496	1114	42	2202	1	200	0	9272
Odiya	11039	7680	1679	66	3187	2	271	0	23924
Punjabi	10215	6382	1822	99	3355	3	312	0	22188
Sanskrit	8396	6470	1048	28	2873	2	241	0	19058
Tamil	8130	3066	1821	98	3353	3	312	0	16783
Telugu	6944	1843	1819	98	3350	0	312	0	14366
Urdu	10424	6816	1822	98	3356	3	313	0	22832

Table 3.2: Linkage Statistics for English to Indian Language WordNets. D stands for Direct links, and H stands for Hypernymy links

(D) along with the synset linkages which have been marked as hypernymy linkages (H).

The statistics show our progress in updating IndoWordnet as a resource. The relatively large number of linkages also show that the Indian wordnets have matured considerably.

Many concepts in the Indian languages are specific to the Indian culture. Thus, their corresponding variant is not available in the Princeton WordNet (and is not likely to be included anytime). Thus, one needs to maintain the translation/transliteration of such notions from Indian languages to the English language as a separate bilingual mapping

[1]. A similar issue arises in case of proper nouns, which should be present in an Indian lexicon but they are not present in Princeton WordNet. They are also handled using bilingual mappings [Singh et al., 2016b]. Some of the synsets in Indian languages are too fine-grained and have a common representation in the English language. This is why we use the principle of Hypernymy linkages for linking such concepts. We reserve a set of synset id numbers later for language-specific concepts and create them to include in these wordnets, individually. These are not linked to the Princeton WordNet and hence are not included in our resource.

With this work, we describe two resources released publicly. We discussed the Indian language wordnets that are part of the IndoWordNet project. We enlisted the statistics of the latest version, which we provide as a single bundle. Next, we described the linkage process for creating English-Indian language links using English-Hindi language links. We then enlisted the statistics of the latest version of this linked data, which is also provided along with this work.

3.2.2 Semi-automatic WordNet Linking using Word Embeddings

With this work, we address the following question:

"Can information about the structure of embedding spaces of different languages and the relation among them be used to aid linking of corresponding wordnets?"

Mikolov et al. [2013a]'s observation about the structure of the embedding space of different languages note that there is a linear mapping between embedding spaces. We demonstrate that this is true, at least in the case of English and Hindi WordNets. We propose an approach to link them using word embeddings. Given a synset of the source language, the approach provides a ranked list of target synsets. This makes the overall linking task easy for human experts, as they have to choose from a relatively small set of potential candidates. Our evaluation shows that our technique is able to retrieve a winner synset in the top 10 ranked list for 60% and 70% of all synsets and noun synsets respectively.

[1]Since bilingual mappings are not standardized, we do not release them along with our resources

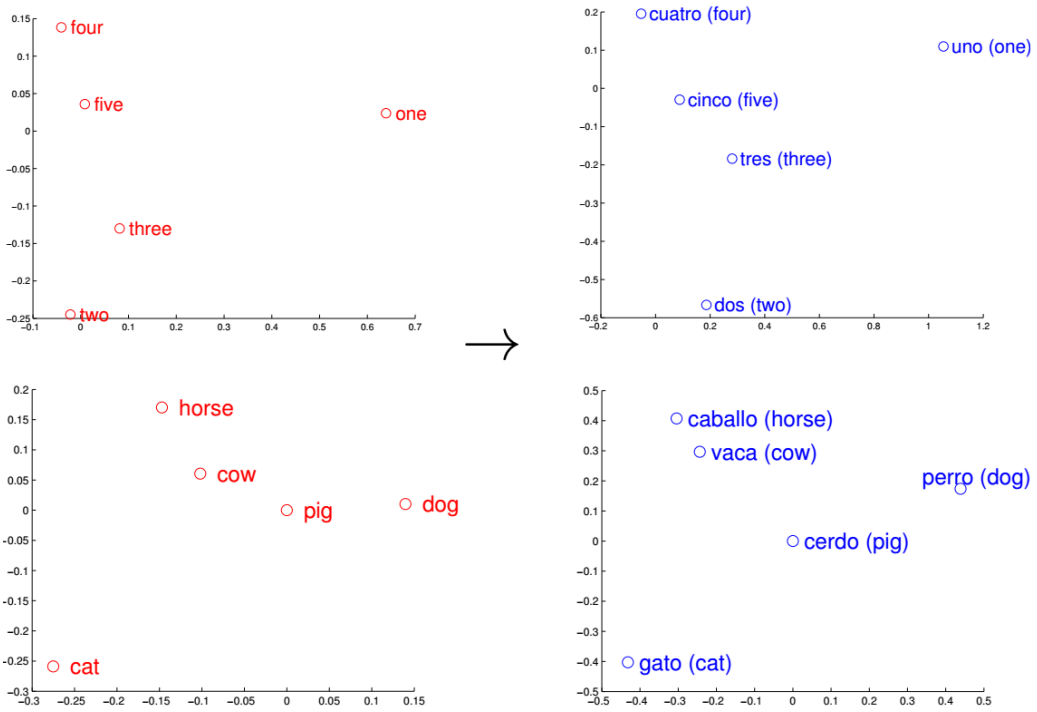


Figure 3.2: Word embeddings of numbers and animals in English (left) and Spanish (right) (taken from [Mikolov et al., 2013a]).

Previously, Joshi et al. [2012] come up with a heuristic-based measure where they use bilingual dictionaries to link two wordnets. They combine scores using various heuristics and generate a list of potential candidates for linked synsets. Singh et al. [2016b] discuss a method to improve the current status of Hindi-English linkage and present a generic methodology, *i.e.,* manually creating bilingual mappings for concepts which are unavailable in either of the languages or not present as a synset in the target wordnet. Their method is beneficial for culture-specific synsets, or for non-existing concepts; but, it is cost and time inefficient, and requires a lot of manual effort on the part of a lexicographer.

Our approach is mainly geared towards reducing effort on the part of the lexicographers. Given wordnets of two different languages E and F with sets of synsets $\{s_E^1, s_E^2, \ldots, s_E^m\}$ and $\{s_F^1, s_F^2, \ldots, s_F^n\}$ respectively, find mappings of the form $< s_E^i, s_F^j >$ which are semantically correct.

We adapted the technique of translating words in Mikolov et al. [2013a] to translate synsets (see fig 3.2). In order to do so, however, we need "synset embeddings". We computed the same by assigning to a synset-id, the average of the "word embeddings" of

its synset-members. To the best of our knowledge, this is a first attempt at solving this problem using word embeddings. The following is a detailed description of the technique.

Let E and F be two languages. Let $|E|$ and $|F|$ be the number of synsets in wordnets of E and F respectively. Let s_E^i and s_F^j be the ith and j^{th} synsets of E and F respectively, with $s_E^i = \{e_\alpha^1, e_\alpha^2, \ldots, e_\alpha^{m_i}\}$ and $s_F^j = \{f_\beta^1, f_\beta^2, \ldots, f_\beta^{n_j}\}$, where e_α^p and f_β^q are words in vocabulary of E and F respectively for $1 \leq p \leq m_i$ and $1 \leq q \leq n_j$, and $1 \leq i \leq |E|$ and $1 \leq j \leq |F|$.

Let $v_{e_\alpha^p}$ be the word embedding corresponding to e_α^p. Then we estimate $v_{s_E^i}$, the embedding for synset s_E^i, as

$$v_{s_E^i} = \frac{1}{m_i} \sum_{p=0}^{m_i} v_{e_\alpha^p} \tag{3.1}$$

Similarly,

$$v_{s_F^j} = \frac{1}{n_j} \sum_{q=0}^{n_j} v_{f_\beta^q} \tag{3.2}$$

Given links of the form $\langle s_E^i, s_F^j \rangle$, we learn W such that the error Err

$$Err = \|W.v_{s_E^i} - v_{s_F^j}\|^2 \tag{3.3}$$

is minimized.

Now, to find a mapping for a new synset s_E^k, one needs to

1. Calculate $v' = W.v_{s_E^k}$
2. Find $v_{s_F^l}$ such that $v_{s_F^l}.v'$ is maximized
3. Create link $\langle s_E^k, s_F^l \rangle$

Our hypothesis is that for a given synset-id, the noise added to its representative embedding by a highly polysemous synset-member will be cancelled out, while the actual information content pertaining to that synset-id will be enhanced, due to contribution from other, relatively less polysemous, synset members.

We applied our technique to link Hindi and English Wordnets. We obtained a dataset of mappings between English and Hindi wordnets from the developers of IndoWordNet. These mappings are of the form $\langle hindi_synset_id, english_synset_id, link_type \rangle$, where $link_type \in$ {DIRECT, HYPERNYMY, *etc.*}. For this experiment, we focused solely on DIRECT links. There are a total of 6,883 such mappings, the distribution among classes of which is mentioned in table 3.3

Class	**Count**
Noun	4757
Adjective	1283
Verb	680
Adverb	143

Table 3.3: Distribution of available links among various classes

For the English language, we used the pre-trained word embeddings published by Google that were trained on the part of Google News Dataset (about 100 billion tokens). These embeddings are of dimension 300 and are created using CBOW model with negative sampling. For the Hindi language, we trained word embeddings on BOJAR HindMonoCorp dataset [Bojar et al., 2014]. Mikolov et al. [2013c] suggests that the input embeddings' dimension should be at least 2.5 to 4 times that of the output dimension. However, we also wanted to check what happens when they are equal. Therefore, we trained two sets of embeddings, one of dimension 300, and the other of dimension 1200.

3.2.2.1 Evaluation and Results

We use the accuracy@n measure, *i.e.,* the prediction is said to be correct if one out of the top n results returned is correct. This is because accuracy@1 is an underestimate of the system's performance, as higher-ranking synonym translations will be counted as mistakes.

Table 3.4 shows the overall accuracy@n of the system, for different values of n. We also performed a per word-class evaluation, along with different settings for the embedding

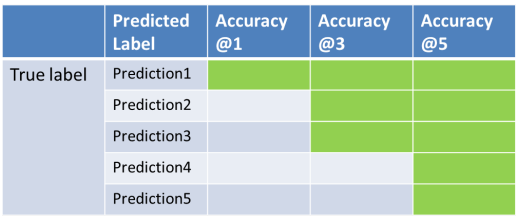


Figure 3.3: Accuracy@n: The green colored cells indicate the predictions considered for exact match for a given accuracy@n

dimensions. Table 3.5 and Table 3.6 shows the accuracy for different word classes [2].

Acc@1	**Acc@3**	**Acc@5**	**Acc@8**	**Acc@10**
0.29	0.45	0.52	0.58	0.60

Table 3.4: Results for the overall setting: Dimension of English embeddings=300, Dimensions of Hindi embeddings=300

Word Class	**Acc@1**	**Acc@3**	**Acc@5**	**Acc@8**	**Acc@10**
Noun	0.35	0.53	0.60	0.65	0.67
Adjective	0.26	0.44	0.50	0.57	0.60
Verb	0.15	0.25	0.29	0.33	0.37
Adverb	0.28	0.51	0.59	0.70	0.73

Table 3.5: Results for the setting: Dimension of English embeddings=300, Dimensions of Hindi embeddings=300

We observe that except for verbs, the approach performs decently. Here we mention some of the reasons for poor performance, as well as possible methods to address them.

- The approach to create synset embeddings is inadequate. The current averaging approach only takes the synset members into account, while ignoring gloss and examples, which could provide additional information. A potential candidate approach for creating synset embeddings should properly utilize the set of French synonyms, gloss, example sentences, and synset relations.

- Synset members are often phrases instead of words. Creating the phrase embeddings is a different problem altogether.

[2] All values reported are the average values obtained from 3-fold cross-validation.

Word Class	Acc@1	Acc@3	Acc@5	Acc@8	Acc@10
Noun	0.35	0.52	0.58	0.63	0.66
Adjective	0.12	0.20	0.24	0.30	0.32
Verb	0.17	0.27	0.32	0.35	0.39
Adverb	0.38	0.52	0.65	0.76	0.80

Table 3.6: Results for the setting: Dimension of English embeddings=300, Dimensions of Hindi embeddings=1200

- Currently, we utilized a word embedding model which gives only one embedding per word. That is one of the reasons for ambiguity. A model which provides one embedding per sense of a word will be more appropriate.

- The linear transformation approach is incorrect. While [Mikolov et al., 2013a] shows the linear relation between English and Spanish languages, this may not be true for all pairs of languages.

- Perhaps, something is fundamentally missing in word embeddings. Probably the presence of only co-occurrence information and lack of other information such as word order, argument frames(for verbs), *etc.* leads to this poor performance.

With this work, we described an approach to link wordnets. It entails creating synset embeddings using the word embeddings of the synset members, and learning a function to map the embedding of a synset from the source language to an embedding in the space of target language, and returning the nearest neighbours as potential candidates for linking. Our evaluation shows that our technique is able to retrieve a winner synset in the top 10 ranked list for 60% and 70% of all synsets and noun synsets, respectively. However, it did not achieve significantly good results for other classes, especially verbs. We discussed the possible reasons for poor performance and suggested mechanisms to address the same.

3.3 Pre-trained Embeddings for Indian Languages

This work was accepted at the SLTU-CCURL Workshop co-located with LREC 2020 Conference. This collaborative effort releases essential resources for our work which requires pre-trained embeddings to be trained on a transliterated corpus. Dense word vectors or 'word embeddings' which encode semantic properties of words have now be-

come integral to NLP tasks like Machine Translation (MT), Question Answering (QA), Word Sense Disambiguation (WSD), and Information Retrieval (IR). In this investigation, we use various existing approaches to create multiple word embeddings for 14 Indian languages. We place these embeddings for all these languages, *viz.,* Assamese, Bengali, Gujarati, Hindi, Kannada, Konkani, Malayalam, Marathi, Nepali, Odiya, Punjabi, Sanskrit, Tamil, and Telugu in a single repository. Relatively newer approaches that emphasize catering to context (BERT, ELMo, *etc.*) have shown significant improvements, but require a large amount of resources to generate usable models. We release pre-trained embeddings generated using both contextual and non-contextual approaches. We also use MUSE and XLM to train cross-lingual embeddings for all pairs of the aforementioned languages. To show the efficacy of our embeddings, we evaluate our embedding models on XPOS, UPOS and NER tasks for all these languages. We release a total of 436 models using 8 different approaches. We hope they are useful for the resource-constrained Indian language NLP.

NLP tasks that benefit from these pre-trained embeddings are very diverse. Tasks ranging from word analogy and spelling correction to more complex ones like Question Answering [Bordes et al., 2014], Machine Translation [Artetxe et al., 2019], and Information Retrieval [Diaz et al., 2016] have reported improvements with the use of well-trained embeddings models. The recent trend of transformer architecture based neural networks has inspired various language models that help train contextualized embeddings [Devlin et al., 2018, Peters et al., 2018, Melamud et al., 2016, Lample and Conneau, 2019]. They report significant improvements over various NLP tasks and release pre-trained embeddings models for many languages. One of the shortcomings of the currently available pre-trained models is the corpora size used for their training. Almost all of these models use Wikimedia corpus to train models which is insufficient for Indian languages as Wikipedia itself lacks a significant number of articles or text in these languages. Although there is no cap or a minimum number of documents/lines which define a usable size of a corpus for training such models, it is generally considered that the more input training data, the better the embedding models.

Acquiring raw corpora to be used as input training data has been a perennial problem for NLP researchers who work with low resource languages. Given a raw corpus, monolingual

word embeddings can be trained for a given language. Additionally, NLP tasks that rely on utilizing common linguistic properties of more than one language need cross-lingual word embeddings, *i.e.,* embeddings for multiple languages projected into a common vector space. These cross-lingual word embeddings have shown to help the task of cross-lingual information extraction [Levy et al., 2017], False Friends and Cognate detection [Merlo and Rodriguez, 2019], and Unsupervised Neural Machine Translation [Artetxe et al., 2018b]. With the recent advent of contextualized embeddings, a significant increase has been observed in the types of word embedding models. It would be convenient if a single repository existed for all such embedding models, especially for low-resource languages. Our work creates such a repository for fourteen Indian languages, keeping this in mind, by training and deploying 436 models with different training algorithms (like word2vec, BERT, *etc.*) and hyperparameters as detailed below. *Our key contributions are:*
(1) We acquire raw monolingual corpora for fourteen languages, including Wikimedia dumps. (2) We train various embedding models and evaluate them. (3) We release these embedding models and evaluation data in a single repository[3]*.*

3.3.1 Dataset Created and Experimental Evaluation

We collect pre-training data for over 14 Indian languages (from a total of 22 scheduled languages in India), including Assamese (as), Bengali (bn), Gujarati (gu), Hindi (hi), Kannada (kn), Konkani (ko), Malayalam (ml), Marathi (mr), Nepali (ne), Odiya (or), Punjabi (pa), Sanskrit (sa), Tamil (ta) and Telugu (te). These languages account for more than 95% of the entire Indian population, with the most widely spoken language, Hindi, alone contributing 43% to the figure[4]. Nonetheless, data that is readily available for computational purposes has been excruciatingly limited, even for these 14 languages.

One of the significant contributions of this work is the accumulation of data in a single repository. This dataset has been collected from various sources, including ILCI corpora [Choudhary and Jha, 2011, Bansal et al., 2013], which contains parallel aligned corpora (including English) with Hindi as the source language in tourism and health domains. As a baseline dataset, we first extract text from Wikipedia dumps[5], and then append the

[3]Repository Link
[4]Wikipedia Reference
[5]As on 15th August, 2019

data from other sources onto it. We added the aforementioned ILCI corpus, and then for Hindi, we add the monolingual corpus from HinMonoCorp 0.5 [Bojar et al., 2014], increasing the corpus size by 44 million sentences. For Hindi, Marathi, Nepali, Bengali, Tamil, and Gujarati, we add crawled corpus of film reviews and news websites[6]. For Sanskrit, we download a raw corpus of prose[7] and add it to our corpus. Further, we describe the preprocessing and tokenization of our data.

3.3.1.1 Preprocessing Collected Data

The corpora collected is intended to be set in a general domain instead of being domain-specific, and hence we start by collecting general domain corpora via Wikimedia dumps. We also add corpora from various crawl sources to respective individual language corpus. All the corpora are then cleaned, with the first step being the removal of HTML tags and links which can occur due to the presence of crawled data. Then, foreign language sentences (including English) are removed from each corpus, so that the final pre-training corpus contains words from only its language. Along with foreign languages, numerals written in any language are also removed. Once these steps are completed, paragraphs in the corpus are split into sentences using sentence end markers such as full stop and question mark. Following this, we also remove any special characters which may have included punctuation marks (example - hyphens, commas, *etc.*).

The statistics for the resulting corpus are listed in Table 3.7.

3.3.1.2 Experiment Setup

There is a prevailing scarcity of standardized benchmarks for testing the efficacy of various word embedding models for resource-poor languages. We conducted experiments across some rare standardized datasets that we could find and created new evaluation tasks as well to test the quality of non-contextual word embeddings. The Named Entity Recognition task, collected from [Murthy et al., 2018], and FIRE 2014 workshop for NER, contains NER tagged data for 5 Indian languages, namely Hindi, Tamil, Bengali, Malayalam, and Marathi. We also use a Universal POS (UPOS), as well as an XPOS (language-specific PoS tags) tagged dataset, available from the Universal Dependency

[6] `https://github.com/goru001`

[7] `http://sanskrit.jnu.ac.in/currentSanskritProse/`

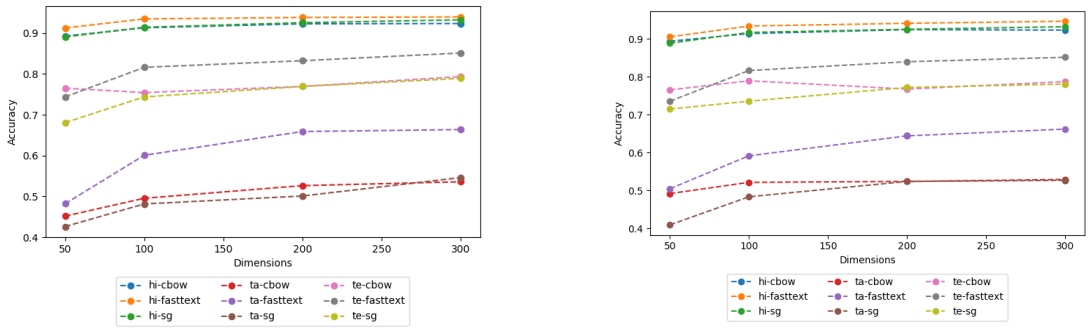
Language	Abbr.	Sentences	Words
Hindi	hin	48,115,256	3,419,909
Bengali	ben	1,563,137	707,473
Telugu	tel	1,019,430	1,255,086
Tamil	tam	881,429	1,407,646
Nepali	nep	705,503	314,408
Sanskrit	san	553,103	448,784
Marathi	mar	519,506	498,475
Punjabi	pan	503,330	247,835
Malayalam	mal	493,234	1,325,212
Gujarati	guj	468,024	182,566
Konkani	knn	246,722	76,899
Oriya	ori	112,472	55,312
Kannada	kan	51,949	30,031
Assamese	asm	50,470	29,827

Table 3.7: Corpus statistics for each Indic language with their ISO 639-3 abbreviations (total number of sentences and words)

(UD) treebank [Nivre et al., 2016], which contains POS tagged data for 4 Indian languages, Hindi, Tamil, Telugu, and Marathi.

For the tasks of NER, UPOS tagging, XPOS tagging, we use the Flair library [Akbik et al., 2018], which embeds our pre-trained embeddings as inputs for training the corresponding tagging models. The tagging models provided by Flair are vanilla BiLSTM-CRF sequence labellers. For the task of word analogy dataset, we simply use the vector addition and subtraction operators to check accuracy, (*i.e.,* $v(\text{France}) - v(\text{Paris}) + v(\text{Berlin})$ should be close to $v(\text{Germany})$).

For contextual word embeddings, we collect the statistics provided at the end of the pre-training phase to gauge the quality of the embeddings - perplexity scores for ELMo, masked language model accuracy for BERT, and so on. We report these values in Table 3.8.



(a) Performance on UPOS tagged dataset

(b) Performance on XPOS tagged dataset

Figure 3.4: Performance of skip-gram, CBOW, and fasttext models on POS tagging task. Plotted graph is Accuracy vs Dimension. Legend is "language"-"model". Note that FastText is the best performer in each case, and learning saturates around 200 dimensions

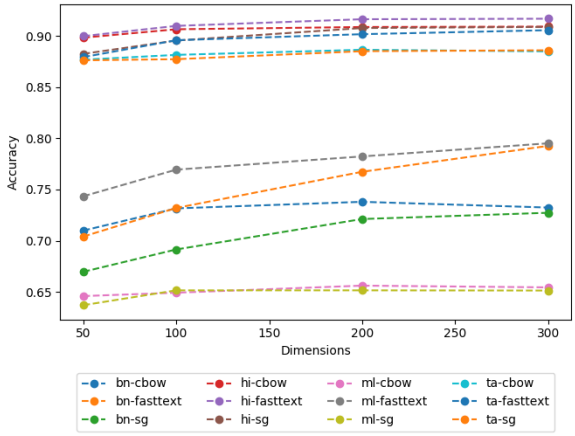


Figure 3.5: Performance of skip-gram, CBOW, and fasttext models on NER tagged dataset

Language	as	bn	gu	hi	ml	mr	kn	ko	ne	or	pa	sa	ta	te
Perplexity	455	354	183	518	1689	522	155368	325	253	975	145	399	781	82

Table 3.8: ELMo prerplexity scores

3.3.2 Models Released and Evaluation Results

Now, We briefly describe the models created using the approaches detailed above.

3.3.2.1 Word2Vec (skip-gram and CBOW)

Word2Vec embeddings [Mikolov et al., 2013c] of dimensions {50, 100, 200, 300} for both skip-gram and CBOW architectures are created using the gensim library [Řehůřek and Sojka, 2010] implementation of Word2Vec. Words with a frequency of less than 2 in the entire corpus are treated as unknown (out-of-vocabulary) words. For other parameters, default settings of gensim are used. There are no pre-trained Word2Vec word embeddings for any of the 14 languages available publicly.

3.3.2.2 FastText

FastText embeddings [Bojanowski et al., 2017b] of dimensions {50, 100, 200, 300} (skip-gram architecture) were created using the gensim library [Řehůřek and Sojka, 2010] implementation of FastText. Words with a frequency of less than 2 in the entire corpus are treated as unknown (out-of-vocabulary) words. For other parameters, default settings of gensim are used. Except for Konkani and Punjabi, the official repository for FastText provides pre-trained word embeddings for the Indian languages. However, we have trained our word embeddings on a much larger corpus than those used by FastText.

3.3.2.3 GloVe

We create GloVe embeddings [Pennington et al., 2014] of dimensions {50, 100, 200, and 300}. Words with occurrence frequency less than 2 are not included in the library. The co-occurrence matrix is created using a symmetric window of size 15. There are no pre-trained word embeddings for any of the 14 languages available with the GloVE embeddings repository[8]. We create these models and provide them with our repository.

3.3.2.4 MUSE

MUSE embeddings are cross-lingual embeddings that can be trained using the fastText embeddings, which we had created previously. Due to resource constraints and the fact that cross-lingual representations require a large amount of data, we choose to train 50-dimensional embeddings for each language pair. We train for all the language pairs (14*14)

[8] `https://nlp.stanford.edu/projects/glove/`

and thus produce 196 models using this approach and provide them in our repository. The training for these models took 2 days over 1 x 2080Ti GPU (12 GB).

3.3.2.5 ELMo

We train ELMo embeddings [Peters et al., 2018] of 512 dimensions. These vectors are learned functions of the internal states of a deep bidirectional language model (biLM). The training time for each language corpus was approximately 1 day on a 12 GB Nvidia GeForce GTX TitanX GPU. The batch size is reduced to 64, and the embedding model was trained on a single GPU. The number of training tokens was set to tokens multiplied by 5. We choose this parameter based on the assumption that each sentence contains an average of 4 tokens. There are no pre-trained word embeddings for any of the 14 languages available on the official repository. We provide these models in our repository.

3.3.2.6 BERT

We train BERT (Bidirectional Encoder Representations from Transformers) embeddings [Devlin et al., 2018] of 300 dimensions. Since BERT can be used to train a single multilingual model, we combine and shuffle corpora of all languages into a single corpus and used this as the pre-training data. We use sentence piece embeddings [Google, 2018] that we trained on the corpus with a vocabulary size of 25000. Pre-training this model was completed in less than 1 day using 3 * 12 GB Tesla K80 GPUs. The official repository for BERT provides a multilingual model of 102 languages, which includes all but 4 (Oriya, Assamese, Sanskrit, Konkani) of the 14 languages. We provide a single multilingual BERT model for all the 14 languages, including these 4 languages.

3.3.2.7 XLM

We train cross-lingual contextual BERT representation language model using the XLM git repository[9]. We train this model for 300-dimensional embeddings and over the standard hyperparameters as described with their work. The corpus vocabulary size of 25000 was chosen. We use a combined corpus of all 14 Indian languages and shuffle the sentences for data preparation of this model. We use the monolingual model (MLM) method to prepare data as described on their Git repository. This model also required the Byte-pair

[9] `https://github.com/facebookresearch/XLM`

encoding representations as input, and we train them using the standard fastBPE implementation as recommended over their Github. The training for this model took 6 days and 23 hours over 3 x V100 GPUs (16 GB each).

3.3.2.8 Evaluation

We evaluate and compare the performance of FastText, Word2Vec, and GloVE embedding models on UPOS and XPOS datasets. The results are shown in the Figure 3.4a and in 3.4b, respectively. The performance of non-contextual word embedding models on NER dataset is shown in Figure 3.5. The perplexity scores for ELMo training are listed in table 3.8. We observe that FastText outperforms both GloVE and Word2Vec models. For Indian languages, the performance of FastText is also an indication of the fact that morphologically rich languages require embedding models with sub-word enriched information. This is clearly depicted in our evaluation.

The overall size of all the aforementioned models was very large to be hosted on a Git repository. We host all of these embeddings in a downloadable ZIP format each on our server, which can be accessed via this link[10].

3.3.3 Discussion

We created a comprehensive set of standard word embeddings for multiple Indian languages. We release a total of 422 embedding models for 14 Indic languages. The models contain 4 varying dimensions (50, 100, 200, and 300) each of GloVE, Skipgram, CBOW, and FastText; 1 each of ELMo for every language; a single model each of BERT and XLM of all languages. They also consist of 182 cross-lingual word embedding models for each pair. However, due to the differences in language properties as well as corpora sizes, the quality of the models vary. Table 3.7 shows the language wise corpus statistics. Evaluation of the models has already been presented in Section 3.3.2.8. An interesting point to note is that even though Tamil and Telugu have comparable corpora sizes, the evaluations of their word embeddings show different results. Telugu models consistently outperform Tamil models on all common tasks. Note that the NER tagged dataset was not available for Telugu, so they could not be compared on this task.

[10]Repository Link

This also serves to highlight the difference between the properties of these two languages. Even though they belong to the same language family, *Dravidian*, and their dataset size is the same, their evaluations show a marked difference. Each language has 3 non-contextual embeddings (word2vec-skipgram, word2vec-cbow and fasttext-skipgram), and a contextual embedding (ElMo). Along with this, we have created multilingual embeddings via BERT. For BERT pre-training, the masked language model accuracy is 31.8%, and next sentence prediction accuracy is 67.9%. Cross-lingual embeddings, on the other hand, have been created using XLM and MUSE.

The recent past has seen tremendous growth in NLP with ElMo, BERT and XLNet being released in quick succession. All such advances have improved the state-of-the-art in various tasks like NER, Question Answering, Machine Translation, *etc.* However, most of these results have been presented predominantly for a single language- English. With the potential that Indian languages computing has, it becomes pertinent to perform research in word embeddings for local, low-resource languages as well. In this work, we present the work done on creating a single repository of corpora for 14 Indian languages. We also discuss the creation of different embedding models in detail. As for our primary contribution, these word embedding models are publicly released.

In the future, we aim to refine these embeddings and do a more exhaustive evaluation over various tasks such as POS tagging for all these languages, NER for all Indian languages, including a word analogy task. Presently evaluations have been carried out on only a few of these tasks. Also, with newer embedding techniques being released in quick successions, we hope to include them in our repository. The model's parameters can be trained further for specific tasks or improving their performance in general. We hope that our work serves as a stepping stone to better embeddings for low-resource Indian languages.

3.4 Summary

In this chapter, we have described distributional semantics-based approaches used for NLP tasks. We provided details of various word embeddings which we study, or use in this thesis. Further, we described our efforts towards creating linked Indian language wordnets and a semi-automatic method to generate linkages between Hindi and English Wordnets. The linked Indian language wordnets help us create datasets for the task of cognate detection among Indian languages. We also described the dataset creation for, and generation of 436 word embedding models[11]. which we release publicly as a resource which can be used for Indian language NLP tasks. Our semi-automatic approach helped generate various linkages which we further utilize for the task of cognate detection amongst Indian and European languages as described in the next chapter.

[11]Repository Link

Chapter 4

Cognate Detection Task: Preliminary Investigations

In the last chapter, we summarised the use of distributional semantics for NLP tasks and described the linked dataset creation for our work. In this chapter, we describe our preliminary investigations for the task of cognate detection. We, first, introduce the cognate detection task briefly at the beginning of this chapter. Later, we discuss experiments which detail the use of monolingual embeddings for the task and conclude it with our work on the creation of challenging datasets for the task of cognate and false friends' detection.

NLP tasks which deal with cross-lingual vocabulary can be improved with the help of Cognate detection. Studies have shown improvements in Machine Translation [Kondrak et al., 2003], Cross-lingual Information Retrieval [Pranav, 2018, Makin et al., 2008, 2007], Computational Phylogenetics [Rama et al., 2018], and many more tasks like Entity Linking [Lawrie et al., 2014], and Drug Discovery [Kondrak and Dorr, 2004]. As described previously, **Cognate Detection is the task of identifying Cognate words from two languages, *i.e.,* L1 and L2, where L1 and L2 are, usually, two different languages.** Cognates within a single language, or **doublets**, may have meanings that are slightly or even totally different. The study of cognates plays a crucial role in applying comparative approaches for historical linguistics, in particular, solving language relatedness and tracking the interaction and evolvement of multiple languages over time. A cognate instance among Indian languages is given as the word group: *putra* (Sanskrit), *putra* (Hindi), *putra* (Marathi) and *puttar* (Punjabi), all of which mean the word "Son".

A cognate etymon does not need to be inherited directly from a proto-language. It can be borrowed from some other language, in which evolution produces cognate forms. For example, the English word dish and the German word Tisch ("table") are cognates because they both come from Latin discus, which relates to their flat surfaces. Cognates may have evolved similar, different or even opposite meanings, but in most cases, there are some similar letters in the word. Some words sound similar, but do not come from the same root; these are called **false cognates**. These are the occurrences where words are completely unrelated words that happen to look similar. For *e.g.,* Greek *kappa* and Japanese *kappa* (transliterated kappa) 'water sprite'. Another example is Portuguese *galinha* 'hen' and English *galena*.

In this chapter, we discuss the creation of a gold-standard dataset for the task of cognate detection and false friends' detection. We shall first discuss our experiments for the detection of cognates among Indian languages and establish that Wordnets can be used for the task of cognate detection. We also explain in detail our efforts to create challenge datasets for the task of cognate and false friends' detection.

4.1 Cognate Identification to Generate Phylogenetic Trees for Indian Languages

This work was accepted at the Young Researcher's Symposium at CoDS-COMAD 2019 Conference. In this preliminary investigation, we detect cognates among a few Indian languages, namely Hindi, Marathi, Punjabi, and Sanskrit for helping build cognate sets for phylogenetic inference. Cognate detection helps phylogenetic inference by helping isolate diachronic sound changes and thus detect the words of a common origin. A cognate set manually annotated with the help of a lexicographer is generally used to infer phylogenetic trees automatically. Our work creates cognate sets of each language pair and infers phylogenetic trees based on a Bayesian framework using the Maximum likelihood method. **We report negative results showing that simple orthographic similarity-based measures are incapable of detection cognates with high precision after the output is validated manually.**

4.1.1 Dataset Creation and Experiments

We create the dataset by extracting word list for Hindi, Marathi, Sanskrit, and Punjabi WordNets. We transliterate the words in the Punjabi wordlist manually. We use the unique words from wordlist extracted from all the individual wordnet databases but maintain them within the ID space. We extract 15000 unique words from every Wordnet and create an aligned wordlist for every language with Sanskrit, the pairs being Sanskrit - Hindi, Sanskrit - Marathi, and Sanskrit - Punjabi. We design our dataset by first creating wordlists for every language pair involved.

We extract unique words from wordnet data publicly available[1]. We align words from every language pair in a comma-separated form for each concept ID, thus ensuring a high probability of detecting cognates. We, also, use the baseline measure XDice and string similarity-based measures to first prepare cognate sets from every individual language pair. Later, we construct more cognate sets with the use of Orthographic cognate detection methods such as alignment of substrings which uses support vector machines and Hidden Markov Models. Among other methodologies, we use the phonetic aspects of the words decomposing them phonetically and aligning them according to phonemes. For validating our cognate sets, we use string similarity measures and use the threshold value of 0.75 arrived at by empirical measures. We use Jaccard, XDice and TF-IDF to validate our cognate sets.

4.1.2 Results and Conclusion

During the validation of cognate sets created by orthographic measures, we decided the threshold of matching at 0.75 for a pair to be cognate words. While arriving at this value, we observed that we could easily form pairs of cognate words which are *Tatsama* words. On the other hand, *Tadbhava* words were hardly detected among the cognate words unless phonetic methodologies are not used. This poses a new challenge as *Tadbhava* word form a large set of cognate words among the Indian languages. This can also be verified intuitively as the former retain their orthographic form and are easy to detect via the string similarity measure and the orthographic measure, but the latter need phonetic measures.

[1] `http://www.cfilt.iitb.ac.in/indowordnet/`

	Hindi-Punjabi	Hindi-Marathi	Hindi-Sanskrit
True Positives	497	621	378
False Positives	301	284	211
Total Detected	798	905	589

Table 4.1: Detected cognates were validated manually with the help of lexicographers, and it was found that simple orthographic similarity-based measures are unable to detect cognates with high precision.

We describe our work on cognate detection for Indian language pairs Hindi - Punjabi, Hindi - Marathi, and Hindi - Sanskrit. We create a wordlist of 15000 unique words from every individual wordnet data and also create cognate word sets for these three languages, and create an online interface to help collect data for phylogenetic inference. In future, we aim to experiment with phonetic and semantic measures to improve upon this work. We also propose to expand our dataset to multiple Indian languages as wordlists in their root form are available publicly via the Indowordnet website. We also aim to experiment with corpus instead of wordlists in their root form as morphological inflection would be a tougher challenge to tackle for detection of cognates in a corpus.

In our next investigation, we perform further experimentation and propose a weighted orthographic similarity-based approach to detect cognates.

4.2 Utilizing Wordnets for Cognate Detection among Indian Languages

In this investigation, we detect cognate word pairs among ten Indian languages with Hindi and use deep learning-based approaches to predict whether a word pair is cognate or not. We identify IndoWordnet as a potential resource to detect cognate word pairs based on orthographic similarity-based methods and train neural network models using the data obtained from it. We identify parallel corpora as another potential resource and perform the same experiments for them.

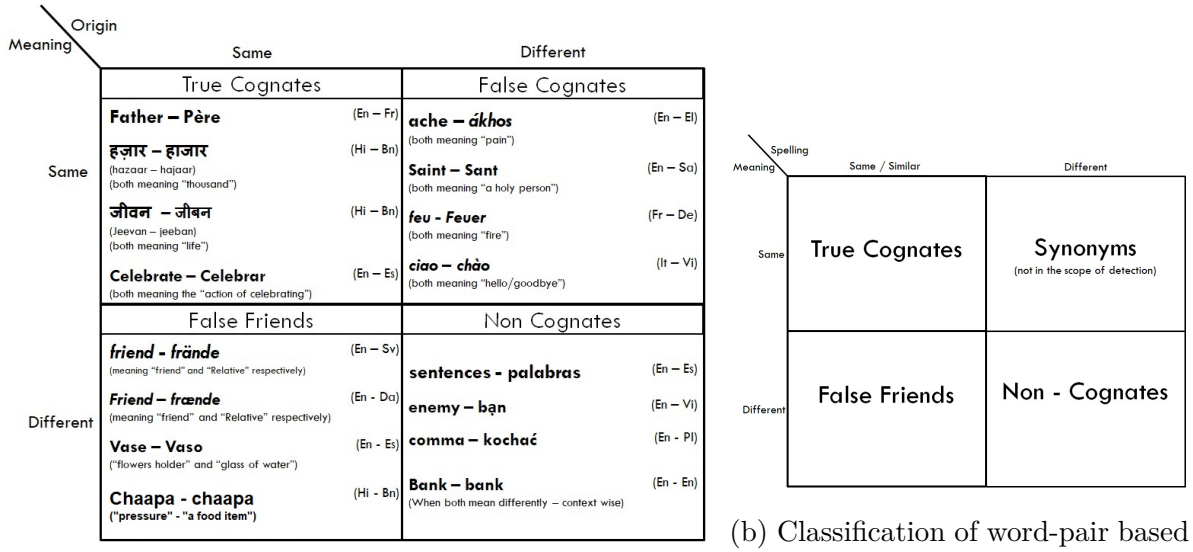
We also validate the contribution of Wordnets through further experimentation and report improved performance of up to 26%. We discuss the nuances of cognate detection

among closely related Indian languages and release the lists of detected cognates as a dataset. We also observe the behaviour of, to an extent, unrelated Indian language pairs and release the lists of detected cognates among them as well.

In many NLP tasks, the orthographic similarity of cognates can compensate for the insufficiency of other kinds of evidence about the translational equivalency of words [Mulloni and Pekar, 2006]. Orthographic similarity-based methods, generally, calculate the similarity score between two words and use the result to build training data for further classification. Cognate detection can also be performed using phonetic features and researchers have previously used consonant class matching (CCM) [Turchin et al., 2010], sound class-based alignment (SCA) [List, 2010], *etc.* to detect cognates in multilingual wordlists. The identification of cognates, here, is based on the comparison of words sound correspondences. Semantic similarity methods have also been deployed to detect cognates among word pairs [Kondrak, 2001]. The measure of semantic similarity uses the context around both word pairs and helps in the identification of a cognate word pair by looking of similarity among the collected contexts.

For our work, we can primarily divide cross-lingual word-pairs into four main categories, *viz.,* **True Cognates, False Cognates, False Friends and Non-Cognates**. In Figure 4.1a, we present this classification with examples from various languages along with their meanings for better understanding. While some false friends are also false cognates, most of them are genuine cognates. *Our primary goal is to be able to identify True Cognates.* Sanskrit (Sa) is known to be the mother of most of the Indian languages. Hindi (Hi), Bengali (Bn), Punjabi (Pa), Marathi (Mr), Gujarati (Gu), Malayalam (Ml), Tamil (Ta) and Telugu (Te) are known to borrow many words from it. Thus, one may observe that *words which belong to the same concept in these languages, if orthographically similar, are True Cognates.* Currently, we include loan words in the dataset used for our work and include them as cognates. Since, eventually, we aim to apply our work to Machine Translation and other NLP applications, we believe that this would help establish a better correlation among source-target language pairs. Also, we do not detect false friends and hence restrict the scope of True cognate detection using this hypothesis to Figure 4.1b.

We utilise the synset information from linked Wordnets to identify words within the same concept and deploy orthographic similarity-based methods to compute similarity scores between them. This helps us identify words with a



(a) Classification of Cross-lingual Word pairs.

(b) Classification of word-pair based on similar form and meaning.

Figure 4.1: An overview of the scope of this work as presented in our investigation presented at GWC 2019.

high similarity score. In case of most of the Indian languages, a sizeable contribution of words/concepts is loaned from the Sanskrit language. In linked IndoWordnet, each concept is aligned to the other based on an 'id' which can be reliably used as a measure to say that the etymological origin is the same, for both the concepts. Hence, words with the same orthographic similarity can be said to be **'True Cognates'**. Using this methodology, we detect highly similar words and use them as training data to build models which can predict whether a word pair is cognate or not. We make the following key contributions via this investigation:

1. We perform cognate detection for eleven Indian Languages.
2. We exploit Indian languages behaviour to obtain a list of true cognates (WNdata from WordNet and PCData from Parallel Corpora).
3. We train neural networks to establish a baseline for cognate detection.
4. We validate the importance of Wordnets as a resource to perform cognate detection.
5. We release our dataset (WNdata + PCdata) of cognate pairs publicly for the language pairs Hi - Mr, Hi - Pa, Hi - Gu, Hi - Bn, Hi - Sa, Hi - Ml, Hi - Ta, Hi - Te, Hi - Ne, and Hi - Ur.

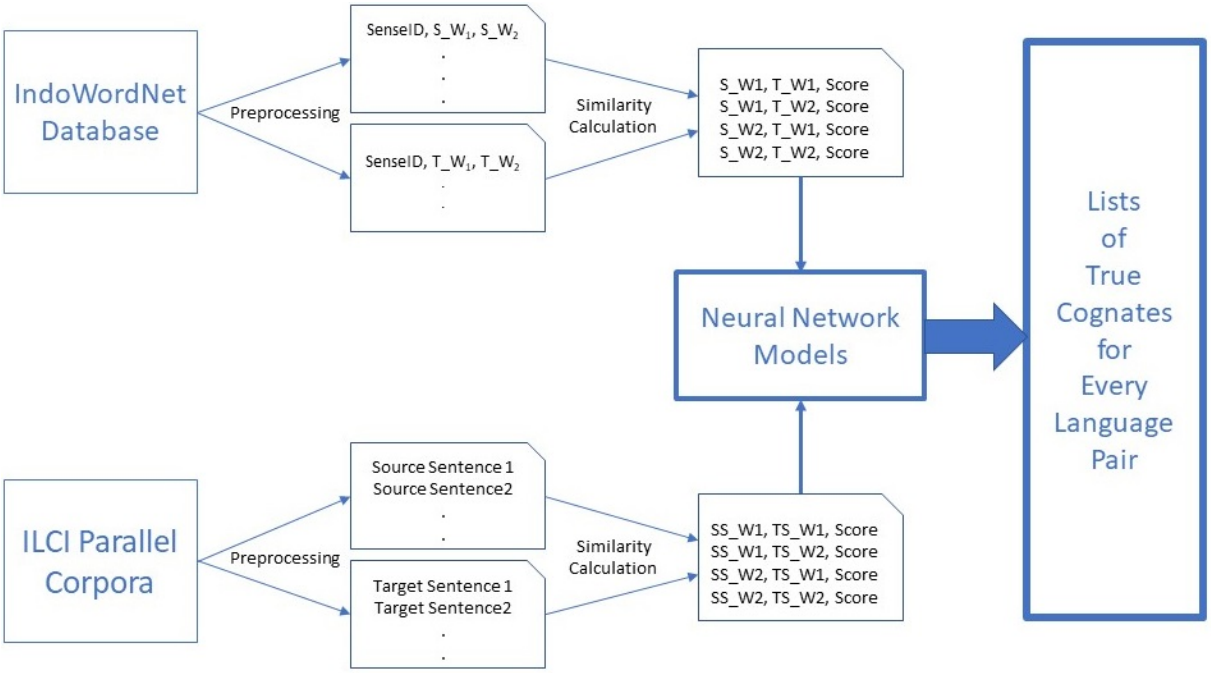


Figure 4.2: Block Diagram for our experimental setup

4.2.1 Script Standardization and Text Normalization

The languages mentioned above share a significant portion of the most spoken languages in India. Although most of them borrow words from Sanskrit, they belong to different language families. Mr, Gu, Bn, Pa, Ne and Ur belong to the Indo-Aryan family of languages; and Ml, Ta, Te belong to the family of Dravidian languages. They also use different scripts to represent themselves textually. For standardisation, we convert all the other written scripts to Devanagari. We perform Unicode transliteration using Indic NLP Library[2] to convert scripts for Bn, Gu, Pa, Ta, Te, Ml, and Ur to Devanagari, for both our datasets. Hi, Mr, Sa, and Ne are already based on the Devanagari script, and hence we only perform text normalisation for both our datasets, for these languages. The whole process is outlined in Figure 4.2.

4.2.2 Dataset Creation

We investigate language pairs for major Indian languages namely Marathi (Mr), Gujarati (Gu), Bengali (Bn), Punjabi (Pa), Sanskrit (Sa), Malayalam (Ml), Tamil (Ta), Telugu (Te), Nepali (Ne) and Urdu (Ur) with Hindi (Hi). We create two datasets as described

[2]https://anoopkunchukuttan.github.io/indic_nlp_library/

below for <source_lang> -<target_lang> where the source language is always Hindi. We describe each step in the subsections below.

Dataset 1: WordNet based dataset

We create this dataset (WNData) by extracting synset data from the IndoWordnet database. We maintain all words, in the concept space, in a comma-separated format. We, then, create word lists by combining all possible permutations of word pairs within each synset. For *e.g.,* If synset ID X on the source side (Hindi) contains words S_1W_1 and S_1W_2, and parallelly on the target side (other Indian languages), synset ID X contains T_1W_1 and T_1W_2, we create a word list such as:

S_1W_1, T_1W_1

S_1W_2, T_1W_1

S_1W_1, T_1W_2

S_1W_2, T_1W_2

To avoid redundancy, we remove duplicate word pairs from this list.

Dataset 2: Parallel Corpora based dataset

We use the ILCI parallel corpora for Indian languages [Jha, 2010b] and create word pairs list by comparing all words in the source side sentence with all words on the target side sentence. Our hypothesis, here, is that words with high orthographic similarity which occur in the same context window (a sentence) would be cognates with a high probability. Due to the unavailability of ILCI parallel corpora for Sa and Ne, we download these corpora from Wikipedia and align it with the Hindi articles from Hindi Wikipedia. We calculate exact word matches to align articles to each other, thus creating comparable corpora and discard unaligned lines from both sides. We, then, create similar word pairs list between Hindi and all the other languages pairs. We removed duplicated word pairs from this list as well and call this data PCData.

We calculate similarity scores for each word on the source side, *i.e.,* Hi by matching it with each word on the target side, *i.e.,* Sa, Bn, Gu, Pa, Mr, Ml, Ne, Ta, Te, and Ur.

Since we match the words from the same concept space or the same context window, we eliminate the possibility of this word pair carrying different meanings, and hence **a**

high orthographic similarity score gives us a strong indication of these words falling under the category of True Cognates. For training neural network models, we then divide the positive and negative labels based on a threshold and follow empirical methods in setting this threshold to 0.5 for both datasets[3]. Using 0.5 as the threshold, we obtained the best training performance and hence chose to use this as the threshold for similarity calculation. The various similarity measures used are described in the next subsection.

4.2.3 Similarity Measures

Normalized Edit Distance Method (NED)

The Normalised Edit Distance approach computes the edit distance [Nerbonne and Heeringa, 1997] for all word pairs in a synset/concept and then provides the output of probable cognate sets with distance and similarity scores. We assign labels for these sets based on the similarity score obtained from the NED method, where the similarity score is (1 - NED score). It is usually defined as a parameterisable metric which is calculated with a set of allowed edit operations. Each of these operations is assigned a cost (possibly infinite). The score is normalised such that 0 equates to no similarity and 1 is an exact match. NED is equal to the minimum number of operations required to transform 'word a' to 'word b'. A more general definition associates non-negative weight functions (insertions, deletions, and substitutions) with the operations.

Cosine Similarity (Cos)

The cosine similarity measure [Salton and Buckley, 1988] is another similarity metric that depends on envisioning preferences as points in space. This measure computes the cosine of the angle between two vectors projected in a multi-dimensional space. In this context, the two vectors are the arrays of character counts of two words. This measure similarity is used in positive space, where the outcome is bounded in [0,1]. For example, in NLP sub-areas of information retrieval and text mining, each term is assigned a different dimension. A document is said to be represented by a vector, where the value in each dimension corresponds to the number of times the said term appears in the document.

[3]We ran experiments with 0.25, 0.60, and 0.75 as well, and choose 0.5 based on training performance

	FFN		RNN	
	D1	D2	D1	D2
Hi-Mr	69.76	85.76	74.76	89.78
Hi-Bn	65.18	81.04	69.18	86.44
Hi-Pa	73.04	78.50	76.04	83.64
Hi-Gu	61.74	79.16	69.84	89.44
Hi-Sa	61.72	85.87	68.92	91.66
Hi-Ml	56.96	74.77	66.96	79.59
Hi-Ta	55.62	61.70	65.62	68.92
Hi-Te	52.78	65.26	62.78	74.83
Hi-Ne	70.20	83.85	80.20	89.63
Hi-Ur	69.99	73.84	76.99	80.12

Table 4.2: Stratified 5-fold Evaluation using Deep Neural Models on both PCData (D1) and WNData (D2)

This similarity score, then, gives a useful measure of how similar two documents are likely to be, in terms of the text in them. This is similar to the cosine, which is 1 (maximum value) when the segments subtend a zero angle and 0 (uncorrelated) when the segments are perpendicular. In this context, the two vectors are the arrays of character counts of two words.

Jaro-Winkler Similarity (JWS)

Jaro-Winkler distance [Winkler, 1990] is a string metric measuring similar to the normalised edit distance deriving itself from Jaro Distance [Jaro, 1989]. It uses a prefix scale P which gives more favourable ratings to strings that match from the beginning, for a set prefix length L. We ensure a normalised score in this case as well. Here, the edit distance between two sequences is calculated using a prefix scale P which gives more favourable ratings to strings that match from the beginning, for a set prefix length L. The lower the Jaro–Winkler distance for two strings is, the more similar the strings are. The score is normalised such that 1 equates to no similarity and 0 is an exact match.

	Corp+WN20		Corp+WN40		Corp+WN60		Corp+WN80		Corp+WN100	
	FFN	RNN	FFN	RNN	FFN	RNN	FFN	RNN	FFN	RNN
Hi-Mr	70.12	74.12	73.56	78.37	76.09	81.56	81.34	85.24	86.90	91.87
Hi-Bn	71.06	73.17	73.29	74.98	77.33	76.28	83.99	81.45	82.18	89.58
Hi-Pa	74.16	75.94	76.02	77.39	76.18	79.04	78.04	81.22	80.66	85.64
Hi-Gu	65.26	70.76	71.21	74.83	75.09	79.95	80.14	84.32	81.85	89.81
Hi-Sa	65.93	74.23	69.25	77.51	74.84	79.92	81.03	86.62	88.13	93.86
Hi-Ml	57.75	59.38	56.31	65.67	58.02	71.19	61.01	75.59	69.11	82.54
Hi-Ta	54.63	60.12	56.69	63.38	57.46	66.17	59.36	67.17	60.41	70.62
Hi-Te	53.21	58.18	56.19	63.90	64.15	67.70	65.19	70.65	66.10	74.92
Hi-Ne	70.78	71.23	74.30	78.11	72.19	83.20	79.70	85.01	84.69	90.95
Hi-Ur	69.94	71.25	70.01	72.35	72.03	76.59	71.07	78.27	73.99	80.99

Table 4.3: Results after combining chunks of WNData with PCData

4.2.4 Models

4.2.4.1 Feed Forward Neural Network (FFN)

In this network, we deal with a word as a whole. Words of the source and target languages reside in separate embedding space. The source word passes through the source embedding layer. The target word passes through the target embedding layer. The outputs of both embedding lookups are concatenated. The resulting representation is passed to a fully connected layer with ReLU activations, followed by a softmax layer.

4.2.4.2 Recurrent Neural Network (RNN)

In this network (see Figure 4.3), we treat a word as a sequence of characters. Characters of the source and the target language reside in separate embedding spaces. The characters of the source word are passed through source embedding layer. The characters of the target word are passed through the target embedding layer. The outputs of both embedding lookups are, then, concatenated. The resulting embedded representation is passed through a recurrent layer. The final hidden state of the recurrent layer is then passed through a fully connected layer with ReLU activation. The resulting output is finally passed through a softmax layer.

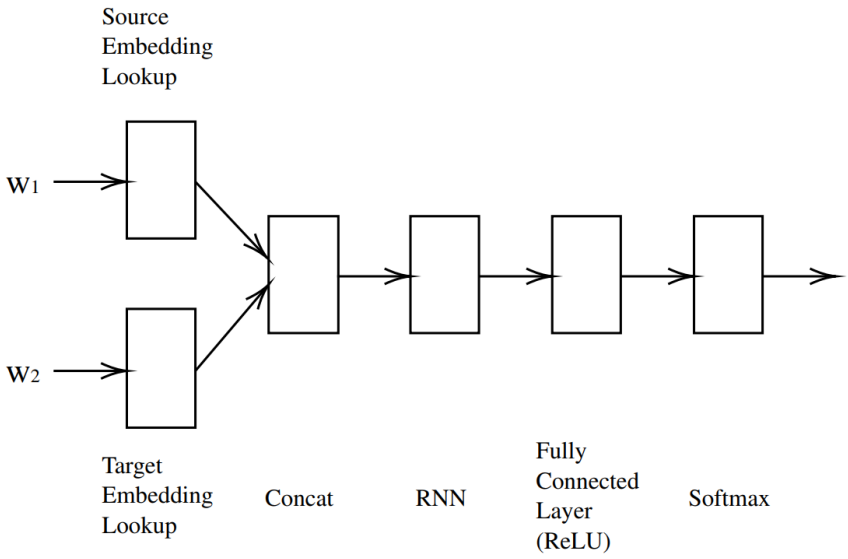


Figure 4.3: Architecture of our Recurrent Neural Network

4.2.5 Results and Discussion

We average the similarity scores obtained using the three methodologies (NED, Cos, and JWS) described above, for each word pair, and then use these as training labels for cognate detection models. We obtain results using the networks described above and report them in Table 4.2. We calculate average scores for both models and both datasets and show the chart in Figure 4.4. We observe that RNN outperforms FFN for both the datasets across all language pairs (see Figure 4.4). We also find that Hi-Sa (see Figure 4.4) has the best cognate detection accuracy among all language pairs (for both RNN and FFN), which is in line with the fact that they are closely related languages when compared to other Indian language pairs. We observe that average scores for WNData are always higher than average scores for PCData for all language pairs (Figure 4.4). Also, in line with our observations above, the overall average of RNN scores for both datasets are even higher than average FFN scores (Figure 4.4).

We perform another set of experiments by combining non-redundant word pairs from both datasets. We add WNData in chunks of 20 per cent to PCData for each language pair and create separate word lists with average similarity scores. We use FFN to train and perform a stratified 5-fold evaluation for each language pair after adding each chunk and show the results in Table 4.3. After evaluating our results for FFN, we perform the same training and evaluation with RNN. **We observe that adding complete WNData to PCdata improves our performance drastically and given us the best results**

for almost all cases. Only in case of Hi-Bn, when using the FFN for training, PCData combined with 80% WNData performs better than 100% Data; possibly due to added sparsity of the additional data. Our hypothesis that adding WNData to PCdata improves the performance holds for all the other cases, including when trained using RNN.

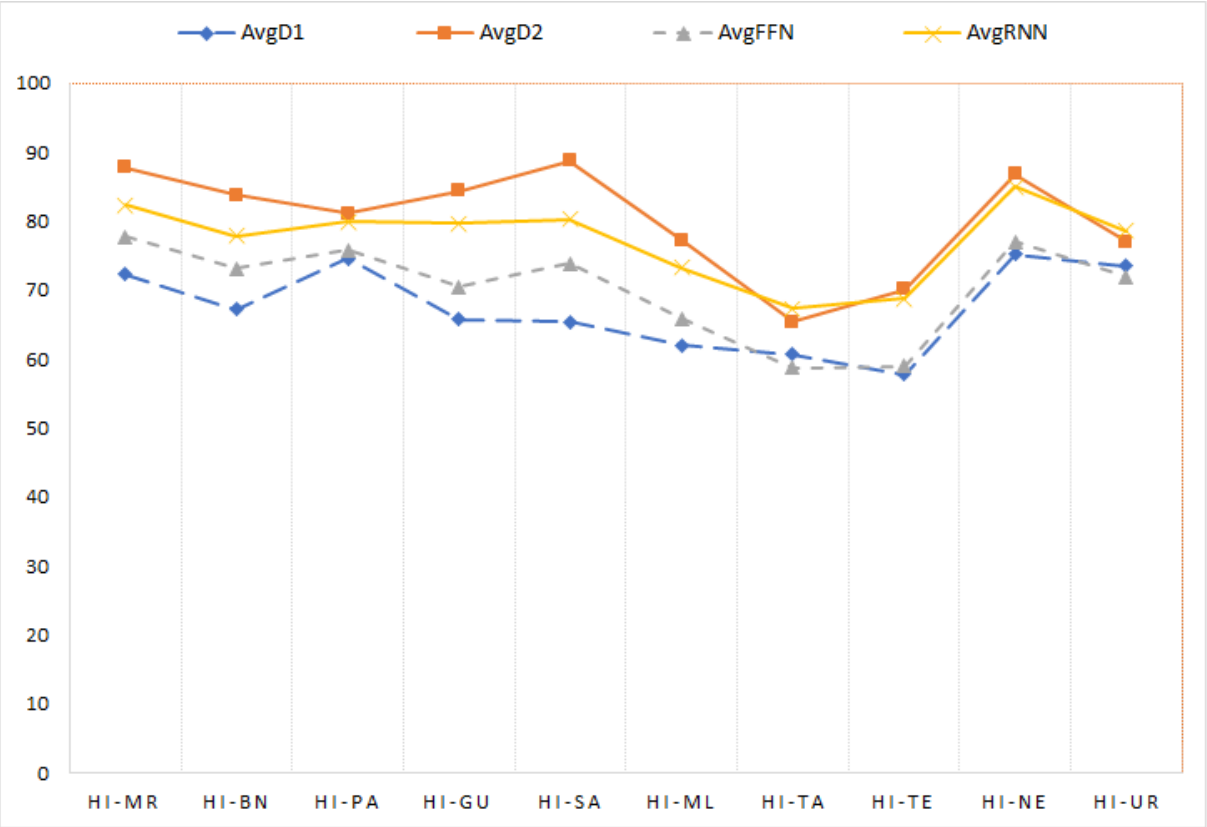


Figure 4.4: Average Results using Neural Network models on both datasets

A parallel corpus is a costly resource to obtain in terms of both time and effort. For resource-scarce languages, parallel corpora cannot easily be crawled. We wanted to validate how crucial Wordnets are as a resource and can they act as a substantial dataset in the absence of parallel corpora. In addition to validating the performance of chunks of WNData combined with PCData, we also calculated the exact matches of word pairs from both the datasets and show the results in Table 4.4. We observed that Hi-Mr had the most matched pairs amongst all the languages. PCData is extracted from parallel corpora and is not stemmed for root words, whereas WNData is extracted from IndoWordnet and only contains root words. Despite many words with morphological inflections, we were able to obtain exactly matching words, amongst the datasets. WNData constitutes a fair chunk of root words used in PCData as well, and this validates the fact that models

	WNPairs	CorpPairs	Matches
Hi-Bn	324537	505721	17402
Hi-Pa	260123	465140	16325
Hi-Mr	322013	555719	17698
Hi-Gu	423030	542311	17005
Hi-Sa	669911	248421	10109
Hi-Ml	353104	315234	12392
Hi-Ta	225705	248207	7112
Hi-Te	369872	431869	7599
Hi-Ne	191701	420176	11264
Hi-Ur	99803	420176	6509

Table 4.4: Total Word Pairs for both datasets and Matches among them

Source Word	**Target Word**	**Meaning**	**Cos**	**NED**	**JWS**
tadanukool	*tadanusaar*	accordingly	0.500	**0.571**	0.482
yogadaan karna	*yogadaan karane*	to contribute	0.631	**0.636**	0.593
duraatma	*dushtaatama*	evil soul	0.629	**0.700**	0.648

Table 4.5: Manual analysis of the similarity scores

trained on WNData can be used to detect cognate word pairs from any standard parallel corpora as well.

It is a well-established fact that Indian languages are spoken just like they are written, and, unlike their western counterparts, are not spoken and spelt differently. Hence, we choose to perform cognate detection using orthographic similarity methods. This very nature of Indian languages allows us to eliminate the need for using aspects of Phonetic similarity to detect true cognates. Most of the Indian languages borrow words from Sanskrit in either of the two forms - *tatsama* or *tadbhava*. When a word is borrowed in *tatsama* form, it retains its spelling, but in case of *tadbhava* form, the spelling undergoes a minor change to complete change. Before averaging the similarity scores, we tried to observe which of the three (NED, JWS, or Cos) scores would perform better for true cognates known to us in *tadbhava* form with minor spelling changes. We analysed individual word

pairs from the data and presented a small sample of our analysis in Table 4.5. We observe that NED consistently outperforms Cos and JWS for cognate word pairs and confirmed that NED based similarity is the most suited metric for cognate detection [Rama et al., 2015]. We also observe that our methodology can handle word pairs without any changes and with minor spelling changes among cognates, the total of which, constitutes a large portion of the cognates among Indian Language pairs.

In this work, we investigate cognate detection for Indian Language pairs (Hi-Bn, Hi-Gu, Hi-Pa, Hi-Mr, Hi-Sa, Hi-Ml, Hi-Ta, Hi-Te, Hi-Ne, and Hi-Ur). A pair of words is said to be Cognates if they are etymologically related; and True Cognates, if they carry the same meaning as well. We know that parallel concepts, bearing the same sense in linked WordNets, are etymologically related. We, then, use the measures of orthographic similarity to find probable Cognates among parallel concepts. We perform the same task for a parallel corpus and then train neural network models on this data to perform automatic cognate detection. We compute a list of True Cognates and release this data along with the data processed previously. We observe that Recurrent Neural Networks are best suited for this task. We observe that Hindi - Sanskrit language pair, being the closest, has the highest percentage of cognates among them. We observe that RNN, which treats the words as a sequence of characters, outperforms FFN for all the language pairs and both the datasets. We validate that Wordnets can play a crucial role in detecting cognates by combining the datasets for improved performance. We observe a minor, but crucial, increase in the performance of our models when chunks of Wordnet data are added to the data generated from the parallel corpora. It confirms that Wordnets are a crucial resource for Cognate Detection task. We also calculate the matches between word pairs from the Wordnet data and the word pairs from the parallel corpora to show that Wordnet data can form a significant part of parallel corpora and thus can be used in the absence of parallel corpora.

In the near future, we would like to use cross-lingual word embeddings, include more Indian languages, and investigate how semantic similarity could also help in cognate detection. We will also investigate the use of phonetic similarity-based methods for Cognate detection. We shall also study how our cognate detection techniques can help infer phylogenetic trees for Indian languages. We would also like to combine the similarity score by providing them weights based on an empirical evaluation of their outputs and extend

our experiments to all the Indian languages.

4.3 Cognate Detection using Siamese Networks

This work was accepted at the Young Researcher's Symposium at CoDS-COMAD 2020 Conference. In this short investigation, we propose the use of word embeddings for detecting cognates. Further, we describe the use of linked Wordnets as a dataset for building candidate cognate lists. We build our word lists using the linked IndoWordnets [Bhattacharyya, 2017] for ten Indian languages namely Hindi (Hi), Bengali (Bn), Gujarati (Gu), Marathi (Ma), Punjabi (Pa), Sanskrit (Sa), Malayalam (Ml), Tamil (Ta), Telugu (Te), Nepali (Ne). We compare words among parallel synsets and store the words which exhibit partial or full lexical similarity. Our word pair list sizes range from 656 (Hi-Ta) to 9472 (Hi-Gu). We obtain monolingual corpora from various sources which range from ~439K lines (Ta) to ~48124K lines (Hi).

4.3.1 Approaches

In all our approaches, we report the results from performing 5-fold cross-validation on WNData. We consider 70% of the data for training, 20% for testing, and the remaining 10% as validation split. The results are calculated over the test split. In the baseline lexical similarity-based approach (LSA), we use a weighted lexical similarity[4] to find out the lexical distance between the *context* of both words (score1) and their respective contexts, (*i.e.,* bag of words based score2). From each set of "bag of words", we compute similarity scores for *every word from the source side* with *every word on the target side* and average them. **The intuition for harnessing a siamese feed forward** network-based approach is that these networks perform a combined mapping of input vectors into a common target space. These networks find a function such that a simple distance in the target space approximates the "semantic" distance or distance in the meaning, from the input space. In the input layer, we provide the embeddings of a word pair. In the output layer, we use cosine similarity and a sigmoid function to predict the class of the word-pair. The network utilises cross-entropy loss as its loss function. *An important contribution of*

[4]A weighted combination of Normalized Edit Distance (NED) and Cosine Similarity (CoS) based on characters, *i.e.,* 0.75*NED + 0.25*CoS

our work is that we perform this classification based on various embedding dimensions. We build embedding models using the sub-word enriched fastText [Bojanowski et al., 2017b] approach. We show reproducible results[5] of our approach in Table 4.6.

LP	Baseline Approach			Our Approach: Siamese Feed-forward Network (SFN)								
	LSA			MEA (200 dim.)			MEA (300 dim.)			MEA (400 dim.)		
	P	**R**	**F**	**P**	**R**	**F**	**P**	**R**	**F**	**P**	**R**	**F**
Hi - Bn	0.39	0.33	0.36	0.80	0.82	0.81	0.81	0.83	**0.82**	0.81	0.80	0.81
Hi - Mr	0.47	0.21	0.29	0.81	0.83	0.82	0.83	0.83	**0.83**	0.82	0.82	0.82
Hi - Gu	0.41	0.16	0.23	0.83	0.84	0.84	0.84	0.86	**0.85**	0.84	0.83	0.84
Hi - Pa	0.29	0.07	0.11	0.78	0.79	0.78	0.82	0.82	**0.82**	0.81	0.80	0.81
Hi - Ml	0.26	0.3	0.28	0.74	0.74	**0.74**	0.73	0.73	0.73	0.73	0.73	0.73
Hi - Te	0.2	0.14	0.16	0.73	0.70	**0.71**	0.70	0.70	0.70	0.70	0.69	0.69
Hi - Ta	0.24	0.17	0.20	0.71	0.71	**0.71**	0.70	0.70	0.70	0.69	0.70	0.70
Hi - Sa	0.41	0.17	0.24	0.82	0.83	0.82	0.81	0.85	**0.83**	0.81	0.81	0.81
Hi - Ne	0.42	0.18	0.25	0.78	0.80	**0.79**	0.78	0.77	0.77	0.78	0.77	0.77

Table 4.6: Results in terms of Precision (P), Recall (R) and F-Score (F) for LSA vs. SFN for various dimension sizes.

In this investigation, we successfully utilise monolingual word embeddings and outperform approaches based on lexical similarity-based metrics. We experiment with various embedding dimensions and show that larger embedding dimensions can be used only when a large corpus size is available to help reduce the ambiguity among the distributional similarity based sense clusters. We establish a use case for the utilisation of word embeddings for the detection of cognates among Indian languages. In future, we would like to utilise cross-lingual word embeddings to project the distribution of senses into a shared space to perform the task of cognate detection.

4.4 Challenge Dataset of Cognates and False Friend Pairs

This work was accepted at the LREC 2020 Conference, and we released manually validated cognate datasets along with a manually validated false friends' dataset which proved to

[5] `http://www.cfilt.iitb.ac.in/cognateSiamese`

	Hindi (Hi)	Marathi (Mr)	Hindi Meaning	Marathi Meaning
Cognate	ank	ank	Number	Number
False Friend	shikshA	shikshA	Education	Punishment

Table 4.7: An example each of a cognate pair and a false friend pair from the closely related Indian languages Hindi (Hi) and Marathi (Mr)

be a challenging dataset when evaluated against the state-of-the-art approaches. In this work, we describe the creation of two cognate datasets for twelve Indian languages, namely Sanskrit, Hindi, Assamese, Oriya, Kannada, Gujarati, Tamil, Telugu, Punjabi, Bengali, Marathi, and Malayalam. We digitise the cognate data from an Indian language cognate dictionary and utilise linked Indian language Wordnets to generate cognate sets. Additionally, we use the Wordnet data to create a False Friends' dataset for eleven language pairs. We also evaluate the efficacy of our dataset using previously available baseline cognate detection approaches. We also perform a manual evaluation with the help of lexicographers and release the curated gold-standard dataset with this work. Dictionaries often include information about cognates and false friends, and there are dictionaries [Hammer and Monod, 1976, Prado, 1993] exclusively devoted to them.

First, we describe the digitisation of one such Cognate dictionary named, "Tatsama Shabda Kosha" and its annotation with linked Wordnet IDs. With the help of a lexicographer, we perform the digitisation of this dictionary. Further, we annotate the cognate sets from the dictionary with Wordnet synset IDs based on manual validation, where the lexicographer checks each Wordnet in the existing linked sense. Based on [Kanojia et al., 2019d]s approach, we use linked Indian Wordnets to generate true cognate data and create another cognate dataset. Additionally, we use the same Wordnet data to produce a list of False Friends and release[6] all the three datasets publicly. Our cognate sets can be utilised for lookup in phrase tables produced during Machine Translation to assess the quality of the translation system in question. They can be utilised as candidate translations for words, and our false friends' list can be utilised by language learners to avoid pitfalls during the acquisition of a second language. False Friend and Cognate detection techniques can use these lists to train automatic cognate detection approaches for Indian languages.

[6]Released Data: Github Link

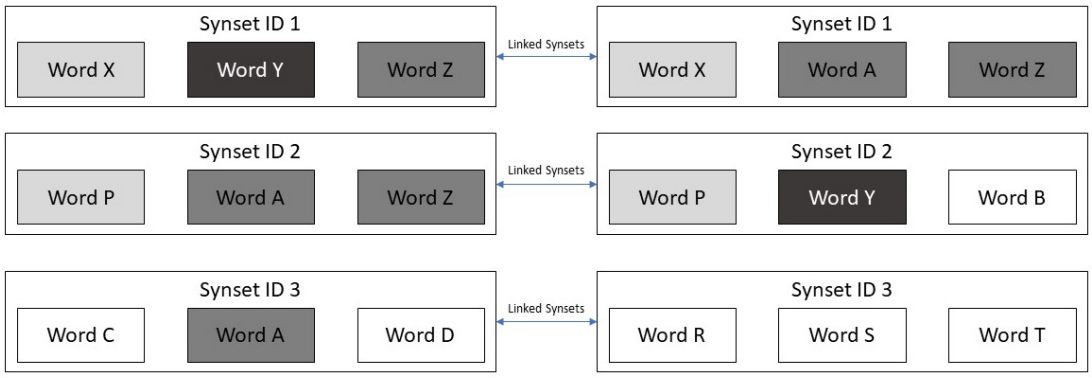


Figure 4.5: The difference between True Cognates (Word X and Word P), False Friends (Word Y) and Partial Cognates (Word A and Word Z) explained for creating our Datasets (D2 and D3).

Wu and Yarowsky [2018] release cognate sets for Romance language family and provide a methodology to complete the cognate chain for related languages. Our work releases similar data for Indian languages. Such a cognate set data has not been released previously for Indian languages, to the best of our knowledge. Additionally, we release lists of false friends' for language pairs. These cognates can be used to challenge the previously established cognate detection approaches further. Kanojia et al. [2019b] perform cognate detection for some Indian languages, but a prominent part of their work includes *manual verification and segratation* of their output into cognates and non-cognates.

Language Pair	**Hi-Bn**	**Hi-Gu**	**Hi-Mr**	**Hi-Pa**	**Hi-Sa**	**Hi-Ml**	**Hi-Ta**	**Hi-Te**	**Hi-As**	**Hi-Kn**	**Hi-Or**
Potential Candidates	50959	81834	47718	25044	33921	18084	5203	16230	14240	12480	54014
Cognates (D2)	15312	17021	15726	14097	21710	9235	3363	936	3478	4103	11894
Percent Agreement	0.9877	0.9849	0.9838	0.9754	0.9617	0.9223	0.9033	0.9553	0.9167	0.9122	0.8833
Cohen's kappa	0.7851	0.7972	0.8628	0.7622	0.7351	0.7046	0.6436	0.7952	0.7591	0.7953	0.8333

Table 4.8: Number of Potential Cognates, Number of cognates retained on both annotators' agreement [Cognates (D2)], Percent agreement among the annotators and Cohen's kappa score for each language pair in our dataset

4.4.1 Dataset Creation

We create three different datasets to help the NLP tasks of cognate and false friends' detection. In this section, we describe the creation of these three datasets for twelve Indian languages, namely Sanskrit, Hindi, Assamese, Oriya, Kannada, Gujarati, Tamil,

Language Pair	Hi-Bn	Hi-Gu	Hi-Mr	Hi-Pa	Hi-Sa	Hi-Ml	Hi-Ta	Hi-Te	Hi-As	Hi-Kn	Hi-Or
Potential Candidates	11128	10378	14430	9062	9285	5192	1018	7149	9374	3384	5011
False Friends (D3)	4380	6204	5826	4489	2193	1076	783	699	3872	926	2602
Percent Agreement	0.8912	0.9122	0.9233	0.9500	0.9018	0.8125	0.9288	0.8492	0.8825	0.9367	0.9133
Cohen's kappa	0.8827	0.8245	0.7815	0.9255	0.9452	0.9064	0.7244	0.8901	0.8432	0.8167	0.9548

Table 4.9: Number of Potential False Friends, Number of False Friend pairs retained on both annotators' agreement [False Friends (D3)], Percent agreement among the annotators and Cohen's kappa score for each language pair in our dataset

Telugu, Punjabi, Bengali, Marathi, and Malayalam.

4.4.1.1 D1 - True Cognate Sets

The first dataset is created with the help of manual annotation. We digitise the book "Tatsama Shabda Kosh" with the help of a lexicographer. The dictionary is a collection of cognates from 15 Indian languages, but focus our work on 12 languages due to the unavailability of Wordnets for the rest of the languages. The lexicographer then also annotates each cognate set with a Wordnet sense ID after manual validation of each cognate in the twelve linked Wordnets. This helps us capture an appropriate sense for the cognate word provided via the dictionary. The annotation was performed manually with the data collected in a CSV format in a text editor. By definition, cognates are supposed to spell and mean the same. Our manual annotation using the Wordnet IDs helps provide an appropriate sense to each cognate set in the dataset and thus can help automatic cognate detection techniques utilise the synset information.

This dataset consists of 1021 cognate sets with a total of 12252 words. The book consisted of a total of 1556 cognate sets, but during manual validation, 535 were found to be partial cognates and have been ignored from this dataset. The percentage share of parts-of-speech categories for Wordnet annotated cognates released is shown in Table 4.10. Partial cognates, as previously explained, mean differently in different contexts. Keeping the applications of our dataset in mind, we ignore the inclusion of partial cognates.

4.4.1.2 D2 - True Cognate Pairs via IndoWornet

In their work, Kanojia et al. [2019d] identify IndoWordnet [Bhattacharyya, 2017] as a potential resource for the task of cognate detection. They utilise deep neural network-

	Nouns	Verbs	Adjectives	Adverbs
D1	78.20	0.06	19.00	0.60
D2	76.35	2.41	20.11	1.10

Table 4.10: The percentage share of parts-of-speech categories in cognate datasets D1 and D2

based approaches to validate their approach for cognate detection. We build this dataset using a simple orthographic similarity-based approach from the IndoWordnet dataset. Our approach combines Normalized Edit Distance (NED) [Nerbonne and Heeringa, 1997] and Cosine Similarity (CoS) [Salton and Buckley, 1988] between words. We compare synset words from every language pair using NED and populate a list of cognate sets where NED score is 0.7 and above. Similarly, we populate another list of cognate sets from every language pair using a shingle (n-gram) based Cosine Similarity with the same threshold. Due to the different methods using which NED and CoS similarity techniques compute scores, both NED and CoS output a different number of word pairs. We choose a common intersection of cognate pairs from among both the lists and populate a final 'potential cognate set' for eleven Indian language pairs. We take the help of two lexicographers and manually validate this output. We are aided by two lexicographers for each of the language pairs of Hindi (source) - (target) 'other Indian languages'[7] Each lexicographer was requested to annotate whether the given word pair is cognate or not, given the context, which contained the definition of the concept and an example sentence. We retain in the final dataset, cognate pairs, which were marked to be true cognates by both annotators. We provide the language pair wise data statistics, per cent agreement, and Cohen's Kappa (IAA) values for the lexicographers' annotation in Table 4.8. The percentage share of POS categories for annotated cognates is shown in Table 4.10.

4.4.1.3 D3 - False Friends' Pairs

The creation of such a False Friends' dataset is another one of our novel contributions in this work. We search for false friend candidate pairs by searching for commonly spelt

[7]We intended to isolate the lexicographers of clues from other language cognate pairs. Hence, we create cognate data in language pairs.

words through the non-parallel synsets. These candidate pairs initially included partial cognates as well, since words which are commonly spelt and belong to different senses, could occur in both the contexts. We further prune this list by ensuring that these commonly spelt words do not occur in parallel synsets and also do not occur in the corresponding linked synset on either the source or the target side. Figure 4.5 explains our heuristic where Word Y is a False Friend among Synsets 1 and 2, Word X and Word P are True Cognates chosen for D2, and Word A / Word Z are ignored because they are partial cognates. Once we find out such unique False Friend pairs, which are exact matches in spelling but do not occur in parallel synsets, on either side, we populate our list of false friend pairs. We populate this list for eleven language pairs where Hindi is always the source language. It should be noted that false friends do not follow transitivity, *i.e.,* if A and B are false friend pairs in languages X and Y, and A and C are false friends in X and Z, it is not necessary that B and C would be false friends. Hence, we populate eleven different false friend lists and take lexicographers to help for each language pair to validate this output manually. Post- manual validation, we choose to retain the false friend pairs, which were annotated as false friends' by both the annotators. We report the statistics for D3 in Table 4.9, which include the number of potential false friend candidates, False friend pairs after manual validation of these potential candidates, Percent agreement among both the annotators and Cohen's Kappa (IAA) score for the annotation performed.

4.4.2 Experiment Setup for Evaluation

We evaluate the challenge posed by our datasets using the tasks of automatic cognate detection and false friend detection, based on previously available approaches. In this section, we describe the task setup and approaches which show the challenges posed by these tasks. We also discuss how our dataset is a challenging dataset for these tasks, and better approaches are needed to tackle the problems posed by a morphologically richer dataset of cognates.

We combine D1 and D2 based on Wordnet Sense IDs and remove duplicates to form a single dataset of true cognates, which we evaluate through the task of cognate detection. We use various approaches to perform the cognate detection task, *viz.,* baseline cognate detection approaches like orthographic similarity-based, phonetic similarity-based, phonetic vectors with Siamese-CNN based proposed by Rama [2016], and deep neural network-

based approaches proposed by Kanojia et al. [2019d]. We use the same hyperparameters and architectures, as discussed in these papers. For the Orthographic similarity-based approach, we use the orthographic similarity between words as a feature. For the Phonetic similarity-based approach, we compute the phonetic similarity between two words using phonetic vectors available via the IndicNLP Library[8]. To classify cognate pairs, we use a simple feed-forward neural network with the respective feature scores passed to a fully connected layer with ReLU activations, followed by a softmax layer (in the first two approaches). We replicate the best reported systems from Rama [2016], *i.e.,* Siamese Convolutional Neural Network with phonetic vectors as features. To replicate Kanojia et al. [2019d]'s approach, we use the Recurrent Neural Network architecture with a combination of Normalized Edit Distance, Cosine Similarity, and Jaro-Winkler Distance as reported in their paper. We have already discussed the manual validation of our datasets in the previous section, which allows us to create a more curated dataset. We use the computational approaches on this curated dataset post manual validation. For the training dataset, we use the data provided by Kanojia et al. [2019d] and create training and validation sets with an 80-20 split. We then test the aforementioned approaches on our dataset.

For the False Friends detection task, since no such dataset is available for Indian languages, we annotate the data created by us with positive labels and divide it into train and test sets. We then add true cognates to the training dataset with negative labels since intuitively, they are the best candidates for misclassification due to common spellings just like false friends do, but in case of true cognates, they also mean the same. We test our false friends' dataset using similar approaches with baseline features like orthographic similarity, and phonetic similarity. We use a simple feed-forward neural network as the classifier with the respective feature scores passed to a fully connected layer with ReLU activations, followed by a softmax layer. To ensure the learning algorithm has 'context' available to decipher the meaning among false friends, we use the notion of distributional semantics and employ a word vectors based approach proposed by Castro et al. [2018]. We use their approach to test the efficacy of our dataset and show better approaches need to be devised for morphologically richer languages. Since the approach proposed by Castro et al. [2018] requires monolingual word embeddings to be used, we train the

[8]https://anoopkunchukuttan.github.io/indic_nlp_library/

monolingual word embeddings using the standard Wikimedia dumps. We, then, train twelve monolingual word embedding models for each Indian language under consideration. In the next section, we discuss the results of our dataset evaluations.

Approaches	Hi-Bn	Hi-As	Hi-Or	Hi-Gu	Hi-Mr	Hi-Pa	Hi-Sa	Hi-Ml	Hi-Ta	Hi-Te	Hi-Kn
Orthographic Similarity	0.36	0.34	0.38	0.25	0.29	0.21	0.24	0.28	0.20	0.16	0.19
Phonetic Similarity	0.42	0.38	0.39	0.29	0.32	0.24	0.25	0.31	0.24	0.22	0.25
Rama et. al. (2016)	0.65	**0.71**	0.61	0.67	**0.72**	0.47	0.53	0.62	**0.53**	**0.65**	0.57
Kanojia et. al. (2019)	**0.68**	**0.71**	**0.62**	**0.75**	**0.72**	**0.73**	**0.72**	**0.66**	**0.53**	0.63	**0.58**

Table 4.11: Results of the Cognate Detection Task (in terms of F-Scores) for D1+D2. We use the same architecture, features and hyperparameters as discussed in the papers for Rama [2016] and Kanojia et al. [2019d]; and observe that these systems do not perform as well on our dataset, as shown in the papers.

4.4.3 Results of Our Evaluation

In table 4.11, we show the results for the cognate detection task. We observe that on our combined cognate dataset, the current approaches do not perform well. These approaches have reported better performance for their own datasets. In most of the cases (Language pairs), Kanojia et al. [2019d]'s approach performs better, but for the Hindi-Telugu language pair, Rama [2016]'s approach performs better. Although both the approaches perform the same for Hi-As, Hi-Mr, and Hi-Ta language pairs, the scores are still lower than what has been previously reported. We believe that these approaches perform well when on a limited dataset, moreover, when the dataset consists of words which are stripped on morphological inflections. NLP tasks such as Machine Translation and Cross-lingual Information Retrieval do not use synthetic data, which is stripped of morphological infor-

Language Pairs	Hi-Bn	Hi-As	Hi-Or	Hi-Gu	Hi-Mr	Hi-Pa	Hi-Sa	Hi-Ml	Hi-Ta	Hi-Te	Hi-Kn
Orthographic Similarity	0.36	0.45	0.49	0.51	0.53	0.44	0.52	0.24	0.29	0.30	0.50
Phonetic Similarity	0.60	**0.66**	**0.67**	0.62	0.59	0.69	0.61	0.54	0.48	0.50	0.57
Castro et. al. (2018)	**0.66**	0.64	0.59	**0.65**	**0.69**	**0.73**	**0.72**	**0.65**	**0.52**	**0.69**	**0.64**

Table 4.12: Results of the False Friends' Detection Task (in terms of F-Scores) for D3. We use the same architecture, features and hyperparameters as discussed in the paper by Castro et al. [2018] and observe that these systems do not perform as well on our False Friends' dataset.

mation. If the cognate detection task has to be a part of a pipeline aiding the NLP tasks, then approaches that perform the task should be robust enough to tackle a dataset such as ours. Hence, we claim our dataset to be a more challenging dataset, which should help develop better approaches.

In table 4.12, we report the results for the task of false friends' detection. We observe that the approach proposed by Castro et al. [2018] does not perform as well as it does for Spanish and Portuguese, as reported previously. We believe that this approach inherently lacks the linguistic intuition which is needed for the false friends' detection task. Please recall that False friends are word pairs that spell the same but do not mean the same. However, for the approach to perform well, monolingual embeddings may not be an appropriate feature. Cross-lingual word embeddings project monolingual word embeddings into a shared space and thus should be able to decipher the 'meaning' or the 'sense' of two different words better when they belong to different languages. Given the recent advancements in word representation models, cross-lingual word embedding based models should be employed for such a task. Let this also be noted that we do not propose a new approach for the task of False friends' detection and hence do not perform any experimentation with cross-lingual word embeddings. However, Merlo and Rodriguez [2019] show that cross-lingual word embeddings obtained using the VecMap [Artetxe et al., 2016] approach have shown promise and can be used to obtain a semantic comparison between two words from different languages.

In this investigation, we describe the creation of a challenging dataset of true cognates which encompasses of cognates from two different sources. First, we digitise a cognate dictionary and annotate it with Wordnet Sense IDs for twelve Indian languages to create Dataset 1 (D1). We also use linked Indian Wordnets to create a true cognate dataset, as described above. For both the datasets, we ensure a quality check with the help of manual validation. We report the per cent agreement and Inter-annotator agreement for D1 and D2. With D2, we retain the cognate pairs, which were marked to be cognates by both the annotators. Two annotators aided us for each language pair. Additionally, we release a curated list of False friends for eleven language pairs where the Hindi language is always the source, and other Indian languages are the target languages. We evaluate the efficacy of all these datasets using previously available approaches for the tasks of Cognate and False Friends' detection. We show that these approaches do not perform

as well on our dataset, given the same hyperparameters and settings as described in their papers. We discuss these results in the previous section. We also believe that this work provides a challenging gold-standard dataset for the tasks for Cognate and False Friends' detection, which can also be used to aid the NLP tasks of Machine Translation, Cross-lingual Information Retrieval, and Computational Phylogenetics. We hope better approaches are developed for these tasks which can perform well on our challenge dataset.

In the near future, we shall include partial cognates in our dataset creation approach and release another dataset on the same repository. Partial cognates mean different given different contexts and can confuse an NLP task. Hence, we believe it is also important to have a challenging dataset for partial cognates as well, which can be evaluated via the bilingual bootstrapping approach described by Frunza [2006]. We would also like to evaluate our dataset on other NLP tasks and report its efficacy in aiding the tasks of MT, CLIR, Cross-lingual Question Answering, *etc.*

4.5 Summary

In this chapter, we have described in detail our contributions to perform the task of cognate detection for Indian languages starting with a preliminary investigation which uses manual validation but establishes a useful negative result. We explained our hypothesis about how linked Wordnets can be used for the task of cognate detection and describe our experiments with their outcomes. We, then, experimented with 'Siamese' networks and showed how the dimensionality of, word embeddings used, plays a crucial role in the classification of cognates. We then described our efforts to create challenge datasets for the task of cognate detection and false friend's using a cognate dictionary and linked Indian language Wordnets. In the next chapter, we shall describe our proposed approaches which use different novel feature sets for the task of cognate detection.

Chapter 5

Cognate Detection Task: Cross-lingual and Other Novel Features

In this thesis so far, we have established motivation for performing the task of cognate detection. The previous chapter detailed our initial investigations towards the same. In this chapter, we discuss a significant portion of our contributions to the detection of cognates among Indian languages. Additionally, we perform an investigation to detect cognates among Indian and European languages which is also described in this chapter. We will, now, describe multiple investigations to detect cognates using various feature sets and conclude this chapter.

5.1 Harnessing Cross-lingual Features for Cognate Detection

This investigation is originally titled as, "Harnessing Cross-lingual Features to Improve Cognate Detection for Low-resource Languages". In this crucial investigation, we demonstrate the use of cross-lingual word embeddings for detecting cognates among fourteen Indian Languages. Our approach introduces the use of context from a knowledge graph to generate improved feature representations for cognate detection. We, then, evaluate the impact of our cognate detection mechanism on neural machine translation (NMT),

as a downstream task. We evaluate our methods to detect cognates on a challenging dataset of twelve Indian languages, namely, Sanskrit, Hindi, Assamese, Oriya, Kannada, Gujarati, Tamil, Telugu, Punjabi, Bengali, Marathi, and Malayalam. Additionally, we create evaluation datasets for two more Indian languages, Konkani and Nepali. We observe an improvement of up to 18% points, in terms of F-score, for cognate detection. Furthermore, we observe that cognates extracted using our method help improve NMT quality by up to 2.76 BLEU.

India is a multilingual, multi-script country with 22 scheduled languages and 12 written script forms primarily belonging to 6 different language families. More than a billion people use these languages as their first language. A significant amount of news and information is found on the web in these languages, which is inaccessible to people of other regions within the country. Most of the Indian language texts found online have several words that have originated from Sanskrit, Persian, and English. While, in many cases, one might argue that such occurrences do not belong to an Indian language, the frequency of such usage indicates a wide acceptance of these foreign language words as Indian language words. In numerous cases, these words also are morphologically altered as per the Indian language morphological rules to generate new variant words. A key question that we try to answer with this work is,

> *"Can semantic information from Cross-lingual models be leveraged to improve cognate detection amongst low-resource languages?"*

We hypothesize that utilizing cross-lingual features by employing existing resources such as wordnets and cross-lingual embeddings should help improve cognate detection. In this work, we utilize the semantic information from cross-lingual word embeddings. Cross-lingual word embeddings can be obtained by training monolingual embeddings for individual languages and then projecting them in a shared space using a bilingual dictionary. In the absence of such a bilingual dictionary for low-resource languages, adversarial training can be used over identical words to generate the projections. We build cross-lingual models for thirteen language pairs with Hindi as the source (L1) and thirteen target Indian languages (L2). We use the context information from a knowledge graph to build the context dictionaries for each pair. The cross-lingual models help us obtain embeddings for the word-pair and the respective context dictionaries, from a shared space. We hypothesize that using this approach shall provide a more accurate semantic measure

for the detection of cognate pairs. The use of orthographic and phonetic similarity-based methods to perform the same task provides us with baselines for a comparative evaluation.

A motivation to investigate this task for low-resource Indian languages stems from the fact that most of the Indian languages borrow cognates or "loan words" from the Sanskrit language. It is, for the most part, considered a historical antecedent of almost all the Indian languages. Indo-Aryan languages like Hindi, Bengali, Gujarati, Punjabi borrow from Sanskrit. They borrow many lexical forms and language properties from Sanskrit. Dravidian languages are highly agglutinative like Sanskrit and morphologically rich, which makes them tough to parse computationally. Marathi and Hindi suffer from the same ailment even though Hindi is not considered as agglutinative as Marathi, but it does exhibit compounding[1] which makes it, yet again, difficult to parse for CLIR and MT systems, and to detect cognates based solely on orthographic similarity. Given that CLIR and MT are usually based on a full-form lexicon, one of the possible issues in the generation of cognates concerns the similarity of words in their root form vs the similarity in their lexical form. For example, the Sanskrit word "*matri*" and the English word "*Mother*" are known cognates from the Proto-Indo-European language family where the root and the meaning are identical, but the lexical form is considerably different. Our approach handles such cases by inculcating the sub-word information while building the embeddings and helps reduce the out-of-vocabulary (OOV) words, which have proven to be a challenge for well-established CLIR systems [Udupa et al., 2009].

Dijkstra et al. [2010] show how cross-linguistic similarity of translation equivalents affects bilingual word recognition, even in tasks manually performed by humans. They discuss how a need for recognizing semantic similarity arises for non-identical cognates, based on the reaction time from human annotators. Similarly, Merlo and Andueza Rodriguez [2019] show that cross-lingual models exhibit the semantic properties of for bilingual lexicons despite their structural simplicities, which leads us to perform our investigation for low-resource Indian languages. All of the previous approaches discussed above, lack the use of an appropriate cross-lingual similarity-based measure and do not work well for Indian languages as shown through this work. This work discusses the quantitative and qualitative results using our approach and then, applies our output to different neural machine translation architectures.

[1]Compounding means when two or more words or signs are joined to make a longer word or sign.

Language Pair	Hi-Bn	Hi-Gu	Hi-Mr	Hi-Pa	Hi-Sa	Hi-Ml	Hi-Ta	Hi-Te	Hi-As	Hi-Kn	Hi-Or	Hi-Ne*	Hi-Ko*
Cognates	15312	17021	15726	14097	21710	9235	3363	936	3478	4103	11894	2560	11295
Non-Cognates	16119	15057	15983	15166	23029	8976	4005	1084	4101	3810	13027	1918	9826

Table 5.1: Number of cognates and non-cognates for each language pair in the dataset. Hi-Ne* and Hi-Ko* were generated via reproducing the previous approach [Kanojia et al., 2020a].

Language	Hi	Bn	Gu	Mr	Pa	Sa	Ml	Ta	Te	Ne	As	Kn	Ko	Or
Corpus Size	48142K	1564K	439K	520K	505K	553K	495K	909K	1023K	706K	504K	159K	214K	744K
STTR (n=1000)	0.5821	0.5437	0.4587	0.6108	0.4314	0.5350	0.7339	0.6411	0.4950	0.4883	0.5968	0.5338	0.5614	0.4160

Table 5.2: Corpus Statistics where corpus size is approximate number of lines, and STTR is the moving average type-token ratio on a windows of 1000 sentences.

5.1.1 Dataset and Experimental Setup

In this section, we describe our primary dataset for the cognate detection task. We also describe the datasets used for building cross-lingual word embedding models, and the parallel corpora used for the Neural Machine Translation (NMT).

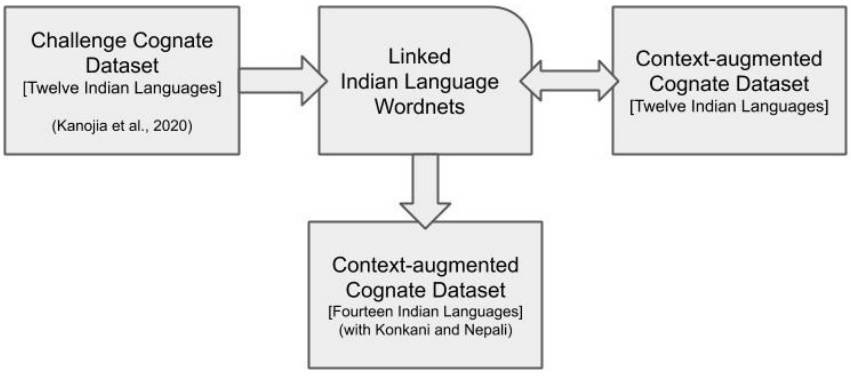


Figure 5.1: Dataset Augmentation with Context and Two Language Pairs using IndoWordnet.

For our experiments, we use the publicly released challenge dataset [Kanojia et al., 2020a] of cognates. This dataset provides labelled cognate and non-cognate pairs for twelve Indian languages namely, Sanskrit (Sa), Hindi (Hi), Assamese (As), Oriya (Or), Kannada (Kn), Gujarati (Gu), Tamil (Ta), Telugu (Te), Punjabi (Pa), Bengali (Be), Marathi (Mr), and Malayalam (Ml). We reproduce their approach to add two more languages, Konkani (Ko) and Nepali (Ne), to this dataset. For building context dictio-

naries, we use linked Indian language wordnets [Bhattacharyya, 2017] and concatenate the concept definition and example sentences. We remove stop words from the context dictionaries and append them with their respective word pairs. The lexical overlap between the language pairs ranges from 13% (for Hi-Te) to only 23% (Hi-Mr). Figure 5.1 shows an accurate description of the dataset creation process. The cognate dataset statistics are described in Table 5.1.

5.1.1.1 Monolingual Corpora for Word Embeddings

The dataset for training cross-lingual models is obtained from various sources. Word embeddings require a large number of monolingual corpora for efficient training of a usable model with high accuracy. We extract corpora for these fourteen Indian Languages from various sources and collect them in a single repository. We extract Wikimedia dumps[2] for all languages and add ILCI corpora for these languages to each of them. For Hindi, Marathi, Nepali, Bengali, Tamil, and Gujarati we add crawled corpus of film reviews and news websites[3] to their corpus. For Hindi, we also add HinMonoCorp 0.5 [Bojar et al., 2014] to our corpus adding approximately 44 million sentences. For Sanskrit, we download a raw corpus of proses[4] and add it to our corpus. Training corpus statistics (approximate number of total lines) are shown in Table 5.2.

5.1.1.2 Parallel Corpora for NMT

To validate the application of cognates for the Machine Translation task, we choose the Neural Machine Translation setting and use the Indian Languages Corpora Initiative (ILCI) Phase 1 corpus [Jha, 2010a]. This corpus contains approximately 50K parallel sentences across 11 languages (English and 10 Indian Languages), from health and tourism domains. For every language pair, the parallel corpus was split up into a training set of 46,277 sentences, a test set of 2000 sentences and development set of 500 sentences. The train, test and development splits were ensured to be parallel across all language pairs involved. The language pair intersection for our cognate detection work and this parallel corpus limited our MT experimentation to the following languages namely, Hindi (Hi), Punjabi (Pa), Bengali (Bn), Gujarati (Gu), Marathi (Mr), Tamil (Ta), Telugu (Te)

[2]Wikimedia Dumps; as on April 22, 2020

[3]Additional Monolingual Corpus

[4]JNU Sanskrit Proses Corpus

and Malayalam (Ml). We keep Hi as the source and remaining languages as the target languages for our experiments. We describe the experimental setup for our task below.

5.1.1.3 Cross-lingual Word Embedding Methodologies

Using the monolingual corpora described above, we build monolingual word embeddings using the FastText library[5] [Bojanowski et al., 2017b] since it takes sub-word information into account, which is beneficial for a task such as ours where sub-words play an important role, and spelling variations can lead to different meanings. We do not use BERT [Devlin et al., 2018] or ELMo [Peters et al., 2018] for word embeddings as their pre-trained models are not trained on transliterated corpora. We choose FastText to train Skipgram word embedding models (100 dimensions) for each language using the following hyperparameters - 15 epochs with 0.1 as the learning rate. We use two characters (bi-gram) as the size of each sub-word for capturing the maximum number of sub-words.

We use three different methodologies for training the cross-lingual word embedding models on all the language pairs with Hindi as a pivot language (Hi-Mr, Hi-Bn and so on). The **first methodology** uses the supervised method of MUSE [Lample et al., 2018][6] which utilizes a manually curated bilingual lexicon[7] for alignments. We use Hindi as a pivot language due to the ease of computation and availability of resources (Corpora and WordNet size). We use the monolingual models described above and train 13 cross-lingual word embedding models (thirteen language pairs over 100 dimensions) using this approach.

The **second cross-lingual methodology** uses VecMap [Artetxe et al., 2018a], which utilizes the monolingual models created above. VecMap uses an optional normalization feature while it builds the mappings between any two monolingual models. It performs orthogonal transformation and maps semantically related words, similar to MUSE, which was used in our first approach for building cross-lingual models. Additionally, it also reduces the dimensions of the embeddings models, which, is optional. We train it using the same hyperparameters as described above, for consistency while evaluating. We used the supervised approach for training these models as well, and the training dictionary was similar to the one provided to the MUSE method. We obtain

[5]FastText - GitHub

[6]MUSE - GitHub

[7]Bilingual Lexicon

thirteen models, one for each language pair, using VecMap. The **third methodology** utilizes contextual embeddings which have shown to outperform the conventional word embeddings based models for many tasks [Devlin et al., 2018]. We choose the most recent methodology for building a single cross-lingual model for all the languages. XLM-R [Conneau et al., 2019] uses previously proposed approaches of XLM [Lample and Conneau, 2019] and RoBERTa [Liu et al., 2019] to attain a very high performing cross-lingual model, especially for low-resource languages. We use our transliterated corpora described above and concatenate it into a single large corpus required for training the model. We then use the unsupervised training method of XLM-R and train a model over six days and a couple more hours with a reduced batch size, which allowed us to be able to train the model under a week's time. The model was then used to output vectors akin to the other cross-lingual models trained above for each word and their contexts. In this approach, however, we did not need a dictionary for the cross-lingual mapping strategy, unlike the two previous approaches. XLM-R generated a single model for all the languages trained on two V100 GPUs over almost seven days time.

To put it more concisely, we trained cross-lingual models using three different methodologies (MUSE, VecMap and XLM-R) where the cross-lingual mapping obtained for MUSE and VecMap were generated via the monolingual embeddings, as described above. We obtained thirteen models using each of these two methods. A single cross-lingual model was, however, trained using XLM-R and used for the third cross-lingual approach whose training methodology has been described above.

5.1.2 Approaches

We use various approaches to perform the cognate detection task, *viz.,* baseline cognate detection approaches like orthographic similarity-based, phonetic similarity-based, phonetic vectors with Siamese-CNN based proposed by Rama [2016], and Recurrent neural network-based approach proposed by Kanojia et al. [2019d]. We use the same hyperparameters and architectures, as discussed in these papers. We describe each of these feature sets in this section.

5.1.2.1 Weighted Lexical Similarity (WLS)

The Normalized Edit Distance (NED) approach computes the edit distance [Nerbonne and Heeringa, 1997] for all word pairs in our dataset. Each of the operations has unit cost (except that substitution of a character by itself has zero cost), so NED is equal to the minimum number of operations to transform 'word a' to 'word b'. We use a similarity score provided by NED, which is calculated as (1 - NED Score). We combine NED with q-gram distance [Shannon, 1948] for a better similarity score. The q-grams ('n-grams') are simply substrings of length q. This distance measure has been applied previously for various spelling correction approaches [Owolabi and McGregor, 1988, Kohonen, 1978]. We provide different weights to both the measures and observe that providing 75% weight to NED and 25% weight to QD provides us with the best F-scores over this dataset. We compared F-scores over different weight balances, *i.e.,* 50-50%, 25-75%, and chose to report 75-25%, after empirical evaluation. Additional results with the weightages are provided in the appendix. For any word pair with words p and q, it is as follows:

$$WLS_{pq} = (NED_{pq} * 0.75) + (QD_{pq} * 0.25) \tag{5.1}$$

Now that this approach can be used to compute a score between each word pair, we use it to find two scores, which are used as features - 'word-pair similarity' and 'contextual similarity'. Each candidate word-pair generates a score, *i.e.,* score1, and the average of scores among all words in the context dictionary generates another score, *i.e.,* score2, which are normalized as follows:

$$\begin{aligned} S_1 &= score1/\left(score1 + score2\right) \\ S_2 &= score2/\left(score1 + score2\right) \end{aligned} \tag{5.2}$$

We use S_1 and S_2 as features for this orthographic similarity-based baseline approach.

5.1.2.2 Phonetic Vectors and Similarity (PVS)

The IndicNLP Library provides phonetic features based vector for each character in various Indian language scripts. We utilize this library to compute a feature vector for each word by computing an average over character vectors. We compute vectors for both words in the candidate cognate pairs (PV_S and PV_T) and also compute contextual vectors (PCV_S and PCV_T) by averaging the vectors for all the context dictionaries on each

side (source and target), generating a total of four vectors. We also calculate the cosine similarity among PV_S and PV_T, and among PCV_S and PCV_T to generate two similarity scores (P_{S1}, and P_{S2}) which are normalized using (2) and, additionally, used as features during classification. It should be noted that using phonetic vectors and their similarity scores has already been proposed in the previous literature [Rama, 2016] for a cognate detection task, and we do not claim this approach to be our novel contribution.

5.1.2.3 Cross-lingual Vectors & Similarity

As described above, we train cross-lingual embedding models by aligning two disjoint monolingual vector spaces through linear transformations, using a small bilingual dictionary for supervision [Doval et al., 2018, Artetxe et al., 2017]. The first two approaches for training cross-lingual methods use this dictionary for supervision. In our novel approach, we propose the use of vectors from the cross-lingual embedding models trained on Indian language pairs. We obtain vectors for word-pairs (WV_S and WV_T) and averaged context vectors (CV_S and CV_T) for the context dictionary, to create feature sets. We obtain vectors for each candidate pair and their context using all the three cross-lingual methodologies.

Additionally, we use angular cosine similarity [Cer et al., 2018] scores for word pairs and their contexts. Angular similarity distinguishes nearly parallel vectors much better as small changes in vector values yield considerable distances. For each word pair vector and it's context vectors, we compute the 'word-pair similarity' and 'contextual similarity'. We use *arccos* to obtain angular cosine similarity (asim) among vectors 'u' and 'v', as shown below:

$$asim(u, v) = \left(1 - arccos\left(\frac{u.v}{\|u\|\|v\|}\right)/\pi\right) \tag{5.3}$$

Each candidate word-pair generates a score, *i.e.,* score1, and the average of scores among all words in the context dictionary generates another score, *i.e.,* score2, which are also normalized using (2).

5.1.2.4 Classification Methodology

We pose the task of detecting cognates as a binary classification problem. We employ both classical machine learning-based models and a simple feed-forward neural network. To compare our work with the previously proposed approaches, we replicate the best-reported systems from Rama [2016], *i.e.,* Siamese Convolutional Neural Network with phonetic vectors as features and also replicate Kanojia et al. [2019d]'s approach which uses a Recurrent Neural Network architecture with a weighted lexical similarity (WLS) as a feature set. The input to our classifiers is the feature sets described above for each candidate pair. The candidates are the complete data described in Table 5.1. Cognates from Table 5.1 are labelled positive, and non-cognates are labelled negative. *We perform 5-fold stratified cross-validation, which divides the data into train and test folds, randomly.* An architecture diagram for our classification approach is shown in Figure 5.2.

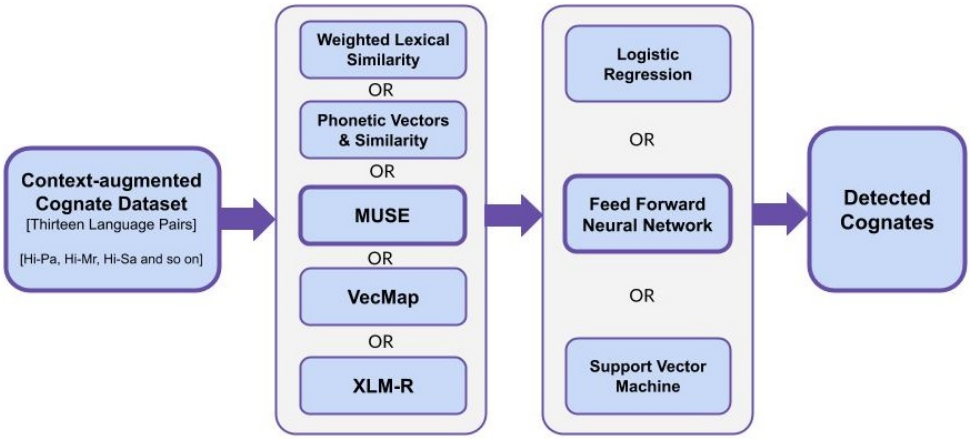


Figure 5.2: Cognate Detection task with different feature sets and classification approaches.

Among the classical machine learning models, we use Support Vector Machines (SVM) and Logistic Regression (LR). We experiment with the use of both linear SVMs and kernel SVMs (Gaussian and Polynomial). We perform a grid-search to find the best hyper-parameter value for C over the range of 0.01 to 1000. We deploy the Feed Forward Neural Network (FFNN) with one hidden layer. We perform cross-validation with different settings for activation function (tanh, hardtanh, sigmoid and relu) and the hidden layer dimension in the network (30, 50, 100, and 150). We use binary cross-entropy as the optimization algorithm. Finally, we choose the hyper-parameter configuration with the best validation accuracy. We train the model with the selected configuration with an

initial learning rate of 0.4, and we halve the learning rate when the error on the validation split increases. We stop the training once the learning rate falls below 0.001. We perform our experiments with the feature sets (Orthographic (WLS), Phonetic (PVS), and three different cross-lingual embeddings based feature sets) described above for all the thirteen language pairs. We also perform an ablation test with various feature sets and report the results for the best feature combination in the next section. The results of our classification task can be seen in Table 5.3 and are discussed in the next section, in detail.

5.1.2.5 Cognate-aware Neural Machine Translation (NMT) Task

For the NMT task, we use the OpenNMT-Py toolkit [Klein et al., 2017] to perform our experiments. We use a Bidirectional RNN Encoder-Decoder architecture with attention [Bahdanau et al., 2014]. We choose three stacked LSTM [Hochreiter and Schmidhuber, 1997b] layers in the encoder and decoder. The hidden-size of the model was 500 units. We optimize using stochastic gradient descent at an initial learning rate of 1, and a batch-size of 1024 units. Training is done for 150,000 steps of which the initial 8,000 steps are for learning rate warm-up. We use Byte-pair encoding (BPE) [Sennrich et al., 2015] merge operations, initially, in an endeavour to find the best baseline model with an optimal number of merge operations. We observe that performing 2500 merge operations provided us with best BLEU [Papineni et al., 2002] scores, for most of the language pairs. We report the best results here, and a complete set of merge operation results in the supplementary. We call this the NMT-BPE Baseline.

To validate our hypothesis that our approach can help the NMT task, we *inject the cognates detected using our approach* to the parallel corpus for their respective language pairs, as single word sentences. Lexical Dictionaries have previously been used to improve the MT task [Arthur et al., 2016, Han et al., 2019]. However, a decent improvement in their BLEU scores is observed when their lexicon sizes are approximately around 1M tokens [Arthur et al., 2016]. Our detected cognate list size varies from 930 cognates (Hi-Te) to 15834 (Hi-Mr). Due to the addition of more parallel instances to the corpus, the vocabulary size for NMT increases. Hence, we experiment further by varying the BPE merges, in a close range, to the optimal merge point obtained earlier. We report the results of the best optimal merge setting, for both NMT-BPE Baseline model and the

cognate injected NMT-BPE model, in the section below. A more detailed set of results for all the merge operations is available in a separate table (Table 5.5).

5.1.3 Results and Discussion

	Baseline Approaches									Cross-lingual Embeddings based Approaches									Ablation Test		
LP	WLS w/ FFNN			PVS w/ Siamese CNN (Rama, 2016)			WLS w/ RNN (Kanojia et al., 2019)			XLM-R w/ FFNN			MUSE w/ FFNN			VecMap w/ FFNN			MUSE + WLS w/ FFNN		
	P	R	F	P	R	F	P	R	F	P	R	F	P	R	F	P	R	F	P	R	F
Hi-Bn	0.51	0.28	0.36	0.68	0.62	0.65	0.67	0.69	0.68	0.81	0.76	**0.78**	0.77	0.75	0.76	0.72	0.74	0.73	0.80	0.75	0.77
Hi-As	0.48	0.26	0.34	0.72	0.71	0.71	0.72	0.70	0.71	0.70	0.72	0.71	0.80	0.75	**0.77**	0.74	0.73	0.73	0.84	0.75	**0.79**
Hi-Or	0.51	0.30	0.38	0.65	0.58	0.61	0.66	0.58	0.62	0.65	0.61	0.63	0.72	0.68	**0.70**	0.67	0.70	0.68	0.81	0.69	**0.75**
Hi-Gu	0.43	0.16	0.23	0.70	0.65	0.67	0.81	0.71	0.76	0.80	0.73	0.76	0.80	0.84	**0.82**	0.77	0.74	0.75	0.83	0.85	**0.84**
Hi-Ne	0.50	0.16	0.24	0.72	0.84	0.78	0.78	0.73	0.75	0.75	0.75	0.75	0.86	0.83	**0.84**	0.78	0.73	0.75	0.86	0.83	**0.84**
Hi-Mr	0.51	0.20	0.29	0.70	0.68	0.69	0.74	0.70	0.72	0.76	0.71	**0.73**	0.70	0.73	0.71	0.71	0.71	0.71	0.72	0.73	0.72
Hi-Ko	0.47	0.24	0.32	0.63	0.63	0.63	0.63	0.59	0.61	0.66	0.58	0.62	0.69	0.73	**0.71**	0.61	0.60	0.60	0.70	0.75	**0.72**
Hi-Pa	0.28	0.17	0.21	0.51	0.44	0.47	0.76	0.72	0.74	0.75	0.71	0.73	0.83	0.78	**0.80**	0.71	0.74	0.72	0.83	0.78	**0.80**
Hi-Sa	0.34	0.19	0.24	0.55	0.51	0.53	0.73	0.71	0.72	0.75	0.70	0.72	0.77	0.76	**0.76**	0.73	0.71	0.72	0.80	0.77	**0.78**
Hi-Ml	0.49	0.20	0.28	0.59	0.66	0.62	0.66	0.66	0.66	0.72	0.63	0.67	0.76	0.71	**0.73**	0.69	0.71	0.70	0.77	0.71	**0.74**
Hi-Ta	0.22	0.19	0.20	0.49	0.58	0.53	0.49	0.58	0.53	0.63	0.51	0.56	0.72	0.68	**0.70**	0.66	0.72	0.69	0.72	0.70	**0.71**
Hi-Te	0.18	0.15	0.16	0.60	0.71	0.65	0.62	0.71	0.66	0.65	0.70	0.67	0.70	0.72	**0.71**	0.67	0.67	0.67	0.73	0.72	**0.72**
Hi-Kn	0.19	0.18	0.18	0.54	0.60	0.57	0.58	0.60	0.59	0.60	0.58	0.59	0.69	0.73	**0.71**	0.65	0.64	0.64	0.70	0.73	**0.71**

Table 5.3: Results of the cognate detection task, in terms of weighted F-scores (5-fold) with baseline features and previous approaches, and our approaches using Cross-lingual similarity based features, for all the language pairs (LP).

From table 5.3, among the baseline approaches, we observe high precision but shallow recall scores when Weighted Lexical Similarity (WLS) based features are used. In fact, for language pairs which contain the Dravidian languages (Hi-Ml, Hi-Ta, Hi-Te, and Hi-Kn), even the precision scores are observed to be very low. The classifiers are not able to predict a significant amount of positively labelled cognate pairs, correctly. Even simple lexical variants such as "*Aag* (Fire)" (Hindi) and "*Agni* (Fire)" (Telugu) were classified incorrectly, as non-cognates. Phonetic vectors paired with a Siamese CNN [Rama, 2016], however, mitigate such misclassifications and are seen to perform well with much higher recall, for all the language pairs. Kanojia et al. [2019d]'s approach, however, outperforms the phonetic vectors based approach. We observe marginal improvements in F-scores for

almost all the language pairs (except Hi-Ko and Hi-Ne) when their RNN based approach is used. As for our approaches, SVM and Logistic Regression based classification methodologies were consistently outperformed by the FFNN method. Hence, we report precision (P), recall (R), and F-scores (F) for only FFNN based approaches in Table 5.3.

Our cross-lingual similarity-based approaches, however, significantly outperform all the baseline approaches. We observe a stark improvement in both precision and recall scores for all the language pairs. The cross-lingual approach, which uses the vectors from VecMap based models, fails to outperform both MUSE and XLM-R based models. XLM-R model exclusively achieves the best f-score for two language pairs (Hi-Bn and Hi-Mr). We believe its performance can be attributed to the closeness of the language pairs as they belong to the same language family (Indo-Aryan). Moreover, XLM-R is a transformer architecture-based model which requires a relatively larger corpora size, and a decent amount of corpus was available to build word embedding models for these target languages (Table 5.2). For other language pairs, we attribute the improvement in performance over the baseline approach to the cross-lingual models, in use. The cross-lingual models built above are used to provide vectors for calculating the similarity between words and contexts, bringing in the notion of semantic similarity for the task of cognate detection. It should be noted that by the definition of cognates, they are semantically similar despite the lexical variance. We observe that MUSE based feature representations paired with FFNN, obtain the best F-scores. This observation stands true even when the target language belongs to the Dravidian language family, where our baseline approaches lack severely in performance. For example, "*mkarand-maKarantam* (pollen)" (Hi-Ta), a cognate pair was classified correctly only using the MUSE based approach.

Additionally, we perform an ablation test with our feature sets for further experimentation. We observe that the combination of WLS and vectors from the MUSE model performs further better. Improvement is observed for eight language pairs out of thirteen ranging from 1% point (Hi-Ko, Hi-Ml, Hi-Ta, Hi-Te) to 5% points (Hi-Or). It should be noted that this is the only combination where no degradation in performance was observed for any language pair and hence, is reported in Table 5.3. Any other combination (MUSE + VecMap, MUSE + XLM-R, MUSE + PVS, and so on) degrades the performance of the cognate detection task, on at least one language pair.

The average improvement observed by using our best model (MUSE + WLS) over

Approaches / LP	Hi-Pa	Hi-Bn	Hi-Gu	Hi-Mr	Hi-Ta	Hi-Te	Hi-Ml
NMT-BPE Baseline	62.79	28.75	52.17	31.66	13.78	19.18	10.4
Cognate-aware NMT-BPE	**65.55**	**29.43**	**52.39**	**32.41**	**13.85**	**19.58**	**11.18**

Table 5.4: Results of the Cognate-aware Neural Machine Translation Task, in terms of BLEU scores, for the language pairs (LP) with available parallel data.

the strongest baseline approach [Kanojia et al., 2019d] is 9% points with the highest being 18% points (Hi-Ta). Over the weakest baseline approach (WLS), our best model obtains an average improvement of 50%, peaking at 61% points (Hi-Or).

We present the results of the Cognate-aware NMT task in Table 5.4. In the NMT task for Hi-Pa language pair, an improvement of 2.76 BLEU is observed, where 15001 cognates were detected including the misclassified pairs. Amongst a consistent improvement for all the language pairs, even when 930 cognate pairs (Hi-Te) are added, an improvement of 0.4 BLEU can be seen. The maximum number of cognate pair size injected into the NMT pipeline is 15834 pairs for the Hi-Mr language pair. Interestingly, we do not observe the most significant improvement for Hi-Mr despite the largest number of cognates injected. We believe this is because Marathi is a morphologically rich language and exhibits agglutination[8].

In this investigation, we harness cross-lingual embeddings to improve the task of cognate detection for thirteen Indian language pairs. We propose the use of a linked knowledge graph to augment a publicly released cognate dataset with a context dictionary. We reproduce the proposed approach and add two additional language pairs to the same dataset and perform experiments using various approaches for a comparative evaluation. We reproduce the previously proposed approaches [Rama, 2016, Kanojia et al., 2019d] for this task to perform a further evaluation. We obtain monolingual Indian language corpora for all the fourteen languages (Section 5.1.1), from various sources to build monolingual models and use a bilingual dictionary to supervise the task of cross-lingual models generation (MUSE and VecMap), for thirteen language pairs (Hi-Mr, Hi-Ta and so on). We also train a single cross-lingual model using the contextual embedding based

[8]Agglutination is a linguistic process in which complex words are formed by stringing together morphemes without changing them in spelling or phonetics.

NMT-BPE Baseline							
	Hi-Gu	Hi-Pa	Hi-Bn	Hi-Te	Hi-Ta	Hi-Ml	Hi-Mr
0k	49.47	**62.79**	28.51	**19.18**	12.86	**10.4**	31.21
2.5k	**52.17**	60.77	**28.75**	19.03	**13.78**	10.25	**31.66**
5k	50.9	59.95	28.16	18.83	13.01	9.9	31.33
7.5k	50.25	59.64	27.59	18.49	12.33	9.35	29.74
10k	49.61	59.17	26.48	17.87	11.64	9.37	29.77
15k	46.94	57.68	24.63	17.29	10.95	8.3	28.14
20k	44.68	57.24	23.6	16.58	9.68	7.56	25.4
Word-level	39.44	51.34	23.1	13.62	8.8	6.11	23.41
Cognate-aware NMT-BPE							
	Hi-Gu	Hi-Pa	Hi-Bn	Hi-Te	Hi-Ta	Hi-Ml	Hi-Mr
0k	-	62.72	-	19.26	-	10.87	-
2.5k	**52.39**	**65.55**	**29.43**	**19.58**	**13.85**	**11.18**	**32.41**
5k	51.46	65.08	29.33	-	13.04	-	31.9
7.5k	-	65.12	-	-	-	-	-

Table 5.5: Complete set of results for all merge operations performed for the NMT Task

approach (XLM-R).

Our experiments use three different approaches to generate better feature representations for the cognate detection task, and all of them show improvements over previously proposed approaches. We observe consistent improvements in terms of precision, recall and F-scores. We also perform an ablation study and show that augmenting WLS baseline feature with MUSE based features provide us with the best results. Over the strongest baseline, this model shows improvements up to 18% points, in terms of F-score. Our best F-score is observed for the Hi-Gu and Hi-Ne language pairs (0.84) which can still be improved and warrant further investigation into the task. Additionally, we use the detected cognate pairs and use a simple approach to inject them into the neural machine translation pipeline. Our Cognate-aware NMT-BPE results also show a consistent improvement for all the Indian language pairs. Furthermore, we release this augmented dataset, along with our code and cross-lingual models for further research.

In future, we aim further to investigate the performance of contextual embeddings for this task. Recent trends show that contextual embeddings based models outperform

conventional word embeddings for most tasks. We, however, do not observe this and attribute this primarily due to the dataset size used to train the contextual embeddings. We shall add more data to our monolingual corpora and perform more experiments using XLM-R. We also aim to investigate the task of cognate detection for other Indian language pairs, along with Indo-European language pairs, in the near future.

5.2 Cognition-aware Cognate Detection

In this investigation, we propose a novel method for enriching the feature sets, with cognitive features extracted from the gaze behaviour of human readers. We collect gaze behaviour data for a small sample of cognate dataset and show that cognitive features extracted, do help the task of cognate detection. We also use this gaze behaviour data to predict cognitive features for a larger sample and show that predicted cognitive features also significantly improve the performance of the cognate detection task. We report improvements of up to 15% using collected gaze features, and 16% using the predicted gaze features. Furthermore, we release the collected gaze behaviour data along with our code, and cross-lingual models, publicly. Merlo and Andueza Rodriguez [2019] perform an investigation to evaluate the use of recently proposed cross-lingual word embeddings and show that these models show a lexical structure matching the bilingual lexicon. This study establishes that the use of cross-lingual word embedding models provides an effective similarity score when compared to their monolingual counterparts, for both cognates and false friends. However, they do not evaluate machine learning algorithms to train models which can distinguish between cognates and false friends. The absence of such an evaluation also perpetrates a doubt whether these cross-lingual similarity scores can provide a significant distinction between cognates and false friends, to machine learning algorithms.

Inspired from their work, we investigate the use of cross-lingual word embeddings and cognitive features to distinguish between cognates and false friends, for the Indian language pair of Hindi and Marathi. Cognitively inspired features have shown to improve various NLP tasks [Mishra et al., 2018]. However, most of their work involves collecting the gaze behaviour data first on a large sample, and then splitting the data into training and testing data, before performing their experiments. While their work does show signif-

icant improvements over baseline approaches, across multiple NLP tasks, collecting gaze behaviour over a large cognate dataset can be costly, both in terms of time and money. Our approach tries to reduce this annotation cost by predicting gaze behaviour data for a larger sample based on the smaller sample of collected gaze behaviour data. Our investigations use three recently proposed cross-lingual word embeddings based approaches to generate features for the task of cognate detection. We also generate cognitive features from the gaze behaviour data of participants and show that gaze behaviour helps the task of cognate detection. Additionally, we use the collected gaze behaviour data and predict gaze-based features for a much larger sample. We believe that our approach of using gaze features will be more applicable only if gaze features can be predicted for unseen samples. We believe that collection of gaze data cannot be performed in all the scenarios and hence hypothesize that predicting such data if it helps improve the task of cognate detection, should be a viable solution.

5.2.1 Motivation

Consider a scenario where an NLP task comes across the false friend pair shown in Table 4.7 above. Orthographic similarity or even phonetic similarity-based techniques will fail to detect the difference between the Hindi meaning of the word "shikshA" and its Marathi counterpart. Here, semantic approaches should be able to detect the distinction in meaning, but monolingual embeddings are trained using a large corpus **from the same language**. In such cases, it becomes imperative that a cross-lingual word embeddings model be utilized. However, Indian languages are known to be low-resource languages compared to English or even many European languages like French, Italian, German, *etc.* Acquiring additional clean data for training cross-lingual models is, yet again, a painful task. In such a scenario then, we ask ourselves,

"Can cognitive features be used to, additionally, help the task of Cognate Detection?"

and furthermore,

"Can the prediction of such features using a small sample help us in further alleviating the need for collecting gaze data?"

5.2.2 Terminology

An *interest area (IA)* is an area of the annotation screen to be processed by the human reader. In our experiments, it is an area where a word is shown to the reader. A *fixation* is an event where the reader focuses within an interest area. A *saccade* is the movement of the eye from one fixation point to another. If the saccades move from an earlier IA to a later IA, such a saccade is called a *progression*. A *regression* is the saccade path when the reader moves back to a previous IA. We also use the terms *reader* and *participant* interchangeably.

5.2.3 Related Literature

Current literature which uses gaze behaviour to solve downstream NLP tasks has been applied to the NLP tasks of sentiment analysis [Mishra et al., 2018, Barrett et al., 2018, Long et al., 2019], sarcasm detection [Mishra et al., 2016], grammatical error detection [Barrett et al., 2018], hate speech detection [Barrett et al., 2018], named entity recognition [Hollenstein and Zhang, 2019], part-of-speech tagging [Barrett et al., 2016], sentence simplification [Klerke et al., 2016], and readability [González-Garduño and Søgaard, 2018, Singh et al., 2016a]. A comprehensive overview is provided by [Mishra and Bhattacharyya, 2018]. The primary motivation of using cognitive features for NLP tasks is derived from the eye-mind hypothesis [Just and Carpenter, 1980], which establishes a direct correlation between a reader's comprehension of the text with the time taken to read the text. This hypothesis has initiated a large body of work in psycholinguistic research that shows a relationship between text processing and gaze behaviour.

Mathias et al. [2020] describe an approach to score essays in a multi-task learning framework automatically. Their approach relies on collecting gaze behaviour for essay reading for a small set of essays and then predicting the gaze behaviour for the rest of the dataset, in a multi-task learning setup. Similarly, Barrett et al. [2016] use token level averages of cognitive features at run time, to mitigate the need for these features at run time. Singh et al. [2016a] and Long et al. [2019] predict gaze behaviour at the token-level as well. Mishra et al. [2018], González-Garduño and Søgaard [2018], Barrett et al. [2018], and Klerke et al. [2016], use multi-task learning to solve the primary NLP task, where learning the gaze behaviour is an auxiliary task.

For the task of cognate detection, however, the use of cognitive features has not been established previously. The task of cognate detection is cross-lingual, and a reader's cognitive load should vary while trying to comprehend the meaning of concepts, from different languages. Our work tries to exploit the difference noted in terms of time taken and eye-movement patterns in cognates vs false friends, to generate additional features for the automatic cognate detection task. Moreover, for Indian languages such as Hindi and Marathi, where agglutination[9] in Marathi varies the word length, the task becomes tougher, computationally.

5.2.4 Dataset Acquisition & Analysis

We pose the problem of cognate detection as a binary classification task in a supervised setting. For this task, we use a recently released challenging dataset [Kanojia et al., 2020a] of cognates and false friends. We extract the Hindi-Marathi cognate and false friend pairs. The number of cognate and false friend pairs released by this work are 15726, and 5826. For reducing this skew, we select an equal number of cognates, at random, thus producing a balanced dataset for the classification task. We also augment the complete dataset with context information from the IndoWordnet [Bhattacharyya, 2017]. The context information contains a *gloss* and an *example sentence* from the Wordnet data. The dataset released by Kanojia et al. [2020a] contains the Synset IDs for each word pair which helps us locate exact concept information from the dataset. We provide positive labels to cognates and negative labels to false friend pairs obtained from this data, and thus construct what we call "D1".

We extract 100 pairs, at random, from each of the positive and negative labels for collecting gaze behaviour data, to construct what we call "D2". This data, extracted from D1, is used for the purpose of collecting gaze behaviour and annotation. Although we have gold labels for the data extracted, we ask the participants to annotate the data by asking them if the concepts shown on the screen mean the same. The annotation screen provides them with contextual clues obtained from Wordnet data on the screen, as shown in Figure 5.3. The complete dataset statistics are shown in Table 5.6.

[9]Agglutination is a linguistic process due to which complex words are formed by joining together morphemes without changing them in spelling or phonetics.

	Cognates (1)	False Friends (0)
Kanojia et. al. (2020)	15726	5826
D1	**5826**	**5826**
D2	**100**	**100**

Table 5.6: Dataset Statistics for Cognate Detection Task

5.2.4.1 Gaze Data Collection and Annotation

The annotation task required the annotators to read word pairs and clues on the screen. It instructed them to read one pair at a time and then label them with binary labels indicating the similarity in meaning, (*i.e.,* positive/negative). It should be noted that the participants were not instructed to annotate whether a pair is a cognate or a false friend, to rule out the Priming Effect, (*i.e.,* if the same task expected beforehand, processing cognate pairs becomes relatively easier [Sáchez-Casas et al., 1992]). This ensures the ecological validity of our experiment in two ways: (1) The participant does not have any clue so that they can treat cognates with special attention (done by asking them to annotate meaning similarity) (2) Cognate pairs are mixed with false friend pairs and the participant does not have any prior knowledge about whether the next word pair would be a cognate or not. This also ensures that the participants pay attention and do not just skim through the word pair.

The collection of gaze data is conducted by following the standard norms in eye-movement research [Holmqvist et al., 2011]. We use an SR-Research Eyelink-1000 eye-tracker (monocular remote mode, sampling rate 500Hz) records several eye-movement parameters like fixations (an extended stay of gaze) and saccade (quick jumping of gaze between two positions of rest) and pupil size. We request a total of 15 participants to perform the annotation task, out of which only 11 participants could perform it. We could not perform experiments with the remaining 4 participants due to the pandemic. Out of the 11 completed annotations, we discarded the data from 2 participants as their gaze behaviour was erratic (the fixations were too far away from the IAs). The participants are graduates with science and engineering background. They are bilingual speakers who know both Hindi and Marathi. Our participants were given a set of instructions beforehand and were requested to seek clarifications before they proceede with the annotation.

The instructions described the nature of the task as discussed above, annotation input method, and the necessity of head movement minimization during the course of the experiment.

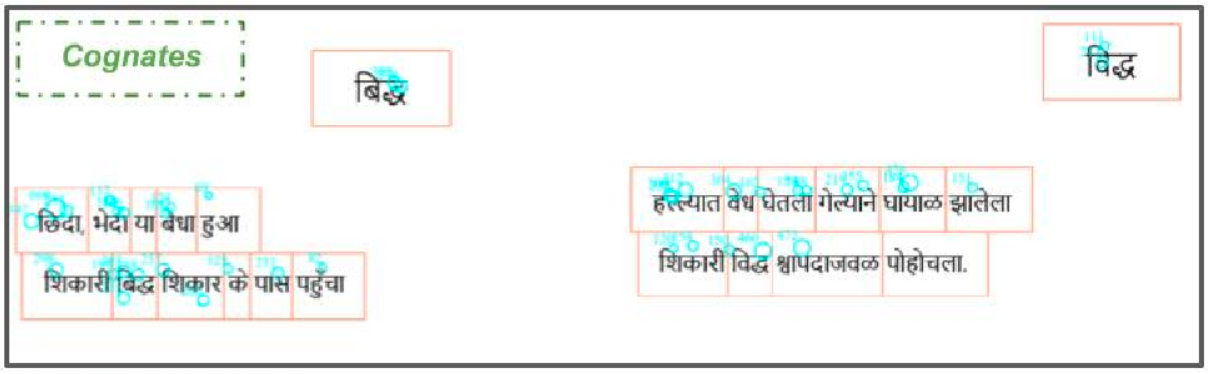


Figure 5.3: Screen capture showing collection of gaze features (via eye tracking) while displaying word pairs along with respective definitions and examples.

	μ_**Pos**	σ_**Pos**	μ_**Neg**	σ_**Neg**	p
P1	9.720	17.867	8.677	4.281	0.028
P2	8.596	10.526	7.619	13.794	0.049
P3	7.770	6.664	7.044	3.900	0.027
P4	9.686	17.729	8.664	4.306	0.031
P5	8.861	8.611	8.099	5.246	0.042
P6	7.854	6.286	7.184	3.442	0.033
P7	8.564	5.499	7.918	3.540	0.033
P8	8.018	5.955	7.340	3.742	0.031
P9	9.720	17.867	8.703	4.305	0.028

Table 5.7: T-test statistics for average fixation duration time per word for Positive labels (Cognates) and Negative labels (False Friends) for participants P1-P9.

5.2.4.2 Gaze Behaviour Data Analysis

The accuracy of similarity annotation by participants lies between 98% to 99.5% for individual annotators. Out of the 1800 annotations (9 annotators over 200 word-pairs), only 40 were mispredicted. The errors in annotation may be attributed to: (a) lack of

patience/attention while reading, (b) issues related to word-pair comprehension, and (c) confusion/indecisiveness caused due to lack of contextual clues. In our analysis, we do not discard the data obtained from these incorrect annotations.

We observe distinct eye-movement patterns for cognate pairs in terms of fixation duration of the human readers. Our analysis shows that fixation duration normalized over word count is relatively larger for cognate pairs. For cognate pairs, we observe that average fixation duration amongst all participants is 1.3 times more than that of false friend pairs. To test the statistical significance, we conduct a two-tailed t-test (assuming unequal variance) to compare the average fixation duration per word for cognate and false friend pairs. The hypothesized mean difference is set to 0, and the error tolerance limit (α) is set to 0.05. The t-test analysis, presented in Table 5.7, shows that for all participants, a statistically significant difference exists between the average fixation duration per word for cognate pairs vs false friend pairs.

5.2.4.3 Cross-lingual Word Embeddings

For this task, we use the cross-lingual word embedding models released by Kumar et al. [2020]. The Hindi-Marathi cross-lingual models released with this work are based on both MUSE [Lample et al., 2018] and XLM[Lample and Conneau, 2019]. Additionally, we build the cross-lingual word embeddings model for Hindi-Marathi using VecMap [Artetxe et al., 2017]. The model uses monolingual corpora released by Kunchukuttan et al. [2020] and a bilingual dictionary[10] required for the supervised method by Artetxe et al. [2017]. These three cross-lingual models provide us with different feature sets the task of cognate detection.

5.2.5 Feature Sets for Cognate Detection

In this section, we discuss the various features used for the task of cognate detection. It is to be noted that *false friends are spelt similarly across languages* but mean differently. Using false friends as data points with negative labels restricts us to the use of semantic similarity based features, as orthographic or phonetic similarity-based measures would fail to detect sufficient distinction between them. Hence, we use the features pro-

[10]Bilingual Lexicon

posed by Rama [2016] and Kanojia et al. [2019b] as baseline features for a comparative evaluation.

5.2.5.1 Phonetic Features

The IndicNLP Library provides phonetic features based vector for each character in various Indian language scripts. We utilize this library to compute a feature vector for each word by computing an average over character vectors. We compute vectors for both words in the candidate cognate pairs (PV_S and PV_T) and also compute contextual vectors (PCV_S and PCV_T) by averaging the vectors for all the contextual clues, generating a total of four vectors. We use these vectors as features for computing the baseline scores using the Siamese Convolutional Neural Network architecture proposed by Rama [2016].

5.2.5.2 Weighted Lexical Similarity

The Normalized Edit Distance (NED) approach computes the edit distance [Nerbonne and Heeringa, 1997] for all word pairs in our dataset. Each of the operations has a unit cost (except that substitution of a character by itself has zero cost), so NED is equal to the minimum number of operations to transform 'word a' to 'word b'. We use a similarity score provided by NED, which is calculated as (1 - NED Score). We combine NED with q-gram distance [Shannon, 1948] for a better similarity score. The q-grams ('n-grams') are simply substrings of length q. This distance measure has been applied previously for various spelling correction approaches [Owolabi and McGregor, 1988, Kohonen, 1978]. This feature set has been proposed by Kanojia et al. [2019b] and we use it with the character-based Recurrent Neural Network architecture proposed by them to compute another set of baseline scores.

5.2.5.3 Cross-lingual Vectors & Similarity

As discussed above, we use the pre-trained cross-lingual embedding models for generating feature vectors for MUSE and XLM based approaches. These models are generated by aligning two disjoint monolingual vector spaces through linear transformations, using a small bilingual dictionary for supervision [Doval et al., 2018, Artetxe et al., 2017]. Additionally, the cross-lingual word embeddings model trained by using Artetxe et al. [2017]'s approach provides us with the third set of feature vectors.

We use these models to obtain vectors for word-pairs (WV_S and WV_T) and averaged context vectors (CV_S and CV_T) from the contextual clues, to create three different feature sets. We obtain vectors for each candidate pair and their context using all the three cross-lingual methodologies. The use of cross-lingual models has been proposed for differentiating between cognates and false friends by Merlo and Andueza Rodriguez [2019], but an evaluation of the cognate detection task has not been performed. We perform this evaluation using our dataset and various classification methods and discuss the results below.

Gaze Feature	**Description**
Average Fixation Duration	The average of all fixation duration across all interest areas present on the screen .
Average Saccade Amplitude	Saccade amplitude is the amplitude of going back and forth measured in terms of duration.
Fixation Count	Counting the number of times user's eyes are fixated on the screen.
Fixation Duration Max	Maximum time for a single fixation on any Interest Area.
Fixation Duration Min	Minimum time for a single fixation on any Interest Area.
IA Count	Interest Area Count (no. of IAs on the screen)
Run Count	Consecutive counts for same Interest Area are ignored in Run Count
Saccade Count	Total counts of Saccades

Table 5.8: Gaze Features used for the task of Cognate Detection

5.2.5.4 Cognitive Features from Gaze Data

Gaze behaviour of participants, characterized by fixations, forward saccades, skips and regressions, can be used as features for NLP tasks [Mishra and Bhattacharyya, 2018]. Since these gaze features relate to the cognitive process in reading [Altmann, 1994], we consider these as features in our model.

From the gaze behaviour data collected, we extract a total of 18 features for each of the 1800 data points. Using supervised feature selection approach, we are able to select eight best features via grid search using Logistic Regression. We use the SelectKBest implementation along with hyperparameter tuning via GridSearchCV, present in the sklearn [Pedregosa et al., 2011] library. Here onwards, we refer to these eight features when we discuss cognitive or gaze-based features in this investigation. These eight best features, along with their description, are listed in Table 5.8.

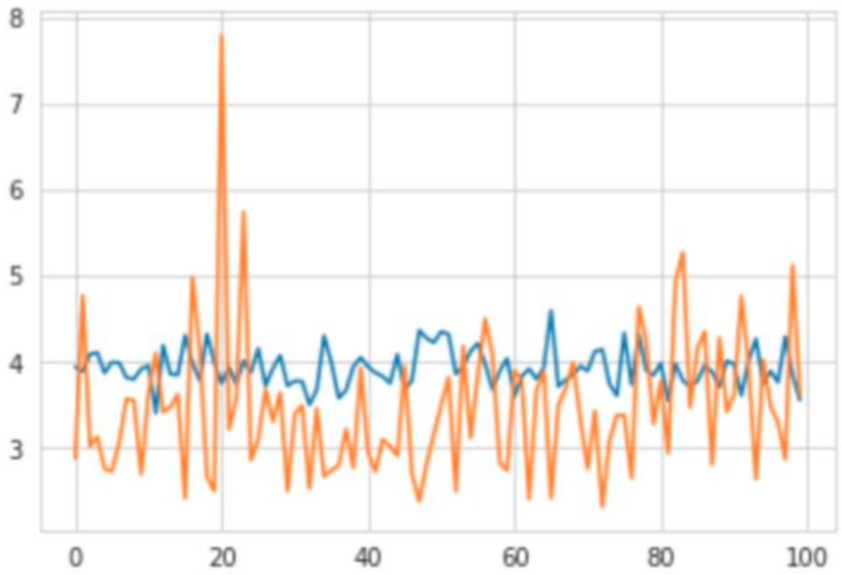


Figure 5.4: Predicted feature values (blue) vs. Gold feature values (orange) for the Saccade Amplitude feature for 100 samples.

5.2.5.5 Gaze Feature Prediction

Collection of gaze data for a large number of samples can be a costly task. We propose a neural model for cognitive features prediction. Our neural model is a feed-forward neural network to perform a regression task and predict gaze features. We collect gaze data for only 200 word-pairs with the help of 9 annotators which provides us with a total of 1800 data points for training and validation. As reported in Table 5.9, the initial results on D1 using different cross-lingual embeddings show that XLM based contextual features perform the best amongst all the cross-lingual models.

As an input to the network, we provide the feature vectors from the XLM model. The output of this network is the predicted gaze features for the D1 dataset, using the gaze features as gold predictions from the D2 dataset. This network contains three linear hidden layers with 128, 64 and 32 dimensions. We use the sigmoid activation function after each layer and also 'dropout' after each sigmoid with a dropout value of 0.2, and use 0.1 as the learning rate. We use the Mean Squared Error (MSE) loss function. A graph comparing the values for the predictions vs the actual values for the average saccade amplitude for 100 samples, can be seen in Figure 5.4.

5.2.6 The Cognate Detection Task

We employ both classical machine learning-based models and a simple feed-forward neural network. To compare our work with the previously proposed approaches, we replicate the best-reported systems from Rama [2016], *i.e.,* Siamese Convolutional Neural Network with phonetic vectors as features and also replicate Kanojia et al. [2019d]'s approach which uses a Recurrent Neural Network architecture with a weighted lexical similarity (WLS) as the feature set. The input to our classifiers is the feature sets described above for each candidate pair.

Among the classical machine learning models, we use Support Vector Machines (SVM) and Logistic Regression (LR). We experiment with the use of both linear SVMs and kernel SVMs (Gaussian and Polynomial). We perform a grid-search to find the best hyper-parameter value for C over the range of 0.01 to 1000. We deploy the Feed Forward Neural Network (FFNN) with one hidden layer. We perform cross-validation with different settings for activation function (tanh, hardtanh, sigmoid and relu) and the hidden layer dimension in the network (30, 50, 100, and 150). We use binary cross-entropy as the optimization algorithm. Finally, we choose the hyper-parameter configuration with the best validation accuracy. We train the model with the selected configuration with an initial learning rate of 0.4, and we halve the learning rate when the error on the validation split increases. We stop the training once the learning rate falls below 0.001. *We perform 5-fold stratified cross-validation, which divides the data into train and test folds, randomly.*

Initially, we perform our experiments with the feature sets from three different cross-lingual embeddings (MUSE, XLM, and VecMap) for the dataset D1. We, then, perform the same task for the smaller dataset D2 by combining cognitive features with individual cross-lingual feature sets. We also observe the performance of standalone gaze features for the D2 dataset. Finally, we evaluate the predicted gaze features on D1 dataset by combining them with cross-lingual features, and as a standalone feature set, using the feed-forward neural network. We report the results of the cognate detection task in the next section and discuss them in detail.

	Phonetic			WLS								
	P	R	F	P	R	F	P	R	F	P	R	F
Rama et. al., 2016 (D1)	0.70	0.68	0.69	-	-	-						
Kanojia et. al., 2019 (D1)	-	-	-	0.74	0.70	0.72						
	XLM			MUSE			VecMap					
Linear SVM (D1)	0.81	0.71	0.76	0.70	0.68	0.69	0.70	0.65	0.67			
LogisticRegression (D1)	0.80	0.75	0.77	0.72	0.74	0.73	0.70	0.73	0.71			
FFNN (D1)	0.80	0.84	0.82	0.81	0.76	0.78	0.77	0.76	0.76			
	XLM+Gaze			MUSE+Gaze			VecMap+Gaze			Gaze		
Linear SVM (D2)	0.81	0.69	0.75	0.72	0.73	0.72	0.70	0.75	0.72	0.77	0.76	0.76
LogisticRegression (D2)	0.84	0.75	0.79	0.76	0.72	0.74	0.81	0.71	0.76	0.80	0.75	0.77
FFNN (D2)	0.83	0.85	**0.84**	0.83	0.78	0.80	0.86	0.83	**0.84**	0.81	0.71	0.76

Table 5.9: Classification results in terms of weighted Precision (P), Recall (R), and F-scores (F) using 5-fold cross-validation using different feature sets as described above.

5.2.7 Results and Dicussion

We report the results of the cognate detection task in Table 5.9. In our initial evaluation on the D1 dataset, cross-lingual model-based features (XLM, MUSE, and VecMap) can be seen to outperform the baseline systems which use phonetic and orthographic features used with their original systems. We observe a stark improvement in both precision and recall scores for cross-lingual models when compared to the baseline systems. The cross-lingual approach, which uses the vectors from VecMap based models, fails to outperform both MUSE and XLM based models. XLM model achieves the best in almost all the settings. We believe its performance can be attributed to the closeness of the language pair, and the use of context from the contextual clues provided. For all standard classifiers, the gaze features are averaged across participants and augmented with cross-lingual features. These results are reported for all the classifiers with D2 dataset. Our feature combinations outperform the baselines with the best F-score improvement of 15% points over Phonetic features. We also report the precision, recall and F-score values when only gaze features are used to predict the labels for our candidate pairs. We observe that standalone gaze features are not as useful as when they are combined with cross-lingual feature vectors.

When gaze features are predicted using the methodology described in Section 4.5, the model performance for FFNN increases slightly. Across, all the feature combinations our model performance either stands the same, or increases by 1% point (Table 5.10). It should be noted that the regression task uses XLM based features to train and predict gaze features for over 10000 samples.

Predicted Gaze Features (D1 + D2)			
XLM+Gaze	0.85	0.85	**0.85**
MUSE+Gaze	0.83	0.85	**0.84**
VecMap+Gaze	0.85	0.78	0.81
Gaze	0.77	0.76	0.76

Table 5.10: Classification results when gaze features are predicted using a neural network (Section 4.5), in terms of weighted Precision (P), Recall (R), and F-scores (F) using 5-fold cross-validation on FFNN.

Our findings show that cross-lingual features outperform baseline features by a significant margin (Table 5.9) and addition of gaze features also helps the task of cognate detection. Furthermore, the predicted gaze features either help improve the classification task, or at least perform at the same level (Table 5.10); but do not degrade the model performance in any of the cases.

In this investigation, we harness cross-lingual embeddings and gaze-based features to improve the task of cognate detection for the Indian language pair of Hindi-Marathi. We create a novel framework that derives insights from human cognition, that manifests over eye movement patterns. We hypothesize that by augmenting cross-lingual features with features obtained from the gaze data, the task of cognate detection can be improved. We use a linked knowledge graph (IndoWordnet) to augment a publicly released cognate dataset with contextual clues. We collect the gaze behaviour data from nine participants over 200 samples and perform the task of cognate detection for both our datasets (with gaze data and without gaze data). We, then, use a neural network to predict gaze features for unseen samples and perform the task of cognate detection to show a slight improvement despite a small sample of training data.

We reproduce the previously proposed baseline approaches and perform experiments using additional features obtained cross-lingual models for a comparative evaluation. The previously proposed approaches [Rama, 2016, Kanojia et al., 2019d] for this task are shown to be outperformed using cross-lingual features and the combination of these features with the gaze data obtained. Our experiments use three different approaches to generate better feature representations for the cognate detection task, and all of them show improvements over previously proposed approaches. We observe consistent improvements in terms of precision, recall and F-scores. Over the stronger baseline, this model shows improvements up to 12% points, and 15% points over the weaker baseline, in terms of F-score. Our best F-score (0.85) is observed for the predicted gaze features with the use of the contextual features from the XLM model. Furthermore, we release this augmented dataset, along with our code and cross-lingual models for further research.

In future, we aim to add more language pairs and leverage contextual information from knowledge graphs using sequence-based neural models, for this task. We also aim to collect gaze behaviour data for them and then model the gaze predictions in a multi-task setting. We also investigate the performance of other multilingual contextual embeddings for this task (*e.g.,* BERT). We also aim to investigate the task of cognate detection for other Indian language pairs, along with Indo-European language pairs, in the near future.

5.3 Cognate Alignment for the Indo-European Language Family

In this short investigation, we create cognate sets among the Indo-European language family. We utilize the Hindi-English concept linkages among the Wordnet data where English is used as a pivot to link concepts among Indian and European languages. The manually validated dataset discussed in section 3.2.1 of this thesis is used as a pivot to generate these cognate sets. We perform this task for 14 Indian languages (discussed in 5.1) and 5 European languages namely Spanish, Portuguese, Italian, French and Polish. Based on the concept linkages obtained via pivot alignment, we align synsets from Indian language wordnet and European Wordnets.

Princeton WordNet or the English WordNet was the first wordnet and inspired the development of many other wordnets. EuroWordNet [Vossen et al., 1997] is linked wordnet comprising of wordnets for European languages, *viz*, Dutch, Italian, Spanish, German, French, Czech and Estonian. Each of these wordnets is structured in the same way as the Princeton WordNet for English [Miller et al., 1990] - synsets (sets of synonymous words) and semantic relations between them. Each wordnet separately captures language-specific information. In addition, the wordnets are linked to an Inter-Lingual-Index, which uses Princeton WordNet as a base. This index enables one to go from concepts in one language to similar concepts in any other language. Such features make this resource helpful in cross-lingual NLP applications like cognate detection and false friends' detection. IndoWordNet [Bhattacharyya, 2010] is linked wordnet comprising of wordnets for major Indian languages, *viz*, Assamese, Bengali, Bodo, Gujarati, Hindi, Kannada, Kashmiri, Konkani, Malayalam, Manipuri, Marathi, Nepali, Oriya, Punjabi, Sanskrit, Tamil, Telugu, and Urdu. These wordnets have been created using the expansion approach with Hindi WordNet as a pivot, which is partially linked to English WordNet.

5.3.1 Dataset Creation

In this section, we describe our primary datasets for the cognate detection task. We use the dataset created by Kanojia et al. [2020a] and utilize the English to Indian language Wordnet linkages to create a pivot-based alignment method. To create cognates sets from the European language Wordnets, we use the data from the Open Multilingual Wordnet (OMW) [Bond and Foster, 2013]. The statistics for Indian language cognate sets as available from Kanojia et al. [2020a] are shown in Table 5.11.

	Hi-Bn	**Hi-Gu**	**Hi-Mr**	**Hi-Pa**	**Hi-Sa**	**Hi-Ml**	**Hi-Ta**	**Hi-Te**	**Hi-As**	**Hi-Kn**	**Hi-Or**
Cognates	15312	17021	15726	14097	21710	9235	3363	936	3478	4103	11894

Table 5.11: Number of Cognates among Indian language pairs from Kanojia et al. [2020c].

We extract concept-aligned words from the OMW, and the total number of synsets and words for each of the five languages are shown in Table 5.12. It should be noted that the OMW does not contain unique synset IDs, the unique key for these datasets is the combination of Sysnset ID and the Part-of-speech category (ID-POS). We provide this

	Esp	Ita	Por	Pol	Fra	Eng
Synsets	38443	34978	43857	33816	59023	116738
Words	57764	63317	74054	52390	102836	207004
Unique Words	36681	41855	54071	45394	55435	161923

Table 5.12: Statistics for the European and the English Wordnets language data from OMW.

unique count along with the total number of unique word count in Table 5.12 along with the total number of words, and the total number of unique words.

The synset linkage discussed in 3.2.1 have been created with the help of lexicographers via manual validation which involves concept matching on a semantic level. Since semantic matching is established for such synsets, we exploit these linkages and generate cognate alignments via further orthographic matching among the aligned synonymous words. The statistics for the linkages between IndoWordnet and English Wordnet are provided in Table 5.13.

	Linked Synsets	Total Words (Hi)	Unique Words (Hi)	Total Words (Eng)	Unique Words (Eng)
Hi-Eng	30554	118981	84946	34300	28656

Table 5.13: Linked Data Statistics for English-Hind Wordnets

We use the source side data (Hindi Cognates) from the challenge dataset [Kanojia et al., 2020a] and try to find existing matches in the linked data discussed in Table 5.13. We determine that a total of 8259 words which have existing cognates in the challenge dataset among Indian languages and also exist in the synset linkages with the English language. For here onwards, we use the weighted lexical similarity-based measure and determine the orthographic similarity among these words based on concept linkages with English and European language synsets. We describe the weighted lexical similarity measure and the total alignments obtained in the section below.

5.3.2 Weighted Lexical Similarity-based Approach (WLS)

The Normalized Edit Distance (NED) approach computes the edit distance [Nerbonne and Heeringa, 1997] for all word pairs in our dataset. Each of the operations has a unit

cost (except that substitution of a character by itself has zero cost), so NED is equal to the minimum number of operations to transform 'word a' to 'word b'. We use a similarity score provided by NED, which is calculated as (1 - NED Score). We combine NED with q-gram distance [Shannon, 1948] for a better similarity score. The q-grams ('n-grams') are simply substrings of length q. This distance measure has been applied previously for various spelling correction approaches [Owolabi and McGregor, 1988, Kohonen, 1978]. This feature has been proposed by Kanojia et al. [2019b].

We use WLS to compute a similarity score among the potential cognate matches among Hindi and English words. Similarly, we further use WLS to compute a similarity score among the potential cognate matches in English and European language words. We apply a relatively stricter threshold of 0.75 on these potential candidates and obtain cognate sets among Hi-Eng-European language triples.

5.3.3 Results and Discussion

Based on the WLS approach, we are able to generate aligned cognate sets for Hi-Eng-European language triples. We present the statistics of each such triple in Table 5.14.

	Hi-Eng-Fra	**Hi-Eng-Esp**	**Hi-Eng-Ita**	**Hi-Eng-Pol**	**Hi-Eng-Por**
Cognate Sets	8750	1694	3186	3343	6311

Table 5.14: Statistics for the aligned cognate sets among Indian and European languages.

Based on the aligned sets above, we observe that due to the lesser number of manually validated synsets, we are unable to obtain a large number of potential cognate matches for the Indo-European language family. Although, with the numbers we obtain, we perform an analysis of the parts-of-speech category to which these cognate sets may belong.

We observe that there are no cognates aligned for verbs or adverbs among the Indian and European languages. Despite the large vocabulary available in wordnets, a large number of synset linkage only belong to nouns and adjective. The aligned synsets do contain a significant number of verb and adverb synsets, but the orthographic matching of the words contained in such synsets does not provide us with even a single candidate match. We believe that such a low number of cognate sets may not be able to affect the outcome of a neural machine translation system at all and shall further look for more

	Noun	Verb	Adjective	Adverb
Hi-Eng-Fra	8190	0	560	0
Hi-Eng-Esp	1557	0	137	0
Hi-Eng-Ita	2877	0	309	0
Hi-Eng-Pol	3322	0	21	0
Hi-Eng-Por	6100	0	211	0

Table 5.15: Distribution of the aligned cognates as per their parts-of-speech categories.

aligned data.

In this short investigation, we used the manually validated English-Hindi Wordnet linkages to generate aligned cognate sets for Hi-Eng-European language triples. We obtain a low number of such aligned pairs, and further analysis of this output also shows that the majority of such pairs belong only to the noun part-of-speech category. The absence of verbs and adverbs from this output is concerning and thus warrants further investigation into different linked datasets such as the Wiktionary or monolingual corpus to look for aligned cognate sets. We aim to extend this study further and also evaluate the use of cross-lingual embeddings as a viable feature to detect similarity among Indo-European cognates.

5.4 Summary

In this chapter, we have described the use of cross-lingual word embeddings for the task of cognate detection. We further discuss another investigation which showed that cognitive psycholinguistics could be used to generate helpful features for the same. We also showed that detected cognates help the downstream NLP task of machine translation. Finally, we conclude the chapter with a description of our efforts to detect cognate among Indian and European languages. We release all the models, datasets and code, which are described in this thesis with links provided while describing the investigations. The next chapter details our investigation to detect false friends among Indian languages.

Chapter 6

False Friends' Detection Task

So far, we have created challenge datasets for the task of cognate and false friends' detection and have detailed our efforts to detect cognates for low-resource Indian languages. In this chapter, we discuss the approaches proposed and our experiments for the detection of false friends. As discussed previously, **False friends are words that are similar in their forms despite having different meanings between two languages L1 and L2**. An example of such an occurrence would be the word Embarrassed, *i.e.,* 'feeling publicly shamed' in English, and *embarazado* 'pregnant' in Spanish.

6.1 Utilizing Cross-Lingual Word Embeddings to Detect False Friends

This work is originally titled, "Utilizing Cross-Lingual Word Embeddings to Detect False Friends among Indian languages". In this investigation, we focus on the task of false friends' detection. We use multilingual gold-standard datasets of false friend pairs [Kanojia et al., 2020a]. Our datasets come from Hindi and one of nine Indian languages, *viz.,* Bengali, Gujarati, Malayalam, Marathi, Nepali, Punjabi, Sanskrit, Tamil, and Telugu. Our classification approach utilizes cross-lingual word embeddings and is evaluated on these datasets extracted from linked Indian language WordNets and raw corpus. We compare our approach with previous baselines that use orthographic, phonetic, or monolingual embeddings based similarity. We report a significant improvement in F-scores, over the orthographic similarity-based baseline, up to 50% points, for all language pairs

and up to 65% points over the SoTA approach.[1]

Cognates can facilitate the foreign language learning process; they have similar meanings and, therefore, they can support the acquisition and learning of a non-native language. However, these words can also have a deceptive meaning as a result of semantic change and dissimilar development in two languages. These deceptive cognate words are commonly known as "false friends" [Domínguez and Nerlich, 2002]. Knowing and utilizing these cognates/false friends can help to improve the performance of computational phylogenetics [Rama et al., 2018] as well as cross-lingual information retrieval [Meng et al., 2001]; which encourages us to investigate this problem for the Indian linguistic area.[2] All the applications of cognate detection depend on an effective method of identifying cognates by computing a score, which reflects the likelihood that the two words are cognates among languages.[3]

Cognate identification approaches can confuse a false friend pair with being a cognate if orthographic similarity-based techniques are relied upon. False friends are especially problematic for language learners as learners tend to overgeneralize and assume that they know the meaning of these misleading words. Hence, we focus on the task of false friends' detection and propose a novel approach, which can identify false friends' from among possible cognate pairs, by utilizing distributional semantics across languages. Current work on false friends' detection has been applied to datasets with very limited language pairs. These datasets are either not available publicly or work with a few language pairs. We create new challenge datasets for this task using rich knowledge graphs, *i.e.,* linked Indian language WordNets and raw monolingual corpora. First, we use orthographic similarity to build candidate false friends' list. Then, with the help of manual annotation, we build a comprehensive gold-standard dataset for Indian language pairs. We also present a salient approach utilizing deep cross-lingual word embeddings that outperform baseline systems, including the most recently proposed methodology, by a significant margin.

[1]We will release the curated [Inter Annotator Agreement (IAA) of up to 0.87] false friends' list along with our evaluation dataset, and models upon publication.

[2]The term linguistic area or Sprachbund [Emeneau, 1956] refers to a group of languages that have become similar in some way as a result of proximity and language contact, even if they belong to different families. The best-known example is the Indian (or South Asian) linguistic area.

[3]Cognates in the same language are referred to as *doublets*.

Our approach concerning the identification of false friends is based on using the notion of semantic similarity from cross-lingual word embeddings. It performs significantly better than baseline systems that employ orthographic similarity, phonetic similarity, and monolingual embeddings based metrics for the task of false friends' detection. We also perform additional ablation experiments and show that a combination of orthographic and cross-lingual similarity-based metrics outperform all available systems and release the gold-standard challenge dataset along with the manually curated output in terms of false friends' lists for nine Indian language pairs.
The key contributions of this work are:

- A robust approach for the detection of false friends evaluated on these datasets, which outperforms previously available approaches.
- Curated list of false friends' for nine Indian language pairs, which can be used to improve downstream NLP tasks.

6.1.1 Past Literature in Brief

For the detection of false friends and true cognates, Chen and Skiena [2016] present an unsupervised transliteration based approach but do not report F-scores for Indian languages. Their results for the detection of false friends are inconclusive not just for eight Indian languages they discuss, but for 50% of languages they report their work on, as stated in their work. For the task of False Friends' detection, they show only a high correlation of English-French and English-Spanish language pairs with manual validation.

Many previous approaches to detect false friends use orthographic similarity [Inkpen et al., 2005, Ljubešić and Fišer, 2013, Torres and Aluísio, 2011, Mitkov et al., 2007, Nakov et al., 2009], which our experiments clearly show to be a weak baseline. The most recent work by Castro et al. [2018], which explores the problem of False Friends' detection, utilizes monolingual embeddings generated using word2vec [Mikolov et al., 2013c] for Spanish and Portuguese. We compare it with our approach and show that their approach performs relatively poorly in the case of morphologically rich Indian language datasets. This also establishes the need for a more robust method, such as ours, which works with language pairs from different language families.

Language Pair	Hi-Bn	Hi-Gu	Hi-Mr	Hi-Pa	Hi-Sa	Hi-Ml	Hi-Ta	Hi-Te	Hi-Ne
Wordnet - D1	10740	15520	9030	4370	6520	5130	1890	4260	5680
Corpus - D2	5160	6110	3660	2670	2910	1180	720	2110	1150

Table 6.1: Approximate Number of False Friends Candidates (word pairs) in D1 and D2 Datasets

6.1.2 The Challenge of False Friends' Detection for Indian Languages

Indian languages share a large number of cognates and false friends due to shared ancestry. Sanskrit is, for the most part, considered a historical antecedent of many Indian languages. This evolutionary relationship among Indo-Aryan languages (Hindi, Marathi, Bengali, Gujarati, Punjabi) and Dravidian languages (Malayalam, Tamil, and Telugu) make them interesting candidates for false friends' and cognate detection studies.

False friends can degrade the performance of any NLP system which learns cross-lingual properties for Indian languages based on characters, orthographic similarity, and phonetic aspects.[4]. A simple use case is when false friend word pairs are translated into each other, which should not happen. Hence, a word-pair lookup list such ours can help remove false friends from phrase tables in both Statistical MT and Unsupervised SMT [Artetxe et al., 2019] systems.

We note that the classical notion of orthographic similarity can be utilized to classify candidate false friends from WordNet [Fellbaum, 2010] data. We identify IndoWordnet [Bhattacharyya, 2017] as a potential resource that can be used to obtain candidate false friends, and leverage the synset information, *i.e.,* words, definition, and example for the semantic concept, to obtain the contextual information. Our approach leverages this contextual information and utilizes the distributional semantics captured via cross-lingual word embeddings to separate false friends in a shared geometric space.

[4]Letters in Indian languages have a close correspondence with the sounds [Singh, 2006a]; failure of orthographic methods to detect false friends indicates that even methods based on phonemes would perform poorly

	Language Pair	Hi-Bn	Hi-Gu	Hi-Mr	Hi-Pa	Hi-Sa	Hi-Ml	Hi-Ta	Hi-Te	Hi-Ne
D1	Number of False Friends	5312	7021	5726	4097	2710	923	663	936	4218
	Percent Agreement	0.9877	0.9849	0.9838	0.9754	0.9617	0.9223	0.9033	0.9553	0.9779
	Cohen's Kappa	0.7851	0.7972	0.8628	0.7622	0.7351	0.7046	0.6436	0.7952	0.8798
D2	Number of False Friends	239	784	832	751	211	129	257	184	644
	Percent Agreement	0.9672	0.9807	0.9785	0.9700	0.8495	0.9500	0.9568	0.9701	0.9490
	Cohen's Kappa	0.8975	0.7677	0.8732	0.8556	0.6828	0.5080	0.8020	0.7698	0.6762

Table 6.2: Statistics of Gold-standard Data, *i.e.,* Number of False Friends retained on both annotators' agreement (Section 6.1.3.3) along with the percent agreement and Cohen's Kappa scores.

Language	Hi	Bn	Gu	Mr	Pa	Sa	Ml	Ta	Te	Ne
Corpus Size	48142K	1564K	439K	520K	505K	553K	495K	909K	1023K	706K
STTR (n=1000)	0.5821	0.5437	0.4587	0.6108	0.4314	0.5350	0.7339	0.6411	0.4950	0.4883

Table 6.3: Word Embedding Corpus Statistics where corpus size is approximate number of sentences, and STTR is the moving average type-token ratio on a window of 1000 sentences. (Section 6.1.3.4)

6.1.3 Datasets and Experimental Setup

We now explain our datasets for ten Indian languages namely Bengali (Bn), Gujarati (Gu), Marathi (Mr), Malayalam (Ml), Nepali (Ne), Punjabi (Pa), Sanskrit (Sa), Tamil (Ta), Telugu (Te) and Hindi (Hi).

6.1.3.1 Wordnet Dataset (D1)

The IndoWordnet project has linked wordnets that can be used to obtain words expressing the same concepts or words which express completely different concepts. Measures such as Normalized Edit Distance (NED), Cosine Similarity (CoS), *etc.* reveal *orthographic similarity* between words. We hypothesize that a pair of words extracted from a linked IndoWordNet concept with high orthographic similarity is a True Cognate candidate. Similarly, **word pairs, which belong to different concepts, can be candidate false friends**. We use the dataset released by Kanojia et al. [2020a]. We report the total number of candidates obtained for this dataset in Table 6.1.

6.1.3.2 Corpus Dataset (D2)

Although, Wordnets are available for many languages[5], they are not yet available for all languages. Also, accurately linking concepts in WordNets is a tedious manual exercise. To ensure that our approaches are language or resource agnostic (in terms of WordNet data), we also apply our approach to raw corpora for the aforementioned languages. We use the ILCI Corpus [Jha, 2010b] available for these languages and extract candidates for the same language pairs. We compare each word from the source side corpus (Hi) to each word from the target side corpus (other Indian languages) and retain words-pairs using a similar approach (intersection between NED and CoS candidate outputs), as used for D1. We report the statistics of the number of candidates formed in Table 6.1.

6.1.3.3 Gold-standard Datasets Construction

After the construction of candidate false friends' lists (D2) as described above, we get them manually validated by two human annotators (for each language pair) and *only retain the word-pairs which were marked as false friends by both our annotators.* It should be noted that the annotators were provided with context information[6] along with the candidate false friends to remove any ambiguity. The percentage agreement between both our annotators for this manual evaluation is provided in Table 6.2 along with the Cohen's Kappa [McHugh, 2012] coefficient score. It should be noted that the number of false friends reported for D2, in Table 6.2, is low because many candidate false friends with high orthographic similarity were found to be duplicated by our lexicographers, and we only retain unique false friend pairs. Further, we provide the dataset and training details of word embedding based models.

6.1.3.4 Word Embeddings Dataset and Training Methodology

Word embeddings require a large amount of raw corpus for the training of a usable model with high accuracy. We extract corpora for these ten Indian Languages from various sources and collect them in a single repository. We extract Wikipedia article texts from WikiMedia dumps[7] for all languages and append the ILCI corpora to each of

[5]Open Multilingual Wordnet

[6]D2 - Complete Sentences

[7]WikiMedia Dumps

them. For Hindi, Marathi, Nepali, Bengali, Tamil, and Gujarati, we add crawled corpus of film reviews and news websites[8]. For Hindi, we also add HinMonoCorp 0.5 [Bojar et al., 2014] to our corpus, adding 44 million sentences. For Sanskrit, we download a raw corpus of prose[9] and add it to our corpus. We report corpus statistics in Table 6.3.

We build monolingual word embeddings using the FastText library[10] [Bojanowski et al., 2017b] since it takes sub-word information into account, which is beneficial for a task such as ours where sub-words play an important role, and spelling variations can lead to different meanings. We do not use BERT [Devlin et al., 2018] or ELMo [Peters et al., 2018] for word embeddings as their pre-trained models are not trained on transliterated corpora. We choose FastText to train Skipgram word embedding models (50 and 100 dimensions) for each language using the following hyperparameters - 15 epochs with 0.1 as the learning rate. We use two characters as the size of each sub-word for capturing the maximum number of sub-words. Further, we train cross-lingual word embedding models using three different strategies for all the language pairs with Hindi as a pivot language (Hi-Mr, Hi-Bn and so on). The first strategy uses the supervised method of MUSE [Lample et al., 2018][11] which utilizes a manually curated bilingual lexicon[12] for alignments. We use Hindi as a pivot language due to the ease of computation and availability of resources (Corpora and WordNet size). We use the monolingual models described above and train 18 cross-lingual word embedding models (nine language pairs over 50 dimensions, and nine language pairs over 100 dimensions) using this approach. In the next section, we describe the various baselines, state-of-the-art, and our novel approach, along with the classification methods used for the task of false friends' detection. We build the models using MUSE and use them to extract vectors and similarity scores among false friend candidates.

The second cross-lingual strategy uses VecMap [Artetxe et al., 2018a], which utilizes the monolingual models created above. VecMap uses an optional normalization feature while it builds the mappings between any two monolingual models. It performs orthog-

[8]Corpus Source
[9]Sanskrit Prose
[10]FastText - GitHub
[11]MUSE - GitHub
[12]Bilingual Lexicon

onal transformation and maps semantically related words, similar to MUSE, which was used in our first approach for building cross-lingual models. Additionally, it also reduces the dimensions of the embeddings models, which, is optional. We train it using the same hyperparameters as described above, for consistency while evaluating. We used the supervised approach for training these models as well, and the training dictionary was similar to the one provided to the MUSE method.

The third strategy utilizes contextual embeddings which have shown to outperform the conventional word embeddings based models for many tasks [Devlin et al., 2018]. We choose the most recent methodology for building a single cross-lingual model for all the languages. XLM-R [Conneau et al., 2019] uses previously proposed approaches of XLM [Lample and Conneau, 2019] and RoBERTa [Liu et al., 2019] to attain a very high performing cross-lingual model, especially for low-resource languages. We use our transliterated corpora described above and concatenate it into a single large corpus required for training the model. We then use the unsupervised training method of XLM-R and train a model over six days and a couple more hours with a reduced batch size, which allowed us to be able to train the model under a week. The model was then used to output vectors akin to the other cross-lingual models trained above for each word and their contexts. In this approach, however, we did not need a dictionary for the cross-lingual mapping strategy, unlike the two previous approaches. XLM-R generated a single model for all the languages trained on two V100 GPUs over almost seven days.

To put it more concisely, we trained cross-lingual models using three different strategies (MUSE, VecMap and XLM-R) where the cross-lingual mapping obtained for MUSE and VecMap were generated via the monolingual embeddings, as described above. We obtained nine models using each of these two methods. A single cross-lingual model was, however, trained using XLM-R and used for the third cross-lingual approach whose training methodology has been described above.

6.1.4 Approaches

False friends exhibit partial or full lexical similarity but are distant in terms of the meaning or sense. To perform the task of false friends' detection, the feature set can

vary between orthographic, phonetic, and semantic features. In this section, we describe various feature sets that we use with standard classification approaches over our gold-standard dataset described previously.

6.1.4.1 Weighted Lexical Similarity (WLS)

The Normalized Edit Distance (also known as Levenshtein Distance) approach computes the edit distance [Nerbonne and Heeringa, 1997] for all word pairs in both our datasets to form word pairs set. Each of the operations has a unit cost (except that substitution of a character by itself has zero cost), so NED is equal to the minimum number of operations to transform 'word a' to 'word b'. We use a similarity score provided by NED, which is calculated as (1 - NED Score). We combine NED with q-gram distance [Shannon, 1948] for a better similarity score. The q-grams ('n-grams') are simply substrings of length q. This distance measure has been applied previously for various spelling correction approaches [Owolabi and McGregor, 1988, Kohonen, 1978].

We observe that NED and QD output a different number of candidates as they compute the orthographic similarity in different ways. We provide different weights to both the measures and observe that providing 75% weight to NED and 25% weight to QD provides us with the best F-scores over our gold-standard dataset. We compared F-scores over different weight balances, *i.e.,* 50-50%, 25-75%, and chose to report 75-25%, after empirical evaluation. For any word pair with words p and q, it can be put as follows:

$$WLS_{pq} = (NED_{pq} * 0.75) + (QD_{pq} * 0.25)$$

Now that this approach can be used to compute a score between each word pair, we use it to find two scores, which are used as features - 'word-pair similarity' and 'contextual similarity'.

We use the information provided with the wordnet synset, *i.e.,* Gloss or Definition of the concept, and Example sentence of the concept (sense) to form a 'bag of words'. From each bag of words (stop words removed), we compute the WLS score. We average the similarity score of all comparisons and obtain 'contextual similarity'. The intuition behind this baseline method is that given both words belong to different concepts, WLS similarity

between the words used to describe them should be relatively lower for false friends. The first score (score1) being the 'word-pair similarity', and the second score (score2) being the 'contextual similarity'.

$$S_1 = score1/\left(score1 + score2\right) \tag{6.1}$$

$$S_2 = score2/\left(score1 + score2\right) \tag{6.2}$$

We then normalize these scores using (1) and (2) and compute the difference between S_1 and S_2. We use S_1 and S_2 as features for classification, as described ahead.

6.1.4.2 Phonetic Vectors and Similarity (PVS)

The IndicNLP Library provides phonetic features based vector for each character in various Indian language scripts. We utilize this library to compute a feature vector for each word by computing an average over character vectors. We compute vectors for both words in the candidate false friend pairs (PV_S and PV_T) and also compute contextual vectors (PCV_S and PCV_T) by averaging the vectors for all the contextual 'bag-of-words' on each side (source and target), generating a total of four vectors. We also calculate the cosine similarity among PV_S and PV_T, and among PCV_S and PCV_T to generate two similarity scores (P_{S1}, and P_{S2}) which can be additionally used as features during classification.

It should be noted that using phonetic vectors and their similarity scores has already been proposed in the previous literature [Rama, 2016] for a cognate detection task, and we do not claim this approach to be our novel contribution. We perform experimentation using this approach to draw a comparison between the use of phonetic features and semantic features, which are reported ahead.

6.1.4.3 Monolingual Vectors & Similarity (MVS)

The most recent approach for the task of false friends' detection was proposed by Castro et al. [2018]. We modify their approach by using FastText based models and the addition of contextual vectors, for comparison. We obtain word-pair vectors (MV_S and MV_T) along with averaged contextual vectors (MCV_S and MCV_T) to be used as features. They use monolingual embeddings and extract only word vectors as features to be used

during the classification task. We have already described the training of FastText based monolingual embedding models. We report the results of this modified state-of-the-art (SoTA) approach (with 100-dimensional models) using the same classification methods, as used by our novel approach. Additionally, we use our corpora to train models using word2vec, with hyperparameters as specified by them. We use their implementation from their code repository[13] to compare their output, over our datasets.

6.1.4.4 Cross-lingual Vectors & Similarity

Cross-lingual word embeddings are becoming increasingly crucial in multilingual NLP. It has been recently reported that these embeddings can be effectively learned by aligning two disjoint monolingual vector spaces through linear transformations, using no more than a small bilingual dictionary for supervision [Doval et al., 2018, Artetxe et al., 2017]. The first two approaches for training cross-lingual methods use this dictionary for supervision.

In our novel approach, we propose the use of vectors from the cross-lingual embedding models trained for Indian language pairs. We obtain vectors for word-pairs (WV_S and WV_T) and averaged context vectors (CV_S and CV_T) to create feature sets. We obtain vectors for each word and its context using all the three cross-lingual approaches for our experiment and discuss the results later in this work. We also use angular cosine similarity [Cer et al., 2018] scores for word pairs and their contexts. Angular similarity distinguishes nearly parallel vectors much better as small changes in vector values yield considerable distances. For each word pair vector and its context vectors, we compute the 'word-pair similarity' and 'contextual similarity'. We use *arccos* to obtain angular cosine similarity (asim) among vectors 'u' and 'v', as shown below:

$$asim(u, v) = \left(1 - arccos\left(\frac{u.v}{\|u\|\|v\|}\right)/\pi\right)$$

Each false friend candidate pair generates a score, *i.e.,* score1 and the average of scores among all words in the context generate another score, *i.e.,* score2, which are normalized using (1) and (2), same as above. For D1, Word pair contexts are built using wordnet information and removing stop words, thus resulting in a 'bag of words'. For D2, we consider the sentence window to be context after removing the stop words. We now have

[13]SoTA - Code Repository

two scores (S_1 and S_2) and four vectors (WV_S, WV_T, CV_S, and CV_T) for each word pair, which are used as features for classification. We describe our classification approaches in the next section.

		Baselines									A Stronger Baseline		
		WLS			PVS			SoTA			MVS		
		P	R	F	P	R	F	P	R	F	P	R	F
D1	Hi-Bn	0.61	0.71	0.66	0.62	0.57	0.59	0.61	0.55	0.58	0.61	0.59	0.60
	Hi-Gu	0.52	0.72	0.60	0.52	0.72	0.60	0.64	0.58	0.61	0.68	0.59	0.63
	Hi-Ml	0.65	0.81	0.72	0.66	0.64	0.65	0.32	0.42	0.36	0.66	0.76	0.71
	Hi-Mr	0.56	0.75	0.64	0.56	0.75	0.64	0.58	0.49	0.53	0.56	0.75	0.64
	Hi-Ne	0.35	0.59	0.44	0.59	0.55	0.57	0.59	0.33	0.42	0.68	0.60	0.64
	Hi-Pa	0.49	0.70	0.58	0.50	0.69	0.58	0.41	0.34	0.37	0.66	0.67	0.66
	Hi-Sa	0.44	0.66	0.53	0.63	0.64	0.63	0.17	0.28	0.21	0.68	0.56	0.61
	Hi-Ta	0.68	0.74	0.71	0.65	0.81	0.72	0.39	0.52	0.45	0.69	0.70	0.69
	Hi-Te	0.47	0.69	0.56	0.66	0.64	0.65	0.55	0.49	0.52	0.47	0.69	0.56
D2	Hi-Bn	0.66	0.29	0.40	0.30	0.54	0.38	0.55	0.35	0.43	0.84	0.53	0.65
	Hi-Gu	0.32	0.48	0.38	0.52	0.72	0.60	0.49	0.65	0.56	0.71	0.62	0.66
	Hi-Ml	0.29	0.22	0.25	0.31	0.50	0.38	0.29	0.38	0.33	0.69	0.61	0.65
	Hi-Mr	0.22	0.57	0.32	0.56	0.75	0.64	0.61	0.55	0.58	0.69	0.66	0.67
	Hi-Ne	0.09	0.28	0.14	0.35	0.59	0.44	0.52	0.41	0.46	0.55	0.56	0.55
	Hi-Pa	0.10	0.54	0.17	0.49	0.70	0.58	0.31	0.39	0.35	0.65	0.52	0.58
	Hi-Sa	0.07	0.69	0.13	0.44	0.66	0.53	0.27	0.18	0.22	0.28	0.21	0.24
	Hi-Ta	0.09	0.58	0.16	0.61	0.61	0.61	0.49	0.32	0.39	0.61	0.54	0.57
	Hi-Te	0.31	0.38	0.34	0.47	0.69	0.56	0.52	0.59	0.55	0.57	0.57	0.57
	es-pt	-	-	-	-	-	-	0.81	0.73	0.77	-	-	-

Table 6.4: Baseline Results of the false friends' detection task, in terms of weighted F-scores for Weighted Lexical Similarity (WLS), Phonetic Vectors and Simialirty (PVS), State-of-the-art (Castro et al., 2018) approach, and Monolingual Similarity (MVS, *i.e.,* SoTA w/ FastText) over all the language pairs, and for both the datasets (D1 and D2).

6.1.4.5 Classification Approaches

We pose the problem of detecting a *false friend* as a binary classification problem. We employ both classical machine learning-based models and a deep learning-based model to detect false friends. We also perform experiments using the SoTA approach. The input to these models is the feature sets described above for each candidate pair. The candidates are the complete data described in Table 6.1. The gold-standard data from Table 6.2 is labelled positive, and rest of the candidate pairs from Table 6.1 are labelled negative. The lexical overlap between the language pairs ranges from 13% (for hi-te) to 23% (hi-mr). We perform 5-fold stratified cross-validation, which divides the data among train and test folds, randomly. Among the classical machine learning models, we use Support Vector Machines (SVM) and Logistic Regression (LR). We experiment with the use of both linear SVMs and kernel SVMs (Gaussian and Polynomial). We perform a grid-search to find the best hyper-parameter value for C over the range of 0.01 to 1000.

We also deploy a simple Feed Forward Neural Network (FFNN) with one hidden layer. We perform cross-validation with different settings for activation function (tanh, hard-tanh, sigmoid and relu) and the hidden layer dimension in the network (30, 50, 100, and 150). We use binary cross-entropy as the optimization algorithm. Finally, we choose the hyper-parameter configuration with the best validation accuracy. We train the model with the selected configuration with an initial learning rate of 0.4, and we halve the learning rate when the error on the validation split increases. We stop the training once the learning rate falls below 0.001.

We perform our experiments with the feature sets (Orthographic, Phonetic, Monolingual Embeddings based, and three different cross-lingual embeddings based) described above for all the nine language pairs and on both datasets. We also perform an ablation test with various feature sets and report the results for the best combination in the next section.

6.1.5 Results and Discussion

Cross-lingual similarity and vectors based features, as shown in Table 6.5, achieve the best results in the false friends' detection task (for both SVM and FFNN). The results

		Our Approaches Utilizing Cross-lingual Vectors and Similarity												Best Ablation		
		MUSE w/ SVM			MUSE w/ FFNN			XLM-R w/ FFNN			VecMap w/ FFNN			MUSE + WLS w/ FFNN		
		P	R	F	P	R	F	P	R	F	P	R	F	P	R	F
D1	Hi-Bn	0.83	0.64	0.72	0.84	0.71	0.77	0.90	0.79	**0.84**	0.82	0.71	0.76	0.84	0.71	0.77
	Hi-Gu	0.73	0.64	0.68	0.91	0.88	**0.89**	0.89	0.83	0.86	0.72	0.76	0.74	0.92	0.87	**0.89**
	Hi-Ml	0.82	0.73	0.77	0.86	0.83	0.84	0.88	0.79	0.83	0.82	0.79	0.81	0.89	0.85	**0.87**
	Hi-Mr	0.73	0.71	0.72	0.93	0.90	0.91	0.91	0.90	0.90	0.72	0.78	0.75	0.93	0.92	**0.92**
	Hi-Ne	0.65	0.66	0.65	0.81	0.74	0.77	0.81	0.90	**0.85**	0.63	0.60	0.62	0.86	0.70	0.77
	Hi-Pa	0.88	0.81	0.84	0.89	0.80	0.84	0.89	0.88	**0.89**	0.84	0.85	0.84	0.91	0.87	**0.89**
	Hi-Sa	0.56	0.75	0.64	0.89	0.84	**0.86**	0.90	0.81	0.85	0.53	0.72	0.61	0.89	0.84	**0.86**
	Hi-Ta	0.65	0.81	**0.72**	0.65	0.81	**0.72**	0.69	0.71	0.70	0.61	0.79	0.69	0.76	0.69	**0.72**
	Hi-Te	0.67	0.68	0.67	0.85	0.72	**0.78**	0.67	0.70	0.69	0.63	0.68	0.65	0.80	0.77	**0.78**
D2	Hi-Bn	0.85	0.60	0.70	0.77	0.72	0.74	0.81	0.82	**0.81**	0.68	0.71	0.69	0.76	0.72	0.74
	Hi-Gu	0.73	0.65	0.69	0.72	0.69	0.70	0.70	0.66	0.68	0.74	0.69	0.71	0.73	0.73	**0.73**
	Hi-Ml	0.76	0.62	0.68	0.74	0.63	0.68	0.71	0.64	0.67	0.72	0.62	0.67	0.70	0.68	**0.69**
	Hi-Mr	0.74	0.69	**0.71**	0.71	0.71	**0.71**	0.74	0.69	**0.71**	0.74	0.61	0.67	0.70	0.70	0.70
	Hi-Ne	0.55	0.60	0.57	0.71	0.62	0.66	0.76	0.68	**0.72**	0.53	0.65	0.58	0.64	0.66	0.65
	Hi-Pa	0.65	0.59	0.62	0.66	0.65	0.65	0.68	0.66	**0.67**	0.62	0.67	0.64	0.65	0.66	0.65
	Hi-Sa	0.26	0.39	0.31	0.65	0.63	**0.64**	0.68	0.55	0.61	0.29	0.38	0.33	0.65	0.64	**0.64**
	Hi-Ta	0.63	0.58	0.60	0.79	0.66	**0.72**	0.60	0.55	0.57	0.60	0.62	0.61	0.77	0.67	**0.72**
	Hi-Te	0.75	0.56	0.64	0.79	0.59	**0.68**	0.72	0.54	0.62	0.72	0.59	0.65	0.61	0.56	0.58
	Es-Pt	0.85	0.77	0.81	0.89	0.80	0.84	0.78	0.87	0.82	0.80	0.82	0.81	0.89	0.84	**0.86**

Table 6.5: Results of the false friends' detection task using our approaches, in terms of weighted F-scores for Cross-lingual Similarity (CLS, *i.e.,* MUSE), the combination of Cross-lingual and Weighted Lexical Similarity (CLS+WLS) based features, Contextual Embeddings based features (XLM-R), and VecMap based features, overall language pairs, and both the datasets (D1 and D2).

obtained using baseline features (Orthographic, Phonetic, and so on) are reported in Table 6.4. Our models can be seen to outperform them with a significant margin. The cross-lingual model training strategy which uses the vectors from VecMap based models fails to outperform both MUSE and XLM-R based models. XLM-R model exclusively achieves the best f-score for six language pairs out of eighteen (D1 + D2). The models generated

using our work above are used to provide vectors for calculating the similarity between words and contexts, bringing in the notion of semantic similarity for the task of false friends' detection. It should be noted that by the definition of false friend candidates, they should be spelt similarly or identically, but such words should differ semantically. We observed that vectors obtained using MUSE models were providing us with the best results in most of the cases, outperforming the other cross-lingual strategies. Hence, we perform further experiments by combining the MUSE features with other feature sets (MUSE + WLS, MUSE + PVS, MUSE + and so on).

After performing an ablation test with a different feature set combinations, we observe that a combination of the orthographic similarity-based measures (WLS) and cross-lingual similarity (MUSE) performs even better, which indicates that orthographic and semantic similarity should be combined to detect linguistic entities such as false friends. Our results resonate with the linguistic identity of false friends, which defines them to be word pairs with high orthographic similarity but low semantic similarity. For baseline approaches in Table 6.4, we report the values for Linear SVM with the value of C as 10, which outperforms all other classification approaches. For the best method (MUSE), we report the values of both SVM and FFNN. We report the results for cross-lingual models with 100-dimensions as they were considerably better than results obtained using features from the 50-dimensional models. We compare our results with the state-of-the-art approach [Castro et al., 2018] over the same hyperparameters as described in their work and show that monolingual vectors do not perform as well, for both word2vec and FastText based (MVS) models.

We also replicate an experiment for Spanish (Es) - Portuguese (Pt) language pair on their dataset, but with our approach, and report an improvement of 13% points. On both our datasets created by us, we observe a significant improvement over their approach. Notably, our best improvements over the SoTA approach, are 65% points (hi-sa) on D1 and 42% points (hi-sa) over D2, for Hi-Mr and Hi-Sa language pairs, respectively. A further explanation for our results can be attributed to the fact that D1 is based on Wordnets, which are a rich knowledge repository and contain a clear sense distinction. It seems amiss in the results for our approach on D2, as can be seen in Table 6.5. Although, using our approach, a significant margin of improvement can be seen even if Wordnets

are not used as a dataset, or for providing contexts.

We perform manual error analysis by presenting a random 50%age of examples to two other human annotators and asking them to classify the errors into various categories. The annotators were provided context, in the form of Wordnet gloss and example (for D1 output) and the complete sentence from the corpus (for D2 output), and then requested to mark positive and negative labels for our output; where positive implied that the word-pair is a false friend pair, indeed. We report the true per cent (when both annotators agreed upon the word pair being a false friend), per cent agreement, and Cohen's Kappa values from this evaluation in Table 6.6. Our methods yield a substantial agreement, with an IAA (Cohen's Kappa) of up to 0.87, demonstrating the general soundness of our approach. We report high 'true per cent' scores among many language pairs, but there were some erroneous classifications. Our annotators report that our candidate construction approach results in a large list, some of which are *verb-cognate object pairs*. A cognate object (or cognate accusative) is a verb's object that is etymologically related to the verb (*e.g.*, sleep-slept). Our approach classified some of them as false friends, which we find as a source of erroneous classification. We plan to look into POS wise classification of false friends' to mitigate this issue possibly. Figure 6.1 shows the difference between best F-scores for each language pair, among D1 and D2 datasets. We report these differences to substantiate our claim that our approach works for both datasets and, is indeed, Wordnet data independent. Despite minimal contextual information obtained from the sentence of the raw corpus where the target word lies, our approach performs well, and the difference between D1 and D2 values range from, as low as 0.03, to 0.22, the average being 0.14. Indian languages are considered low-resource languages, which is reflected in the size of monolingual corpora (except Hi), in Table 6.3. If the size of monolingual corpora is sufficient (*approx.* 1 million sentences) to train cross-lingual embeddings, *i.e.,* for Hi-Bn, Hi-Ne, Hi-Ta, & Hi-Te, our approach works with minimal contextual information as can be seen in Figure 6.1.

We also show anecdotal examples (one each) of a false friend pair detected using our approach from each language pair in Table 6.7, along with the sense definitions in both the languages. We observe that our approach can detect the semantic shift between fine-grained senses to detect false friends. For example, in the Hi-Ne language pair, the word

	Language Pair	Hi-Bn	Hi-Gu	Hi-Mr	Hi-Pa	Hi-Sa	Hi-Ml	Hi-Ta	Hi-Te	Hi-Ne
D1	True Percent (%)	87.28	95.62	92.54	96.59	90.73	81.21	67.36	86.09	86.81
	Percent Agreement(%)	93.78	98.62	97.84	98.57	96.33	93.01	87.81	96.48	98.01
	Cohen's Kappa	0.6423	0.8054	0.8188	0.7277	0.7336	0.7309	0.6876	0.8350	0.9070
D2	True Percent (%)	70.70	86.15	87.41	83.37	69.36	82.88	86.58	89.22	83.24
	Percent Agreement(%)	91.83	94.71	96.77	94.96	88.99	95.21	96.64	95.59	96.25
	Cohen's Kappa	0.7836	0.7347	0.8732	0.8556	0.7084	0.8093	0.8381	0.7188	0.8518

Table 6.6: Evaluation of our output (list of false friends) in terms of percent agreement and Cohen's Kappa scores.

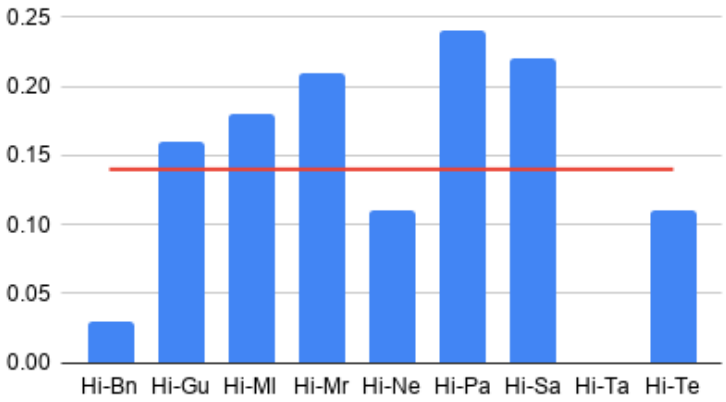


Figure 6.1: Difference among best systems for D1 & D2

'*anantataa*' which means '*uncountable*' in Hindi but takes the sense of '*large*' Nepali. This word is borrowed from the Sanskrit language into Hindi, where the only sense of this word in Sanskrit and Hindi is 'uncountable', but interestingly, it develops a new sense of 'large' in Nepali. We also observe that for the language pair Hi-Mr, the word '*shiksha*', which means '*education*' in Hindi takes on the sense of '*punishment*' in Marathi.

In this investigation, we introduce a novel approach to detect false friends by using deep cross-lingual word embeddings with the highest attained F-score of 0.92 for the Hi-Mr language pair. We create gold-standard challenge dataset (D2) for the task of false friends' detection over nine Indian language pairs and release them publicly. We use various previously proposed approaches like lexical similarity-based, phonetic similarity-based, monolingual embeddings based SoTA approach, and modified monolingual embeddings based (MVS) for comparison. We perform our experiments on these nine language pairs and report significant improvements over the baseline orthographic similarity-based approach,

LP	Source Word	Target Word	Source Sense Definition	Target Sense Definition
Hi - Bn	अनाचार	अनाचार	दूसरों के साथ बलपूर्वक किया जानेवाला वह अनुचित व्यवहार जिससे उन्हें बहुत कष्ट हो	धर्मेर बिरुद्ध काज
	anaachaara	anaachaara	doosro ke saatha balpoorvaka kiyaa jaanewaalaa weha anuchita vyavahaara jis-se unhe bahut kashta ho	dharmer biruddha kaaja
	misconduct	blasphemous	behavious which affects others in a negative manner	against religion
Hi - Mr	शिक्षा	शिक्षा	विद्या, संगीत आदि पढ़ाने या सिखाने की क्रिया	अपराधाबद्दल भोगावे लागणारे शारीरिक कष्ट वा आर्थिक नुकसान
	shikshaa	shikshaa	vidyaa, sangeeta aadi padhaane yaa sikhaane ki kriyaa	ya aparaadhaa-baddala bhogave laganaare shariirika kashta vaa aarthika nuksaana
	education	punishment	the activities of educating or instructing	the act of punishing
Hi - Pa	खालिक	खालिक	धर्मग्रंथों द्वारा मान्य वह सर्वोच्च सत्ता जिसे सृष्टि का स्वामी माना जाता है	निरमाण करन जां बणाउण वाला
	khaalika	khaalika	dharmagrantho dwaaraa maanya waha sarvoccha sattaa jise srushti kaa swaamii maanaa jaataa hai	nirmaana karana jan banaaun vaalaa
	God	Movie Producer	the entity of worship in monotheistic religions	someone who produces movies
Hi - Ml	शिष्ट	शिष्टं	उच्च आचार-विचार रखने और भले आदमियों का-सा व्यवहार करने वाला	ഒളിച്ചുള്ള (ഏഴ്ങനേയാ എതുവിധമോ മുളുവനും തന്നെ നഷ്ടമായിപ്പോയതിന്റെ ബാക്കിയുള്ളത്)
	shishta	shistam	uccha achaara-vichaara rakhne aur bhale aadmiyon kaa-saa vyavahaara karne walaa	oḷiccuḷḷa (eṅṅaneyēā ētuvidhamēā muḷuvanuṁ tanne naṣṭamāyippēāyatinṟe bākkiyuḷḷat).
	well-behaved	remainder	socially or conventionally correct	something left after other parts have been taken away
Hi - Te	योजन	योजन	धर्मग्रंथों द्वारा मान्य वह सर्वोच्च सत्ता जिसे सृष्टि का स्वामी माना जाता है	రాముడి భార్య
	yojana	yojana	dharmagrantho dwaaraa maanya waha sarvoccha sattaa jise srushti kaa swaamii maanaa jaataa hai	raamudi bhaarya
	God	(Named Entity)	the entity of worship in monotheistic religions	The wife of Rama, *i.e.,* A proper noun referring to Sita
Hi - Gu	सरस	सरस	पानी का बड़ा कुंड	જોવામાં સારું અને સુંદર લાગનારું
	sarasa	sarasa	paanii kaa badaa kunda	jovaamaan saarun ane sundara laagnaaru
	reservoir	beautiful	water reservoir	having great beauty and splendor
Hi - Ta	तिमि	तिमि	एक रोग जिसमें रात के समय दिखाई नहीं पड़ता	கடலில் வசிக்கக்கூடிய மீனின் வடிவத்தையுடைய பெரிய நீர்வாழ் உயிரினம்
	timi	timi	eka roga jismein raata ke samaya dikhaayii nahii padtaa	Kaṭalil vacikkakkūṭiya mīṉiṉ vaṭivattaiyuṭaiya periya nīrvāḻ uyiriṉam
	Night-blindness	whale	Disease which causes inability to see clearly in dim light or night	any of the larger cetacean mammals having a streamlined body
Hi - Ne	अनन्तता	अनन्तता	अनगिनत होने की अवस्था या भाव	अधिक हुने अवस्था या भाव
	anantataa	anantataa	anaginata hone ki avasthaa yaa bhaava	adhika hune avasthaa yaa bhaava
	uncountable	large	too many to be counted	the state of being more than full
Hi - Sa	शस्य	शस्य	कुछ पौधों से उत्पन्न होने वाले दाने	यः सततकर्तव्यकर्मानुष्ठाता प्रकृताचारशीलः तथा च न्यायपथावलम्बी अस्ति
	shasya	shasya	kucha paudhon se utpanna hone waale daane	yah satata-kartavaya-karmaanushthaataa prakrut--aachaarasheelah tathaa cha nyaaya-patha-avalambii asti
	grain	good	dry seedlike fruit produced by the cereal grasses: e.g. wheat, barley,	having desirable or positive qualities

Table 6.7: An example of detected False Friends along with sense definitions, from each language pair. The pronunciation and translation of each table entry are provided in the table.

which uses lexical similarity to detect the distance between words and their context. We also show significant improvements over the state-of-the-art approach and report gains of up to 27% points (Hi-Mr) and 40% points (Hi-Sa). We perform a manual error analysis of our output and report a high IAA of up to 0.87. We show that our approach is language and resource agnostic (in terms of Wordnet data) as consistent improvements are shown over all the language pairs, and over both datasets. *We release all the cross-lingual models, our code and the curated list of false friends' for further applications to NLP.*

In the future, we would like to perform experiments for the task of cognate detection using our proposed approach, and the same cross-lingual models. We would also like to add more corpora to cross-lingual word embedding models and use contextual sentence embeddings. We would also like to apply our method to other languages, including, but not limited to, English and all the "Romance Languages" (French, Italian, Romanian, and so on).

6.2 Summary

In this chapter, we described our efforts to improve the task of false friends' by utilizing features from cross-lingual word embeddings. We described in detail the creation of a gold standard dataset and then proposed classification approaches. Our evaluations show that cross-lingual word embeddings helped improve the task with a significant margin. We conclude this chapter with anecdotal examples of false friends detected from each language pair in our dataset.

Chapter 7

Computational Phylogenetics and Language Typology

In this thesis, so far we have discussed relatedness among languages and how vocabulary is shared among two or more languages. We established a motivation to perform the task of cognate detection and false friends' detection. We described the creation of datasets required for the task of cognate detection, *viz.,* creation of linked wordnet dataset, creation of pre-trained embeddings and the challenge datasets for the task of cognate and false friends' detection. We provided a detailed description of our investigations with regards to automatic cognate detection and false friends' detection. In this chapter, however, we start the second part of this thesis which details our investigations in the application of distributional semantics for computational phylogenetics for variant manuscripts and language typology for Indian languages. We begin by providing a history of textual criticism and stemmatology in the section below.

7.1 A Brief History of Textual Criticism

Textual criticism provides us with the principles for the scholarly editing of the texts of cultural heritage. In the western world, the tradition and practice of collecting, tending, and the act of preserving records was first instituted in the Hellenistic period[1]ri. The great library at Alexandria was the foremost collection of manuscripts of the ancient times. A school of textual scholarship was established at Alexandria, but its systematic principles,

[1]Helleistic Period: Reference

for the most part, have not survived. The next few ages were oblivious of this textual discipline. Instead, the scriptoria[2] of the centres of medieval age learning were affected by the pragmatics of the copyist. Scribes interpreted texts as they copied them, and as they did so they often compared variant source document exemplars[3] and, in the process, altered texts in transmission. Such interpretive criticism of variant readings remained a standard for the philologists of the medieval age. They laid the early foundations of modern textual scholarship and textual criticism. Their first attempts were the classical and medieval texts in the Latin language and the Greek language, but by the eighteenth century, scholarly editing was practised equally on vernacular texts [Groden and Kreiswirth, 1994].

Stemmatology started viewing textual criticism as "an articulation of a series of principles and rules for editing" based on genealogy and aimed at reconstructing a "text" [4]. At first, it was manuscript oriented, but later it was also adopted in biblical studies once rationalism had started questioning the belief that "a written text" or "scripture" was literally given by God, and thus, opened up an understanding of the historicity of the Bible through textual scholarship. The word 'stemmata' in classical Latin means "family trees" which shows that this meaning shows the relationships of the surviving witnesses. The family tree is also known as a cladogram, and hence the term cladistics, which means the tracing or classification of a family tree.

Such a classification when based on stemmatology or stemmatics is based on the logic that "community of error means community of origin". This logic implies that if two or more scribes observe a common set of errors in variants of a text, they are possibly dealing with variants which have originated from a common intermediate source, called a "hyparchetype". When all the intermediate variants are placed in the family tree based on this logic, it leads the stemmatologist to what is known as the "stemma". The process is known as "recension". After creating the stemma, the next step, called "selection" examines variants from the closest hyparchetypes to the archetype. The process of selection retains only the closest of the hyperachetypes and others hyperarchetypes are pruned from

[2]A room where scripts or texts are written or copied.

[3]Something that serves as a model; in this case, a particular version of the text being copied

[4]A Text here refers to a historical piece of written text, *viz.,* Panini's Grammer, The Bible, *etc.*

the stemma. After this step, if the stemmatologist still believes that there are errors due to lack of sources to back some of the hyperarchetypes, then the process of "examination" is applied to find corruptions and corrections can be made based on conjecture expertise and experience of the stemmatologist. This whole process helps obtain a single root also known as the "archetype". It helps create a revised version of the text which claims to be the "root of the text" also known as the critical edition of the text. An essential assumption that stemmatology makes is that *each variant or "witness" is derived from one, and only one, predecessor*. If a scribe refers to more than one source when creating their copy, then, the new copy will not fall into a single branch of the family tree. In the stemmatic method, a variant that is derived from more than one source is said to be 'contaminated'.

Stemmatology helped preserve much history and thereby immense cultural knowledge. Although in order to disseminate this cultural knowledge effectively, its digitization is a necessary step. By employing new computational methods unavailable to previous generations, Burrows [1987] showed that literary scholars had been ignoring most of the available evidence, as expressed in the opening sentence of the book, 'It is a truth not generally acknowledged that, in most discussions of works of English fiction, we proceed as if a third, two-fifths, a half of our material were not really there'. They showed that the function words — the 100 or so words that comprise articles, conjunctions, prepositions, and other linguistic 'glue' holding our sentences together — are just as amenable to literary criticism as the more visible, rarer lexical words. Since the publication of his work, many more texts have been digitized, and many more algorithms have been invented to process them in various ways. The digitization of manuscript data from old texts led to computational methods which could automate the stemmatics of text to a certain extent.

7.2 Computational Phylogenetics

Computational phylogenetics helps perform phylogenetic analysis where the eventual goal is to create a phylogenetic tree or phylogeny using computational methods and algorithms. It performs phylogenetic analysis by representing a hypothesis about the evolutionary relationships among a set of genes, species, or taxa. When the phylogenetic analysis is applied to variants of the same text, it helps perform cladistics using similar-

ity measures discussed in the chapters above. The similarity measures (orthographic or phonetic) can be used to obtain distances among manuscript variants and then generate a distance matrix. This distance matrix can then be used to generate a phylogenetic tree for the text, which can help build the critical edition of the text with further manual analysis by expert philologists. We have introduced computational phylogenetics in detail in Chapter 1.

Let us, now, dive straight into the short investigation which helps generate a typological tree for Indian languages with the help of cross-lingual word embeddings.

7.3 Harnessing Cross-lingual Word Embeddings to Infer Typological Trees

Establishing language relatedness by inferring phylogenetic trees has been a topic of interest in the area of diachronic linguistics. However, existing methods face meaning conflation deficiency due to the usage of lexical similarity-based measures. In this investigation, we utilize Wordnets to create inter-language distances using our novel method of computing 'Synset Distance'. We also conduct a pilot study on using deep cross-lingual word embeddings to compute inter-language distances and provide an effective distance matrix to infer phylogenetic trees. We conduct experiments on ten Indian languages and infer phylogenetic trees using this approach. For comparison, we also compare baseline approaches which use lexical similarity-based measures to infer phylogenetic trees, with our methodology. We show that our methodology produces better trees which club closely related languages together compared to the baseline methods.

Quantitative methods of studying language evolution require data collection and comparison, similar to the biological study of human evolution. While the study of human evolution requires a comparison of physical or genetic characteristics of biological species, language evolution requires the study of linguistic data. Under the purview of 'historical linguistics' also known as 'diachronic linguistics', establishing relationships among languages which have been in contact has been a topic of interest [Chevillet, 2000]. However, this sub-area has been much less explored in the computational linguistics community. Most of the previous work is focused on the reconstruction of phylogenetic trees for a

particular language family using manually curated word lists [Gray and Atkinson, 2003, Atkinson et al., 2005, Atkinson and Gray, 2006, Nakhleh et al., 2005] or using synthetic data [Barbançon et al., 2013]. Phylogenetic reconstruction and analysis are based on a data matrix where the rows represent the languages to be studied, and the columns represent linguistic features or characteristics [Nichols and Warnow, 2008]. Moreover, the methods inspired from glottochronology[5] take a boolean matrix as input, which denotes the change in the state of the 'characters'[6] to infer the phylogenetic trees.

Rama and Singh [2009] use corpus-based measures to compute the distance matrix containing inter-language distances and construct phylogenetic trees for a linguistic area[7]. Corpus-based measures can calculate the inter-language distance, but they use feature n-grams and cognate identification methods which loosely take into account the semantics of a word. It is well known that word meaning can be represented with a range of senses/concepts. The methods above do not take into account the 'concepts' or 'synsets' in a language and measure the inter-language distance only based on associated words pairs. Recently, an increasing boom on large-scale pre-trained word embedding models *e.g.,* FastText [Bojanowski et al., 2017b], ELMo [Peters et al., 2018], BERT [Devlin et al., 2018] have attracted considerable attention in the field of NLP. Inspired by the above works, this work proposes to use deep cross-lingual word embeddings [Lample et al., 2018] to find the inter-language distance based on 'concepts'. We introduce the term 'Synset Distance', *i.e.,* the distance between two concepts or senses calculated using the Wordnet [Fellbaum, 2012] data and use it to find the inter-language distance. We, then, use the inter-language distances among Indian language wordnets [Bhattacharyya, 2017] to infer phylogenetic trees and compare them with our baseline methods which are based on calculating the 'Synset Distance' using lexical similarity-based measures. The contributions of this investigation are as follows:

1. We hypothesize 'synset distance' based on wordnet data and utilize it to calculate inter-language distance.

[5]Study of the rate of change occurring in the vocabularies of languages.

[6]The 'characters' can be lexical, morphological or phonological

[7]The term linguistic area or Sprachbund [Emeneau, 1956] refers to a group of languages that have become similar in some way as a result of proximity and language contact, even if they belong to different families. The best-known example is the Indian (or South Asian) linguistic area.

2. We use deep cross-lingual word embeddings to calculate 'synset distance' and compute a distance matrix containing inter-language distances. Finally, we utilize this distance matrix to infer phylogenetic trees.

3. We release our dataset of synset distances along with the already public IndoWordnet data which can be used to help the cause of computational phylogenetics in future.

Among the many available methods [Huelsenbeck, 1995] to construct phylogenetic trees, UPGMA [Gronau and Moran, 2007] is widely used in historical linguistics. It assumes a constant rate of evolution and is not considered to be a perfect method for inferring relationships among the variants unless this assumption has been tested and justified for the data set being taken into account. The UPGMA method constructs phylogenetic trees based on a distance matrix which can be computed in various ways. Symmetric cross-entropy is one of the methods which is purely a letter n-gram based measure similar to the one used by Singh [2006b] for language and encoding identification. Singh and Surana [2007] used corpus-based measures to show that corpus can be used for a comparative study of languages. They used both character n-gram distances and surface similarity [Singh, 2006a] to identify the potential cognates[8], which in turn are being used to estimate the hypothesized inter-language distance. Rama and Singh [2009] also used measures based on cognate identification, and feature n-grams to infer this matrix. Ellison and Kirby [2006] discussed establishing a probability distribution for every language through intralexical comparison using confusion probabilities and estimate distances using KLdivergence and Rao's distance [Atkinson and Mitchell, 1981]. To the best of our knowledge, no one has utilized cross-lingual word embeddings to construct the distance matrix of inter-language distances. Also, wordnets, which are manually curated knowledge graphs of concepts and are publicly available for many languages [Bond et al., 2016][9] have been very underutilized in this regard. Synsets in a Wordnet are exhaustive in terms of words that can be used to represent a concept and thus provide sufficient representation

[8]Potential cognates are words of different languages which are similar in form and therefore are likely to be cognates. They might include some 'false friends', *i.e.,* words which are not etymologically inherited. It is worthwhile to experiment (using statistical techniques) on potential cognates, even without removing the 'false friends' because a large percentage of them are cognates in the linguistic sense.

[9]Collaborative Wordnets aligned to English

for it. Moreover, example sentences and glosses available along with the words can also be utilized to create better synset embeddings for our purpose.

7.3.1 Dataset and Experimental Setup

7.3.1.1 Wordnet Data

We build our dataset using IndoWordnet[10] data and investigate language pairs for Indian languages namely Hindi (Hi), Marathi (Mr), Konkani (Ko), Gujarati (Gu), Bengali (Bn), Oriya (Or), Assamese (As), Punjabi (Pa), Sanskrit (Sa), Tamil (Ta), Telugu(Te), Malayam (Ml), Kannada (Kn), and Nepali (Ne). Since Indian language wordnets are aligned based on synset IDs, we locate parallel concepts using the unique ID of synsets from two or more languages. We create word pairs for `<source_lang>-<target_lang>` for every language pair. We perform Unicode transliteration using Indic NLP Library[11] to convert scripts for As, Or, Kn, Bn, Gu, Pa, Ta, Te, and Ml to Devanagari, for standardization. Hi, Mr, Sa, and Ne are already based on Devanagari.

7.3.1.2 Training Data

We obtain the ILCI parallel corpora [Jha, 2010b] and add text parsed from Wikimedia[12] dumps. We normalize the scripts of the combined corpora and convert them to Devanagari. To ensure the presence of Wordnet words in these corpora, we add the Wordnet glosses and examples of these languages to their corpora. Thus, reducing the chances of out-of-vocabulary words when looking in the embedding space. The size of our monolingual corpora varies from 5 million lines for Marathi to 17 million lines for Hindi.

7.3.2 Methodology

We use two different methodologies to construct the language distance matrix required by the UPGMA method. For establishing a baseline, we use the first methodology based on lexical similarity-based measures. Later, we use our approach to compute the language distance matrix and compare the results. We describe both these methodologies below.

[10]IWN Data Download

[11]Indic NLP Library

[12]Wikimedia Dumps; as on March 22, 2019

7.3.2.1 Baseline Methods

We use the lexical similarity[13] based measures and employ different metrics to calculate the distance between each word pair in our dataset (from 7.3.1.1). We average over these distances and hypothesize it as the 'Synset Distance'. Similarly, we calculate the synset distance for all the parallel synsets in every language pair. Later, we average over all the synset distances and calculate the inter-language distance between two languages. We use the following similarity metrics:

Normalized Edit Distance Method (NED)

The Normalized Edit Distance approach computes the edit distance [Nerbonne and Heeringa, 1997] for all word pairs in a synset/concept and then provides as output the average distance between all word pairs or 'Synset Distance'.

Cosine Distance (CoD)

The cosine similarity measure [Salton and Buckley, 1988] is another similarity metric that measures the cosine of the angle between two vectors projected in a multi-dimensional space. Here, the two vectors are the arrays of character counts of two words. We calculate the cosine distance as (1 - Cosine Similarity).

Jaro-Winkler Distance (JWD)

Jaro-Winkler distance is a metric which measures the edit distance between two sequences. It uses a prefix scale P which gives more favourable ratings to strings that match from the beginning, for a set prefix length L.

7.3.2.2 Distance Matrix Computation

The above similarity metrics use different ways to compute similarity between each word pair and hence produces varying distance matrices. For computational purposes, we provide all the metrics equal weightages initially, and compute the distance matrix

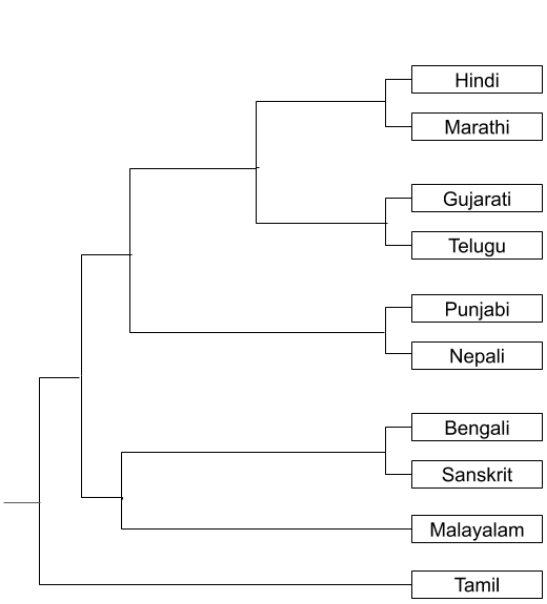
[13]String Similarity Library

using the average score of all three methods. For languages p and q, **the average inter-language distance** is defined as:

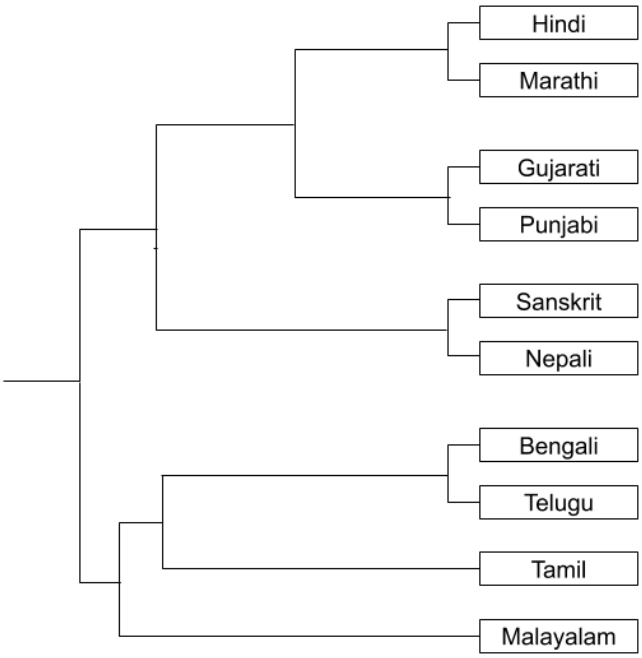
$$LD_{pq} = \frac{(NED_{pq} + CoD_{pq} + JWD_{pq})}{3} \tag{7.1}$$

We experiment over weightages and later provide different weightages to each method. Empirically, we find best results by setting the weight of NED to 50%, CoD to 25%, and JWD to 25%. For languages p and q, **the weighted average inter-language distance** is defined as:

$$LD_{pq} = (NED_{pq} * 0.5) + (CoD_{pq} * 0.25) + (JWD_{pq} * 0.25) \tag{7.2}$$



(a) Baseline: Average



(b) Baseline: Weighted Avg.

Figure 7.1: Trees Generated via UPGMA using Baseline method

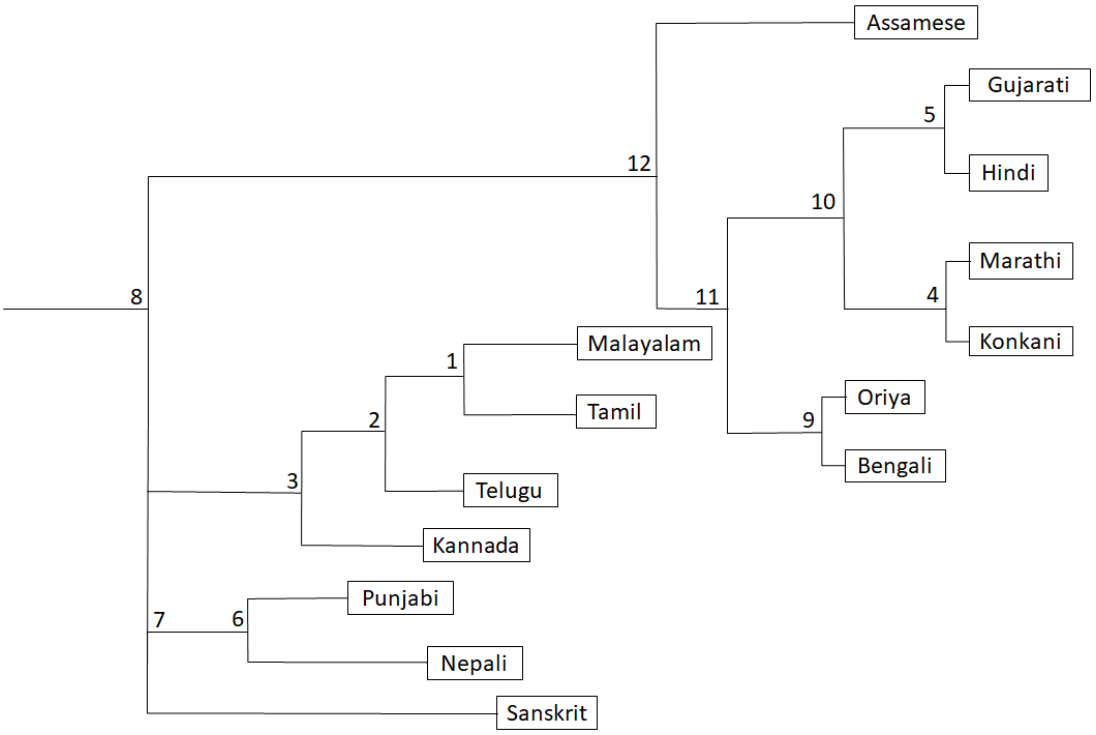


Figure 7.2: Resultant Tree Using Cross-lingual Embeddings

7.3.2.3 Our Approach using Cross-lingual Embeddings

We construct an efficient cross-lingual sense representation method using the data described in Subsection 7.3.1.2. First, we train monolingual word embeddings from this combined corpora using the FastText library. We train the monolingual word embeddings with 300 dimensions with 0.1 as the learning rate. The size of the context window is limited to 5 since FastText also uses sub-word information, and we train our model over 5 epochs. We use the default loss function (*softmax*) for training.

We calculate the cosine distance between all word pairs belonging to the same synset in the common embedding space shared by two languages. Thus, the average over the word pair distances gives us 'Synset Distance'. Similar to the baseline method, we average over all synset distances to find out the inter-language distance for each language pair and compute the distance matrix. Since angular cosine distance distinguishes nearly parallel vectors better [Cer et al., 2018], we also use angular cosine distance and calculate the inter-language distance for each language pair, in a similar fashion.

7.3.2.4 Typological Tree Construction

We use the UPGMA method to construct phylogenetic trees for all the language pairs. The input to the UPGMA method is the distance matrix created via the methodologies described above. We use the implementation of UPGMA provided by PHYLIP [Felsenstein, 1993] and generate baseline trees for NED, CoD, JWD, Average, and Weighted Average distance matrices. We also generate trees for distance matrices obtained using our approach of cosine distances and angular cosine distances from cross-lingual word embeddings.

7.3.3 Results and Discussion

As a baseline approach, we use a weighted lexical similarity measure to calculate the distance matrix. *The average of word-pair distances provides us 'synset distance' and further averaging of parallel synset distances provides us with a baseline inter-language distance.* **Our novel approach computes the angular cosine distance [Cer et al., 2018] between all word pairs belonging to the same synset in the common embedding space shared by two languages.** Thus, the average over the word-pair distances, and further 'synset distances' provides us with a more effective 'inter-language distance'. We use the UPGMA method to construct a phylogenetic tree (Figure 7.2) of all the language pairs. There are three subgroups of the languages which are visible in all the trees generated using our approach namely, Indo-Aryan (Hindi and Marathi; which share a lot of common words and even cognates [Kanojia et al., 2019b]), Indo-Aryan (Bengali, Gujarati, Punjabi, and Sanskrit), and Dravidian languages (Tamil, Malayalam and Telugu). Sanskrit, being the historical antecedent of most of the Indo-Aryan languages, is shown to have been accurately placed a level above the trees generated by our approach. Nepali is placed incorrectly probably due to the small size of Nepali Wordnet, and probably also due to the lack of a sizeable Nepali corpus for training data. There are apparent similarities in the trees which are generated by all the methods. All the methods group Hindi and Marathi with each other. Using the baseline approach, Telugu is not grouped with Dravidian languages which is incorrect becuase it is closer to Dravidian languages in terms of shared vocabulary. However, it belongs to the subgroup of Dravidian languages in the baseline weighted average, which shows the merit of the empirical work

done. Overall, it can be seen that Dravidian languages which borrow much lesser from the Indo-Aryan language family are in a different sub-group when our approach is used.

We introduce a new method of calculating the distance matrix used to create phylogenetic trees through UPGMA. We introduce 'Synset Distance' and successfully use Wordnets to calculate inter-language distances for Indian languages namely Hindi, Marathi, Gujarati, Bengali, Punjabi, Sanskrit, Tamil, Telugu, Malayam, and another Indo-Aryan language Nepali. We train deep cross-lingual word embeddings for every language pair and employ measures like cosine distance and angular cosine distance to compute distance matrices. We produce the same distance matrices using baseline lexical similarity-based measures like Normalized Levenshtein Distance, Cosine Distance, and Jaro-Winkler Distance as well. We use the UPGMA method to generate phylogenetic trees to find closeness in the languages and show that our approach produces better trees than the baseline methods. These experiments also demonstrate that our technique can be applied even to linguistic areas, not just language families. In future, we would like to include more languages and increase the size of the raw corpus used to create embeddings. We would also like to use other pre-trained cross-lingual embeddings and provide further substance to our claim.

Continuing upon our hypothesis for Inter-language distance, we apply the same logic to compute inter-manuscript distances and, describe below, our work in computational phylogenetics for variant manuscripts in Sanskrit. The experiments which now use monolingual embeddings for the Sanskrit language are detailed in the sections below in this chapter.

7.4 Utilizing Word Embeddings based Features for Phylogenetic Tree Generation

This work was accepted at the 6th International Sanskrit Computational Linguistics Symposium.

Tracing the root of a text, *i.e.,* the original version of the text, by inferring phylogenetic trees has been a topic of interest in philological studies. However, existing methods face meaning conflation deficiency due to the usage of lexical similarity-based measures which feed the distance matrix to clustering algorithms. In this investigation, we utilize word embeddings as features to compute the distances among manuscripts. We conduct this pilot study on using word embeddings to compute inter-manuscript distances and provide an effective distance matrix to infer phylogenetic trees. We conduct experiments on the historical Sanskrit text known as Kāśikāvṛtti (KV) and infer phylogenetic trees using this approach. For comparison, we also develop baseline methods using lexical distance-based measures to infer phylogenetic trees for KV. We show that our methodology produces better trees which club closely related manuscripts together compared to the baseline methods.

Phylogenetics is defined as the task of creating a "Phylogenetic Tree" that represents a hypothesis about the evolutionary relations of a set of genes, species, or any other taxa. It is the study of evolutionary history and relationships among various taxa. A Taxon represents a group of one or more manuscripts written in Sanskrit in our case, where we analyze how the manuscripts are related to each other. These relationships are discovered through phylogenetic methods that compute observed heritable traits in a manuscript, such as **spelling errors, variations in text, text deletion, the morphology of the text, etc.** under a model of the evolution of these traits. The result of these analyses is a phylogeny (also known as a phylogenetic tree) – a diagrammatic hypothesis about the history of the evolutionary relationships of a group of manuscripts (usually belonging to the same text). Computational historical linguistics, which involves the development of methods for estimating evolutionary histories of languages and, of models of language evolution, is another research problem based on phylogenetics. Phylogenetic methods are designed to recover the "true" evolutionary tree as often as possible. They do not guarantee to do so with high probability under reasonable conditions. Some which offer this guarantee vary considerably in their requirements [Warnow et al., 2001]. To rigorously establish the validity of such a phylogenetic approach, a fundamental question that must be addressed is whether the models in use are *identifiable*.

Our work is based on an earlier published sample edition of the KV on Aṣṭādhyāyī text (AST) 2.2.6 [Kulkarni, 2009]. This edition was prepared using seventy manuscripts written in several scripts and collected from various parts of the world. This earlier work did not utilize the computational method to establish inter-relations between manuscripts. [Kulkarni and Kahrs, 2018] also published a manually drawn tree based on the edition mentioned above. In this work, we apply the computational methods on the same data mentioned above and automatically infer phylogenetic trees that show the inter-relations between manuscripts. The question that we try to answer with this work is:

"Can word embeddings with sub-word information help build more accurate phylogenetic trees from multiple versions of a text?"

To the best of our knowledge, no one has utilized word embeddings to construct the distance matrix for inter-manuscript distances. We deploy lexical similarity-based methods as a baseline for inter-manuscript distance and compare the tree with the trees generated via our approach, *i.e.,* using word-embeddings to construct the distance matrix for the clustering methods (UPGMA and Neighbour Joining).
We contribute the following through this work:

- We hypothesize inter-manuscript distance and create efficient distance matrices for phylogenetic tree construction.
- We build baseline methodology using lexical similarity-based measures for comparison with our approach and generate phylogenetic trees.
- We construct a distance matrix through a word embeddings based approach as a novel contribution and show that the trees generated are better than the baseline method.

7.4.1 Dataset and Experiment Setup

We collect the following data for performing our experiments and tree construction.

7.4.1.1 KV Dataset

For distance matrix generation, we focus on specific portions of the KV. We collect seventy different versions of the KV on AST 2.2.6. We perform cleaning and manual

analysis with the help of philologists. These versions were available in different parts of the country from where we accumulated them in a single repository. We observe different kinds of changes in these versions and describe them in Section 7.4.7.

7.4.1.2 Raw Corpus for obtaining Word embeddings

We obtain raw monolingual Sanskrit corpus from various sources. We download the Sanskrit Wikimedia dump and collate all the articles as a single corpus. We, also, add Glosses and Example sentences from the Sanskrit Wordnet to this corpus. We obtain raw corpus from other sources available online[14]. We perform cleaning for this corpus by removing any other ASCII characters apart from the Devanagari script. The final cleaned corpus used for creating embeddings contains 5,38,323 lines. Eventually, We use binarized vectors to compute the distance between two words.

7.4.2 Experimental Setup

The Neighbor-Joining method and the UPGMA method are both distance-based methods, as described in Section 7.4.3. They require a distance matrix, which specifies the distance between the Taxa being used to populate the phylogeny. We also describe the methodologies used to obtain these matrices in Section 7.4.3. For our experiments, we divide the KV data into different functional units. The functional unit division in KV depends on the type of sūtra. The sūtra that we use for our experiments, namely AST 2.2.6, is of the type *vidhi*.
The functional unit division of this type is as follows:

- vidhi: This type of sūtra prescribes either a verbal element or an operation. The KV on this sūtra contains the following functional parts (sūtra AST 2.2.6):

    1. The sentence explaining the meaning of the words in the sūtra.
    2. Examples

These functional units help us understand the text in a better manner, and for computational purposes, they create separate divisions in the text so that the versions are compared to each other in an efficient manner. We compare each functional unit only

[14]Available on the School of Sanskrit and Indic Studies, J.N.U. and NLP for Sanskrit from GitHub

with its counterpart from the versions. For example, in AST 2.2.6 dataset, we compare the examples from one version only with the examples of the other version.

For training the word embeddings based model, we use Gensim[15]. We choose FastText [Bojanowski et al., 2017b] for training the word embeddings and obtaining vectors as it utilizes subword-level information within the text. Sanskrit is an agglutinative language which is also highly morphological. To capture the morphology and semantics within each word, we also need to take into account the sub-word level information. We train the models with the following hyperparameters. We create these models based on 100 and 50 dimensions due to a limited amount of the corpus collected[16]. The rest of the parameters were the same for both the models. We restrict the context window to 5 and use 0.1 as the learning rate. The maximum length of word n-gram we use is one word. We retain the sampling threshold at a default 0.0001. We use softmax as the loss function and train the models for five epochs[17].

7.4.3 Methodology

In this section, we describe the various methodologies used for calculating the inter-manuscript distances and tree construction.

We use two approaches for constructing the inter-manuscript distances. The baseline approach utilizes various lexical similarity-based measures, and later, we also provide weights to them, using empirical approaches, to increase their efficiency. In our approach, we use word-embedding based models and compute distances using vectors obtained from them. Since angular cosine distance distinguishes nearly parallel vectors better [Cer et al., 2018], we also include this in our approach, apart from cosine distance to generate more trees and discuss the outcome in Section 7.4.6.

[15]Gensim Source

[16]The standard number of dimensions for word embeddings, given a large corpus, is 300

[17]More epochs usually lead to a better learned/trained model; we retain the best epoch output with a minimum loss to be utilized for our work

7.4.3.1 Lexical Distance-based measures: A Baseline Approach

We use the following lexical similarity-based measures to compute the distances among manuscripts:

Normalized Edit Distance Method (NED)

The Normalized Edit Distance (also known as Levenshtein Distance) approach computes the edit distance [Nerbonne and Heeringa, 1997] for all word pairs in a functional unit of the text and then provides as output the average distance between all word pairs (we term it as 'Unit Distance'). In each of the operations has a unit cost (except that substitution of a character by itself has zero cost), so NED is equal to the minimum number of operations required to transform 'word a' to 'word b'. A general definition associates non-negative weight functions (insertions, deletions, and substitutions) with the operations.

Cosine Distance (CoD)

The cosine similarity measure [Salton and Buckley, 1988] is another similarity metric that depends on envisioning preferences as points in space. It measures the cosine of the angle between two vectors projected in a multi-dimensional space. The cosine similarity is particularly used in positive space, where the outcome is neatly bounded in [0,1]. The name derives from the term "direction cosine": in this case, unit vectors are maximally "similar" if they are parallel and maximally "dissimilar" if they are orthogonal (perpendicular). This is analogous to the cosine, which is 1 (maximum value) when the segments subtend a zero angle and 0 (uncorrelated) when the segments are perpendicular. In this context, the two vectors are the arrays of character counts of two words. We calculate the cosine distance as (1 - Cosine Similarity).

Jaro-Winkler Distance (JWD)

Jaro-Winkler distance [Winkler, 1990] is a string metric measuring similar to the normalized edit distance deriving itself from Jaro Distance [Jaro, 1989]. Here, the edit distance between two sequences is calculated using a prefix scale P which gives more favourable ratings to strings that match from the beginning, for a set prefix length L. The

lower the Jaro–Winkler distance for two strings is, the more similar the strings are. The score is normalized such that 1 equates to no similarity, and 0 is an exact match.

7.4.4 Distance Matrix Computation

The above similarity metrics use different ways to compute the distance between each word pair and hence, produce varying distance matrices. We compute the distance between a sūtra by averaging over each 'Unit Distance' present in a sūtra. We computer these distances between all the manuscript pairs. Thus, we generate three inter-manuscript distance matrices based on the methods described above.

Since all the matrices above use different ways to compute distances, we performed another set of experiments for coming up with a more homogenous approach. For computational purposes, we provide all the metrics equal weightages initially, and compute a single distance matrix using the average score of all three methods. So, for manuscripts p and q, **the average inter-manuscript distance** is defined as:

$$LD_{pq} = \frac{(NED_{pq} + CoD_{pq} + JWD_{pq})}{3} \tag{7.3}$$

We, also, experiment over weightages and later provide different weightages to each method. Empirically, we find best results by setting the weight of NED to 50%, CoD to 25%, and JWD to 25%. For masnucripts p and q, **the weighted average inter-manuscript distance** is defined as:

$$LD_{pq} = (NED_{pq} * 0.5) + (CoD_{pq} * 0.25) + (JWD_{pq} * 0.25) \tag{7.4}$$

Using the baseline methodology, we create a total of 5 matrices for the text in the AST 2.2.6 dataset.

7.4.4.1 Word embeddings based distance measures: Our Approach

We calculate the cosine distance between all word pairs belonging to the same functional unit from the embedding space. Thus, the average over the word pair distances

gives us 'Unit Distance'. Similar to the baseline method, we average over all unit distances to find out the inter-manuscript distance for each manuscript pair and compute the distance matrix. Since angular cosine distance distinguishes nearly parallel vectors better [Cer et al., 2018], we also use angular cosine distance and calculate the inter-manuscript distance for each manuscript pair, in a similar fashion. We perform this experiment using two different models described in the experimental setup.

Thus, for each dataset, our approach generates four matrices, *i.e.,* a matrix which utilizes Cosine Distance from the model with 100 dimensions, another which utilizes Cosine Distance from the model with 50 dimensions and another pair of matrices with Angular Cosine Distance from the models with 100 and 50 dimensions each. Using this approach, we create a total of four matrices. Using all of the methodologies described above (both baseline and our approach), we create a total of 9 matrices for the text in AST 2.2.6 dataset.

7.4.5 Tree generation using Distance-based methods

We choose two distance-based methods for our work, namely, the Neighbor-joining method and the UPGMA method. We further describe these methods below, along with the reasons for choosing these methods.

Distance analysis compares two aligned manuscripts at a time and builds a matrix of all possible sequence pairs. During each comparison, the number of changes (base substitutions and insertion/deletion events) is counted and presented as a proportion of the overall sequence length. These final estimates of the difference between all possible pairs of manuscripts are known as pairwise distances. A variety of distance algorithms are available to calculate the pairwise distance (between versions), for example, Proportional (p) distances. We use the baseline approach and our approach to compute these pairwise distances. Once the pairwise distances are calculated, they must be arranged into a tree. There are many ways to "arrange" the Taxa according to their distances. One way to cluster or optimize the distances is to join Taxa together according to their increasing differences, as embodied by their distances. Other ways use various coefficients to measure how well the branch lengths of the tree reflects the original pairwise distances.

Distance-matrix methods of phylogenetic analysis explicitly rely on a measure of "genetic distance" between the manuscripts being classified, and therefore they require an MSA (multiple sequence alignment) as an input. Distance is often defined as the fraction of mismatches at aligned positions, with gaps either ignored or counted as mismatches [David, 2001]. The main disadvantage of distance-matrix methods is their inability to efficiently use information about local high-variation regions that appear across multiple subtrees [Felsenstein, 2004]. Distance methods attempt to construct an all-to-all matrix from the sequence query set describing the distance between each sequence pair. From this is constructed a phylogenetic tree that places closely related manuscripts under the same interior node and whose branch lengths closely reproduce the observed distances between manuscripts. Distance-matrix methods may produce either rooted or unrooted trees, depending on the algorithm used to calculate them. They are frequently used as the basis for progressive and iterative types of multiple sequence alignment. The distance-based methods which we use are describe below.

UPGMA Method

The Unweighted Pair Group Method with Arithmetic mean (UPGMA) method [Sokal and Rohlf, 1962] produces rooted trees and requires a constant-rate assumption, *i.e.,* they assume an ultrametric tree in which the distances from the root to every branch tip are equal. At each step, the two of the nearest clusters are combined into an upper-level cluster. The distance between any two clusters $\mathcal{A}$ and $\mathcal{B}$, each of size, *i.e.,* cardinality, $|\mathcal{A}|$ and $|\mathcal{B}|$, is taken to be the average of all distances $D(x, y)$ between pairs of objects x in $\mathcal{A}$ and y in $\mathcal{B}$, that is, the mean distance between elements of each cluster. At each step, the updated distance between the joined clusters and a new cluster X is given by the proportional averaging of the distance between A given X and the distance between B given X.

We use the UPGMA method to construct phylogenetic trees for all the manuscript pairs. The input to the UPGMA method is the distance matrix created via the methodologies described above. We use the implementation of UPGMA provided by PHYLIP [Felsenstein, 1993] and generate baseline trees for NED, CoD, JWD, Average, and Weighted Average distance matrices. We also generate trees for distance matrices obtained using

our approach of cosine distances and angular cosine distances from word embeddings space.

Neighbor Joining Method

Neighbour-Joining [Saitou and Nei, 1987] is a bottom-up (agglomerative) clustering method for the creation of phylogenetic trees. It applies general data clustering techniques to sequence analysis and uses genetic distance as a clustering metric. The simple version of the neighbor-joining method produces unrooted trees, but it does not assume a constant rate of evolution, (*i.e.,* a constant timeline) across lineages. Neighbor-joining may be viewed as a greedy algorithm for optimizing according to the 'balanced minimum evolution' (BME) criterion. For each topology, the tree length (sum of branch lengths) is a weighted sum of the distances in the distance matrix, with the computed weights depending on the topology. The optimal topology (as per BME) is the one which minimizes this length. At each step, it greedily joins the pair of taxa, which provides the most significant decrease in the estimated length of the tree. This process is not guaranteed to find the topology, which is optimal by the BME criterion.

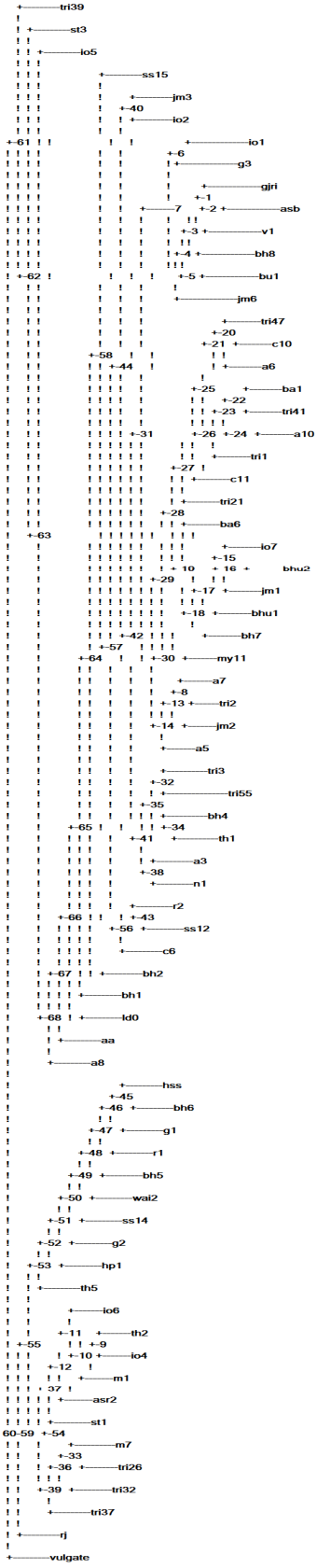
Similarly, we use the neighbor-joining method to construct phylogenetic trees for all the manuscript pairs. The input to this method is also the distance matrix created via the methodologies described above. We use the implementation of neighbor-joining provided by PHYLIP [Felsenstein, 1993] and generate all the trees from the matrices described above.

7.4.6 Results

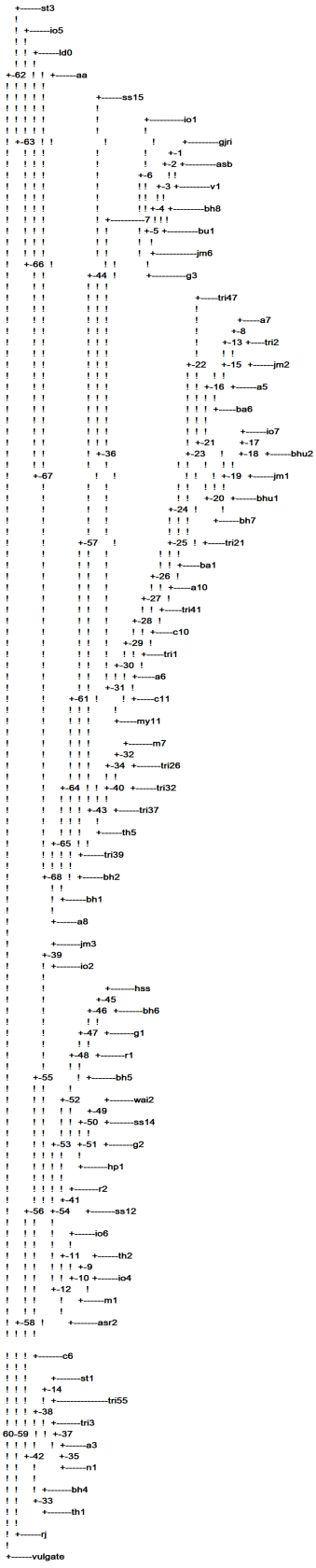
We generate trees using both the neighbor-joining and the UPGMA methods for all the matrices described above and compare them with the trees manually created by our philologists. With the help of two philologists, we annotate the group of sub-trees correctly clubbed, as output by both methodologies. The basis of this evaluation was the expert knowledge of our philologists who have studied the KV and are aware of the origin, groupings, and a vague timeline of all these manuscript versions. Their findings indicate that the trees generated via our approach of using word embeddings were closest to the manually created trees and required *fewer corrections* among the subgroupings to

be accurate. Although, among the baseline approaches, the weighted average methodology also reached closer to the manually created phylogenetic tree, yet it was *still a few corrections behind*. We cannot present the complete set of trees (9 x UPGMA and 9 x Neighbour Joining) here hence show the best tree generated by the baseline method in Figure 7.3a for the text in 2.2.6 dataset. Please refer to Appendix B for all the eighteen trees generated on various datasets. We obtain this tree using our novel approach of using a word-embeddings based model and using Neighbour-joining as the tree generation methodology. In Figure 7.3b for the text in 2.2.6 dataset, we also show the tree obtained by the weighted average lexical similarity measure, which was also generated using the Neighbour-joining method.

Among the word embeddings based approach, the trees generated via cosine distance are reported to be more accurate than the trees generated via angular cosine distance, as per our philologists. We compared the matrices generated by both cosine distance and angular cosine distance and found out that the distance values did not have much difference. This is probably due to the lack of a large raw monolingual corpus for the creation of word embeddings for Sanskrit. Despite being one of the most ancient languages, the availability of the resource for Sanskrit is scarce, which motivates us further to keep exploring this area. We discuss the results of our work and the merits of our methodology in the next section. We also provide justifications of our philologists' views in the forthcoming section.



(a) Distance matrix computed using the word-embeddings based method



(b) Distance matrix computed using the lexical similarity-based method.

7.4.7 Discussion

We discussed the functional units of the AST 2.2.6 dataset in the section above in brief and generated results based on the comparison of each unit. The division of KV data for the AST 2.2.6 text is shown in Table 7.1.

2.2.6.	नञ्
2.2.6.1	नञ् समर्थेन सुबन्तेन सह समस्यते तत्पुरुषश्च समासो भवति।
2.2.6.2	न ब्राह्मणो अब्राह्मणः। अवृषलः।।

Table 7.1: Example of Functional Unit based Division for sūtra AST 2.2.6

As can be seen in Figure 7.3a above, the sub-groupings for manuscripts have been done more accurately. Manuscripts io1, g3, gjri, asb, v1, bh8, bu1 and jm6 have been grouped since they do not contain a common functional unit. The same can be said about the tree in Figure 7.3a, but it does not group bh1 and ld0 in the same sub-group, which should not have been the case.

Differences among the manuscript variants in this edition Kulkarni [2009] are mainly divided into four categories. The apparatus of this edition contains the mention of the following types of variants:

Omission (Om.): absence of a word.

Addition (Add.): presence of an additional word

Change of word (CW): lexical changes in the word due to morphological inflection, or due to the opinion of the scribe who created the manuscript variant.

Change in the place of a word (CPW): change in the positioning of a word among the functional unit in a text.

We develop both the baseline approach and word embeddings based approach keeping these variants in mind. Our approaches handle these variants in the following manner:

Ommission (Om.)

Ommission reflects the omitted portion of the text derived after comparing the critical edition with the manuscripts of the text. Our approaches calculate the distances between all word pairs of each functional unit, on both sides. When we perform the comparison between an omitted word on one side and do not find its counterpart on the other side, it results in a higher penalty and a greater distance like it should for an omitted word.

Addition (Add.)

Addition refers to the added portion of the text as available in the manuscripts. It can be one or more words depending on the variant. When we average of all the distances between all word pairs, and in the comparisons made, do not find the added words; it results in a high penalty a more significant overall distance like it should for an added portion.

Change of a word (CW)

CW refers to a change of word, in the manuscript, in comparison with the critical edition, *i.e.,* a word may undergo some morphological inflection or takes some other form but retains a semantic notion. In such a case, the baseline approach measures the lexical changes in a word but penalize this change relatively lower in magnitude. In our approach, since the semantic notion is maintained, the embeddings would provide with nearby vectors and thus also penalize relatively lower in magnitude, which is what should be done for such a variance.

Change in the place of a word (CPW)

CPW refers to the change in the place of the word in the manuscript in comparison with the critical edition. CPW implies that the words in question exist in the manuscript but changes its place. This is not the case with the previous three types of changes. Our methodology counters this variance when we average over all the word pairs. Since the word is indeed present in the functional unit of the text, we should be able to find its occurrence on the other side, and thus this would result in a penalty of lower magnitude in terms of distance. We discuss these approaches with our philologists, and their views

are in accordance with what our methodology does in penalizing computing distances.

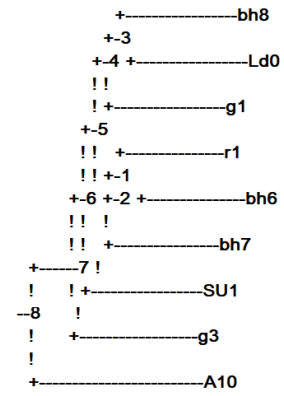


Figure 7.4: Phylogenetic Tree for the dated manuscripts generated using our method

Availibility of the timeline

Ancient Sanskrit text and its manuscripts are scarcely found dated. The unavailability of a timeline (or a temporal reference of versions) of how these texts evolved is a primary reason phylogenetic methods are needed to derive the root version (or the critical edition). We also note that some manuscripts among all the versions are dated, which do help identify the accuracy of a generated tree. Among the seventy versions of KV, we currently have the temporal references for eleven versions. We also generate phylogenetic trees for these versions using the neighbor-joining method based on the distance matrix computed using the word embeddings based approach they provided us with the best trees for AST 2.2.6. We depict this tree in Figure 7.4. In this tree, we have not yet implemented a method to refer to the timeline which is available. We plan to refine and generate such sub-trees based on the temporal references available to implement more accurate sub-trees of this type.

In this work, we presented a novel word embeddings based approach to create inter-manuscript distances and hypothesize functional units as a part of the text. We devised a baseline approach for drawing a comparison from our approach, which is based on lexical distance-based measures. We collect manuscript versions from different sources and accumulate them in a single repository and compute the inter-manuscript distance between all manuscript pairs, thus formulating a distance matrix for each approach. We collect raw

Sanskrit corpus from various sources and create a word embeddings model using the state-of-the-art library. We release this word embeddings model publicly for the use of other researchers looking to explore this area. Also, we compute inter-manuscript distances using this model and generate trees for both using both the baseline and this approach. We compare the trees manually, evaluate them with the help of expert philologists where we go on to show that the trees generated via word embeddings based models were better in subgrouping and required the least number of corrections to reach the state of manually drawn trees. We discuss the merits of our approach with examples and provide justifications of our results. Our approach clearly outperforms the baseline method and thus should help the researchers in this area to create better, more accurate phylogenetic trees soon.

In future, we would like to extend our dataset of the KV text to complete all the containing sūtras and perform the same experiments for all such portions of the KV text. We plan to divide each of such portions of text into functional units and perform the same experiment for the text. We aim to include the other material like text commentaries and earlier texts as a part of the experiment in the future, as they provide essential references to the text.

7.5 Inducing Timeline in Embeddings based Phylogenetic Tree Generation Method

It is observed that more and more scholars in the field of Indology are inclined to prepare critical editions of texts in Sanskrit and other Indian languages. These texts were transmitted in the oral as well as the written form for several centuries. Earlier, the community of Indological scholars used paper-based methodologies to develop critical editions. Now, with the availability of digital technology, scholars are using various tools for this purpose. Phylogeny tools are getting more popular in this regard as they provide tools based historical-scientific criteria for generating trees which show the interrelation among the sources used and thereby the transmission that happened. Maas [2009a] use the phylogenetic tools to compare variants and generate a tree for a portion of Carakasaṃhitā Vimānasthāna. Similarly, Phillips-Rodriguez et al. [2009] use compu-

tational phylogenetics to infer a tree for thirty-one collated variants of the Mahābhārata text. Kanojia et al. [2019a] introduce the use of word embeddings and perform phylogenetics using clustering-based approaches to show that the use of semantic similarity can help phylogenetic approaches to generate better trees. This approach used seventy manuscripts of the text known as "Kāśikāvṛtti" (KV) and successfully generated phylogenetic trees demonstrating the interrelation of manuscripts and thereby the transmission of this text. It also complemented the manually drawn tree [Kanojia et al., 2020b] of the same data and explained the evolution this text underwent in the 19th century, as demonstrated manually earlier by Kulkarni [2002d]. We note that only a handful of the manuscripts used [Kulkarni, 2002d, Kanojia et al., 2019a, 2020b] give information regarding the period, when they were copied. The timeline information for KV manuscripts is shown in Table 7.2.

Manuscript	g1	bh6	wai2	a10	ss15	bh7	ld0	R1	hp1	jm6	R2	ss3	jm2	io1	io6	g2	g3
Year	1408	1486	1533	1547	1596	1660	1693	1732	1756	1769	1832	1838	1852	1864	1869	1870	1887

Table 7.2: Approximated timeline data of the 17 manuscript variants which we use introduce as an added factor to compute our distance matrices.

Most of the manuscripts do not provide this information. Partial availability of timeline becomes a challenge, therefore, for a philologist to construct even a manual tree showing the interrelation of manuscripts. Availability of the timeline data for manuscript variants has been an issue for computational phylogeny-based approaches. Despite partial availability of such data, this has not been introduced with the heuristics which are currently being used by philologists. Ideally, a completely accurate timeline data should produce accurate phylogenetic trees, but such data is not available in its entirety, almost all of the times. What is available, is partial timeline data, which is based on an approximation of when the taxa evolved. Due to the availability of only partial and approximate data, this factor has been so far unimplemented throughout, when inferring phylogenetic trees. However, phylogenetic methods comprise of a few heuristics which assume a molecular clock of evolution at different rates [Zuckerkandl et al., 1960, Thorpe, 1982, Clarkson et al., 2017, Brown and Smith, 2018], but as far as our knowledge is concerned, no heuristic currently available utilizes the partial and approximated data. We hypothesize that if an approximated timeline information is induced in the distance matrix, it should be

able to help generate more accurate trees in terms of parent-child relationships. The key question that we try to answer with this work is,

"**Can approximated timeline data be utilized to generate accurate phylogenetic trees?**"

In this work, we show how partial and approximate timeline information can help a distributed similarity-based approach to compute a distance matrix. This distance matrix can, thus, be utilized to generate more accurate phylogenetic trees. We perform our experiments of two datasets KV AST 2.2.6 and KV AST 1.1.3 to show that our approach is dataset agnostic. Previous approaches [Kanojia et al., 2019a] generate a phylogenetic tree without using the timeline data. We compare our results on the same dataset with the added information of the timeline and show that introducing this 'factor', indeed, helps generate more accurate phylogenetic trees. Also, we further perform experiments on another dataset (KV AST 1.1.3) and show similar results.

7.5.1 Dataset Used

We utilize the following datasets to perform experiments using the baseline approaches and also using our novel approach.

7.5.2 Primary Datasets - AST 2.2.6 and AST 1.1.3

For our experiments, we use two primary datasets from KV. Kanojia et al. [2019a] perform their experiments on AST 2.2.6 and show that word embeddings can be utilized to generate an efficient distance matrix for clustering methods. We perform our experiment on their dataset for comparison. We also use Textual History Tool [Kanojia et al., 2019c] and digitize AST 1.1.3, with the help of philologists. We were able to obtain fifty-nine variants of AST 1.1.3 from various sources and digitized them. In the process of this manual digitization, we ensure the additional notes from the early scribes are ignored and stored in a separate column in our database. We avoid storing them with the main text of the variant since most of them contain the opinion of scribes who copied them. These do not belong to the text of the KV and only add noise while performing a comparison among the variants. We are, thus, able to ensure the clean digitization of AST 1.1.3 and use this as another dataset to compare the results of our approach with Kanojia et al.

[2019a]. We perform our experiments and infer trees using both these datasets and show them in Section 7.5.9. We also observe different kinds of variances in all the versions which we also discuss in Section 7.5.9.

7.5.3 Sanskrit Corpus & Monolingual Embeddings

We obtain monolingual Sanskrit corpora from various sources to build an efficient word embeddings based model. We download and collate articles from the Sanskrit Wikimedia dump[18]. We add more monolingual corpus to this dataset; a Prose corpus[19] and another monolingual corpus[20]. We, further, add KV text from the vulgate of AST 2.2.6 and AST 1.1.3 to this corpus. We choose only to add 'vulgate' variant to avoid duplication of the KV data in our corpus. We remove any other ASCII characters apart from the ones which belong to the Devanagari script. The final cleaned corpus contains approximately 6 million lines. We use this corpus to create the word embeddings models as described below, in this section. We use FastText to train a monolingual word embeddings model with the following hyperparameters. We train a 50 dimension due to a limited amount of the corpus collected[21]. We use the default learning rate of 0.1 and restrict the context window for each word to 5. We also restrict the maximum length of word n-gram to 1. We retain the sampling threshold at a default 0.0001. We use softmax as the loss function and train the models for five epochs[22]. The similarity obtained from these models in terms of the angular cosine function is normalized between $[0, 1]$. We simply subtract the similarity values from 1 to obtain the "distance value" for each word-pair comparison.
The functional unit divisions are as follows:

- vidhi: This type of sūtra prescribes either a verbal element or an operation. The KV on this sūtra contains the following functional parts (sūtra AST 2.2.6):

    1. The sentence explaining the meaning of the words in the sūtra.

[18]Wikimedia Downloads

[19]Available on the School of Sanskrit and Indic Studies, J.N.U.

[20]NLP for Sanskrit from GitHub

[21]The usual number of dimensions for word embeddings model, given a significantly large enough corpus, is 300

[22]The higher the number of epochs, better the learned model; we retain the best epoch output with a minimum loss to be utilized for our work

2. Examples

- meta-rules: This type of sūtra introduces the meta-rules within the system of AST. The KV on this sūtra contains the following functional parts (sūtra AST 1.1.3):

  1. The sentence explaining the type of meta-rule.
  2. The explanation of the words in the sūtra.
  3. Examples
  4. Discussion on an important word in the sūtra.
  5. Counterexamples.

These functional units help us understand the text in a better manner, and for computational purposes, they create separate divisions in the text so that the versions are efficiently compared to each other's corresponding functional units. We compare each functional unit only with its counterpart from the versions. For e.g., In AST 2.2.6 dataset, we compare the examples from one version only with the examples of the other version. Similarly, in the dataset from AST 1.1.3, we compare the explanation of words of the sūtra only with an explanation of words from the other version. We explain this approach of dividing the sūtra into different functional units for obtaining an effective manuscript distance in the next section.

7.5.4 Approaches

We describe the approaches used for calculating the inter-manuscript distances and tree construction.

7.5.5 Inter-Manuscript Distances

We use three approaches for constructing the inter-manuscript distances. The baseline approach utilizes a weighted combination of lexical similarity-based measures and the approach proposed by [Kanojia et al., 2019a] to construct distance matrices. We use an approach similar to [Kanojia et al., 2019a] but add the timeline data to our primary datasets. With each of the 21 variants, we add the year (approx.) associated with it to the dataset. The variants, for which, we do not have any year data; we add a 0 instead of

the year value. With this novel approach, we are able to compare the year of one variant to another with the use of word embeddings based model and the distances among these years help us provide a time-aware distance matrix, by using angular cosine distance, which helps distinguish nearly parallel vectors better [Cer et al., 2018].

7.5.6 Lexical Distance-based measures: A Baseline Approach

We use the following lexical distance-based measures to compute the distances among manuscripts:

Normalized Edit Distance Method (NED)

As described in Section 7.4.3.

Jaro-Winkler Distance (JWD)

As described in Section 7.4.3.

Q-gram Cosine Distance (CoD)

The cosine similarity measure [Salton and Buckley, 1988] is another similarity metric that depends on envisioning preferences as points in space. We construct these vectors by matching the q-grams of one word against the other and use padding where necessary. The cosine similarity is used in positive space, and the score is bound between [0,1]. The cosine similarity is 1 (maximum value) when the segments subtend a zero angle and 0 (uncorrelated) when the segments are perpendicular. In this context, the two vectors are the arrays of character counts of two words. We calculate the cosine distance as (1 - Cosine Similarity).

Distance Matrix Computation

The above similarity metrics use different ways to compute the distance between each word pair and hence, produce varying distance matrices. We compute the distance between a sūtra by averaging over each 'Unit Distance' present in a sūtra. We computer these distances between all the manuscript pairs. Thus, we generate three inter-manuscript distance matrices based on the methods described above.

Since all the matrices above use different ways to compute distances, we performed another set of experiments for coming up with a more homogenous approach. We rely on the empirical measures [Kanojia et al., 2019b,a] for providing weightage(s) to these measures. For manuscripts p and q, **the weighted inter-manuscript distance** is the same as defined in our work above.

7.5.7 Word embeddings based distance measures: A Stronger Baseline

We calculate the angular cosine distance between all word pairs belonging to the same functional unit from the embedding space, as described by Kanojia et al. [2019a]. Thus, the average over the word pair distances gives us 'Unit Distance'. Similar to the baseline method, we average over all unit distances to find out the inter-manuscript distance for each manuscript pair and compute the distance matrix. Thus, for each dataset, this approach generates one matrix.

7.5.8 Time-aware Distance Matrix: Our Novel Approach

We describe above how we add the approximate year value of a variant to the dataset. A sample key-value pair (where the key is the manuscript label) of our dataset for the variant "bh1", from AST 2.2.6, looks like

[bh1;1890;नञ्; नञ् समर्थेन सुबन्तेन सह समस्यते तत्पुरुषश्च समासो भवति।; न ब्राह्मणो अब्राह्मणः। अवृषलः।।]

["Manuscript Label";"Year";"Functional Unit 1";"Functional Unit 2";...;"Functional Unit N"].

We compare each value with its counterpart in a variant and compute a distance score. These distance scores are then averaged to compute a sūtra distance, which in our case is manuscript distance as we compare only one sūtra. This approach can be generalized, and when multiple sūtras are being compared, we can average over sūtra distances to compute 'Inter-manuscript distance'. These distances are represented in a distance matrix which contains the distance between each pair of variants.

Using all of the methodologies described above (both baselines, and our novel approach), we create distance matrices and then use the following algorithms to infer phylogenetic trees.

Fitch-Margoliash Method

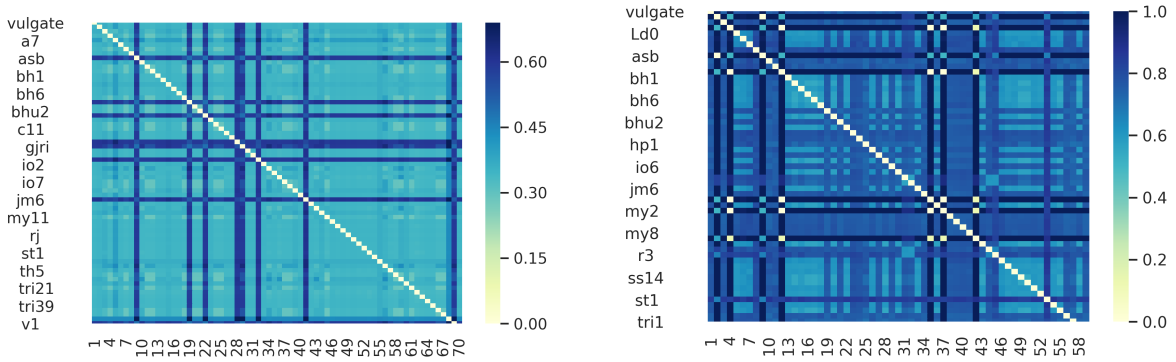
Lespinats et al. [2011] describe how molecular sequences can be viewed as elements in a multi-dimensional space. We represent words in a multi-dimensional space with the use of word embeddings. They use the Fitch-Margoliash method and show how phylogenetics can deal with the features of high-dimensional spaces like the concentration of measured phenomenon. The Fitch–Margoliash method [Fitch and Margoliash, 1967] uses a weighted least squares method for clustering, which is based on the distance matrix. The closely related are nearby sequences are awarded a higher weight in the tree inference process, to correct for a possible high inaccuracy in measuring distances between distantly related sequences. The input distances used to infer trees using this method must be normalized to prevent inaccuracies. The linearity criterion for distances in this method requires that the expected values of the branch lengths must equal the expected value of the sum of the two branch distances. The least-squares criterion applied to these distances is accurate but less efficient than the neighbor-joining method. An improvement that corrects for correlations between distances that arise from many closely related sequences in the dataset can also be applied at an increased computational cost, in terms of time. To be able to find the optimal least-squares tree with any correction factor is NP-complete [Day, 1987]. We use the implementation of the Fitch-Margoliash method provided by PHYLIP [Felsenstein, 1993] and generate all the trees from the matrices described above.

7.5.9 Results and Discussion

In this section, we present and discuss the results of the distance matrix computation, and the trees generated using the approaches described above. We also discuss the results in brief and explain how our approach helps include the available timeline data into trees generated. Figure 7.5 shows the distance matrix heatmaps for our primary datasets, AST 2.2.6 and AST 1.1.3, in (a) and (b), respectively. Similarly, Figure 7.6 shows the heatmaps generated using the approach proposed by Kanojia et al. [2019a]. Their approach does

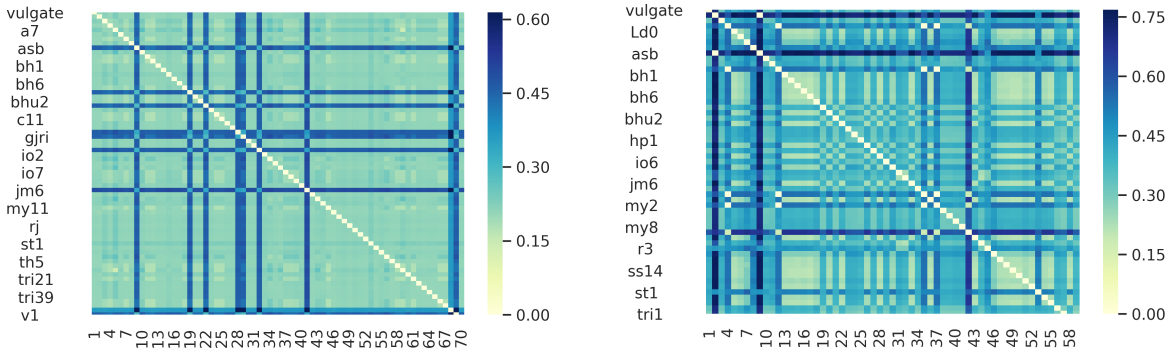
not take timeline data into account and uses the neighbor-joining method. The neighbor-joining method is fast but less efficient when it comes to accurately depicting the distances among the leaf nodes and clades (group/family of nodes). The distance matrix heatmap generates using our approach is shown in Figure 7.7 for both the datasets. The darker blue shades in the heatmaps represent a higher distance among the nodes. Among the three figures for each dataset, it can be clearly seen that our approach reduces the distances among the variants without changing the inherent representation/relationship among the taxa. The darkest heatmaps for the baseline approach show how distance these variants are to each other when only lexical similarity-based approaches are used. The use of semantic similarity-based approach proposed by Kanojia et al. [2019a] improves the distance matrix as claimed in their work. Our approach, however, further improves the distance matrix as seen in the heatmap. We use these matrices to generate trees using each approach described above.

Trees generated using the lexical similarity-based baseline are shown in Figure 7.8(a) and Figure 7.8(b), for AST 2.2.6 and AST 1.1.3, respectively. Both these trees show the nodes to be very distant from each other as lexical similarity-based approaches cannot capture the semantic similarity among the sūtras despite the variances. The distance matrix heatmaps for this approach also show the nodes to be far apart from each other, which is also evident in the generated trees. Although some of the groupings inferred in these trees are correct, as per our philologists, but it is **still very inaccurate in terms of grouping most of the clades.** Figure 7.9(a) and 7.9(b) show the trees generated by the stronger baseline which groups most of the clades correctly but does not take the timeline data into account. As a result, in the tree generated for AST 2.2.6, manuscript variant **g1** is shown to be a child node of **wai2**, which is shown to be one of the child nodes of **hp1**. Similarly, in the tree generated for AST 1.1.3, manuscript variant **bh1** is shown to be the parent of variants **st3**, **hss**, **ss13**, and **ss16** which should not be the case.



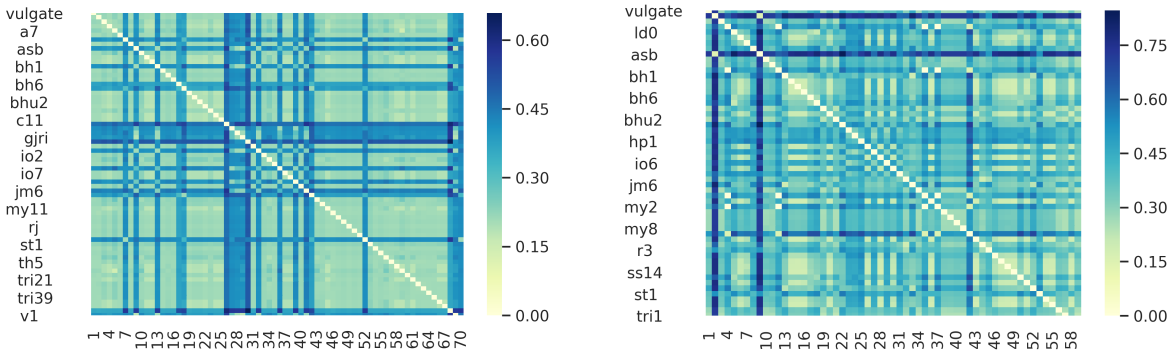
(a) 226 baseline (b) 113 baseline

Figure 7.5: Heatmap representations of the distance matrices computed using the weighted lexical similarity baseline, for AST 2.2.6 and AST 1.1.3 datasets



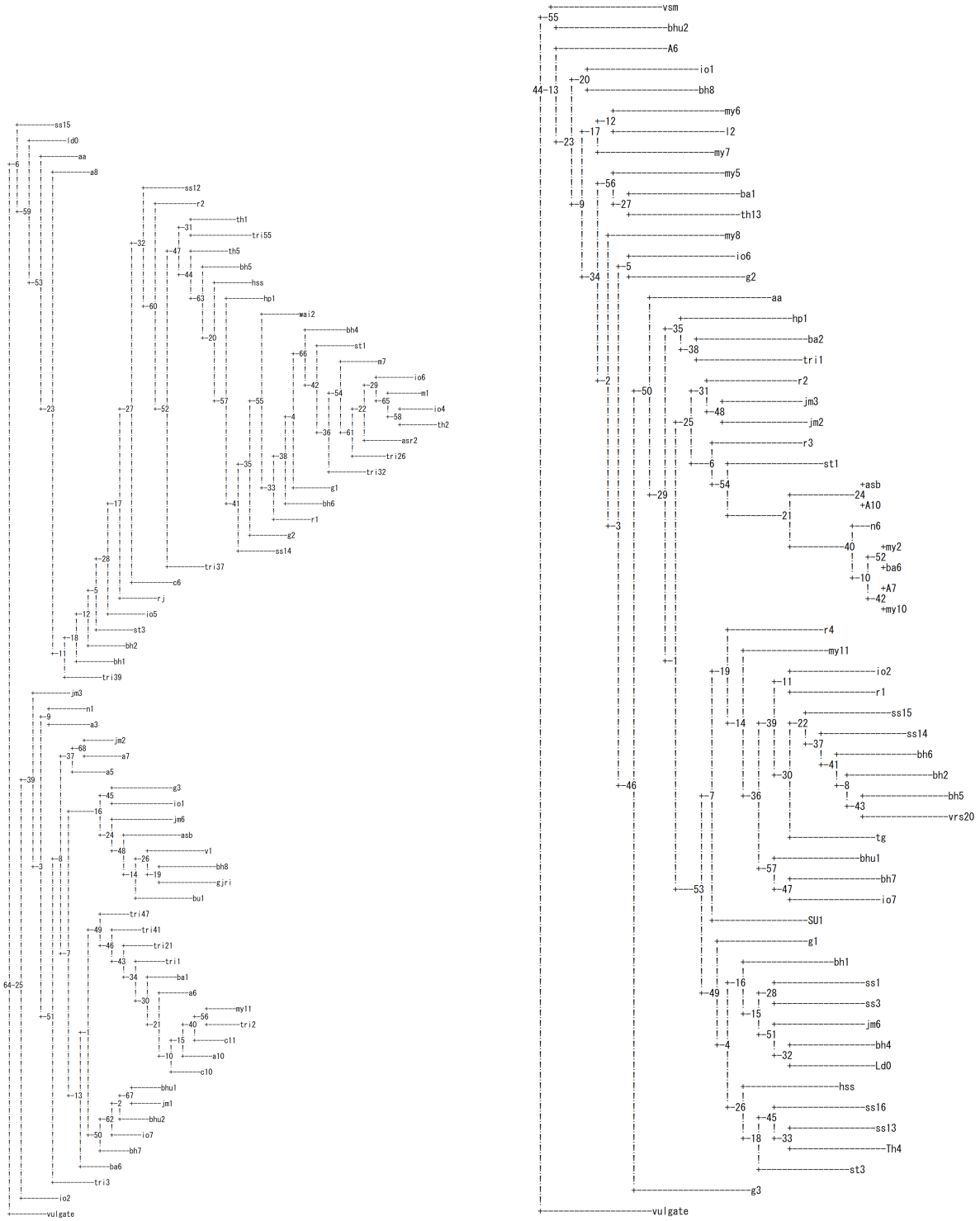
(a) 226 without time (b) 113 without time

Figure 7.6: Heatmap representations of the distance matrices computed using the stronger baseline approach, for AST 2.2.6 and AST 1.1.3 datasets



(a) 226 with time (b) 113 With time

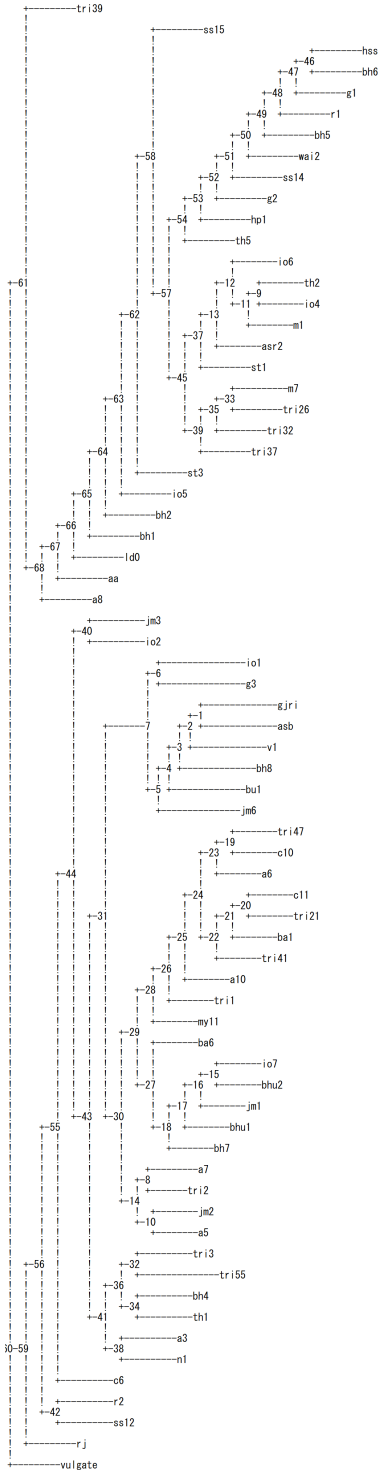
Figure 7.7: Heatmap representations of the distance matrices computed using our novel approach, for AST 2.2.6 and AST 1.1.3 datasets



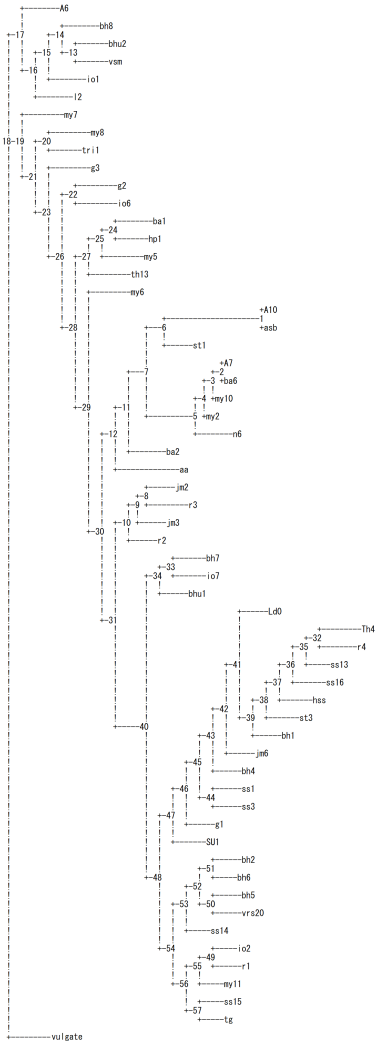
(a) AST 2.2.6: Baseline Approach

(b) AST 1.1.3: Baseline Approach

Figure 7.8: Trees generated using the weighted lexical similarity based baseline approach, for AST 2.2.6 and AST 1.1.3 datasets

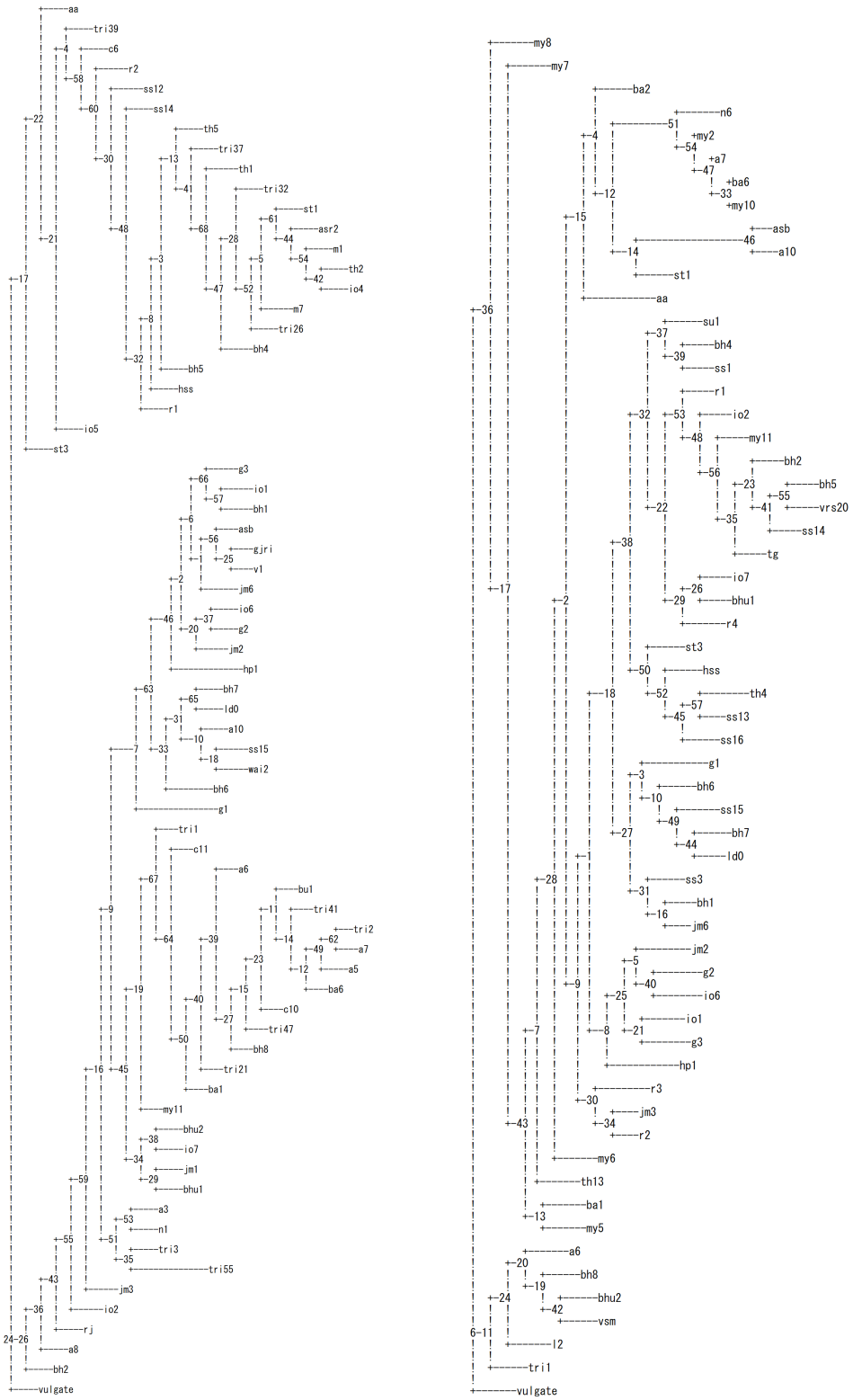


(a) AST 2.2.6: Stronger Baseline Approach



(b) AST 1.1.3: Stronger Baseline Approach

Figure 7.9: Trees generated using the stronger baseline approach, for AST 2.2.6 and AST 1.1.3 datasets



(a) AST 2.2.6: Our Approach (b) AST 1.1.3: Our Approach

Figure 7.10: Trees generated using the our approach, for AST 2.2.6 and AST 1.1.3 datasets

Using our approach, however, the trees generated do not adhere to these inaccuracies. Figure 7.10(a) and Figure 7.10(b) show the trees generated for AST 2.2.6, and AST 1.1.3, respectively. These trees have been generated using our approach which adds the timeline data to our primary datasets and uses the Fitch-Margoliash methods to infer more accurate phylogenetic trees. Our tree (AST 2.2.6), shows the variant **g1** to be a parent node for **bh6**, and **wai2**. It also show variants **io1**, **gjri**, **asb**, **v1** in the same clade; among other correct clades, it also shows **bhu1**, **io7**, and **jm1**, grouped together. Similar accurate parent-child relationships can also be found in the other tree generated using our approach [Figure 7.10(b)].

We discuss the functional units of the AST 2.2.6 dataset in the section above in brief and generate results based on the comparison of each unit. The division of KV data for the AST 2.2.6 text is shown in Table 7.3. Differences among the manuscript variants in this edition [Kulkarni, 2009] are mainly divided into four categories.

year	<approximate year, if available>
2.2.6.	नञ्
2.2.6.1	नञ् समर्थेन सुबन्तेन सह समस्यते तत्पुरुषश्च समासो भवति।
2.2.6.2	न ब्राह्मणो अब्राह्मणः। अवृषलः।।

Table 7.3: Example of Functional Unit based Division for sūtra AST 2.2.6

year	<approximate year, if available>
1.1.3.	इको गुणवृद्धी
1.1.3.1	परिभाषा इयं स्थानिनियमार्था । अनियमप्रसङ्गे नियमो विधीयते ।
1.1.3.2	वृद्धिगुणौ स्वसंज्ञया शिष्यमाणौ इकः एव स्थाने वेदितव्यौ । वक्ष्यति सार्वधातुकार्द्धधातुकयोः अङ्गस्य गुण इति। स इकः एव स्थाने वेदितव्यः।
1.1.3.3	तरति नयति भवति । वृद्धिः खल्वपि अकार्षीत् अहार्षीत् अचैषीत् अनैषीत् अलावीत् अस्तावीत्।
1.1.3.4	गुणवृद्धी यत्र स्वसंज्ञया विधीयेते तत्र इकः इति एतदुपस्थितं द्रष्टव्यम् । किं कृतं भवति ? द्वितीया षष्ठी प्रादुर्भाव्यते । मिदिमृजिपुगन्तलघूपधच्छिदृशिक्षिप्रक्षुद्रेष्वङ्गेनेग् विशेष्यते । जुसि सार्वधातुकादिगुणेषु इकाङ्गं विशेष्यते । मेद्यति अबिभयुः ।
1.1.3.5	इकः इति किम् ? आत्सन्ध्यक्षरव्यञ्जनानां मा भूत् यानम् ग्लायति उम्भिता । पुनर्गुणवृद्धिग्रहणं स्वसंज्ञया विधाने नियमार्थम् इह मा भूत् द्यौः पन्थाः सः इमम् इति ॥

Table 7.4: Example of Functional Unit based Division for sūtra AST 1.1.3

The apparatus of this edition contains the mention of the following types of variants:

Omission (Om.): absence of a word.

Addition (Add.): presence of an additional word

Change of word (CW): lexical changes in the word due to morphological inflexion, or due to the opinion of the scribe who created the manuscript variant.

Change in the place of a word (CPW): change in the positioning of a word among the functional unit in a text.

Our approach uses a distance matrix, as explained in the chapter above, keeping these variances in mind. Our method relies on the approach proposed by Kanojia et al. [2019a] and they have described in detail how this approach handles these variances.

In this work, we presented a novel approach to generate more accurate phylogenetic trees. This approach uses word embeddings to create inter-manuscript distances and utilize functional units as a part of the text. We use the approaches proposed by Kanojia et al. [2019a], as baselines to compare our results. These baselines are based on a weighted lexical similarity and word embeddings. The latter, however, does not use the timeline data which we inculcate in our primary datasets, for this work. We also use the Fitch-Margoliash method instead of the Neighbor-Joining method, unlike the baseline approaches. We collect manuscript versions from different sources and accumulate them in a single repository and compute the inter-manuscript distance between all manuscript pairs for the baseline approaches. We, then, add the available timeline data (for 17 variants) as additional functional units for re-computing the inter-manuscript distances to hopefully improve the distance matrices produced. We collect raw Sanskrit corpus from various sources and create a word embeddings model as described above. We release this word embeddings model publicly for the use of other researchers looking to explore this area. This should also help other researchers reproduce our results. We, then, compute the inter-manuscript distances using this model, utilizing the angular cosine distance metric and generate trees for both primary datasets using our approach.

Upon manual evaluation with the help of expert philologists, we present our finding that the trees generated via word embeddings based models were better in subgrouping and required less number of corrections to reach the state of manually drawn trees. We also observe that the **trees generated using our approach required the least number**

of corrections, and indeed **depicted the parent-child relations accurately**. We discuss the merits of our approach with examples and provide justifications of our results above. Our approach outperforms the baseline approaches and thus should help the researchers in this area to create better, more accurate phylogenetic trees, in the near future. Thus, we conclude by saying that, we successfully generated better trees using partial and approximated timeline information using word embeddings based approach. Using this approach, we could also figure out the approximate time information of those manuscript sources which do not have such information.

In future, we would like to extend our dataset of the KV text to complete all the sūtras and perform the similar experiments for all such divisions of the KV text. We also plan to include our work in the Textual History Tool [Kanojia et al., 2019c] which should be able to present a graphical interface to the philologists thus helping them digitize the data with the help of the phylogenetic tree output. We plan to divide each of such portions of text into functional units and perform the same experiment for the text. We aim to include the other material like text commentaries and earlier texts as a part of the experiment in the future, as they provide important references to the text.

7.6 Utilizing Multiple Factors in Texts to Generate Phylogenetic Trees

Texts in Indian history can be said to have travelled across a large time span. In the process they get commentaries written on them whose aim is to quote and explain them; they get quoted in many other texts either for the purpose of their refutation or to hail them as an authority in a dispute. These texts also get copied by the scribes in various scripts available in various regions. Sometimes, these copies can be said to be in touch with these commentaries and/or quotations. Sometimes, they can also be said to be influenced by the commentaries and/or quotations. All these factors make the transmission of the text a very complex phenomenon. In addition to this complexity, some modern editors bring out the printed editions of such texts with minimal basis, without consulting all these factors. Such editions add confusion to the chaotic situation that already exists.

A philologist who aims to bring out an authentic critical edition of the text should aim to consult all the material, as mentioned above, related to that text. This way, the philologist should be able to explain the historical transmission of the text and shall be able to point to the historically earlier version of the text. The text of the KV is believed to have been written in the 7th century CE. It got a direct commentary written on it in 8th century CE called Nyāsa (Ny) and another direct commentary was written on it in 11th century CE called Padamañjarī (Pm). These two commentaries quote and explain the text of the KV as they received. They also become a vital source for the transmission of the KV during this period. Kulkarni [2012] studies this aspect with respect to the ganapathas and showed that words in the ganas could be shown to have entered the gana in pre-Ny, post-Ny/pre-Pm, and post-Pm stages. The text of the KV got quoted in various texts of the Paninian grammatical tradition. Kulkarni [2002c] collected more than a thousand such quotations. The copied manuscripts of the KV are available from 15th century onwards. The text of the KV got printed for the first time in 1876 AD, and since, more than 10 printed editions have been published. None of them, however, can be said to have utilized the exhaustive resources mentioned above.

In the modern age, the use of digital technology to facilitate the editor to consult all textual material to prepare a critical edition is possible. The digital technology can also provide inputs in showing the interrelation amongst these sources by mapping the available evidence which can be represented in a tree format. The use of the phylogenetic method in this regard is proving to be very efficacious. In this regard, Kanojia et al. [2019c] created a comprehensive tool called the Textual History Tool (THT), which provides all the historical material related to a text at one place. It also generates trees showing interrelation amongst sources and displays them on an online interface. Their approach [Kanojia et al., 2019a] in generating these trees is based on the 'distributional hypothesis'. The distributional hypothesis is the basis for word embeddings, *i.e., words that are used in the same context and occur in the same contexts, tend to belong to similar meanings*. The underlying idea being that "a word is characterized by the company it keeps" [Firth, 1957]. This hypothesis suggests that the more semantically similar two words, the more distributionally similar they should be, and thus the more they should occur in similar linguistic contexts. This aforementioned approach, however, takes into

account only the manuscript evidence. We hypothesize that the trees generated using other factors, as mentioned above, would make the evidence exhaustive and make the tool more robust. We, in this work, attempt to remove this desideratum by generating trees while taking into account, factors like commentaries, testimonia, as well as printed editions. We observe that these factors play a crucial role in deciding the hierarchical relationship among variants as we discuss our findings later in this work. The key question that we try to answer with this work is,

"**Can these 'factors' be augmented within a framework to generate more insightful phylogenetic trees?**"

7.6.1 Dataset and Experiment Setup

We collect the following data for performing our experiments and tree construction.

7.6.1.1 KV Dataset

With the help of philologists, we digitize fifty-nine different versions of the KV on AST 1.1.3. We avoid the digitization of additional notes and comments from the scribes present on the variants for the effective functional unit division. By ignoring additional comments and noise, we are able to compare variants with each other based only on the text in the manuscript. Manual digitization of our variant texts also helps ensure clean dataset for AST 1.1.3.

We obtain these versions from different parts of the country and accumulate them in a single repository. We perform our experiments and infer trees using both these datasets and show them in Section 7.6.4. We also observe various kinds of variances in all the versions, which are also described in Section 7.6.4.

7.6.1.2 Commentaries Dataset

We also digitize two direct commentaries on the text of the KV. These commentaries are known as **Nyāsa (Ny) and Padamañjarī (Pm)**. We also request our philologists to provide two additional datasets from the aforementioned commentaries. Those words which are directly indicated in the manuscript text constitute of our additional dataset **NyDirect** and **PmDirect** and the words which are indirectly referred to in the

manuscripts constitute the other additional dataset from these commentaries. We refer to them as **NyIndirect** and **PmIndirect**. This additional information or more 'factors' help us generate separate phylogenetic trees and gather more insight into how they relate to the variants.

7.6.1.3 Testimonia

Unfortunately, we do not find any references to KV AST 1.1.3 in other texts. Testimonia is what we call the portions of a KV sūtra being quoted in other texts. KV 1.1.3, to be specific, has no such quotations in other texts, as far as our knowledge is concerned. We include this as a factor in our study because our work is generalizable, and if such a factor is available for other sūtras, we would be able to include it in further studies.

7.6.1.4 Printed Editions

We use seven different printed editions of the text of the KV. These printed editions are the work of modern scholars, as stated above, and claim to be an edition with additional insights depending upon the number of sources; these scholars take into account when collating the variants for their work. We digitize AST 1.1.3 from seven such variants known as Balashastri Edition (balaPE - 1876), Narela Edition (narelaPE - 1979), Suddhi Edition (suddhiPE - 1965), Raghuveer Edition (raghuPE - 1997), Ishwarchandra Edition (ishwarPE - 2004), Bahalgadh Edition (bahalPE - 1997), Chaukhamba Edition (chaukhPE - 1996). The acronyms provided to these editions are used as their labels while computationally augmenting them with the manuscript variant data. They are treated as separate variants available for the text of AST 1.1.3, which helps them be a part of this phylogenetic tree generation study.

7.6.1.5 Sanskrit Monolingual Corpus

One needs a large amount of raw monolingual corpus to build an efficient word embeddings model. This model can, then, be used to obtain distances between manuscript variants using both cosine distance and angular cosine distance. One can obtain the vector representation for a word from the said the model, and then choose to use either of the distance metrics to compute a distance score among two words. Manuscript variants are generated copies of the same text over a long period of time and contain different types

of variances. Despite these variances, manuscript variants essentially contain the essence from the same text and try to convey a similar thought. Hence, to be able to distinguish between *nearly the same* text, we need a metric which distinguishes nearly parallel vectors more accurately. We choose angular cosine distance as it has shown to do precisely that in previous studies [Cer et al., 2018]. This corpus contains data collected from various sources. We obtain Sanskrit Wikimedia dump[23] and collate the articles present in it to form a base corpus. To this base corpus, we add gloss and example sentences extracted from the Sanskrit Wordnet. We further add to it another Prose corpus[24] and another dataset[25] to ensure sufficient quantity.

Our word embeddings model is the crux of our approach and to ensure that this model is familiar with the AST text, we also add KV text from AST 1.1.3 to our corpus. Although, we only choose to add 'vulgate' variant text to it and not all the variants, to avoid adding too many similar sentences. We obtain this dataset from various sources as described above and to ensure its cleanliness; we remove any other ASCII characters apart from the ones which belong to the Devanagari script. The final cleaned corpus contains approximately 6 million lines. Eventually, we use this corpus to create the word embeddings models as described below in the experimental setup.

7.6.1.6 Experimental Setup

A word embeddings model provides the vector for a word. In distributed semantics, the basic approach is to collect distributional information in high-dimensional vectors and to define distributional or semantic similarity in terms of vector similarity. Different kinds of similarities can be extracted depending on which type of distributional information is used to collect the vectors: topical similarities can be extracted by populating the vectors with information on which text regions the linguistic items occur in; paradigmatic similarities can be extracted by populating the vectors with information on which other linguistic items the items co-occur with. We are concerned with paradigmatic similarity since the semantic similarity of two words can be computed by determining the agreement of their lexical neighbourhoods [Ruge, 1992].

[23]Wikimedia Dumps

[24]Available on the School of Sanskrit and Indic Studies, J.N.U.

[25]NLP for Sanskrit from GitHub

We use Gensim[26] to train our embeddings model by using the FastText [Bojanowski et al., 2017b] approach. We choose FastText for training the word embeddings and obtaining vectors since it is known to utilize subword-level information from amongst the words. To capture the morphology and semantics from within Sanskrit text, we need to take sub-word information into account. We train the models with the hyperparameters as described below. Previous empirical studies [Kanojia et al., 2020c] have shown that the amount of data collected by us for building a word embeddings model may still not be enough. In such cases, reducing the number of dimensions for a vector can be a possible solution, which we do. We create these 50-dimension models as we could only acquire a limited amount of the corpora. The learning rate hyperparameter was left at the default value of 0.1, and we restrict the context window for each word to 5. We also restrict the maximum length of word n-gram to 1, which helps us train the model faster. We set the sampling threshold at a default value of 0.0001. We use softmax as the loss function and train the models for five epochs[27]. The similarity obtained from these models in terms of the angular cosine function is normalized between $[0, 1]$. **We simply subtract the similarity values from 1 to obtain distance for each word-pair comparison.**

Amongst the tree generating approaches, we use the Neighbor-Joining method and the Fitch-Margoliash method. The neigbor-joining method was previously used by Kanojia et al. [2019a] and had proved to be effective when compared to the UPGMA method, in their study. We display the results of only the Fitch-Margoliash method in this version of the work as there are many different factors, and we generate five different trees using a single method. We choose to display the results for Fitch-Margoliash as they appear to be more coherent in terms of variant clades. We can display the other results as supplementary material in the camera-ready version of this work if accepted. All three of these methods are distance-based methods, as described in Section 7.6.2. As an input, they need a distance matrix, which specifies the distance between the taxa being used to populate the phylogeny. Our approach, much similar to [Kanojia et al., 2019a], generates this distance matrix by using word embeddings. We also describe this methodology used to obtain these matrices in Section 7.6.2. For our experiments, we divide each KV data

[26]Gensim Source

[27]More epochs usually lead to a better learned/trained model; we retain the best epoch output with a minimum loss to be utilized for our work

into different functional units. The functional unit division in KV depends on the type of sūtra. The sūtra that we use for our experiments, namely AST 1.1.3, is of the type *meta-rules*. The functional unit division is as follows:

- meta-rules: This type of sūtra introduces the meta-rules within the system of AST. The KV on this sūtra contains the following functional parts (sūtra AST 1.1.3):

    1. The sentence explaining the type of meta-rule.
    2. The explanation of the words in the sūtra.
    3. Examples
    4. Discussion on important words in the sūtra.
    5. Counterexamples.

These functional units help us understand the text in a better manner, and for computational purposes, they create separate divisions in the text so that the versions are compared to each other efficiently. We compare each functional unit only with its counterpart from the variants. For *e.g.,* In the dataset from AST 1.1.3, we compare the explanation of words of the sūtra only with an explanation of words, from the other variant. We explain this approach of dividing the sūtra into different functional units for obtaining an effective manuscript distance below.

7.6.2 Approaches

In this section, we describe the various approaches used for computing our distance matrix and tree construction.

7.6.3 Computing the Inter-Manuscript Distances

We use the approach described by Kanojia et al. [2019a] for computing inter-manuscript distances. We utilize the word embeddings based model created as described above, and simply obtain the angular cosine distances for each word pair comparison, in a functional unit. The approach averages each word-pair distance to compute a functional unit distance, and further averages all functional unit distances to obtain the manuscript variant distance. In case, we would have used multiple sūtras in the KV dataset, averaging the

functional unit distances should have provided us with a sūtra distance, and further averaging would lead us to manuscript distance. However, since we use only one sūtra as a dataset, we can obtain manuscript variant distance, in our case, by averaging the functional unit distances.

7.6.3.1 Tree Generating Methodologies

We now describe the methodologies used to generate phylogenetic trees using the distance matrix-based approach. This analysis compares two aligned manuscripts at a time and builds a matrix of all possible sequence pairs. For each comparison, the number of changes (substitutions/insertion/deletion events) is counted and presented as a proportion of the overall sequence length. These final estimates of the difference between all possible pairs of manuscripts are known as pairwise distances. Such analyses explicitly rely on a measure of "genetic distance" between the manuscripts which are being classified, and therefore they require a multiple sequence alignment (MSA) as an input. Their distance is often defined as the fraction of mismatches at aligned positions, with gaps either ignored or counted as mismatches [David, 2001]. A disadvantage of using distance-matrix methods is their inability to efficiently use information about local high-variation regions that appear across multiple subtrees [Felsenstein, 2004]. Distance matrix-based methods attempt to construct an all-to-all matrix from the sequence query set, which describe the distance between each sequence pair. They construct a phylogenetic tree from these distances that places closely related manuscripts under the same interior node, and their branch lengths closely reproduce the observed distances between manuscripts. Further, we describe the methods we choose to use for generation of trees for our work.

Neighbor Joining Method

Neighbor-joining [Saitou and Nei, 1987] is a method that belongs to the clustering based methodologies, but it does not require the data to be 'ultrametric', *i.e.,* it does not require that all lineages have diverged by equal amounts. This methodology is suited for manuscripts comprising lineages with vastly varying rates of evolution, such as ours. This method can be combined with methods that allow correction for superimposed substitutions. As the neighbor-joining algorithm aims to represent the data in the form of an "additive" tree, it may assign a negative length to the branch. We, however, do not allow

the algorithm to compute negative branch lengths for our manuscript variant comparisons. Here, the interpretation of branch lengths as an estimated number of substitutions gets into difficulties. When this occurs, it is advised to set the branch length to zero and transfer the difference to the adjacent branch length so that the total distance between an adjacent pair of terminal nodes remains unaffected.

For each topology, the tree length (sum of branch lengths) is a weighted sum of the distances in the distance matrix computed, with the weights depending on the topology. The optimal topology (as per BME) is the one which minimizes this length. At each step, it greedily joins the pair of taxa, which provides the most significant decrease in the estimated length of the tree. We do not show the trees generated using neighbor-joining method in this work and aim to show them as additional supplementary material only. This is because **on manual evaluation, we find that the Fitch-Margoliash method generates relatively more accurate trees, and we do not want to unnecessarily include large tree images, primarily when the trees generated from the Fitch-Margoliash method can be used to extract necessary insights about our factors as discussed.**

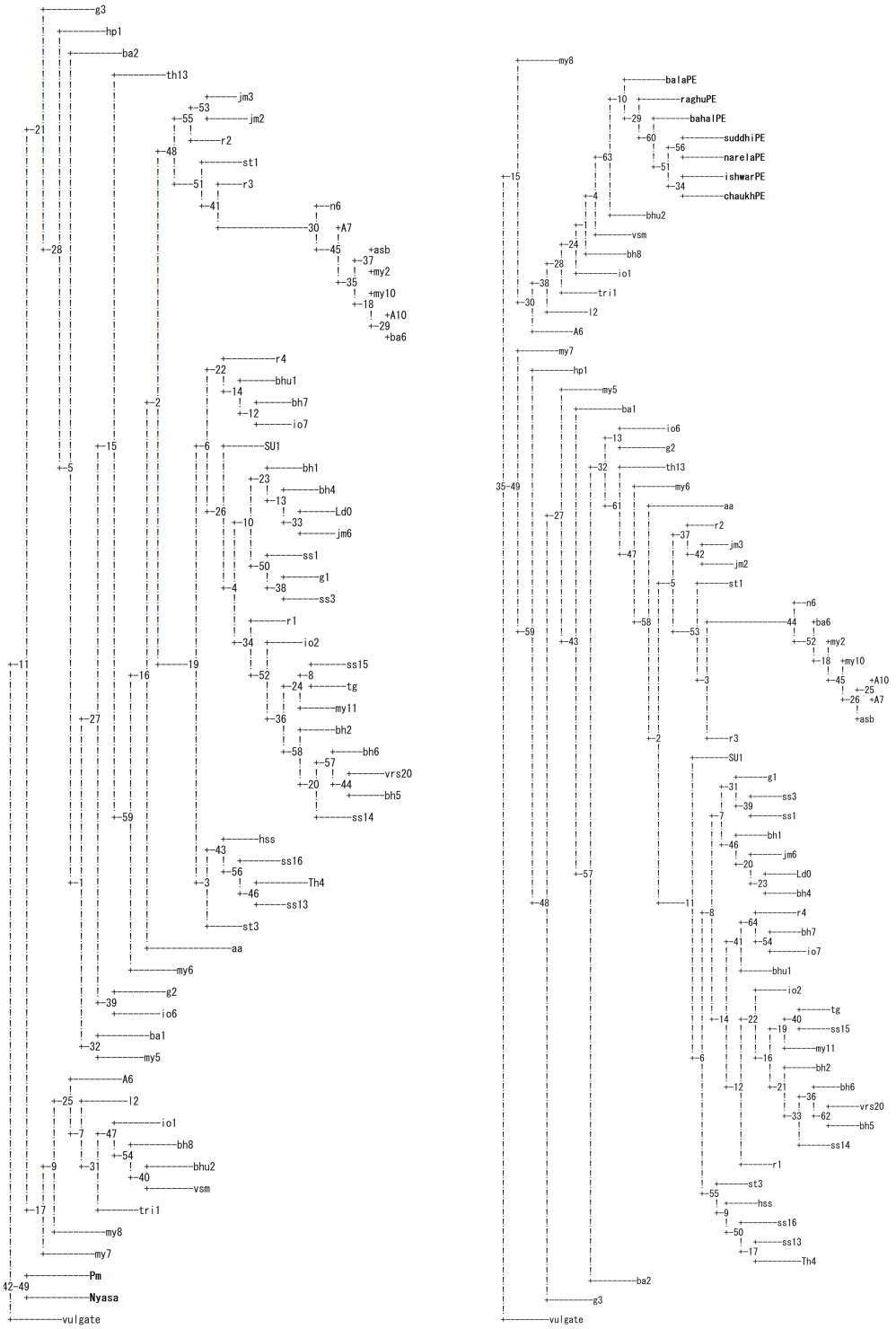
Fitch-Margoliash Method

Fitch-Margoliash method is based on the least-squares principle for phylogenetic tree reconstruction. The evolutionary distances between the manuscripts are determined by the Jukes-Cantor model when sequences (instead of distances) of the same length are entered. *However, we use the distances pre-computed via the word embeddings model.* This method is based on an optimality criterion that selects the tree with a minimum amount of residual. The algorithm estimates the total branch length (distance) and clusters per taxa pair to determine the unrooted tree with minimum distance. **This method does not assume a constant rate of evolution, which is a more accurate way of dealing with manuscript data that can evolve at any rate.** An optimized tree can be selected out of several possible trees in this method. One of the demerits of using this method is the underestimation of very long evolutionary distances because it ignores homoplasies, *i.e.,* absence of a character in the common ancestor. However, it is being shared by a group of related species originating from the common ancestor. This is a more

accurate method for generating trees for manuscripts as compared to the Neighbor-joining method as the latter is a more efficient way or a fast way of generating trees, whereas, for a large amount of data, **Fitch-Margoliash can consume much time but generates more accurate relationship among the taxa, which in our case, are manuscript variants and the factors we aim to utilize for a better insight.**

7.6.4 Results and Discussion

In this section, we display our resultant tree generated using the approaches and methods described above. We introduce the aforementioned factors with our manuscript variants dataset one-by-one and in combination, but our manuscript variants remain consistent. In figure 7.11(a), we show the tree generated for AST 1.1.3 when only commentaries (Ny and Pm) are introduced as the additional factor with the manuscript variants. In figure 7.11(b), we present the tree generated when only printed editions are introduced as an additional factor.



(a) AST 1.1.3 with Commentaries (Ny and Pm)

(b) AST 1.1.3 with Printed Editions

Figure 7.11: Trees generated with two different factors, for only commentaries (a) and only printed editions (b), respectively

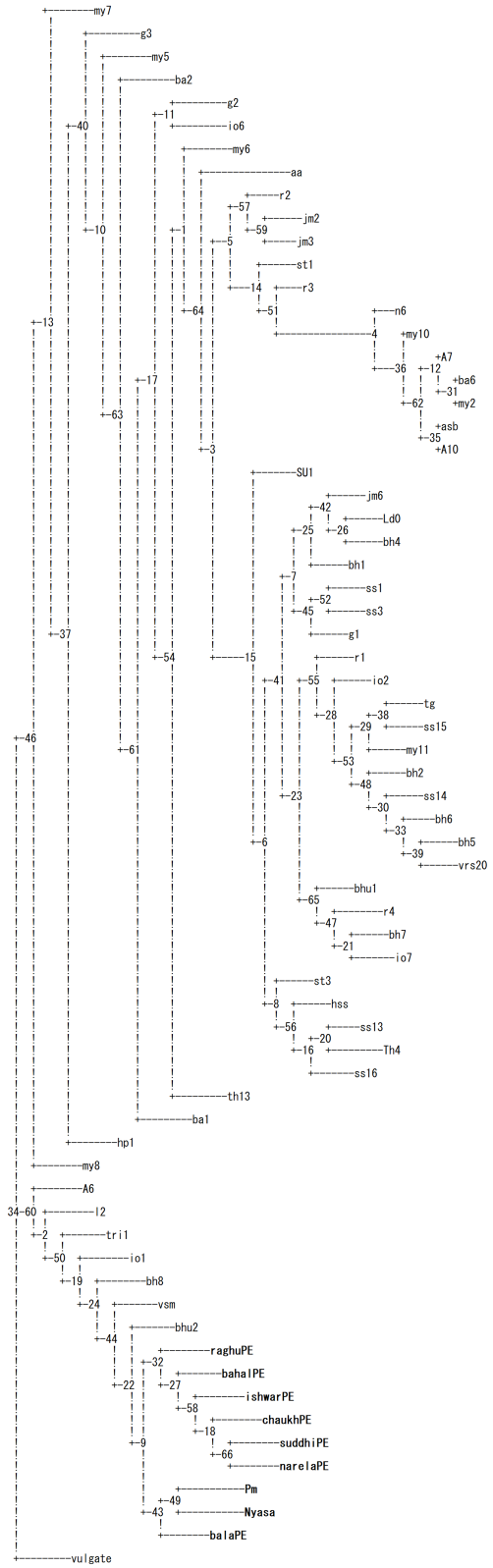
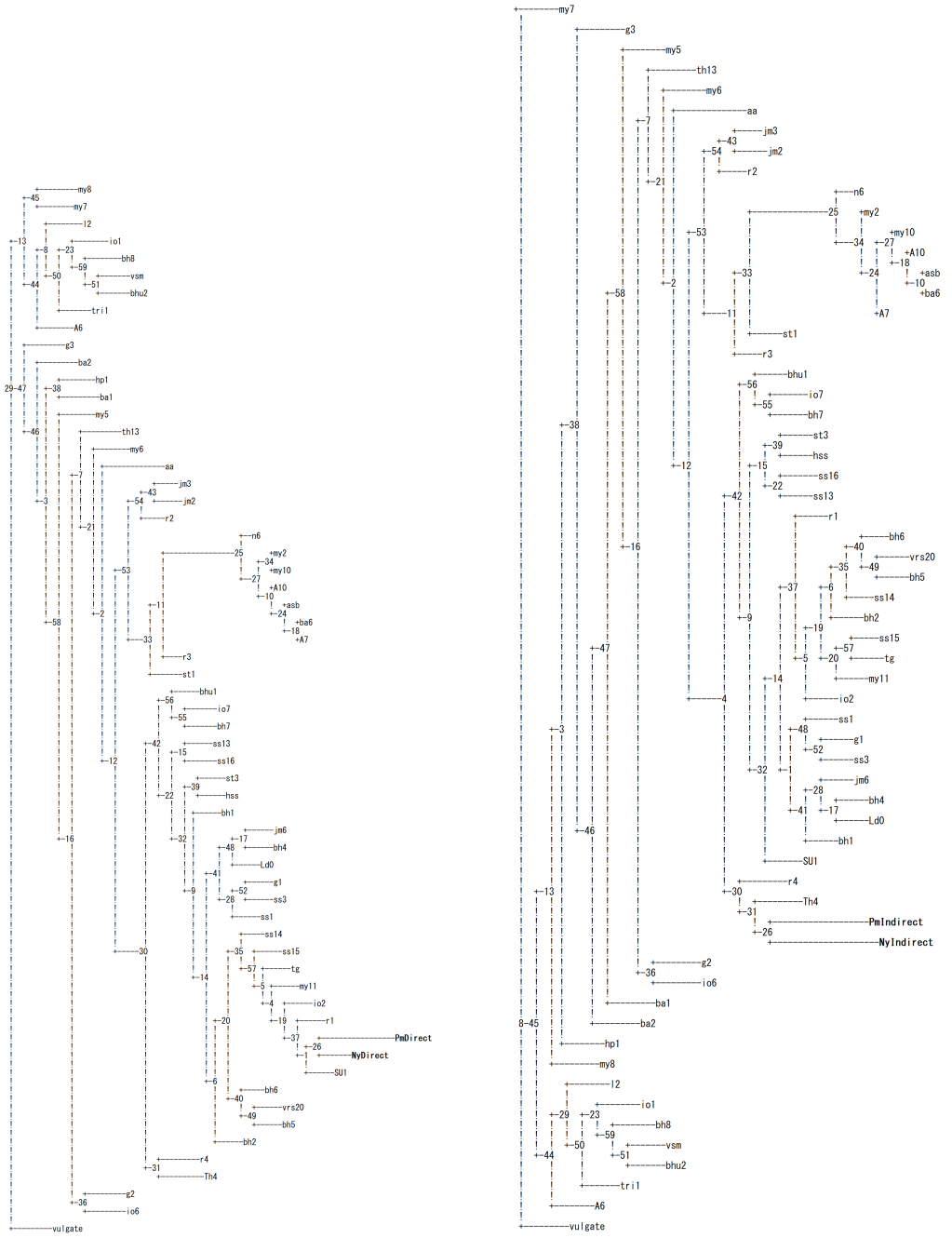


Figure 7.12: Tree generated by combining the datasets for commentaries and printed editions with the manuscript variants



(a) AST 1.1.3 with **directly** quoted words from Commentaries (Ny and Pm)

(b) AST 1.1.3 with **indirectly** quoted words from Commentaries (Ny and Pm)

Figure 7.13: Trees generated via further bifurcation of the commentary text into directly quoted words (Ny and Pm), and indirectly quoted words (Ny and Pm), respectively.



(a) AST 1.1.3 with **directly and indirectly quoted** words from the **Ny** Commentary

(b) AST 1.1.3 with **directly and indirectly quoted** words from the **Pm** Commentary

Figure 7.14: Trees generated via further bifurcation of the commentariusing es Ny (directly and indirectly quoted words) and Pm (directly and indirectly quoted words), respectively.

Since our trees are large and contain more than 59 leaf nodes for any generated tree diagram, we have boldfaced the labels for any additional factors for slightly better visibility in the Figures. From figure 7.11(a), we can observe that the commentaries Pm and Nyāsa leaf nodes are closest to the edition we consider 'vulgate'. We also observe that most of the clades formed in this tree belong to the same family of variants, as per our philologist's opinion. For *e.g.,* manuscript variants **asb**, **my2**, **my10**, **A10** and **ba5** belong to the same family and are clubbed together. The clubbing of the commentary texts Nyāsa and Pm with our 'vulgate' is a clear indication of interrelation between the 'vulgate' and the editions of the Ny and Pm. Given they are the earliest commentaries on the text of the KV, this should be treated as a piece of strong evidence towards the reconstruction of this text.

From figure 7.11(b), we observe that the printed editions concerned are not very close to our 'vulgate' edition. Although, Balashashtri printed edition is the closest to 'vulgate' when compared to any other editions from this study. This also goes to show that these printed editions are closer to some manuscript variants as compared to the 'vulgate' namely **bhu2**, **vsm**, **bh8**, **io1**, **tri1**, as shown. With the help of our philologists, we also confirm that the printed editions refer heavily from these variants. This is also an indication of the fact that these editions considered some texts important which may be missing from this analysis. We observe that this analysis considers only printed editions and the manuscript variants. Printed editions, as we know, are the works of modern scholars who, in all probability, take the oldest commentaries into consideration when coming up with an edition. Hence, we combine the datasets of printed editions and commentaries with the variants and generate another tree, as shown in Figure 7.12.

From figure 7.12, we observe that when both the factors are combined, we are able to obtain a clearer picture of phylogenetic reconstruction that went on to arrive at the printed editions. We see that Balashashtri printed edition (balaPE) is closest to the 'vulgate' now that the missing link of the commentaries is present during the tree generation. We see that the commentaries which earlier ranked closest to our 'vulgate' text, now, rank after balaPE. This probably happens because the text of balaPE and our vulgate are very similar. A phylogenetic tree should ideally place the variants hierarchically; but please keep in mind, that these additional factors are not our conventional variants, but some

of them are carefully constructed editions which have been arrived at, via the study and reconstruction/collation from various sources including the commentaries. Hence, it is not a surprise that a printed edition ranks closer to our 'vulgate' even compared to the oldest and direct commentaries. Figure 7.12 also shows that same family of manuscript variants are shown to be closest to the rest of the printed editions, *i.e.,* raghuPE, bahalPE, ishwarPE and so on, which implies these editions borrow heavily from the nearest family of variants, as far as AST 1.1.3 is concerned. This concludes our primary observations regarding commentaries and printed editions.

Additionally, when we bifurcate the Ny and Pm commentary data into directly quoted words, and indirectly quoted words, we generate 4 more trees, *i.e.,* AST 1.1.3 with directly quoted words for Ny (Figure 7.13(a)) and PM (Figure 7.13(b)) and AST 1.1.3 with directly and indirectly quoted words of the Ny commentary (Figure 7.13(a)) and with directly and indirectly quoted words of the Pm commentary (Figure 7.13(b)). From the first two figures which depict the directly quotes words for Ny/Pm and indirectly quoted words for Ny/PM, we observe that the Ny commentary ranks closer to manuscript variants **r1**, **io2**, **my11**, and so on; similarly, the Pm commentary text ranks closer to **Th4**, and **r4**. Interestingly, we also see that the length of the tree edges for both directly quoted nodes is shorter than the length of the tree edge for indirectly quoted nodes. This provides us with a piece of evidence that directly quoted word should be given more weightage while phylogenetic reconstruction and such bifurcation are beneficial for philologists.

It is noted that functional unit 1.1.3.4 has close resemblance with the text on the same sūtra in the Vyākaraṇa Mahābhāṣya of Patañjali. We note that both Ny and Pm do not have any direct evidence nor indirect evidence to support the existence of this functional unit as received by them. We also know that a group of 32 manuscripts also do not provide any evidence to the existence of this functional unit in the text of the KV. It means that only 27 manuscript sources posses evidence for the existence of this functional unit. Our trees show (in Figure 7.14(a) and (b)) that the group of 32 manuscripts in both the trees is aligned with the evidence presented by the Ny and the Pm. Primary amongst these 32 manuscripts are **g1**, **bh6**, **ld0**, **bh7**, *etc.* Thus, with the help of the combination of factors in the phylogenetic trees, we are able to show two streams of transmission clearly. One following the Mahabhashya tradition, and the other following the other tradition.

We present our insights into the utilization of additional factors for the generation of phylogenetic trees. Using a previously proposed approach which uses the semantic properties of a text to compute a distance matrix, we augment these additional factors into the aforementioned distance matrix. We, then, improve upon the said approach and use a different, more accurate method for phylogenetic tree generation and discuss our insights. We observe that from amongst recently published printed editions, the Balashashtri printed edition ranks closest to our 'vulgate'. This edition is also very close to the direct commentaries on the text of the KV, namely, Ny and Pm. Our approach and trees generated using that can also reveal the sources of the printed editions. We further bifurcate the commentaries dataset into directly quoted and indirectly quoted words and discuss our findings in the section above. Our approach successfully presents textual evidence comprehensively to the reader or the editor and is clearly able to support the hypothesis of multi-source transmission of the text of the KV. In this way, textual evidence related to the texts with complex history can be comprehensively presented via phylogenetic reconstruction; thereby explaining the complex transmission of that text in a better scientific manner.

In future, we would like to perform similar experiments with more sūtras of the text of the KV. We would like to present a comprehensive picture by also adding sūtras which have quotations present in other texts, thus including additional factors, *i.e.,* Testimonia/Earlier Texts, for such phylogenetic studies. We aim to inculcate this factor based tree generation methodology with the Textual History Tool so that philologists can benefit from this kind of reconstruction while digitizing other texts.

7.7 Summary

In this chapter, we have discussed various investigations which describe our work on computational phylogenetics and a short investigation on generating typological tree for Indian languages. We conclude with observations that cross-lingual embeddings can be successfully used to generate typological trees in the first study and show that closely-related languages are clubbed together. Later, we described the use of monolingual word embeddings to generate accurate distance matrices for multiple investigations. We, first, showed the results of our pilot study, which established the use of these embeddings for phylogenetics tree generation. We further show how embeddings can also be used to generate more accurate trees if timeline information is inculcated in the distance matrix. Finally, we showed how embeddings of various commentaries and printed editions can be used to analyze phylogenies generated using our method. Our qualitative analysis showed that commentaries, printed editions and quotations are placed at appropriate levels in the phylogenetic tree. This can help philologists arrive at an informed decision when finalizing a critical edition for a text. Our efforts to create an online tool which uses our phylogenetic approaches are described in the appendix of this thesis. With this chapter, we conclude the details of our investigations. We conclude the thesis in the next chapter with the hope that the reader can use our research, released datasets, code and other evidences from this thesis, for further research.

Chapter 8

Conclusion & Future Work

This chapter concludes this thesis on investigations into distributional semantics for the task of cognate detection and computational phylogenetics. In this chapter, we summarize the previous chapters, draw conclusions and discuss possible future directions.

8.1 Summary

With advancements in NLP research, there was a need to understand the machine-readable human language data. Humans understand natural languages due to years of acquired background knowledge and context, which computational models are trying to acquire as is evident by the significantly growing NLP research in the sub-area of semantics. The ability of a machine to understand human languages is primarily due to the development of distributional semantics and its applications to various NLP tasks. The computational linguistics community draws insights from natural languages and aims to improve the said machine-ability with statistical and rule-based methods alike. We first discussed one of the key aspects of multilingual or cross-lingual learning, *i.e.,* the relatedness of languages and the vocabulary shared among them. While we established a motivation to perform the investigations described in this thesis, we also described the tasks associated, *viz.,* automatic cognate detection, false friends' detection, and computational phylogenetics. We then described the previous research in these areas with a literature survey which detailed the datasets, approaches, and issues related to these tasks. We observed that the orthographic and phonetic feature-based approaches lacked an understanding of word-pairs in question. We also observed that there was a lack of

concentrated efforts to perform cognate detection for Indian languages which led us to investigate the use of various semantic feature-based approaches, for the Indian linguistic area. Our literature survey in phylogenetics led us to investigate the use of embeddings for phylogenetic tree generation. Our investigations are divided into four parts: understanding the phenomenon of distributional semantics, performing cognate detection with the help of various feature sets and deep learning-based approaches, performing false friends' detection with the help of cross-lingual features, and utilizing word embedding based features for computational phylogenetics. Each of these is summarized below:

Understanding Distributional Semantics We study and describe the various approaches in distributional semantics used to create embedding models which can be used for our research. This includes monolingual embeddings which are used for computational phylogenetics and cognate detection. It also entails a significant study of methods which help generate cross-lingual and bilingual embedding models. To understand the components of the aforementioned tasks, we perform the following studies:

- We describe the creation of a manually validated linked Indian language Wordnet data created with the efforts of lexicographers and release the data publicly. We also describe our efforts to link Indian Wordnet with the English wordnet. Our observations include the applications of this linked knowledge graph for various NLP tasks.
- In the second study, we discuss a semi-automated method to link wordnets using word embeddings. We observe that our method helps reduce the manual effort on the part of lexicographers by creating synset-embeddings for each synset and thus helps the linkage of concepts in two different languages.
- In the third study, we create various types of embedding models for Indian languages which are based on approaches proposed to generate monolingual, cross-lingual and multilingual embeddings. We create a total of 436 models and release them publicly[1]. We also evaluate these models based on the Named Entity Recognition task, model perplexity, and the Part-of-speech tagging task to further discuss the quality of the models released.

[1]Repository Link

Cognate Detection Task We presented seven investigations towards the task of cognate detection, which compare previously proposed approaches to cognate detection with our research:

- In the first study, we perform the task of cognate detection for a small dataset in Indian languages and observe that orthographic similarity-based approaches fail to detect cognates among Indian languages. With the help of manual validation, the low precision scores obtained using orthographic similarity implored us to further the research in another direction.
- In the second study, however, we proposed a combination of orthographic similarity measures using empirical measures and observed that using a deep learning-based approach helps improve task performance even with orthographic features. We hypothesized that the linked Indian Wordnets could be used for the task of cognate detection and showed it via our experiments.
- In the third study, we discussed how the dimensionality of word embedding models plays a crucial role in the cognate detection task. We explored the use of a shared-layer based Siamese Feed Forward Neural Network and showed how the dimensions of embedding models should be kept in a check if one has a dearth of lexical resources needed to create them.
- In the fourth study, we discuss the creation of challenge datasets for the task of cognate and false friends' detection. We digitize a cognate dictionary with the help of lexicographers and use the linked Wordnet data to generate potential cognates and false friend word-pairs. With the help of bilingual annotators, we manually validate the data and release a curated list of cognates and false friends'. We also evaluate these datasets on previously proposed approaches and show that these datasets prove to be a challenge for the said approaches showing a significant reduction in their performance.
- In the fifth study, we use cross-lingual features from the embedding models released previously and show that they help the task of cognate detection for Indian languages. We also induce the detected cognates in a Machine Translation pipeline and show that detected cognates help improve the MT task with an improvement in BLEU scores.

- In the sixth study, we perform the task of gaze-data collection for a small set of cognate and false friend pairs in Indian languages. With the help of collected gaze-data, we generate additional features which help improve the task performance. We also train a neural network to predict the gaze-data for a large number of cognate and false friend pairs show that predicted gaze-features also marginally improve the task performance.
- In this final study, we perform the task of cognate alignment for Indian and European languages and show that Wordnet linkages obtained earlier in our research are able to help us generate cognate sets for the Indo-European language family. The resulting low number of cognate sets warrant further investigation for this study.

False Friends' Detection Task Cognates can confuse false friends while the task of automatic cognate detection is being performed.

- In this study, we perform the task of False Friends' detection using cross-lingual features and show that a combination of cross-lingual features can help the task of detecting false friends'. We perform a manual analysis of the detected false friends' and show that our approach is able to make semantic distinctions between fine-grained senses across languages for the false friend word-pairs.
- We release this curated false friends list along with the models and code for further research.

Computational Phylogenetics and Language Typology In this part of the thesis, we perform different studies which are show relatedness among Indian languages and also show how variant manuscripts can be clubbed together to generate accurate phylogenies based on distributional semantics.

- In the first study, we hypothesize inter-language distance based on Wordnet synsets. We show how closely related languages can be clubbed together using a distance matrix for the task of a typological tree generation. Our observations include the grouping of Indo-Aryan languages and Dravidian languages in an accurate manner when cross-lingual features are used.

- In the second study, we create a word embedding model for the Sanskrit language and utilize it to generate a distance matrix for phylogenetic tree generation. We show that orthographic similarity-based distance matrix generation is shown to infer trees which do not club closely related manuscript variants. With the help of expert philologists, our observations show that phylogenetic trees generated with the help of word embeddings are more accurate and club related variants together.

- In this third study, we describe our efforts to inculcate the timeline data for manuscript variants, wherever available, to improve the phylogenetic tree generation further. We show that by including the timeline data as an additional functional unit, the trees generated are able to show an improved descendance of the text into manuscript variants.

- In this fourth study, we observe how various factors like printed editions, quotations and commentaries, when included in our proposed approach, can help the manual phylogenetic analysis to be performed by the philologists. Our observations include the recently found commentary and sub-commentary data which can help generate a critical edition of the text with a more fine-grained analysis by the philologists.

- In an additional study in the appendix, we describe the creation and public release of an online tool called the Textual History Tool. This tool helps philologists enter manuscript data with an online interface and then depicts the information in various modes created. It uses our proposed approaches to generate phylogenetic trees on the backend and shows them on the online interface for further manual phylogenetic analysis to be done by the philologists. We describe all the possible features of the tool impending some future work which can help with a more fine-grained data addition. We also describe the additional work done towards improving the tool features which allow it to capture the data of commentaries and sub-commentaries for a text. Our observations include a proposed classification of text which can be generally applied to various manuscripts.

8.2 Conclusion

We start this work by formally defining our research problem in terms of cognates, false friends, partial cognate, and non-cognates. We define the tasks of automatic cognate detection and false friends' detection and establish a motivation for performing these tasks in the context of downstream NLP tasks. The existence of linked Indian language Wordnets was crucial for this research. We capitalize on the ongoing effort and utilize it for establishing our hypothesis. For solving our problem, we present our hypotheses that Indian language Wordnets can be utilized to prepare datasets for cognate and false friend detection. With the help of manual validation, we prepare the datasets for both the tasks and describe them in this thesis. Additionally, we report our work on creating Wordnet linkages and the training of embedding models.

Our work present novel insights into the task of cognate detection and false friends' detection. Our work reported in this thesis is built around the notion of cross-lingual similarity and language relatedness. We present our work which utilizes cross-lingual word embeddings and aids in the detection of cognates and false friends among the candidate pairs selected from two different datasets, respectively for both the tasks. We report our work with fourteen Indian languages namely Marathi (Mr), Konkani (Ko), Punjabi (Pa), Gujarati (Gu), Bengali (Bn), Oriya (Or), Assamese (As), Sanskrit (Sa), Nepali (Ne), Malayalam (Ml), Tamil (Ta), Kannada (Ka), Telugu (Te) and Hindi (Hi). Our proposed approach for the cognate detection task also shows improvement over the previous state-of-the-art approaches. We show how it can be applied to improve the task of Machine Translation using a straightforward method. We also generate additional features for the task of cognate detection using gaze-data captured on a small set of cognate and false friend pairs. We use our approach and also perform cognate detection for the task of the Indo-European language family.

For the task of False friends detection, we report our work on Indian language pairs using the same notion. Moreover, we devise a simple approach to implement, which requires only a monolingual corpus from each language to build the embeddings models. The cross-lingual projections can be built via either a bilingual dictionary or via adversarial training. We report results for our approaches with significant improvements over both all baseline approaches. We attribute these results to particular merit of our approach,

i.e., it captures the 'contextual similarity' of words better, using the word embeddings based model enriched by sub-word level information, in contrast to the baseline approach, which only considers the character substitutions as a metric to compute similarity or the neural classification which probably suffers from a lack of data. For the False friends' detection task, cross-lingual similarity and vectors based features to achieve the best results. After performing an ablation test with different feature set combinations, we observe that a combination of the orthographic similarity-based measures and cross-lingual similarity performs even better, which indicates that orthographic and semantic similarity should be combined to detect linguistic entities such as false friends. Our results resonate with the linguistic identity of false friends, which defines them to be word pairs with high orthographic similarity but low semantic similarity. We also discuss the reduced dimensionality of embedding models when low-resource languages are concerned.

We described how language relatedness could be used to improve NLP tasks and showed relations among languages in terms of a shared vocabulary. In the later part of this thesis, we discuss language typology and conceptualize inter-language distance based on linked knowledge graphs. We use the same notion to hypothesize inter-manuscript distance and generate distance matrices which help infer phylogenetic trees. Dividing the manuscript data into various functional units helps us compare manuscript variants in a better manner, thus improving our phylogenetic tree generation. Manual analysis with the help of philologists shows us that embeddings based tree generation seems to help the task of phylogeny creation. Further, we experiment with the induction of timeline data into the distance matrix based on the functional unit division proposed by us earlier. The availability of timeline data for manuscript variants is rare and can hinder the phylogenetic analysis being performed. With our approach, we are able to use whatever data is available and help the philologists with further analysis and the building of a critical edition for the text in question. All of the above state work in computational phylogenetics can be applied in a general fashion to any manuscript text in question. For the manuscript data we have, we go a step ahead and perform a fine-grained analysis on how various factors can also help the phylogenetic analysis is they are included in our tree generation approach. The current performance of our approaches is significantly better than previous approaches.

In conclusion, this thesis looks at handling multiple problems in the domain of lan-

guage relatedness and exploits distributional semantics to propose various approaches. The relatedness among shared vocabulary can be captured using cross-lingual features, and relatedness among manuscripts can be captured with monolingual embedding based features. The proposed approaches can be used to help the downstream NLP tasks and improve their performance. Hopefully, the outcome of our research can be used to further the investigations in cross-lingual domain of NLP.

8.3 Future Work

Our investigations in these areas point to several directions of future work:

Short-term Future Work : The following work items could make incremental additions to the findings of this thesis:

- We aim to perform an immediate evaluation of cross-lingual embeddings for the potential cognates among the Indo-European language family.
- Our cross-lingual approaches to cognate and false friends' detection can be improved by training the embeddings using the byte-pair encoding method.
- Integration of word etymology based dictionary can also provide us with etymological evidence for the task of cognate and false friends' detection.
- Based on our experiments in cognition-aware cognate detection, we can train a better approach to generate gaze-data in a multi-task setting for cognate detection and gaze-data prediction.
- We can perform the phylogenetic tree generation for other manuscript data based on the same approaches proposed by us.
- We should also look at how cognates are affected by language change over time in various areas of the world.

Mid-term Future Work : The following future work items could be additionally helpful:

- We can collect additional gaze-data for the task of cognate detection among other Indian language pairs and perform the task with a more fine-grained

analysis of how different languages speaker generate gaze-data for the task of cognate detection.

- We can also perform phylogenetic tree evaluation based on an automated method which we can devise based on further research. We should be able to compare the trees with a manually drawn gold-standard tree using an automated method.
- We should be able to perform the task of false friends' detection by creating a dataset of false friends' for other language families.
- We can also perform experiments on how cognates and false friends' are affected by various cultural changes.
- Improvement of the depiction of trees on the online tool created by us, with the use of Javascript-based libraries. Instead of using ASCII text generated by scripts, we could show the trees using nodes and edges.

Long-term Future Work : The following new directions are possible in the area of distributional semantics:

- In the long term, we envision improvements in how cross-lingual features are generated. Instead of using a simple projection layer to generate cross-lingual embeddings, we should be able to come up with a linguistically inspired approach to solve the problems of low-quality cross-lingual embeddings.
- This can help solve the downstream NLP tasks in a more general fashion instead of performing individual tasks and fine-tuning models for a given task.
- A special focussed research on how computational phylogenetics can help mitigate the manual phylogenetic analysis completely should also be undertaken.

This thesis presented our linguistically-motivated investigations using distributional semantics in the area of natural language processing. We discussed the relatedness of languages and how cognate words and shared vocabulary can help NLP tasks. We also investigated computational phylogenetics using the notion of semantic relatedness. The philosophy of building approaches which can help NLP using semantic relatedness is a useful direction provided in this thesis for future work in the area of NLP and AI.

Bibliography

Alan Akbik, Duncan Blythe, and Roland Vollgraf. Contextual string embeddings for sequence labeling. In *COLING 2018, 27th International Conference on Computational Linguistics*, pages 1638–1649, 2018.

Yaser Al-Onaizan, Jan Curin, Michael Jahr, Kevin Knight, John Lafferty, Dan Melamed, Franz-Josef Och, David Purdy, Noah A Smith, and David Yarowsky. Statistical machine translation. In *Final Report, JHU Summer Workshop*, volume 30, 1999.

Gerry TM Altmann. Regression-contingent analyses of eye movements during sentence processing: Reply to rayner and sereno. *Memory & Cognition*, 22(3):286–290, 1994.

Mark Amengual. Interlingual influence in bilingual speech: Cognate status effect in a continuum of bilingualism. *Bilingualism*, 15(3):517, 2012.

Mikel Artetxe, Gorka Labaka, and Eneko Agirre. Learning principled bilingual mappings of word embeddings while preserving monolingual invariance. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2289–2294, Austin, Texas, November 2016. Association for Computational Linguistics. doi: 10.18653/v1/D16-1250. URL `https://www.aclweb.org/anthology/D16-1250`.

Mikel Artetxe, Gorka Labaka, and Eneko Agirre. Learning bilingual word embeddings with (almost) no bilingual data. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 451–462, Vancouver, Canada, July 2017. Association for Computational Linguistics. doi: 10.18653/v1/P17-1042. URL `https://www.aclweb.org/anthology/P17-1042`.

Mikel Artetxe, Gorka Labaka, and Eneko Agirre. A robust self-learning method for fully unsupervised cross-lingual mappings of word embeddings. In *Proceedings of the 56th*

Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), volume 1, pages 789–798, 2018a.

Mikel Artetxe, Gorka Labaka, and Eneko Agirre. Unsupervised statistical machine translation. *arXiv preprint arXiv:1809.01272*, 2018b.

Mikel Artetxe, Gorka Labaka, and Eneko Agirre. An effective approach to unsupervised machine translation. *arXiv preprint arXiv:1902.01313*, 2019.

Philip Arthur, Graham Neubig, and Satoshi Nakamura. Incorporating discrete translation lexicons into neural machine translation. *arXiv preprint arXiv:1606.02006*, 2016.

Colin Atkinson and Ann FS Mitchell. Rao's distance measure. *Sankhyā: The Indian Journal of Statistics, Series A*, pages 345–365, 1981.

Quentin Atkinson, Geoff Nicholls, David Welch, and Russell Gray. From words to dates: water into wine, mathemagic or phylogenetic inference? *Transactions of the Philological Society*, 103(2):193–219, 2005.

Quentin D Atkinson and Russell D Gray. How old is the indo-european language family? illumination or more moths to the flame. *Phylogenetic methods and the prehistory of languages*, 91:109, 2006.

Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. Neural machine translation by jointly learning to align and translate, 2014. URL `http://arxiv.org/abs/1409.0473`. cite arxiv:1409.0473Comment: Accepted at ICLR 2015 as oral presentation.

Akanksha Bansal, Esha Banerjee, and Girish Nath Jha. Corpora creation for indian language technologies–the ilci project. In *the sixth Proceedings of Language Technology Conference (LTC '13)*, 2013.

François Barbançon, Steven N Evans, Luay Nakhleh, Don Ringe, and Tandy Warnow. An experimental study comparing linguistic phylogenetic reconstruction methods. *Diachronica*, 30(2):143–170, 2013.

Maria Barrett, Joachim Bingel, Frank Keller, and Anders Søgaard. Weakly supervised part-of-speech tagging using eye-tracking data. In *Proceedings of the 54th Annual Meet-*

ing of the Association for Computational Linguistics (Volume 2: Short Papers), pages 579–584, Berlin, Germany, August 2016. Association for Computational Linguistics.

Maria Barrett, Joachim Bingel, Nora Hollenstein, Marek Rei, and Anders Søgaard. Sequence classification with human attention. In *Proceedings of the 22nd Conference on Computational Natural Language Learning*, pages 302–312, Brussels, Belgium, October 2018. Association for Computational Linguistics. doi: 10.18653/v1/K18-1030. URL `https://www.aclweb.org/anthology/K18-1030`.

William H Baxter and Alexis Manaster Ramer. Beyond lumping and splitting: probabilistic issues in historical linguistics. *Time depth in historical linguistics*, 1:167–188, 2000.

Y. Bengio, P. Simard, and P. Frasconi. Learning Long-term Dependencies with Gradient Descent is Difficult. *IEEE Transactions on Neural Networks*, 5(2):157–166, March 1994. ISSN 1045-9227. doi: 10.1109/72.279181. URL `http://dx.doi.org/10.1109/72.279181`.

Yoshua Bengio, Réjean Ducharme, Pascal Vincent, and Christian Janvin. A neural probabilistic language model. *J. Mach. Learn. Res.*, 3:1137–1155, March 2003. ISSN 1532-4435.

Aditya Bhargava and Grzegorz Kondrak. Multiple word alignment with profile hidden markov models. In *Proceedings of Human Language Technologies: The 2009 Annual Conference of the North American Chapter of the Association for Computational Linguistics, Companion Volume: Student Research Workshop and Doctoral Consortium*, pages 43–48. Association for Computational Linguistics, 2009.

Pushpak Bhattacharyya. Indowordnet. In *In Proc. of LREC-10*. Citeseer, 2010.

Pushpak Bhattacharyya. Indowordnet. In *The WordNet in Indian Languages*, pages 1–18. Springer, 2017.

Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. Enriching Word Vectors with Subword Information. *Transactions of the Association for Computational Linguistics, TACL*, 5, 2017a.

Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. Enriching word vectors with subword information. *Transactions of the Association for Computational Linguistics*, 5:135–146, 2017b. ISSN 2307-387X.

Ondrej Bojar, Vojtech Diatka, Pavel Rychlỳ, Pavel Stranák, Vít Suchomel, Ales Tamchyna, and Daniel Zeman. Hindencorp-hindi-english and hindi-only corpus for machine translation. In *LREC*, pages 3550–3555, 2014.

Francis Bond and Ryan Foster. Linking and extending an open multilingual wordnet. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1352–1362, 2013.

Francis Bond, Piek Vossen, John P McCrae, and Christiane Fellbaum. Cili: the collaborative interlingual index. In *Proceedings of the Global WordNet Conference*, volume 2016, 2016.

Antoine Bordes, Sumit Chopra, and Jason Weston. Question answering with subgraph embeddings. *arXiv preprint arXiv:1406.3676*, 2014.

Cecil H Brown, Eric W Holman, Søren Wichmann, and Viveka Velupillai. Automated classification of the world's languages: a description of the method and preliminary results. *STUF-Language Typology and Universals Sprachtypologie und Universalienforschung*, 61(4):285–308, 2008.

Joseph W Brown and Stephen A Smith. The past sure is tense: on interpreting phylogenetic divergence time estimates. *Systematic Biology*, 67(2):340–353, 2018.

John Frederick Burrows. *Computation into criticism: A study of Jane Austen's novels and an experiment in method*. Clarendon Pr, 1987.

Santiago Castro, Jairo Bonanata, and Aiala Rosá. A high coverage method for automatic false friends detection for spanish and portuguese. In *Proceedings of the Fifth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial 2018)*, pages 29–36, 2018.

Daniel Cer, Yinfei Yang, Sheng-yi Kong, Nan Hua, Nicole Limtiaco, Rhomni St John, Noah Constant, Mario Guajardo-Cespedes, Steve Yuan, Chris Tar, et al. Universal sentence encoder. *arXiv preprint arXiv:1803.11175*, 2018.

Devendra Singh Chaplot, Sudha Bhingardive, and Pushpak Bhattacharyya. Indowordnet visualizer: A graphical user interface for browsing and exploring wordnets of indian languages. In *Proceedings of the Seventh Global Wordnet Conference*, pages 338–345, 2014.

Yanqing Chen and Steven Skiena. False-friend detection and entity matching via unsupervised transliteration. *arXiv preprint arXiv:1611.06722*, 2016.

F Chevillet. L. campbell.-" historical linguistics: An introduction"(book review). *Études Anglaises*, 53(1):59, 2000.

Kyunghyun Cho, Bart van Merrienboer, Çaglar Gülçehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing, EMNLP 2014*, 2014.

Narayan Choudhary and Girish Nath Jha. Creating multilingual parallel corpora in indian languages. In *Language and Technology Conference*, pages 527–537. Springer, 2011.

Alina Maria Ciobanu and Liviu P Dinu. Automatic detection of cognates using orthographic alignment. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, volume 2, pages 99–105, 2014.

Alina Maria Ciobanu and Liviu P Dinu. Automatic discrimination between cognates and borrowings. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, volume 2, pages 431–437, 2015.

James J Clarkson, Steven Dodsworth, and Mark W Chase. Time-calibrated phylogenetic trees establish a lag between polyploidisation and diversification in nicotiana (solanaceae). *Plant Systematics and Evolution*, 303(8):1001–1012, 2017.

Jacob Cohen. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological bulletin*, 70(4):213, 1968.

Ronan Collobert and Jason Weston. A unified architecture for natural language processing: deep neural networks with multitask learning. In William W. Cohen, Andrew McCallum, and Sam T. Roweis, editors, *ICML*, volume 307 of *ACM International Conference Proceeding Series*, pages 160–167. ACM, 2008. ISBN 978-1-60558-205-4.

Ronan Collobert, Jason Weston, Léon Bottou, Michael Karlen, Koray Kavukcuoglu, and Pavel Kuksa. Natural Language Processing (Almost) from Scratch. *Journal of Machine Learning Research*, 12, 2011.

Montserrat Comesaña, Ana Paula Soares, Rosa Sánchez-Casas, and Cátia Lima. Lexical and semantic representations in the acquisition of l2 cognate and non-cognate words: Evidence from two learning methods in children. *British Journal of Psychology*, 103 (3):378–392, 2012.

Alexis Conneau and Guillaume Lample. Cross-lingual language model pretraining. In *Advances in Neural Information Processing Systems*, pages 7059–7069, 2019.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. *arXiv preprint arXiv:1911.02116*, 2019.

Michael A Covington. An algorithm to align words for historical comparison. *Computational linguistics*, 22(4):481–496, 1996.

Marc Csernel and François Patte. Critical edition of sanskrit texts. In *Sanskrit Computational Linguistics*, pages 358–379. Springer, 2007.

W Mount David. Bioinformatics: sequence and genome analysis. *Bioinformatics*, 28, 2001.

William HE Day. Computational complexity of inferring phylogenies from dissimilarity matrices. *Bulletin of mathematical biology*, 49(4):461–467, 1987.

Annette MB De Groot and Gerard LJ Nas. Lexical representation of cognates and noncognates in compound bilinguals. *Journal of memory and language*, 30(1):90–123, 1991.

Johannes Dellert. Evaluating cross-linguistic polysemies as a model of semantic change for cognate finding. In *Workshop on semantic technologies for research in the humanities and social sciences (STRiX)*, 2014.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.

Paramveer S. Dhillon, Dean P. Foster, and Lyle H. Ungar. Eigenwords: Spectral Word Embeddings. *Journal of Machine Learning Research*, 16:3035–3078, 2015. URL `http://jmlr.org/papers/v16/dhillon15a.html`.

Fernando Diaz, Bhaskar Mitra, and Nick Craswell. Query expansion with locally-trained word embeddings. *arXiv preprint arXiv:1605.07891*, 2016.

Ton Dijkstra, Koji Miwa, Bianca Brummelhuis, Maya Sappelli, and Harald Baayen. How cross-language similarity and task demands affect cognate recognition. *Journal of Memory and language*, 62(3):284–301, 2010.

Aron B Dolgopolsky. Gipoteza drevnejšego rodstva jazykovych semej severnoj evrazii s verojatnostej točky zrenija [a probabilistic hypothesis concerning the oldest relationships among the language families of northern eurasia]. *Voprosy jazykoznanija*, 2:53–63, 1964.

Pedro J Chamizo Domínguez and Brigitte Nerlich. False friends: their origin and semantics in some selected languages. *Journal of pragmatics*, 34(12):1833–1849, 2002.

Yerai Doval, Jose Camacho-Collados, Luis Espinosa-Anke, and Steven Schockaert. Improving cross-lingual word embeddings by meeting in the middle. *arXiv preprint arXiv:1808.08780*, 2018.

Michael Dunn. Indo-european lexical cognacy database (ielex). *Nijmegen: Max Planck Institute for Psycholinguistics*, 2012.

Michael Dunn, Robert Foley, Stephen Levinson, Ger Reesink, and Angela Terrill. Statistical reasoning in the evaluation of typological diversity in island melanesia. *Oceanic Linguistics*, pages 388–403, 2007.

T Mark Ellison and Simon Kirby. Measuring language divergence by intra-lexical comparison. In *Proceedings of the 21st International Conference on Computational Linguistics and the 44th annual meeting of the Association for Computational Linguistics*, pages 273–280. Association for Computational Linguistics, 2006.

Murray B Emeneau. India as a lingustic area. *Language*, 32(1):3–16, 1956.

Harry Falk. *Schrift im alten Indien: ein Forschungsbericht mit Anmerkungen*, volume 56. Gunter Narr Verlag, 1993.

Christiane Fellbaum. *WordNet*. Wiley Online Library, 1998.

Christiane Fellbaum. Wordnet. *Theory and applications of ontology: computer applications*, pages 231–243, 2010.

Christiane Fellbaum. Wordnet. *The encyclopedia of applied linguistics*, 2012.

Joseph Felsenstein. *PHYLIP (phylogeny inference package), version 3.5 c*. Joseph Felsenstein., 1993.

Joseph Felsenstein. The troubled growth of statistical phylogenetics. *Systematic Biology*, pages 465–467, 2001.

Joseph Felsenstein. *Inferring phylogenies*, volume 2. Sinauer Associates Sunderland, 2004.

John Rupert Firth. Applications of general linguistics. *Transactions of the Philological Society*, 56(1):1–14, 1957.

Walter M Fitch. Toward defining the course of evolution: minimum change for a specific tree topology. *Systematic Biology*, 20(4):406–416, 1971.

Walter M Fitch and Emanuel Margoliash. Construction of phylogenetic trees. *Science*, 155(3760):279–284, 1967.

Joseph L Fleiss and Jacob Cohen. The equivalence of weighted kappa and the intraclass correlation coefficient as measures of reliability. *Educational and psychological measurement*, 33(3):613–619, 1973.

Oana Frunza and Diana Inkpen. Semi-supervised learning of partial cognates using bilingual bootstrapping. In *Proceedings of the 21st International Conference on Computational Linguistics and the 44th ACL*, pages 441–448. ACL, 2006.

Oana Frunza and Diana Inkpen. Identification and disambiguation of cognates, false friends, and partial cognates using machine learning techniques. *International Journal of Linguistics*, 1(1):1–37, 2009.

Oana Magdalena Frunza. *Automatic identification of cognates, false friends, and partial cognates*. PhD thesis, University of Ottawa (Canada), 2006.

Ana V González-Garduño and Anders Søgaard. Learning to predict readability using eye-movement data from natives and learners. In *Thirty-Second AAAI Conference on Artificial Intelligence*, 2018.

Ana Valeria González-Garduño and Anders Søgaard. Using gaze to predict text readability. In *Proceedings of the 12th Workshop on Innovative Use of NLP for Building Educational Applications*, pages 438–443, Copenhagen, Denmark, September 2017. Association for Computational Linguistics.

Julio Gonzalo, Felisa Verdejo, Irina Chugur, and Juan Cigarran. Indexing with wordnet synsets can improve text retrieval. *arXiv preprint cmp-lg/9808002*, 1998.

Google. Sentence piece embeddings. `https://github.com/google/sentencepiece`, 2018.

Alan Grafen. The phylogenetic regression. *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, 326(1233):119–157, 1989.

Russell D Gray and Quentin D Atkinson. Language-tree divergence times support the anatolian theory of indo-european origin. *Nature*, 426(6965):435, 2003.

Simon J Greenhill. Levenshtein distances fail to identify language relationships accurately. *Computational Linguistics*, 37(4):689–698, 2011.

Simon J Greenhill, Robert Blust, and Russell D Gray. The austronesian basic vocabulary database: from bioinformatics to lexomics. *Evolutionary Bioinformatics*, 4:EBO–S893, 2008.

Kevin R Gregg. Krashen's monitor and occam's razor. *Applied linguistics*, 5(2):79–100, 1984.

Michael Groden and Martin Kreiswirth. *The Johns Hopkins guide to literary theory & criticism*. Johns Hopkins Univ. Press, 1994.

Ilan Gronau and Shlomo Moran. Optimal implementations of upgma and other common clustering algorithms. *Information Processing Letters*, 104(6):205–210, 2007.

Kilem L Gwet. *Handbook of inter-rater reliability: The definitive guide to measuring the extent of agreement among raters*. Advanced Analytics, LLC, 2014.

Petra Hammer and Madeleine Monod. English-french cognate dictionary. In *English-French Cognate Dictionary*. ERIC, 1976.

Dong Han, Junhui Li, Yachao Li, Min Zhang, and Guodong Zhou. Explicitly modeling word translations in neural machine translation. *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)*, 19(1):1–17, 2019.

Jürgen Hanneder. *Text genealogy, textual criticism and editorial technique: guest ed. Jürgen Hanneder...* Verlag der Österr. Akad. der Wiss., 2010.

Zellig Harris. Distributional Structure. *Word*, 10(23):146–162, 1954.

Bradley Hauer and Grzegorz Kondrak. Clustering semantically equivalent words into cognate sets in multilingual lists. In *Proceedings of 5th international joint conference on natural language processing*, pages 865–873, 2011.

Margot Haynes. Patterns and perils of guessing in second language reading. *on TESOL*, 83:163–176, 1984.

Timothy Hayward and André Moulin. False friends invigorated. In *Lexeter'83 Proceedings*, pages 190–198, 1984.

John M Henderson and Fernanda Ferreira. Eye movement control during reading: Fixation measures reflect foveal but not parafoveal processing difficulty. *Canadian Journal of Experimental Psychology/Revue canadienne de psychologie expérimentale*, 47(2):201, 1993.

Sepp Hochreiter and Jürgen Schmidhuber. Long Short-Term Memory. *Neural Computation*, 9(8):1735–1780, November 1997a. ISSN 0899-7667. doi: 10.1162/neco.1997.9.8.1735. URL http://dx.doi.org/10.1162/neco.1997.9.8.1735.

Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Comput.*, 9(8):1735–1780, November 1997b. ISSN 0899-7667. doi: 10.1162/neco.1997.9.8.1735. URL http://dx.doi.org/10.1162/neco.1997.9.8.1735.

Nora Hollenstein and Ce Zhang. Entity recognition at first sight: Improving NER with eye movement information. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 1–10, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1001. URL https://www.aclweb.org/anthology/N19-1001.

Eric W Holman, Cecil H Brown, Søren Wichmann, André Müller, Viveka Velupillai, Harald Hammarström, Sebastian Sauppe, Hagen Jung, Dik Bakker, Pamela Brown, et al. Automated dating of the world's language families based on lexical similarity. *Current Anthropology*, 52(6):000–000, 2011.

Kenneth Holmqvist, Marcus Nyström, Richard Andersson, Richard Dewhurst, Halszka Jarodzka, and Joost Van de Weijer. *Eye tracking: A comprehensive guide to methods and measures*. OUP Oxford, 2011.

Eduard Hovy. Combining and standardizing large-scale, practical ontologies for machine translation and other uses. In *Proceedings of the 1st International Conference on Language Resources and Evaluation (LREC)*, pages 535–542, 1998.

Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. *arXiv preprint arXiv:1801.06146*, 2018.

Eric H. Huang, Richard Socher, Christopher D. Manning, and Andrew Y. Ng. Improving Word Representations via Global Context and Multiple Word Prototypes. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2012.

Jian-Xiong Huang, Jian Zhang, Yong Shen, Ju-yu Lian, Hong-lin Cao, Wan-hui Ye, Linfang Wu, and Yue Bin. Different relationships between temporal phylogenetic turnover

and phylogenetic similarity and in two forests were detected by a new null model. *PLOS ONE*, 9(4):1–7, 04 2014. doi: 10.1371/journal.pone.0095703. URL `https://doi.org/10.1371/journal.pone.0095703`.

John P Huelsenbeck. Performance of phylogenetic methods in simulation. *Systematic biology*, 44(1):17–48, 1995.

John P Huelsenbeck, Fredrik Ronquist, Rasmus Nielsen, and Jonathan P Bollback. Bayesian inference of phylogeny and its impact on evolutionary biology. *science*, 294 (5550):2310–2314, 2001.

W John Hutchins. The georgetown-ibm experiment demonstrated in january 1954. In *Conference of the Association for Machine Translation in the Americas*, pages 102–114. Springer, 2004.

Diana Inkpen, Oana Frunza, and Grzegorz Kondrak. Automatic identification of cognates and false friends in french and english. In *Proceedings of the International Conference RANLP*, volume 9, pages 251–257, 2005.

Gerhard Jäger. Phylogenetic inference from word lists using weighted alignment with empirically determined weights. In *Quantifying Language Dynamics*, pages 155–204. Brill, 2014.

Gerhard Jäger. Support for linguistic macrofamilies from weighted sequence alignment. *Proceedings of the National Academy of Sciences*, 112(41):12752–12757, 2015.

Gerhard Jäger, Johann-Mattis List, and Pavel Sofroniev. Using support vector machines and state-of-the-art algorithms for phonetic alignment to identify cognates in multi-lingual wordlists. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers*, volume 1, pages 1205–1216, 2017.

Matthew A Jaro. Advances in record-linkage methodology as applied to matching the 1985 census of tampa, florida. *Journal of the American Statistical Association*, 84(406): 414–420, 1989.

Girish Nath Jha. The TDIL program and the Indian langauge corpora intitiative (ILCI). In *Proceedings of the Seventh International Conference on Language Resources and Evaluation (LREC'10)*, Valletta, Malta, May 2010a. European Language Resources Association (ELRA). URL `http://www.lrec-conf.org/proceedings/lrec2010/pdf/874_Paper.pdf`.

Girish Nath Jha. The tdil program and the indian langauge corpora intitiative (ilci). In *LREC*, 2010b.

Sittichai Jiampojamarn, Colin Cherry, and Grzegorz Kondrak. Integrating joint n-gram features into a discriminative training framework. In *Human Language Technologies: The 2010 Annual Conference of the North American Chapter of the Association for Computational Linguistics*, pages 697–700. Association for Computational Linguistics, 2010.

Karen Sparck Jones. Natural Language Processing: A Historical Review. In *Natural Language Processing: A Historical Review*, 1994.

Aditya Joshi, Vaibhav Tripathi, Kevin Patel, Pushpak Bhattacharyya, and Mark Carman. Are word embedding-based features useful for sarcasm detection? In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1006–1011, Austin, Texas, November 2016. Association for Computational Linguistics.

Salil Joshi, Arindam Chatterjee, Arun Karthikeyan Karra, and Pushpak Bhattacharyya. Eating your own cooking: automatically linking wordnet synsets of two languages. *COLING 2012*, 2012.

Marcel A Just and Patricia A Carpenter. A theory of reading: From eye fixations to comprehension. *Psychological review*, 87(4):329, 1980.

Diptesh Kanojia, Abhijeet Dubey, Malhar Kulkarni, Pushpak Bhattacharyya, and Gholemreza Haffari. Utilizing word embeddings based features for phylogenetic tree generation of sanskrit texts. In *Proceedings of the 6th International Sanskrit Computational Linguistics Symposium*, pages 152–165, 2019a.

Diptesh Kanojia, Malhar Kulkarni, Pushpak Bhattacharyya, and Gholamreza Haffari. Cognate identification to improve phylogenetic trees for indian languages. In *Proceed-*

ings of the ACM India Joint International Conference on Data Science and Management of Data, pages 297–300. ACM, 2019b.

Diptesh Kanojia, Malhar Kulkarni, Pushpak Bhattacharyya, and Eivind Kahrs. An introduction to the textual history tool. In *Proceedings of the 6th International Sanskrit Computational Linguistics Symposium*, pages 166–180, IIT Kharagpur, India, October 2019c. Association for Computational Linguistics. URL `https://www.aclweb.org/anthology/W19-7512`.

Diptesh Kanojia, Kevin Patel, Pushpak Bhattacharyya, Malhar Kulkarni, and Reza Haffari. Utilizing wordnets for cognate detection among indian languages. In *Global Wordnet Conference*, page 404, 2019d.

Diptesh Kanojia, Malhar Kulkarni, Pushpak Bhattacharyya, and Gholamreza Haffari. Challenge dataset of cognates and false friend pairs from indian languages. In *Proceedings of The 12th Language Resources and Evaluation Conference*, pages 3096–3102, 2020a.

Diptesh Kanojia, Malhar Kulkarni, Eivind Kahrs, and Pushpak Bhattacharyya. Materials for a critical edition of the kāśikāvṛtti (full edition). In *Materials for a critical edition of the Kāśikāvṛtti (Full Edition)*, pages 113–133. KKS University, Ramtek and New Bharatiya Book Corporation, Delhi, 2020b. ISBN 978-81-8315-372-0.

Diptesh Kanojia, Sravan Munukutla, Sayali Ghodekar, Pushpak Bhattacharyya, and Malhar Kulkarni. Keep your dimensions on a leash: True cognate detection using siamese deep neural networks. In *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD*, pages 324–325. ACM, 2020c.

Martin Kay. The logic of cognate recognition in historical linguistics. Technical report, RAND CORP SANTA MONICA CALIF, 1964.

Brett Kessler. *The significance of word lists*. Center for the Study of Language and Inf, 2001.

Yoon Kim. Convolutional neural networks for sentence classification. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*,

pages 1746–1751, Doha, Qatar, October 2014. Association for Computational Linguistics.

Guillaume Klein, Yoon Kim, Yuntian Deng, Jean Senellart, and Alexander M Rush. Opennmt: Open-source toolkit for neural machine translation. *arXiv preprint arXiv:1701.02810*, 2017.

Sigrid Klerke, Yoav Goldberg, and Anders Søgaard. Improving sentence compression by learning to predict gaze. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1528–1533, San Diego, California, 2016. Association for Computational Linguistics.

Kevin Knight and Steve K Luk. Building a large-scale knowledge base for machine translation. In *AAAI*, volume 94, pages 773–778, 1994.

Teuvo Kohonen. A very fast associative method for the recognition and correction of misspelt words, based on redundant hash addressing. In *Proceedings of the fourth International Joint Conference on Pattern Recognition, 1978*, 1978.

Grzegorz Kondrak. A new algorithm for the alignment of phonetic sequences. In *Proceedings of the 1st North American chapter of the Association for Computational Linguistics conference*, pages 288–295. Association for Computational Linguistics, 2000.

Grzegorz Kondrak. Identifying cognates by phonetic and semantic similarity. In *Second Meeting of the North American Chapter of the Association for Computational Linguistics*, 2001.

Grzegorz Kondrak and Bonnie Dorr. Identification of confusable drug names: A new approach and evaluation methodology. In *COLING 2004: Proceedings of the 20th International Conference on Computational Linguistics*, pages 952–958, 2004.

Grzegorz Kondrak and Graeme Hirst. Algorithms for language reconstruction, 2002.

Grzegorz Kondrak, Daniel Marcu, and Kevin Knight. Cognates can improve statistical translation models. In *Companion Volume of the Proceedings of HLT-NAACL 2003-Short Papers*, pages 46–48, 2003.

Malhar Kulkarni. On identifying the manuscript(s) at the base of the first printed edition of the Kāśikāvrtti. In *Annals of the Bhandarkar Oriental Institute*, pages 203–212. Bhandarkar Oriental Research Institute, Pune, 2000.

Malhar Kulkarni. A study of quotations of the kāśikāvrtti in the late paninian grammatical tradition., 2002a.

Malhar Kulkarni. Quotations of the kasikavrtti with special reference to the later paninian grammatical literature, 2002b.

Malhar Kulkarni. Quotations of the kashikvrtti in later paninian grammatical texts, 2002c.

Malhar Kulkarni. On a vārttika on p. 2.2. 6 in the kāśikāvrtti. *Annals of the Bhandarkar Oriental Research Institute*, 83:201–205, 2002d.

Malhar Kulkarni. The sharada manuscripts of the kāśikāvrtti. In *Prof. P.G.Lalye Felicitation volume*, pages 353–364. Prof. P.G.Lalye Felicitation Committee, Pune, 2003.

Malhar Kulkarni. Malayalam manuscripts of the kāśikāvrtti: A study, 2006.

Malhar Kulkarni. The sharada manuscripts of the kāśikāvrtti: Part ii. In *In Linguistic Traditions of Kashmir, Essays in memory of Pandit Dinanath Yaksh*, pages 419–428. D.K.Printworld, New Delhi, 2008.

Malhar Kulkarni. A sample of the new edition of the kāśikāvrtti: 2.2.6. In *Bharatiya Vidya*, volume LXV, pages 116–127. Bharatiya Vidya Bhavan, 2009.

Malhar Kulkarni. The malayalam manuscripts of the kāśikāvrtti: A study. In *SamskritVimarshaH: A Journal of Rashtriya Sanskrit Sansthan, World Sanskrit Conference Special*, volume 6, pages 103–112. Rashtriya Sanskrt Sansthan, 2012.

Malhar Kulkarni. Memory: a device in traditional sanskrit learning. *Memory and Human Wellbeing: Interdisciplinary Perspectives*, pages 57–72, 2015. Edited by Yahei Kanayama, Malhar Kulkarni and Toshiya Unebe.

Malhar Kulkarni and Eivind Kahrs. Materials for the critical edition of kāśikāvrtti 1.1, 2018.

Malhar Kulkarni and Eivind Kahrs. Variant readings in the text of the Kāśikāvrtti as noted by the Padamañjarī. *Journal of the Oriental Institute (Vadodara)*, 67:143–159, 2019a.

Malhar Kulkarni and Eivind Kahrs. Some more reflections on the role of the Nyāsa and the Padamañjarī in reconstructing the textual history of the transmission of the Kāśikāvrtti. In *Nagoya Studies in Indian Culture and Buddhism, Sambhasha*, pages 35–48. Department of Indian Studies, Graduate School of Letters, Nagoya University, 2019b.

Saurav Kumar, Saunack Kumar, Diptesh Kanojia, and Pushpak Bhattacharyya. "a passage to India": Pre-trained word embeddings for Indian languages. In *Proceedings of the 1st Joint Workshop on Spoken Language Technologies for Under-resourced languages (SLTU) and Collaboration and Computing for Under-Resourced Languages (CCURL)*, pages 352–357, Marseille, France, May 2020. European Language Resources association. ISBN 979-10-95546-35-1. URL `https://www.aclweb.org/anthology/2020.sltu-1.49`.

Anoop Kunchukuttan, Ratish Puduppully, and Pushpak Bhattacharyya. Brahmi-Net: A transliteration and script conversion system for languages of the Indian subcontinent. In *Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations*. Association for Computational Linguistics, 2015.

Anoop Kunchukuttan, Divyanshu Kakwani, Satish Golla, Avik Bhattacharyya, Mitesh M Khapra, Pratyush Kumar, et al. Ai4bharat-indicnlp corpus: Monolingual corpora and word embeddings for indic languages. *arXiv preprint arXiv:2005.00085*, 2020.

Sofie Labat and Els Lefever. A classification-based approach to cognate detection combining orthographic and semantic similarity information. In *Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2019)*, pages 602–610, Varna, Bulgaria, September 2019. INCOMA Ltd. doi: 10.26615/978-954-452-056-4_071. URL `https://www.aclweb.org/anthology/R19-1071`.

Guillaume Lample and Alexis Conneau. Cross-lingual language model pretraining. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Guillaume Lample, Alexis Conneau, Marc'Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. Word translation without parallel data. In *International Conference on Learning Representations*, 2018.

Dawn Lawrie, James Mayfield, Paul Mcnamee, and Douglas W Oard. Cross-language person-entity linking from twenty languages. *Report*, 2014.

Quoc V. Le, Navdeep Jaitly, and Geoffrey E. Hinton. A Simple Way to Initialize Recurrent Networks of Rectified Linear Units. *CoRR*, abs/1504.00941, 2015. URL `http://arxiv.org/abs/1504.00941`.

Sylvain Lespinats, Delphine Grando, Eric Maréchal, Mohamed-Ali Hakimi, Olivier Tenaillon, and Olivier Bastien. How fitch-margoliash algorithm can benefit from multi dimensional scaling. *Evolutionary Bioinformatics*, 7:EBO–S7048, 2011.

Vladimir I Levenshtein. Dvoičnye kody s ispravleniem vypadenij, vstavok i zameščenij simvolov [binary codes with correction of deletions, insertions and replacements]. *Doklady Akademij Nauk SSSR*, 163(4):845–848, 1965.

Omer Levy and Yoav Goldberg. Dependency-based word embeddings. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics, ACL 2014, June 22-27, 2014, Baltimore, MD, USA, Volume 2: Short Papers*, pages 302–308, 2014.

Omer Levy, Minjoon Seo, Eunsol Choi, and Luke Zettlemoyer. Zero-shot relation extraction via reading comprehension. *arXiv preprint arXiv:1706.04115*, 2017.

Johann-Mattis List. Sca: phonetic alignment based on sound classes. In *New directions in logic, language and computation*, pages 32–51. Springer, 2010.

Johann-Mattis List. Lexstat: Automatic detection of cognates in multilingual wordlists. In *Proceedings of the EACL 2012 Joint Workshop of LINGVIS & UNCLH*, pages 117–125. Association for Computational Linguistics, 2012.

Johann-Mattis List. *Sequence comparison in historical linguistics*. PhD thesis, düsseldorf university press, 2014.

Johann-Mattis List, Simon J Greenhill, and Russell D Gray. The potential of automatic word comparison for historical linguistics. *PloS one*, 12(1):e0170046, 2017a.

Johann-Mattis List, Mary Walworth, Simon Greenhill, Tiago Tresoldi, and Robert Forkel. Sequence comparison in computational historical linguistics phonetic alignments and cognate detection with lingpy 2.6. *Journal of Language Evolution*, 2017b.

Pengfei Liu, Shafiq R Joty, and Helen M Meng. Fine-grained opinion mining with recurrent neural networks and word embeddings. In *EMNLP*, pages 1433–1443, 2015.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

Nikola Ljubešić and Darja Fišer. Identifying false friends between closely related languages. In *Proceedings of the 4th Biennial International Workshop on Balto-Slavic Natural Language Processing*, pages 69–77, 2013.

Yunfei Long, Rong Xiang, Qin Lu, Chu-Ren Huang, and Minglei Li. Improving attention model based on cognition grounded data for sentiment analysis. *IEEE Transactions on Affective Computing*, 2019.

Philip A. Maas. On what to do with a stemma - towards a critical edition of the carakasamhitm vimanasthana 8. *Indian Journal of History of Science*, pages 163–185, 2009a.

Philipp A Maas. Computer aided stemmatics—the case of fifty-two text versions of carakasamhitā vimānasthāna 8.67-157. *Wiener Zeitschrift für die Kunde Südasiens/Vienna Journal of South Asian Studies*, 52:63–119, 2009b.

Philipp A Maas. On what became of the carakasamhitā after drdhabala's revision. *eJournal of Indian Medicine*, 3(1):1–22, 2010.

Wesley Mackay and Grzegorz Kondrak. Computing word similarity and identifying cognates with pair hidden Markov models. In *Proceedings of the Ninth Conference on Computational Natural Language Learning (CoNLL-2005)*, pages 40–47, Ann Arbor, Michigan, June 2005. Association for Computational Linguistics. URL `https://www.aclweb.org/anthology/W05-0606`.

Ranbeer Makin, Nikita Pandey, Prasad Pingali, and Vasudeva Varma. Approximate string matching techniques for effective clir among indian languages. In *International Workshop on Fuzzy Logic and Applications*, pages 430–437. Springer, 2007.

Ranbeer Makin, Nikita Pandey, Prasad Pingali, and Vasudeva Varma. Experiments in cross-lingual ir among indian languages. *Advances in Multilingual and Multimodal Information Retrieval. Springer Berlin/Heidelberg*, 2008.

Gideon S Mann and David Yarowsky. Multipath translation lexicon induction via bridge languages. In *Proceedings of the 2nd NAACL-HLT*, pages 1–8. ACL, 2001.

Sandeep Mathias, Rudra Murthy, Diptesh Kanojia, Abhijit Mishra, and Pushpak Bhattacharyya. Happy are those who grade without seeing: A multi-task learning approach to grade essays using gaze behaviour. *arXiv preprint arXiv:2005.12078*, 2020.

Mary L McHugh. Interrater reliability: the kappa statistic. *Biochemia medica: Biochemia medica*, 22(3):276–282, 2012.

I Dan Melamed. Bitext maps and alignment via pattern recognition. *Computational Linguistics*, 25(1):107–130, 1999.

Oren Melamud, Jacob Goldberger, and Ido Dagan. context2vec: Learning generic context embedding with bidirectional lstm. In *Proceedings of the 20th SIGNLL conference on computational natural language learning*, pages 51–61, 2016.

Helen M Meng, Wai-Kit Lo, Berlin Chen, and Karen Tang. Generating phonetic cognates to handle named entities in english-chinese cross-language spoken document retrieval. In *IEEE Workshop on Automatic Speech Recognition and Understanding, 2001. ASRU'01.*, pages 311–314. IEEE, 2001.

Paola Merlo and Maria Andueza Rodriguez. Cross-lingual word embeddings and the structure of the human bilingual lexicon. In *Proceedings of the 23rd Conference on Computational Natural Language Learning (CoNLL)*, pages 110–120, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/K19-1011. URL `https://www.aclweb.org/anthology/K19-1011`.

Paola Merlo and Maria Andueza Rodriguez. Cross-lingual word embeddings and the structure of the human bilingual lexicon. In *Proceedings of the 23rd Conference on Computational Natural Language Learning (CoNLL)*, pages 110–120, 2019.

Tomas Mikolov, Quoc V. Le, and Ilya Sutskever. Exploiting similarities among languages for machine translation. *CoRR*, abs/1309.4168, 2013a. URL `http://arxiv.org/abs/1309.4168`.

Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. In *Advances in neural information processing systems*, pages 3111–3119, 2013b.

Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. In *Advances in neural information processing systems*, pages 3111–3119, 2013c.

Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic Regularities in Continuous Space Word Representations. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 746–751, Atlanta, Georgia, June 2013d. Association for Computational Linguistics. URL `https://www.aclweb.org/anthology/N13-1090`.

George A Miller, Richard Beckwith, Christiane Fellbaum, Derek Gross, and Katherine J Miller. Introduction to wordnet: An on-line lexical database. *International journal of lexicography*, 3(4):235–244, 1990.

Abhijit Mishra and Pushpak Bhattacharyya. *Cognitively Inspired Natural Language Processing: An Investigation Based on Eye-tracking*. Springer, 2018.

Abhijit Mishra, Diptesh Kanojia, Seema Nagar, Kuntal Dey, and Pushpak Bhattacharyya. Harnessing cognitive features for sarcasm detection. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1095–1104, Berlin, Germany, August 2016. Association for Computational Linguistics. doi: 10.18653/v1/P16-1104. URL `https://www.aclweb.org/anthology/P16-1104`.

Abhijit Mishra, Srikanth Tamilselvam, Riddhiman Dasgupta, Seema Nagar, and Kuntal Dey. Cognition-cognizant sentiment analysis with multitask subjectivity summarization based on annotators' gaze behavior. In *Thirty-Second AAAI Conference on Artificial Intelligence*, 2018.

Ruslan Mitkov, Viktor Pekar, Dimitar Blagoev, and Andrea Mulloni. Methods for extracting and classifying pairs of cognates and false friends. *Machine translation*, 21(1): 29, 2007.

Bernard ME Moret, Usman Roshan, and Tandy Warnow. Sequence-length requirements for phylogenetic methods. In *International Workshop on Algorithms in Bioinformatics*, pages 343–356. Springer, 2002.

David W Mount. Bioinformatics: sequence and genome analysis. 2004, 2004.

Andrea Mulloni and Viktor Pekar. Automatic detection of orthographics cues for cognate recognition. In *LREC*, pages 2387–2390, 2006.

Andrea Mulloni, V Pekar, R Mitkov, and D Blagoev. Semantic evidence for automatic identification of cognates. In *Proceedings of the RANLP'2007 workshop: Acquisition and management of multilingual lexicons*, pages 49–54, 2007.

Rudra Murthy, Mitesh M Khapra, and Pushpak Bhattacharyya. Improving ner tagging performance in low-resource languages via multilingual learning. *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)*, 18(2):9, 2018.

Luay Nakhleh, Don Ringe, and Tandy Warnow. Perfect phylogenetic networks: A new methodology for reconstructing the evolutionary history of natural languages. *Language*, pages 382–420, 2005.

Svetlin Nakov, Preslav Nakov, and Elena Paskaleva. Unsupervised extraction of false friends from parallel bi-texts using the web as a corpus. In *Proceedings of the International Conference RANLP-2009*, pages 292–298, 2009.

Roberto Navigli and Simone Paolo Ponzetto. Babelnet: Building a very large multilingual semantic network. In *Proceedings of the 48th annual meeting of the association for computational linguistics*, pages 216–225. Association for Computational Linguistics, 2010.

Roberto Navigli and Simone Paolo Ponzetto. Babelnet: The automatic construction, evaluation and application of a wide-coverage multilingual semantic network. *Artificial Intelligence*, 193:217–250, 2012.

John Nerbonne and Wilbert Heeringa. Measuring dialect distance phonetically. In *Computational Phonology: Third Meeting of the ACL Special Interest Group in Computational Phonology*, 1997.

Johanna Nichols and Tandy Warnow. Tutorial on computational linguistic phylogeny. *Language and Linguistics Compass*, 2(5):760–820, 2008.

Joakim Nivre, Marie-Catherine De Marneffe, Filip Ginter, Yoav Goldberg, Jan Hajic, Christopher D Manning, Ryan McDonald, Slav Petrov, Sampo Pyysalo, Natalia Silveira, et al. Universal dependencies v1: A multilingual treebank collection. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*, pages 1659–1666, 2016.

Olumide Owolabi and DR McGregor. Fast approximate string matching. *Software: Practice and Experience*, 18(4):387–393, 1988.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting on association for computational linguistics*, pages 311–318. Association for Computational Linguistics, 2002.

Karl Pearson. Vii. note on regression and inheritance in the case of two parents. *proceedings of the royal society of London*, 58(347-352):240–242, 1895.

Bolette Pedersen, Krister Linden, Kadri Vider, Markus Forsberg, Neeme Kahusk, Jyrki Niemi, Lars Nygaard, Mitchell Seaton, Heili Orav, Lars Borin, et al. Nordic and baltic wordnets aligned and compared through "wordties". *Proceedings of NODALIDA 2013*, 2013.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in python. *the Journal of machine Learning research*, 12:2825–2830, 2011.

Jeffrey Pennington, Richard Socher, and Christopher D. Manning. GloVe: Global Vectors for Word Representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP 2014)*, pages 1532–1543, 2014.

Matthew E Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. Deep contextualized word representations. *arXiv preprint arXiv:1802.05365*, 2018.

Wendy J Phillips-Rodriguez, Christopher J Howe, and Heather F Windram. Some considerations about bifurcation in diagrams representing the written transmission of the mahābhārata. *Wiener Zeitschrift für die Kunde Südasiens/Vienna Journal of South Asian Studies*, 52:29–43, 2009.

Marcial Prado. Ntc's dictionary of spanish false cognates. In *NTC's dictionary of Spanish false cognates*. NTC, 1993.

A Pranav. Alignment analysis of sequential segmentation of lexicons to improve automatic cognate detection. In *Proceedings of ACL 2018, Student Research Workshop*, pages 134–140, 2018.

Jelena Prokić, Martijn Wieling, and John Nerbonne. Multiple sequence alignments in linguistics. In *Proceedings of the EACL 2009 Workshop on Language Technology and Resources for Cultural Heritage, Social Sciences, Humanities, and Education*, pages 18–25. Association for Computational Linguistics, 2009.

Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. Improving language understanding by generative pre-training, 2018.

Taraka Rama. Automatic cognate identification with gap-weighted string subsequences. In *Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1227–1231, 2015.

Taraka Rama. Siamese convolutional networks for cognate identification. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, pages 1018–1027, 2016.

Taraka Rama and Lars Borin. Estimating language relationships from a parallel corpus. a study of the europarl corpus. In *Proceedings of the 18th Nordic Conference of Computational Linguistics (NODALIDA 2011)*, pages 161–167, 2011.

Taraka Rama and Johann-Mattis List. An automated framework for fast cognate detection and bayesian phylogenetic inference in computational historical linguistics. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 2019.

Taraka Rama and Anil Kumar Singh. From bag of languages to family trees from noisy corpus. In *Proceedings of the International Conference RANLP-2009*, pages 355–359, 2009.

Taraka Rama, Lars Borin, GK Mikros, and J Macutek. Comparative evaluation of string similarity measures for automatic language classification., 2015.

Taraka Rama, Johannes Wahle, Pavel Sofroniev, and Gerhard Jäger. Fast and unsupervised methods for multilingual cognate clustering. *arXiv preprint arXiv:1702.04938*, 2017.

Taraka Rama, Johann-Mattis List, Johannes Wahle, and Gerhard Jäger. Are automatic methods for cognate detection good enough for phylogenetic reconstruction in historical linguistics? *arXiv preprint arXiv:1804.05416*, 2018.

Keith Rayner. Eye movements in reading and information processing: 20 years of research. *Psychological bulletin*, 124(3):372, 1998.

Radim Řehůřek and Petr Sojka. Software Framework for Topic Modelling with Large Corpora. In *Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks*, pages 45–50, Valletta, Malta, May 2010. ELRA. `http://is.muni.cz/publication/884893/en`.

Don Ringe, Tandy Warnow, and Ann Taylor. Indo-european and computational cladistics. *Transactions of the philological society*, 100(1):59–129, 2002.

Sebastien Roch. A short proof that phylogenetic tree reconstruction by maximum likelihood is hard. *IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB)*, 3(1):92, 2006.

Frank Rosenblatt. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological Review*, 65(6):386–408, November 1958.

Joshua Rudder. *The Grammar of Romance: A Comparative Introduction to Vulgar Latin & the Romance Languages*. Joshua Rudder, 2012.

Gerda Ruge. Experiments on linguistically-based term associations. *Information Processing & Management*, 28(3):317–332, 1992.

Rosa M Sáchez-Casas, José E García-Albea, and Christopher W Davis. Bilingual lexical processing: Exploring the cognate/non-cognate distinction. *European Journal of Cognitive Psychology*, 4(4):293–310, 1992.

Naruya Saitou and Masatoshi Nei. The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Molecular biology and evolution*, 4(4):406–425, 1987.

Gerard Salton and Christopher Buckley. Term-weighting approaches in automatic text retrieval. *Information processing & management*, 24(5):513–523, 1988.

Adheesh Sathaye. The scribal life of folktales in medieval india. *South Asian History and Culture*, 8(4):430–447, 2017.

Desmond Schmidt and Robert Colomb. A data structure for representing multi-version texts online. *International Journal of Human-Computer Studies*, 67(6):497 – 514, 2009. ISSN 1071-5819. doi: https://doi.org/10.1016/j.ijhcs.2009.02.001. URL `http://www.sciencedirect.com/science/article/pii/S1071581909000214`.

Stefan Schulz, Kornél Markó, Eduardo Sbrissia, Percy Nohama, and Udo Hahn. Cognate mapping-a heuristic strategy for the semi-supervised acquisition of a spanish lexicon from a portuguese seed lexicon. In *COLING 2004: Proceedings of the 20th International Conference on Computational Linguistics*, pages 813–819, 2004.

Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural machine translation of rare words with subword units. *arXiv preprint arXiv:1508.07909*, 2015.

Claude E Shannon. A mathematical theory of communication. *Bell system technical journal*, 27(3):379–423, 1948.

Foster G Simard and G Foster. M. and p. isabelle. 1992. using cognates to align sentences in bilingual corpora. In *Proceedings of the 4 th International Conference on Thoretical and Methodological Issues in Machine Translation (TMI'92)*, pages 67–81, 1992.

Michel Simard, George F Foster, and Pierre Isabelle. Using cognates to align sentences in bilingual corpora. In *Proceedings of the 1993 conference of the Centre for Advanced Studies on Collaborative research: distributed computing-Volume 2*, pages 1071–1082. IBM Press, 1993.

Abhinav Deep Singh, Poojan Mehta, Samar Husain, and Rajkumar Rajakrishnan. Quantifying sentence complexity based on eye-tracking measures. In *Proceedings of the Workshop on Computational Linguistics for Linguistic Complexity (CL4LC)*, pages 202–212, Osaka, Japan, December 2016a. The COLING 2016 Organizing Committee.

Anil Kumar Singh. A computational phonetic model for indian language scripts. In *Constraints on Spelling Changes: Fifth International Workshop on Writing Systems*. Nijmegen, The Netherlands, 2006a.

Anil Kumar Singh. Study of some distance measures for language and encoding identification. In *Proceedings of the Workshop on Linguistic Distances*, pages 63–72. Association for Computational Linguistics, 2006b.

Anil Kumar Singh and Harshit Surana. Can corpus based measures be used for comparative study of languages? In *Proceedings of ninth meeting of the ACL special interest group in computational morphology and phonology*, pages 40–47. Association for Computational Linguistics, 2007.

Meghna Singh, Rajita Shukla, Jaya Jha, Laxmi Kashyap, Diptesh Kanojia, and Pushpak Bhattacharyya. Mapping it differently: A solution to the linking challenges. In *Eighth Global Wordnet Conference*. GWC 2016, 2016b.

Manish Sinha, Mahesh Reddy, and Pushpak Bhattacharyya. An approach towards construction and application of multilingual indo-wordnet. In *3rd Global Wordnet Conference (GWC 06), Jeju Island, Korea*, 2006.

Samuel L. Smith, David H. P. Turban, Steven Hamblin, and Nils Y. Hammerla. Offline

bilingual word vectors, orthogonal transformations and the inverted softmax. *ICLR*, 2017.

Robert R Sokal. Distance as a measure of taxonomic similarity. *Systematic Zoology*, 10 (2):70–79, 1961.

Robert R Sokal and F James Rohlf. The comparison of dendrograms by objective methods. *Taxon*, pages 33–40, 1962.

Charles Spearman. Demonstration of formulae for true measurement of correlation. *The American Journal of Psychology*, pages 161–169, 1907.

Adam St Arnaud et al. Identifying cognate sets across dictionaries of related languages, 2017.

DL Swofford. Phylogenetic analysis using parsimony (and other methods) paup* 4.0. *Sinauer, Sunderland*, 1999.

DL Swofford, GJ Olsen, PJ Waddell, and DM Hillis. Phylogenetic inference, p. 407–514. *Molecular Systematics (second edition). Sinauer Associates, Sunderland, Massachusetts*, 1996.

John P Thorpe. The molecular clock hypothesis: biochemical evolution, genetic differentiation and systematics. *Annual Review of Ecology and Systematics*, 13(1):139–168, 1982.

Jorg Tiedemann. Automatic construction of weighted string similarity measures. In *1999 Joint SIGDAT Conference on Empirical Methods in Natural Language Processing and Very Large Corpora*, 1999.

Lianet Sepúlveda Torres and Sandra Maria Aluísio. Using machine learning methods to avoid the pitfall of cognates and false friends in spanish-portuguese word pairs. In *Proceedings of the 8th Brazilian Symposium in Information and Human Language Technology*, 2011.

Dan Tufis. A cheap and fast way to build useful translation lexicons. In *COLING 2002: The 19th International Conference on Computational Linguistics*, 2002.

Dan TufiŞ, Radu Ion, and Nancy Ide. Fine-grained word sense disambiguation based on parallel corpora, word alignment, word clustering and aligned wordnets. In *Proceedings of the 20th international conference on Computational Linguistics*, page 1312. Association for Computational Linguistics, 2004.

Juhan Tuldava. A mathematical model of the vocabulary-text relation. In *COLING 1980 Volume 1: The 8th International Conference on Computational Linguistics*, 1980.

Peter Turchin, Ilia Peiros, and Murray Gell-Mann. Analyzing genetic connections between languages by matching consonant classes. . : . , pages 117–126, 2010.

Joseph Turian, Lev Ratinov, and Yoshua Bengio. Word Representations: A Simple and General Method for Semi-supervised Learning. In *Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics*, ACL '10. Association for Computational Linguistics, 2010.

Raghavendra Udupa, K Saravanan, Anton Bakalov, and Abhijit Bhole. "they are out there, if you know where to look": Mining transliterations of oov query terms for cross-language information retrieval. In *European Conference on Information Retrieval*, pages 437–448. Springer, 2009.

Kãrumũri V Subbãrão. *South Asian Languages: A Syntactic Typology*. Cambridge University Press., 2012.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, pages 5998–6008, 2017.

Piek Vossen et al. Eurowordnet: a multilingual database for information retrieval. In *Proceedings of the DELOS workshop on Cross-language Information Retrieval*, pages 5–7, 1997.

Tandy Warnow, Bernard ME Moret, and Katherine St John. Absolute convergence: true trees from short sequences. In *Proceedings of the twelfth annual ACM-SIAM symposium on Discrete algorithms*, pages 186–195. Society for Industrial and Applied Mathematics, 2001.

Tandy J Warnow. Constructing phylogenetic trees efficiently using compatibility criteria. *New Zealand Journal of Botany*, 31(3):239–247, 1993.

Martin L West. *Textual criticism and editorial technique applicable to Greek and Latin texts*. Walter de Gruyter, 1973.

William E Winkler. String comparator metrics and enhanced decision rules in the fellegi-sunter model of record linkage., 1990.

Winston Wu and David Yarowsky. Creating large-scale multilingual cognate tables. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, 2018.

Emile Zuckerkandl, Richard T Jones, and Linus Pauling. A comparison of animal hemoglobins by tryptic peptide pattern analysis. *Proceedings of the National Academy of Sciences of the United States of America*, 46(10):1349, 1960.

Appendix A

Efforts Towards the Creation of Textual History Tool

This additional chapter details our work on the creation of the Textual History Tool and an additional effort to improve it. This tool helps philologists add manuscript data to a digital database and utilizes our phylogenetic approaches to display trees on an online interface along with various other functionalities described below.

A.1 An Introduction to the Textual History Tool

This work was accepted at the 6th International Sanskrit Computational Linguistics Symposium, and a demonstration of the tool was provided at the symposium.

The Textual History Tool (THT) captures the historical evolution of a text through various temporal stages, and inter-related data culled from various types of related texts. This tool also provides a historical view of the transmission of a text through the manuscript tradition. This tool provides an online interface which allows philologists to enter manuscript data for a text. It also provides an interface which helps philologists compare the variants in a separate mode. It allows the user to generate phylogenetic trees, for the text, based on distance methods using the data entered in the tool. It also contains the facility to generate critical edition using a semi-supervised approach. This tool also divides the text into meaningful functional units and helps achieve a better comparison among the manuscripts. The text of the Kāśikāvṛtti (KV) and its textual history is mentioned as a

specific example to demonstrate the features of this tool.

Many of the texts which exist in the form of a manuscript have been studied carefully and brought in the print form in various editions. Editions are of various kinds; *"Critical Edition"* being considered authentic as it aims to account for the existing manuscripts' traditions. Validating multiple versions using Document Similarity measures under the purview of computational linguists [Huang et al., 2014] is also possible, and a desirable step to building text version families or *clades*. Tools are required to query, retrieve and visualize information from such knowledge bases. In our work, we discuss such a data entry/visualization tool for computational phylogenetics.

In the twentieth century, before computers came to be used in the effort of preparing the critical edition of a text, philologists used paper-based methods for various purposes, *viz.,* collation, description of manuscripts, inter-relation of manuscripts, apparatus creation, *etc.* With the advent of computers and development in technology, we can now have tools with us, that can facilitate the data entry, storage, and display of the aforementioned functions, all on the same interface. The tool described in this chapter is of the same kind.

"A text is, generally, structured verbal expression of intellectual processes."

Oral transmission led to the development of various *vikṛtis*, *i.e.,* methodologies used to memorize Vedic lore, based on cognitive features. Written transmission is carried out through copies of the text, also known as manuscripts. Historically, manuscripts were written or copied by one or more scribes. Transmission of the text from one source to another generates variants which are significantly different when compared to each other. In terms of expression, the text undergoes various changes in terms of spellings, word replacements, *etc.* Texts are used as the primary sources by scholars in reconstructing the History. The texts assume more significance as a source when it comes to reconstructing the history of an intellectual tradition. These texts represent important stages in the development of thought that contributes to the continuation of the intellectual tradition. What makes the process of reconstruction of intellectual history more complex and therefore, perhaps, more interesting as well as challenging, is the fact that these

texts, themselves, are part of a historical process, also known as transmission, and have evolved in certain typical manners and ways in the course of time. It becomes necessary, therefore, in order to study the history of intellectual tradition, the history of the text used as a primary resource.

In the Indian context, we know that the transmission of texts happened in two major ways: oral and written. Texts like Vedas were transmitted from one generation to another, primarily, orally, and were written down eventually. So is the case of Epic poems like Ramayana and Mahabharata[1]. In the case of Vedas, though, there is no scope of evolution of the text as such, as it was orally transmitted in a regulated manner with components of the texts noted down in great detail up to the level of single letters and accent marks. In the case of Epics, however, the evolution of the text was observed by scholars and traditionally as well, it is believed that Mahabharata, for example, originally consisted of merely 10000 verses which grew in the course of time and has now become a text of one hundred thousand verses (śatasāhasrī saṁhitā).

A.1.1 Functional Divisions of the text of KV

The text of KV, as mentioned above, can be, generally, divided into its functional parts. There are two basic divisions in the text of KV, one that of the sūtra and other of the KV. Within the KV, the text can further be divided according to its functional properties based on the type of sūtra it is commenting upon. We present below the functional divisions in the KV on the saṁjñā sūtra. Functional parts of the KV on vidhi sūtra is described in [Kulkarni, 2012].

- saṁjñā: this type of sūtra introduces a technical term, and hence the KV on this sūtra contains the following functional parts:

    1. Introduction of the words in the sūtra and meaning of the sūtra.
    2. Examples.

[1]When Malhar Kulkarni delivered his lecture on 'Text and Transmission with special reference to Classical Sanskrit Texts' in Almaty, Kazakhstan on 25th August 2015, some members of the audience remarked that there exist texts even in Kazakhstan, which were committed to memory and were handed down from one generation to the next orally. For oral traditions of India, see [Falk, 1993] and for more recent discussions, see [Kulkarni, 2015].

3. Mention of other sūtras in which this technical term appears.

The Textual History Tool is required because, at one go, it can present to a reader, the entire history of a text. A text in the Indian context can have a predecessor text as well as a successor text. It is an outcome of the intellectual activity based on one or more predecessor texts as well as textual traditions. It becomes a part of intellectual discourse and is commented upon by critical scholars within the same tradition. It gets quoted in the successor texts of the same tradition as well as other traditions and disciplines. It gets copied down in written form for various generations across different geographical regions and in different scripts. In this process, the text itself undergoes various stages of evolution, which can be marked as historical landmarks in the development of thought. Capturing the history of this intellectual world, at a glance, is the aim of this tool.

Currently, available tools do not present the historical information in a form which is coherent, and they do not provide an efficient data-entry interface which can help computational phylogenetics. There are multiple toolkits available which perform computational phylogenetics given the data is formatted in their required input format; none of them takes raw manuscript data to automate the complete pipeline, which is the eventual aim of this tool. We allow users to enter raw manuscript data and create functional divisions to ease the task of phylogenetics which is a novel contribution of our work. The critical contribution of this work is: '*Building a comprehensive tool for visualizing the transmission and history of a text - a tool which can,*

(i) Visualize the multiple versions of the same text which also allows data entry for manuscript versions and thus, helping one compare these versions with each other and aids one in adapting them to a graphical model, viz., a phylogenetic tree.
(ii) Visualize the data from earlier texts.
(iii) Visualize the data from testimonia.
(iv) Visualize the data from commentaries.'

Currently, a lot of texts written in Sanskrit are available in the electronic format avail-

able at SARIT[2], GRETIL[3], DCS[4], *etc.* Many of them are in a searchable format. DCS presents texts with various other applied tools like Morphological Analyzer, POS tagger, *etc.* However, no tool presents historical information the way it is needed, *i.e.,* with manuscript versions, which can be compared/edited at the same time. KWIC is an acronym for Key Word In Context (KWIC) and is the most common format for concordance lines. DCS employs KWIC to be used in the concordance functionality it provides on its interface. Some tools for visualization of data are available online. [Csernel and Patte, 2007] discuss the LCS algorithm for preparing a critical edition of Sanskrit texts and provide a method for comparison of Sanskrit manuscripts using XML and HTML formats. BabelNet [Navigli and Ponzetto, 2010] is an important lexical resource as far as computational aspects are concerned. [Navigli and Ponzetto, 2012] design an explorer to visualize its database. It uses the tree layout for visualization, which, in the convention, is similar to the phylogenetic visualization of texts. Visuwords[5] is an online graphical dictionary designed for accessing Princeton WordNet and uses a force-directed graph layout for visualizing the synset structure. Nodebox visualizer[6], on the other hand, provides a very static layout. WordTies [Pedersen et al., 2013] is a WordNet visualizer designed for Nordic and Baltic wordnets. [Chaplot et al., 2014] present such a visualizer for IndoWordNet-which is a lexical resource for Indian language WordNets.

Overlapping textual structures can be accurately modelled either as a minimally redundant directed graph, or, more practically, as an ordered list of pairs, each containing a set of versions and a fragment of text or data [Schmidt and Colomb, 2009]. On a similar note, [Hanneder, 2010] writes about text genealogy and textual criticism. [Maas, 2009b] discusses the textual versions of Carakasaṃhitā Vimānasthāna and uses computer stemmatics to aid them in the construction of a Phylogenetic tree later [Maas, 2010]. [Sathaye, 2017] present an analysis of Vetāla-pañcaviṃśati, in the context of 'fluid' textual dynamics and discuss the differences in oral folklore when compared to written text. [Phillips-Rodriguez et al., 2009] discuss the transmission of the Mahābhārata and the bifurcations within the diagrams about its written transmission. [Kulkarni, 2002d] discuss

[2] http://sarit.indology.info/

[3] http://gretil.sub.uni-goettingen.de/

[4] http://www.sanskrit-linguistics.org/dcs/

[5] https://visuwords.com/

[6] https://www.nodebox.net/code/index.php/WordNetwo

the transmission of KV and conclude that there seems to be no Vt (version) on AST 2.2.6 in the KV. [Kulkarni, 2015] discuss the perspectives on how memory acts as a vital device in the tradition of oral transmission of texts.

The TEI Critical Edition[7] Toolbox is a tool for preparing a digital TEI critical edition which allows one to check for the encoding of the text. It also facilitates the parallel look-up of the manuscript version by visualizing them on a web-based GUI. However, the software is not available for download and offline use, yet. In the current state, it accepts only TEI format XML files but does not allow one to generate versions. A technique for textual criticism is also provided by [West, 1973]. Classical Text Editor[8] allows one to build a critical edition and critical apparatus manually. It also allows one to prepare the phylogenetic trees but does not provide a visualization interface. It allows one to collate the textual versions and edit them on an offline interface. Our work is significantly different from CTE as our online interface allows multiple users to collaborate and enter data for the same text. It allows the users to create functional divisions in the sūtra text being entered and thus helps our novel phylogenetic methodology. In philosophy, our tool is focussed on the entire textual history of which manuscripts are an important part. Our tool preserves testimonia, printed editions, commentaries, *etc.*, which the CTE does not. PAUP is a tool for Phylogenetic Analysis based on Maximum Parsimony [Fitch, 1971] and other related methodologies, has been created by [Swofford, 1999] and is available online[9]. To the best of our knowledge, there is no tool which presents a comprehensive picture of the history of a text by presenting various resources useful for the reconstruction of the history of a text like testimonia, commentaries, earlier texts, printed editions, *etc.*

A.1.2 Data Entry Interface

The Data entry interface, based on the user login, allows the user to start with the creation of a new manuscript, or takes them back directly to the last entry they made in a previous manuscript they were working on. At any point, a user can choose to start a new manuscript creation. In such a case, the tool requests the entry of the manuscript

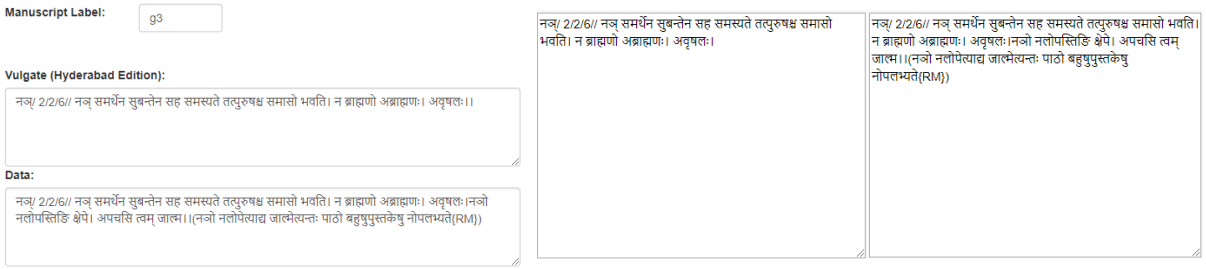
[7] `http://ciham-digital.huma-num.fr/teitoolbox/`

[8] `http://cte.oeaw.ac.at/`

[9] `http://paup.sc.fsu.edu/`

label. Upon the entry of the manuscript label, the tool presents the user with an option to enter the manuscript data in a functional unit division or directly in a text box.

We provide this option because manuscripts are different and may not contain that text or may contain the text in a different form. More importantly, the user can choose to enter text directly if they do not feel the need to divide the text into logical units. In such a case, the tool presents the users with text boxes with next and previous buttons, which allows the user to enter the text and move on the next text entry from the manuscript. In the case where the user chooses to enter the text in a functional unit division, they are presented with a text ID along with a text entry field for data. Such fields can be added or removed by the user as per the manuscript text. The user is allowed to create multiple logical divisions, and even leave a functional unit entry empty if the manuscript data requires them to do so. The tool requests the user to enter vulgate data which can be a fundamental building block for manuscript data for phylogenetic analysis, if the vulgate data is not present the user can ignore the request, and the phylogenetic analysis can then be carried out without it; although they can enter vulgate data at any point later in time. The data entry interface also allows a user to enter commentaries and quotations into the database. These optional entries can allow a philologist to evaluate the phylogenetic tree constructed, and can also aid the tree construction.



(a) View Mode Snapshot

(b) Comapre Mode Snapshot

Figure A.1: **Screenshot from the Textual History Tool**

A.1.3 View Mode

In this mode, the user can view the manuscript version on the interface based on the label. They can select a label from the list labels in the database or search for a label

and view the sūtra entries, one at a time; this mode also provides the option to correct an entry based on user privileges. We have added the functionality of viewing the sūtras in the form of functional unit division if they were created with one. This can also be used to instantaneously compare the current version with the Vulgate text, which appears on the top in view mode for each manuscript (if present in the database). A snapshot of the said mode is shown in Figure A.1a

A.1.4 Compare Mode

It allows a user to view different manuscript versions on the interface based on user selection. The data from Vulgate, if present in the database, is always shown on top for a base comparison. This mode does not facilitate editing of the manuscript versions but allows one to compare versions, the outcome of which can be utilized during a manual analysis later. It allows the user to select one to four versions for comparison. A snapshot of this mode is shown in Figure A.1b

A.1.5 Phylogenetic Tree Mode

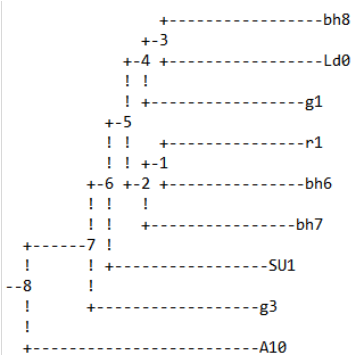


Figure A.2: A sample tree produced in the Phylogenetic Tree Mode

This mode is a novel contribution of our work, where based on functional unit distances, a distance matrix can be created. These functional units are part of a text, and thus the user has a choice for selecting one or more texts wherein the functional unit division has been created in the Data Entry mode described in a subsection above. We use two different approaches to create a distance matrix. The baseline approach, which uses the notion of lexical similarity, uses Cosine Distance, Jaro-Winkler Distance, and Normalized

Edit Distance to compute these distances. The second approach utilizes word-embeddings learned from Sanskrit corpora, which are stored in a model. These approaches are further detailed in Section A.1.6.2.

Eventually, the distance matrix is used to cluster similar manuscripts in the same sub-group, and then the tree can be created using one of the distance-based methods, *viz.,* Neighbor-Joining or UPGMA. These methodologies are also explained in detail here. The tree visualization is shown on the interface in the form of manuscript labels being shown as leaf nodes, which can be seen in Figure A.2. The user is allowed to view the tree on the interface as well as download it in PDF format for further analysis.

A.1.6 Technical Development Details

This section provides a detailed technical description of the tool interface frontend and backend. Along with the interface description, it also entails the methodologies used to create the distance matrix, which is used for tree generation in the Phylogenetic Tree mode. The tool architecture is shown as a diagram in Figure A.3.

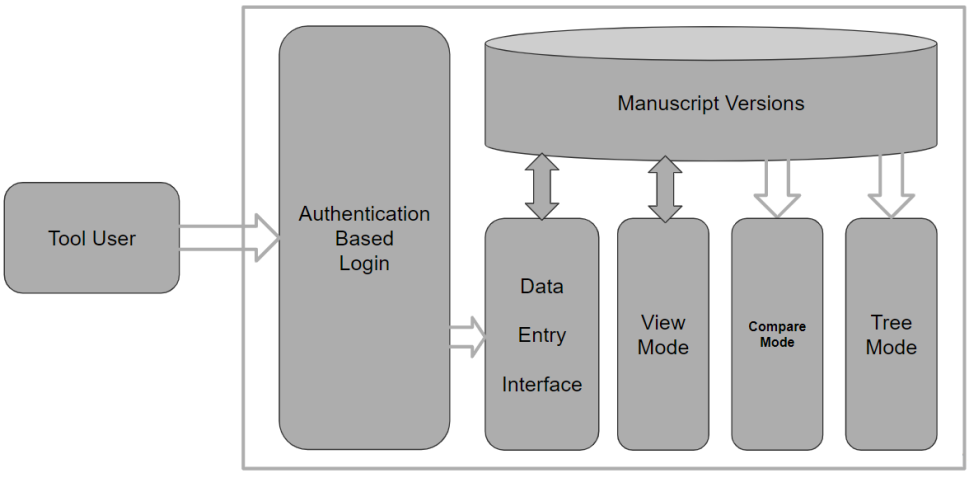


Figure A.3: The architecture of Textual History Tool

A.1.6.1 Tool Interface

The tool is built as an online web-based interface hosted locally on an Apache Server. It is built using PHP, Javascript and utilizes jQuery for querying the backend. The tool backend utilizes MySQL to store the manuscript data in a relational database format

efficiently. MySQL queries from the tool frontend are sanitized before they are sent towards the backend to escape injection attacks. The tool consists of an authentication interface that is based on username/password-based login. The users have to be approved by an administrator after registration, which is available on the login page. The tool users can be granted different privileges based on their usage and expertise in the area. The tool source code can be downloaded and stored offline for local usage.

A.1.6.2 Methodologies for Distance Computation

The phylogenetic tree mode utilizes distance matrix creation based on code written in Python, which can be run for selected manuscripts. Our methodology requires as input the distance matrix between manuscript versions to infer the phylogenetic trees. This distance matrix is computed based on the distance among the functional units, which are divisions in the text, as described above. In case of the unavailability of the division of functional units, the matrix can be computed based on the complete text acting as a single functional unit. The computation of this matrix can be done based on lexical similarity-based measures as a baseline method. Our novel approach utilizes word embeddings from a large Sanskrit Corpus-based model, the details of which are present above in Chapter 7. We use the same methodologies as described in the previous chapter to construct trees and display them on the tool interface.

A.1.7 Tool Features and Functionalities

The tool comprises of the following additional features and functionalities as described below:

A.1.7.1 Manuscript Pictures

In addition to the tree generation and other salient features like a comprehensive data entry mode, the tool comprises an additional feature where it enables the user to view the pictures of the manuscript document as proof to substantiate the data. Philologists can attach pictures of the manuscript entry in the data entry mode as an option along with typing the manuscript data for the database entry. This picture (shown in Figure A.4 as a screenshot), if uploaded by the philologist, is shown with the data entered in the view mode (Section A.1.3).

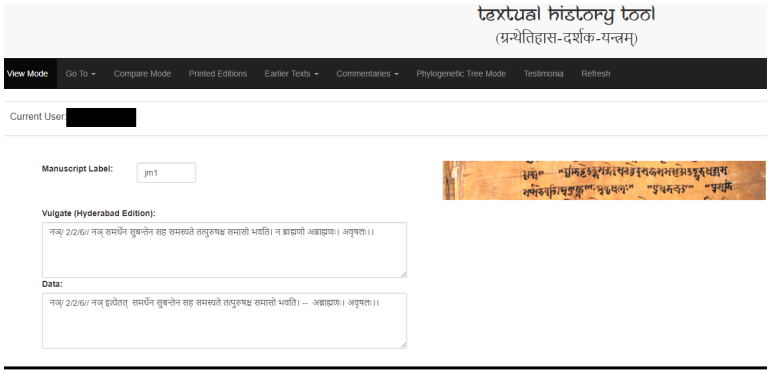


Figure A.4: Screenshot of view mode displaying manuscript picture along with the text in the view mode

A.1.7.2 Critically Edited Text

The tool also allows one to view the critically edited text in the view mode of the tool. The critically edited text allows a user to have a summarized view with additional opinions for the philologists. This helps a user decide which portion of the manuscript they want to consider for creating phylogenetic trees.

A.1.7.3 Critical Apparatus

The critically edited text is usually accompanied by a critical apparatus. The critical apparatus for a text consists of the set of variations made to the critically edited text. These changes are essential to note down as they are an essential part of the preservation of historical texts. These changes allow one to notice the originally written text and how it changed over some time. The tool allows a user to view the critical apparatus in view mode as well.

A.1.7.4 Text Visualizer

Manuscripts can be envisioned as a tree in a hierarchical manner, which helps philologists analyze them conventionally. We propose a different method of viewing the manuscripts based on their distance. This text visualizer of the manuscripts allows one to view the manuscripts as leaf nodes connected using edges where one can manually change the leaf nodes in the visualizer setting. The visualizer uses the database and computes a distance matrix to visualize the graph. The graph is then creating using a javascript-based library, which enlists all the manuscripts in an interactive way where one can manually change

the leaf nodes and create their version of a tree.

Additionally, we also implement the visualizer to depict the relationship between the text and earlier texts. It can also display the inter-relations between the text and its commentaries along with the testimonia. It provides the user with an option to view these visualizations together and also as different visualization. This feature allows the user to gather temporal information from the visualization as the database contains dated entries for the testimonia, commentaries, and some manuscripts. This will help the reader to study the evolution of the text as happened over time.

A.1.7.5 Text Commentaries

There are some direct and indirect commentaries available which comment on the KV text. The two direct commentaries are Nyāsa (Ny) and Padamañjarī (Pm).

The tool allows a user to view these commentaries on each sūtra by providing a button, clicking on which, the commentary available for this sūtra is displayed. This button acts dynamically on the page and is only visible as a clickable button if a commentary is available for the said sūtra which is under view on that page. This option provides additional insight into the text and allows a more holistic view of the work done on the KV text. Another button to view a sub-commentary is also provided. We also provide the option to view a consolidated version of the textual evidence available through the commentaries, as mentioned above.

[Kulkarni, 2002a] mentions the effort on the part of its author to collect information from the Ny and the Pm. This can act as evidence to reconstruct the text of the KV. [Kulkarni and Kahrs, 2019a] enlist the variants of the text of the KV as found in the Pm through more than 300 quotations.

> "There are instances where both the Ny and Pm record the same pratīka. There we can say that both the commentaries received the text of the KV in a similar form. There are also cases when both these commentaries are silent about certain readings. Furthermore, when they remain silent about certain important units of the text, say a vārttika, then it increases the probability

> that that vārttika might not have been there in the original text of the KV as received by these two commentaries. There are also cases when the pratīka recorded by the N and Pm vary. Such cases pose a problem for an editor. In these cases, the problem gets another dimension if the reading of both N and Pm is seen recorded in some number of mss."

[Kulkarni and Kahrs, 2019b] show that the textual evidence available in these two commentaries can be classified under two broad categories: Direct and Indirect. While Direct evidence is clearly visible in the text of the Ny and Pm, indirect evidence can be further classified under two categories: *paroksha* and *atiparoksha*. They, in turn, can further be classified into six and three categories, respectively. This categorization is shown below in Figure A.5. The button in this tool does show all these categories of evidence, thereby displaying the text of the KV as known to these two commentaries.

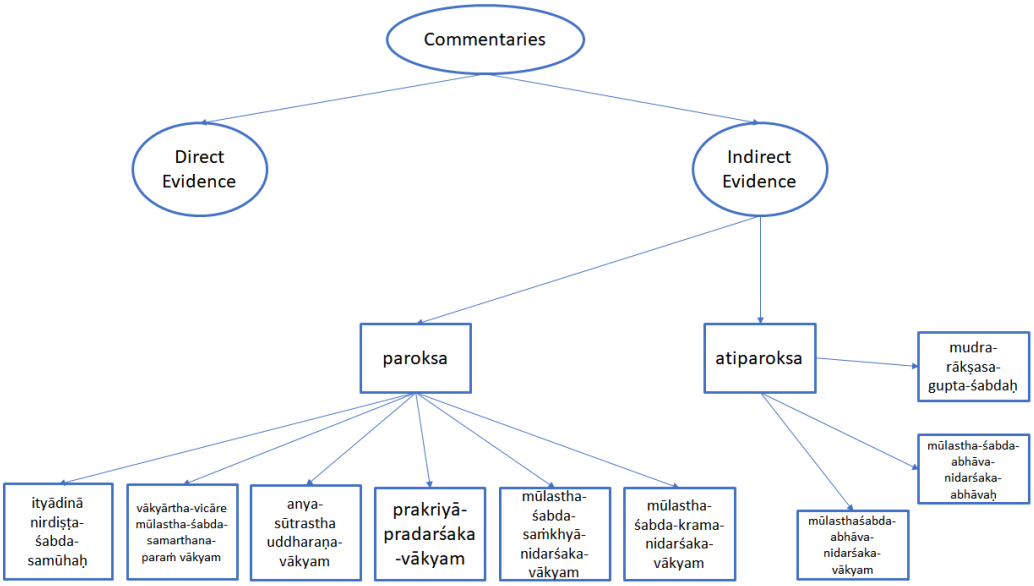


Figure A.5: Classification of evidence from the commentaries on the KV[Kulkarni and Kahrs, 2019b].

Indirect commentaries are the commentaries on the direct commentaries. Tantrapradipa (Tp) is a commentary on Ny. Therefore, it becomes an indirect commentary on the KV. Some portions of Tp which are available, are used in this work. Tp allows us to determine readings in the Ny, thereby indirectly helping reconstruct the text of the KV.

A.1.7.6 Earlier Texts

On the interface, we also provide an option to view the earlier texts. The purpose of this is to provide the reader with the historical view of the text. After clicking on the earlier texts button, the user is provided with an option to choose between "Paninian" and "Non-Paninian" texts. By choosing the option to view "Paninian" texts, the interface shows the earlier texts in the Paninian tradition, in this context, the Vyakarana Mahabhashya (VMbh). This allows the user to see whether there is any historical connection between the KV and the VMbh. It is noted that VMbh is not available on at least more than 2300 sūtras. In those cases, obviously, the tool shows "Text Not Available".

When viewing "Non-Paninian" texts, the interface shows the earlier texts in the Non-Paninian traditions, namely Katantra, Chaandra, *etc.* This allows the user to see whether there is any historical connection between the KV and these traditions. This historical connection is also presented in the text visualizer. The visualizer also provides an option to compare the manuscript version in the database with the earlier texts. This allows the user to study the inter-relation of a particular version of the text of the KV and the earlier Paninian and non-Paninian texts.

A.1.7.7 Testimonia

The text of the KV is quoted in the later texts grammatical as well as non-grammatical. [Kulkarni, 2002a] collected and arranged more than 1000 such quotations chronologically as available from the later Paninian grammatical tradition. [Kulkarni, 2002b] studied one quotation of the KV as found in the Shabdkaustubha and showed the inter-relation of KV manuscripts and Shabdkaustubha. The testimonia button displays all these quotations for the sūtra under study.

A.1.7.8 Printed Editions

The KV was printed for the first time in 1876. [Kulkarni, 2000] traced the manuscript sources of this edition. Ever since then, the text of the KV got printed more than ten times (See Footnote 4). When "Printed Editions" is clicked, the interface displays all the printed editions' text of the sūtra. This historical development in the printed editions is

also presented in the text visualizer. It is hoped that the amount of variation available in the printed editions will serve as a basis to understand the manuscript variants.

A.1.7.9 Reverse Engineering and the Critical edition

This functionality allows a user to create the manuscript versions of the text based on the critical edition and the apparatus. We use the critical edition of the text and apply the variations mentioned in the apparatus to populate the manuscript versions. We believe that this function acts as a validator for the data present in the tool database.

This portion of the tool allows a user to digitize a complete text and its versions through a data entry mode. The data entry mode allows one to partition the text data based on functional units for a more accurate phylogenetic evaluation. The tool also comprises of view mode and compare mode, which can allow a user to view various parts in the text, along with the comparison of the parts in different manuscripts. Based on the data entry and/or division of functional units in the data, the tool also allows one to compute a distance matrix in the backend, which can be further used to compute a phylogenetic tree in the tree mode. The tool comprises of more features like showing manuscript pictures, visualization of manuscripts like a graph, *etc.* This portion of the tool successfully digitizes one specific text, and we hope this can also be applied in a general domain. Utilizing all the features of the tool described above, it enables us to identify the 19th century as an essential stage in the evolution and development of this text, as the manuscripts belonging to this period add 2.2.6.3 to the main text. The justifications for this observation are noted by [Kulkarni, 2002d].

The tool may have its technological advantages but still needs humans to interpret the text. We believe this tool can help the community digitize and view the manuscript data in a format that can be helpful to philologists for drawing further insights from the text. In the future, we would like more functionalities and different tree inferring methods to the tool. Currently, it only supports distance-based methods, as described in the chapter above. We would also like to provide options such as fuzzy matching between the text and the commentaries based on which a portion of the commentary can be aligned to a particular portion of the text. This automation can ease the philologists'

work by automatically showing them alignments between the commentary portions and the main text. We would also like to implement the generation of phylogenetic trees at the micro-level (sūtras) as well as the macro-level (*padas*, *adhyayas* and entire text).

In this work, we describe a tool that captures the historical evolution of a text and allows a user to view the transmission of a text through its history in a comprehensive manner. The tool allows a user to digitize a complete text and its versions through a data entry mode. The data entry mode allows one to partition the text data based on functional units for a more accurate phylogenetic evaluation. The tool also comprises of view mode and compare mode, which can allow a user to view various parts in the text, along with the comparison of the parts in different manuscripts. Based on the data entry and/or division of functional units in the data, the tool also allows one to compute a distance matrix in the backend, which can be further used to compute a phylogenetic tree in the tree mode. The tool comprises of more features like showing manuscript pictures, visualization of manuscripts like a graph, *etc.* In this work, we show how this tool successfully digitizes one specific text, and we hope this can also be applied in a general domain. Utilizing all the features of the tool described above, it enables us to identify the 19th century as an important stage in the evolution and development of this text, as the manuscripts belonging to this period add 2.2.6.3 to the main text. The justifications for this observation are noted by Kulkarni [2002d].

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In the near future, we would like more functionalities and different tree inferring methods to the tool. Currently, it only supports distance-based methods, as described above. We would also like to provide options such as fuzzy matching between the text and the commentaries based on which a portion of the commentary can be aligned to a particular portion of the text. This automation can ease the philologists' work by automatically showing them alignments between the commentary portions and the main text. We would also like to implement the generation of phylogenetic trees at the micro-level (sūtras) as well as the macro-level (*padas*, *adhyayas* and entire text).

A.2 Effective Digitization of Commentaries for Textual History

This work describes additional aspects of a digital tool called the 'Textual History Tool'. It has been accepted at the SSSU 2020 Conference and published under the title, "Strategies of Effective Digitization of Commentaries and Sub commentaries Towards the Construction of Textual History". We describe its various salient features with a particular reference to those of its features that may help the philologist digitize commentaries and sub-commentaries on a text. This tool captures the historical evolution of a text through various temporal stages, and interrelated data culled from various types of related texts. We use the text of the Kāśikāvṛtti (KV) as a sample text, and with the help of philologists, we digitize the commentaries available to us. We digitize the Nyāsa (Ny), the Padamañjarī (Pm) and sub-commentaries on the KV text known as the Tantrapradīpa (Tp), and the Makaranda (Mk). We divide each commentary and sub-commentary into functional units and describe the methodology and motivation behind the functional unit division. Our functional unit division helps generate more accurate phylogenetic trees for the text, based on distance methods using the data entered in the tool.

Texts are invaluable resources of any culture. They provide information that helps a reader understand the intellectual tradition of that culture. In the Indian context, historically speaking, Sanskrit texts were composed and transmitted, orally as well as in written form, for more than at least two thousand years. These texts serve the purpose of sources of Indian culture. Since these Sanskrit texts usually have a long period of transmission, it affects the very nature of the composition of such texts and their transmission. Many texts are composed in terse sūtra style, whereas those in elaborate verse or prose may be prone to loss of meaning comprehension on the part of the contemporary reader. There is another aspect worth mentioning here, since these texts travel over time, they also stand testimony to the evolution and development of the intellectual process.

The fuller comprehension of the terse texts depends on the elaborate explanation of words used in the sūtra. It is necessary to contain the loss of meaning comprehension of texts which travel in the course of history by codifying a methodology to interpret those

texts. When texts travel, they also kickstart a process of thinking based on those texts, which causes the development of thought. All three features mentioned above are the characteristics of what is known as "Commentatorial Texts", popularly known as "Commentaries". Commentaries elaborately explain the wordings of terse sūtras. They codify the various methodologies of interpreting these texts, and they also act as a mirror as far as the process of intellectual development is concerned. When we talk of a specific case like the Paninian tradition, we can see that the commentaries such as the Mahābhāṣya and the Kāśikāvṛtti explain the Paninian sūtras in detail. They also put in place various methodologies like anuvṛtti, adhikāra, ekavākyatā, in order to interpret the sūtras of Panini. They also represent the evolution and development of linguistic thought in India over a period of more than one thousand years. It is therefore essential to have with us a critical text of the commentary, for example, the Kāśikāvṛtti (KV), to accurately perform the above-mentioned functions. Interestingly, in the Indian context, all the above-mentioned features of a commentary (KV) with respect to an original sūtra text, the Aṣṭādhyāyī (AST) are also applicable to that same commentary (KV) and its sub-commentaries (Ny and Pm). It is also worth noting that what is applicable to a commentary (KV) and its sub-commentaries (Ny and Pm), is also applicable to the sub-commentaries (Ny and Pm) and their sub-sub-commentaries (Tp and Mk). We also believe that this recurrence continues even today and will continue in the future course of time and, most importantly, all of this will contribute to the understanding and reconstruction of the source text.

Figure A.6 describes the evolution of commentaries for Panini's Aṣṭādhyāyī (A) and depicts the evolution of sub-commentaries such as the Ny and the Pm which are prominent in terms of the evolution of thought, and in terms of richness of explanation. This is observed in the commentaries and sub-commentaries on it. In our tool, we propose a hierarchy of data entry for the manuscript text and its related texts. We require users to enter the manuscript text. We, then, allow the addition of commentaries on the text for functional units or sūtras that have already been entered into the database. Once the commentaries are in the database, we allow the user to choose the commentary for which a sub-commentary is to be added and allow the user the option to enter the sub-commentary text. This recursive approach allows one to maintain the hierarchy and helps preserve the thought development, which can be seen while entering data in a sequential

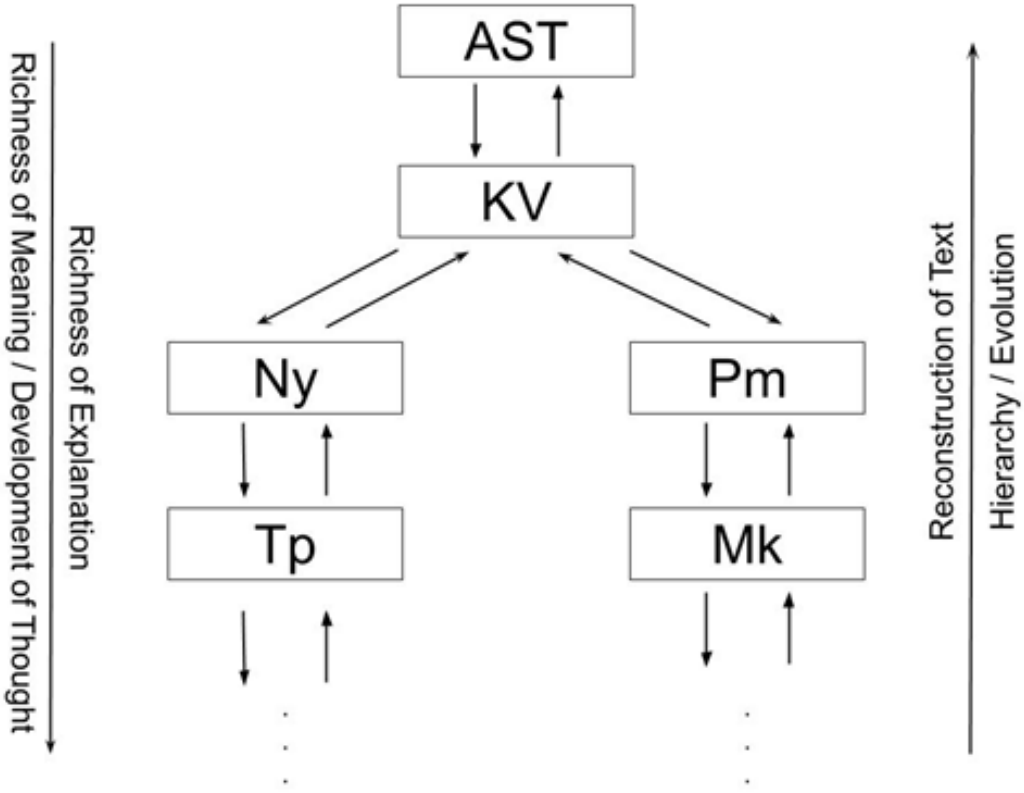


Figure A.6: The hierarchy of evolution vis-a-vis the development of thought.

manner.

The critical contribution of this work is to **describe additional aspects of the textual history tool which embodies the strategies to effectively digitize the commentaries and sub-commentaries towards achieving the goal of text reconstruction and its historical evolution**.

A.2.1 Classification of the Textual Evidence from Commentaries

Textual evidence as available from commentaries can be classified into Direct and Indirect evidence as found through the historical mentions and explanations of the text being discussed. Indirect evidence can further be classified into various categories, as shown in figure A.7.

A.2.2 Reconstruction of the History of the Text

We provide below two detailed examples of how commentaries and sub-commentaries

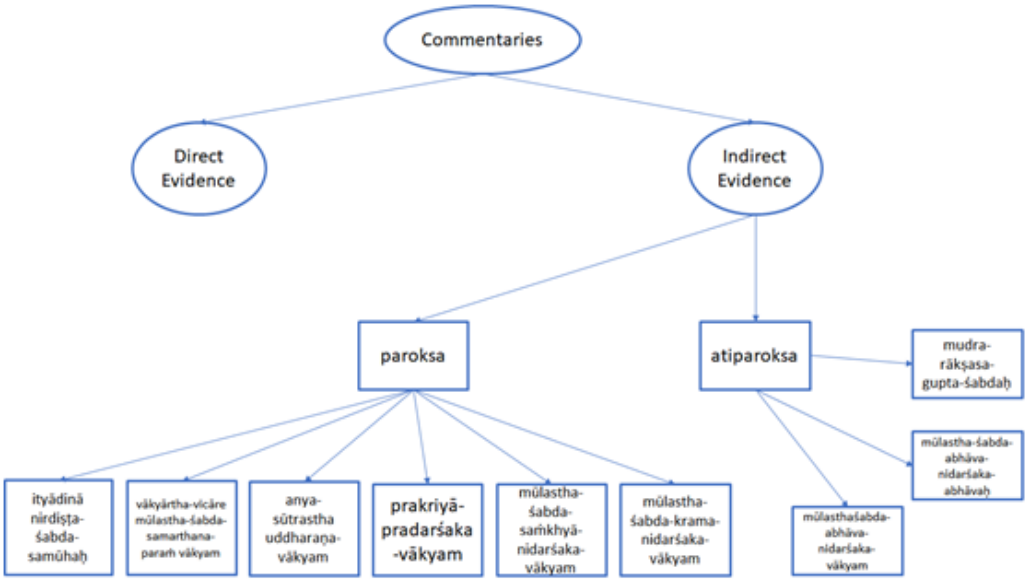


Figure A.7: The classification of evidence found in commentaries as proposed by Kulkarni and Kahrs (2019)

1.1.1.	sūtra	वृद्धिः आत् ऐच्
1.1.1.1.	Introduction and Meaning	सञ्ज्ञात्वेन [विधीयते] प्रत्येकम् आदैचाम् वर्णानाम् सामान्येन तद्भावितानाम् [अतद्भावितानाम्] च तपरकरणम् [ऐजर्थम्] तात् अपि परः तपरः इति खह्वैडकादिषु [त्रिमात्रचतुर्मात्रप्रसङ्गनिवृत्तये]
1.1.1.2.	Examples	आश्वलायनः ऐतिकायनः औपगवः औपमन्यवः शालीयः मालीयः
1.1.1.3.	Other Occurrences of the term	वृद्धिप्रदेशाः सिचि वृद्धिः परस्मैपदेषु इति एवमादयः

Table A.1: Example of a sūtra from KV (sūtra 1.1.1.)

may explain the text better. These example sūtras are taken from printed editions of the Ny and Pm, and the Ny and Tp.

A.2.2.1 Example One

It is observed that amongst the three sections of the KV 1.1.1.3 has got no support in the form of evidence from the Ny as well as the Pm.

In the remaining two sections, there are twenty-five words out of which twenty-four are supported by the evidence found in the Ny and twelve are supported by the evidence found in the Pm. We note that the evidence found in the Ny and the one found in the Pm is uniform and has no variation to offer. Based on this, we can say that the transmission of the two sections of the KV on AST 1.1.1. can be reconstructed in near entirety on the basis of the evidence available to us from the Ny supported by the Pm. A comparison

2.1.22.	sūtra	तत्पुरुषः
2.1.22.1.	Introduction and Meaning	तत्पुरुषः इति संज्ञा ऽधिक्रियते प्राग् बहुव्रीहेः। यानित ऊर्ध्वम् अनुक्रमिष्यामः, तत्पुरुषसंज्ञास्ते वेदितव्याः।
2.1.22.2.	Examples	वक्ष्यति, द्वितीय श्रितातीतपतित इति। कष्टश्रितः। पूर्वाचार्यसंज्ञा चेयं महती, तदङ्गीकरणौपाधेरपि तदीयस्य ,
2.1.22.3.	Other Occurrences of the term	तत्पुरुषप्रदेशाः तत्पुरुषे कृति बहुलम् इत्येवम् आदयः।

Table A.2: Example of a sūtra from the KV (sūtra 2.1.22.)

of this consistent evidence with the manuscript evidence may give rise to the knowledge of uniform transmission of the two sections of the KV on A 1.1.1. However, the same comparison will also reveal the fact that 1.1.1.3, which has no support whatsoever from either the Ny or the Pm, receives concrete support from the manuscripts. This leads to the variation in the transmission of the text of 1.1.1.3 This might suggest the fact that the archetype of the manuscripts that we consult on 1.1.1.3 could be post-Pm.

A.2.2.2 Example Two

In Table A.2, we present the text of the KV on AST 2.1.22. This is also a saṁjñā sūtra like A 1.1.1, so the text of the KV is also divided into three sections as above. We observe that the Ny presents evidence that supports the existence of nine words in the first two sections. We also note that the Tp presents evidence to support only one word out of these nine, but without any variation. So, the textual transmission of these two sections can be said to be uniform, extending itself to even the Tp. Even here, the Ny does not offer any evidence to support the existence of 2.1.22.3, and we also note that the Tp too does not offer any evidence to support the existence of 2.1.22.3. So, we can say that overall, in all the sections of 2.1.22, a uniform transmission is observed as far as the Ny and the Tp are concerned. As far as 2.1.22.3 is concerned, in comparison with comments made above, we can extend the argument and say that our archetype can be said to be post-Tp. So, a collection of evidence from the Tp on similar sections in all the available saṁjñā sūtras can help shed more light on the transmission of this section.

Our work describes an effective method for the digitization of commentaries and sub-commentaries available for a text. We describe a digital tool which provides access to users based on an authentication-based platform and helps them add sub-commentary data in a hierarchical and recursive manner. We highlight the changes made to this Textual

History Tool describe above, and show, with the help of screenshots, how improvements made to the tool functionalities help annotators digitize the commentary data. Using this approach one can effectively digitize the available data and use them to create a phylogenetic tree for the text. This is a novel contribution of our work done with the Textual History Tool. The tree can be created by using the various versions of the text as available in the manuscripts and including commentaries or by only using the commentary data. The tree generation methodologies which have been described by us above, are the same for our work. We also present examples of sub-commentary data described in this section. The trees which are generated via our approach assume that the commentaries present in the form of printed editions are authorized versions of the written text. We aim to use this tool[10]., and the work is done to successfully help digitize many texts and commentaries over them. In future, we would like to include the testimonia addition to the tool. This can help create trees even based on mentions of the text in other texts. These could belong to the same tradition or some other tradition. This can help provide clues to the tree generation algorithms as to when and where the text was mentioned and what form of the text was quoted in the other text. We also aim to extend our work to other texts than the KV and show that the tool has indeed a generic applicability towards text digitization and reconstruction.

A.3 Summary

We have described, in detail, an online tool which can be used by philologists to enter and manipulate manuscript data, for phylogenetic analysis. Such a tool can help philologists perform phylogenetic analysis and create the the critical edition of a text. We also add various functionalities to the tool which help it store commentaries, and printed editions. We discuss the classification of commentaries of the KV text and propose a generic classification which can be helpful in classifying commentaries from such texts.

[10]Tool Link

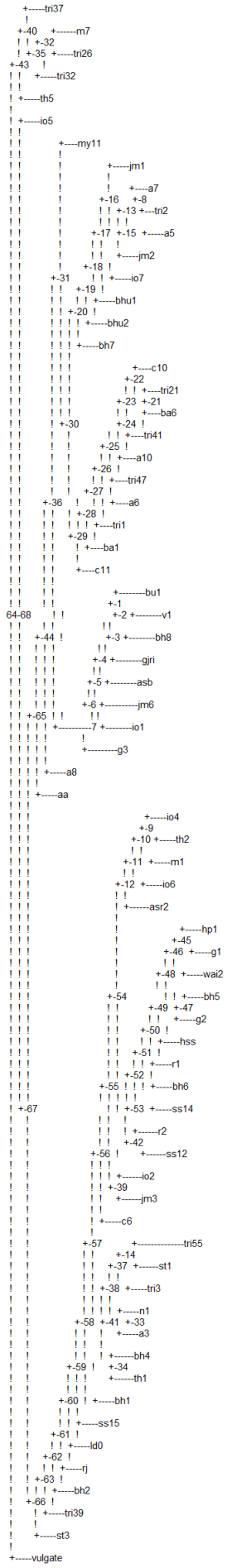
Appendix B

Additional Trees Generated

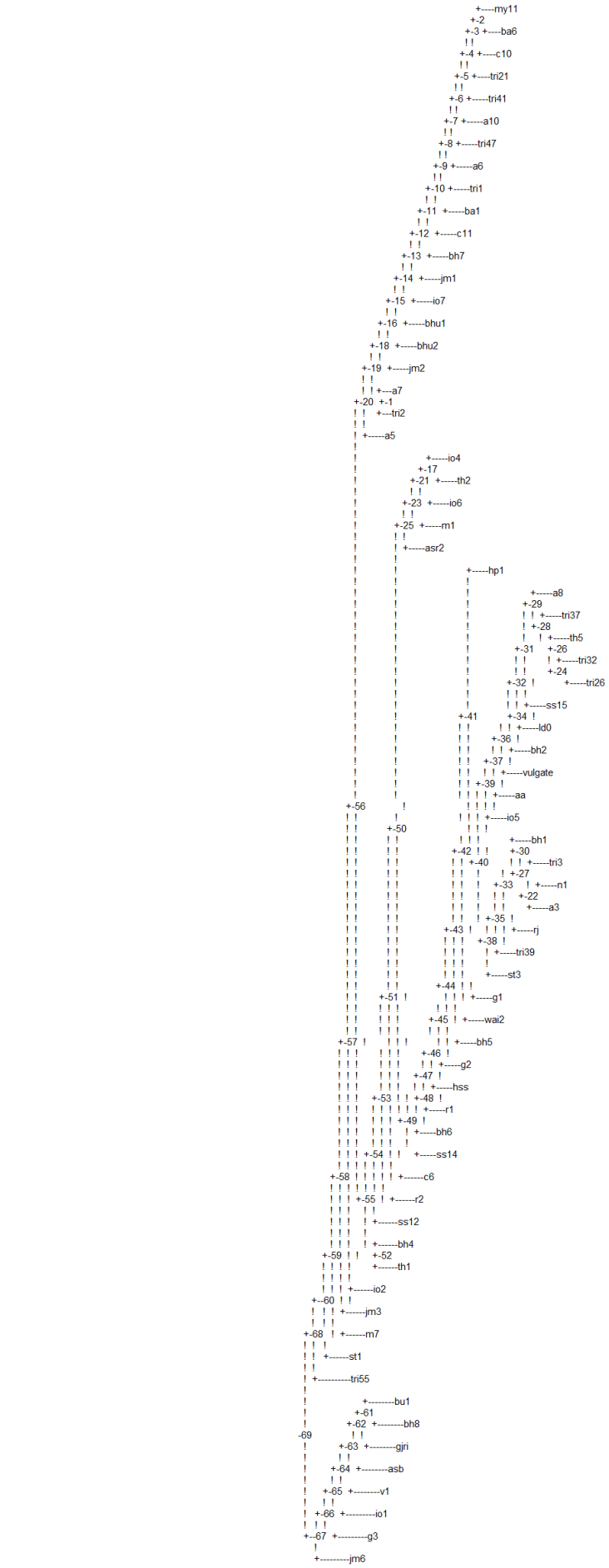
This additional appendix presents all the trees generated during our phylogenetic experiment, referenced from Section 7.4.6. Each tree is shown on a single page for reading clarity. Please see the pages below.

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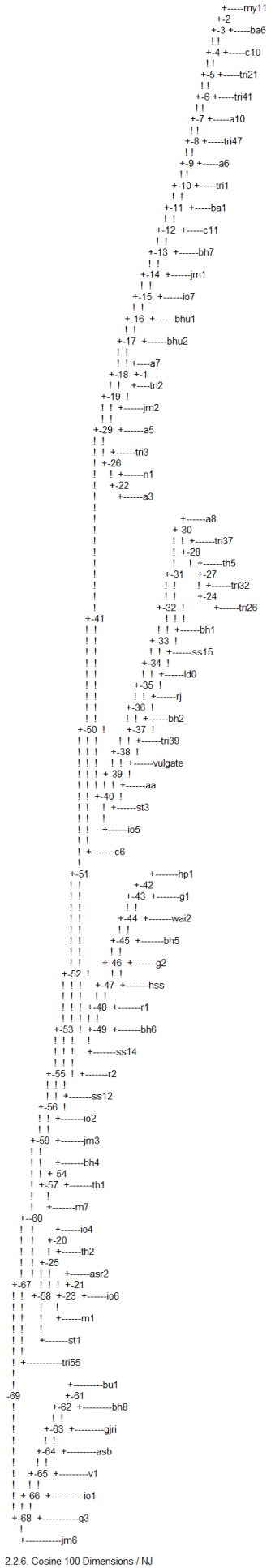
Each tree is shown on a single page for reading clarity. Please see the pages below.



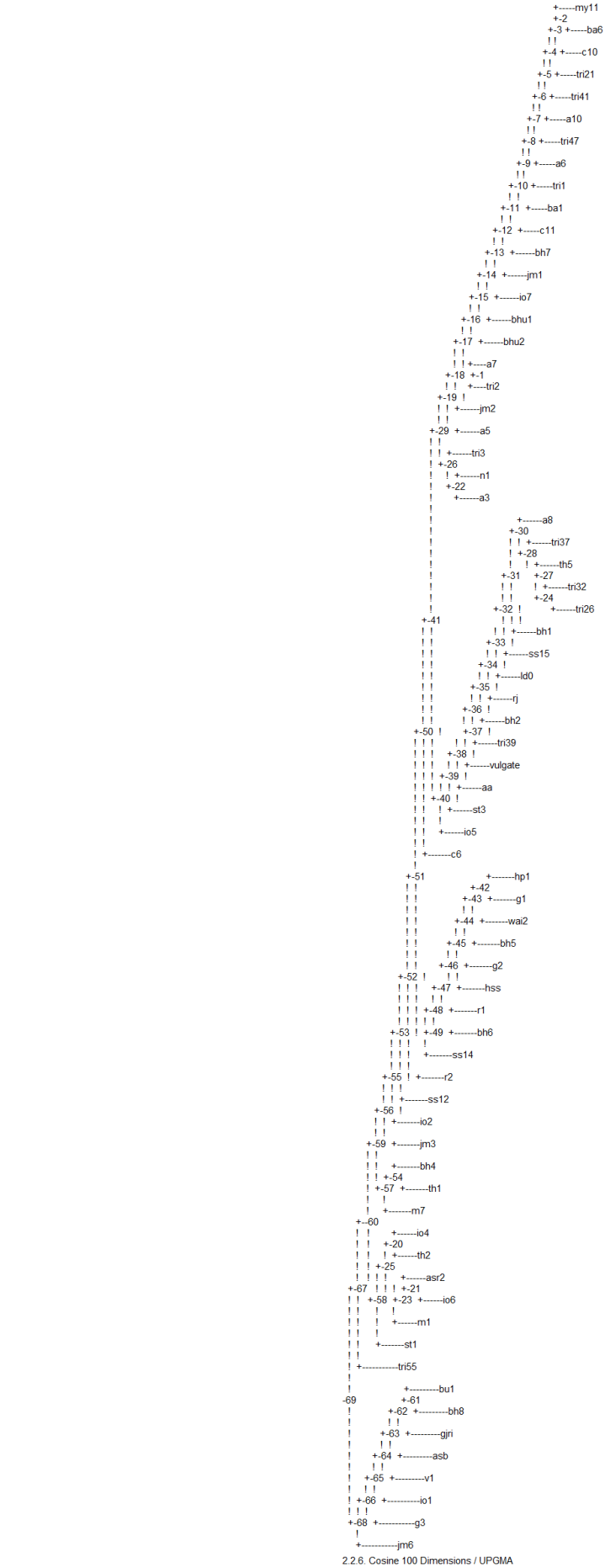
2.2.6. Cosine 50 Dimensions / NJ



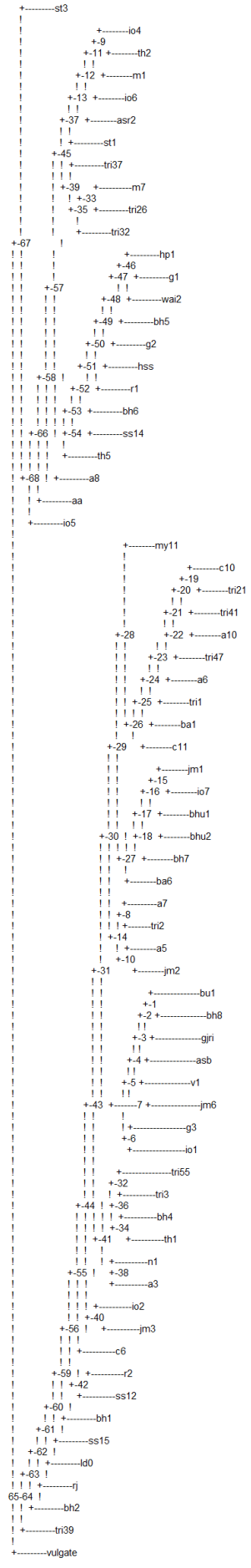
2.2.6. Cosine 50 Dimensions / UPGMA



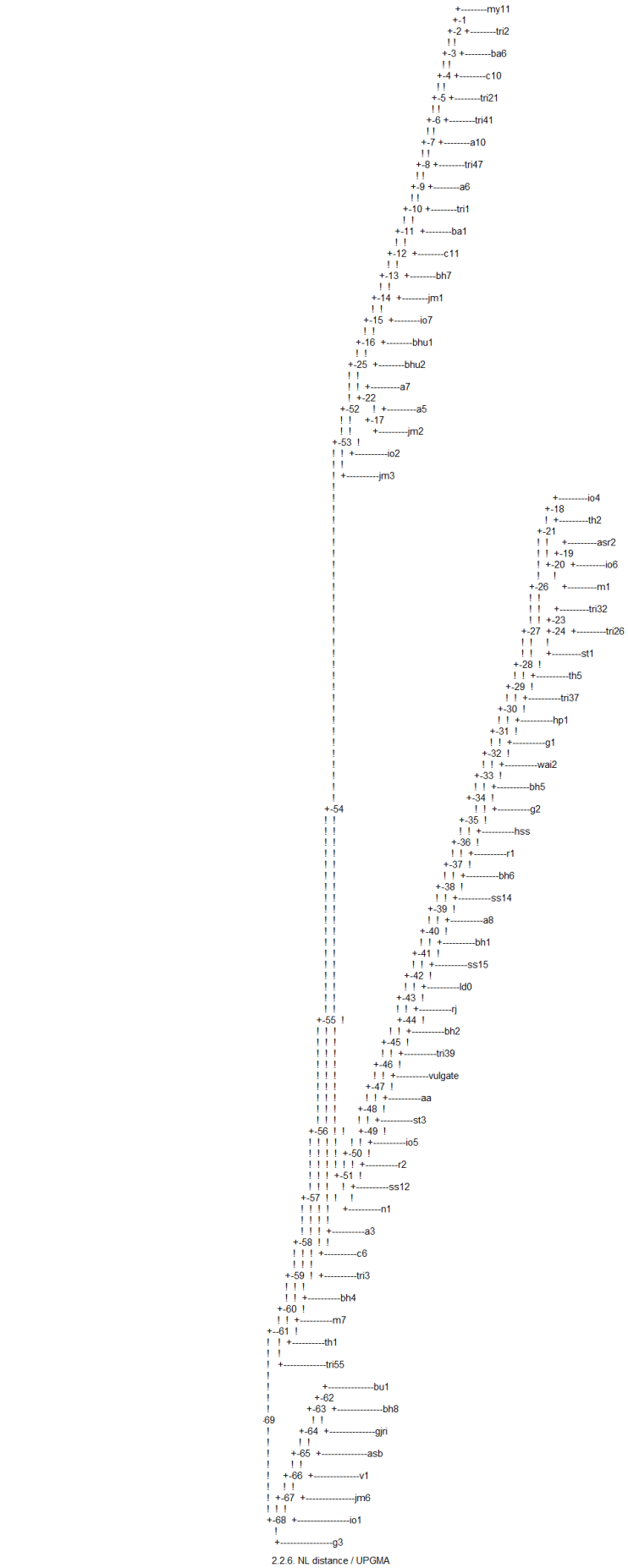
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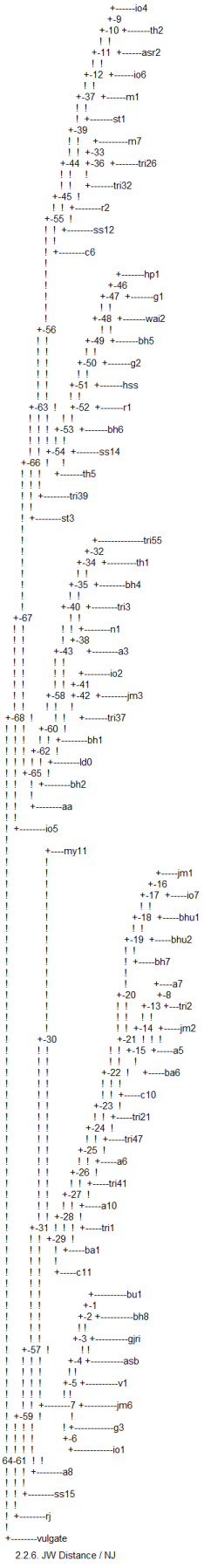
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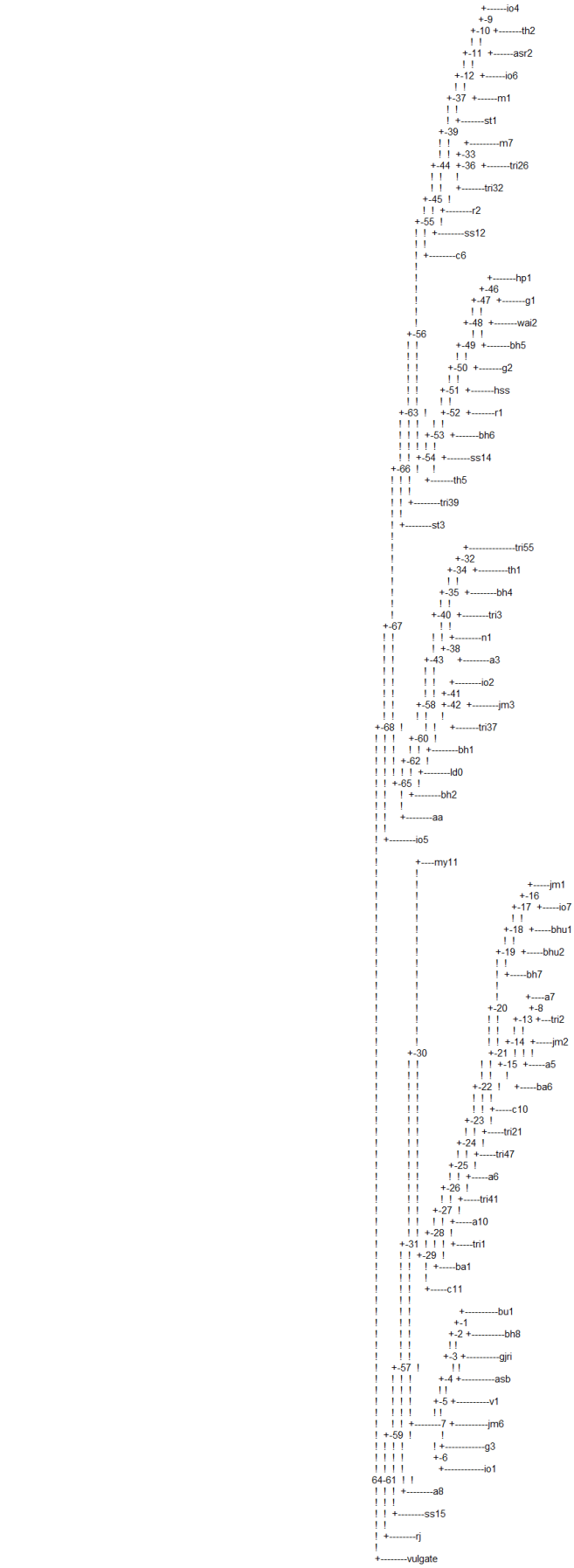
2.2.6. NL distance / NJ



2.2.6. NL distance / UPGMA



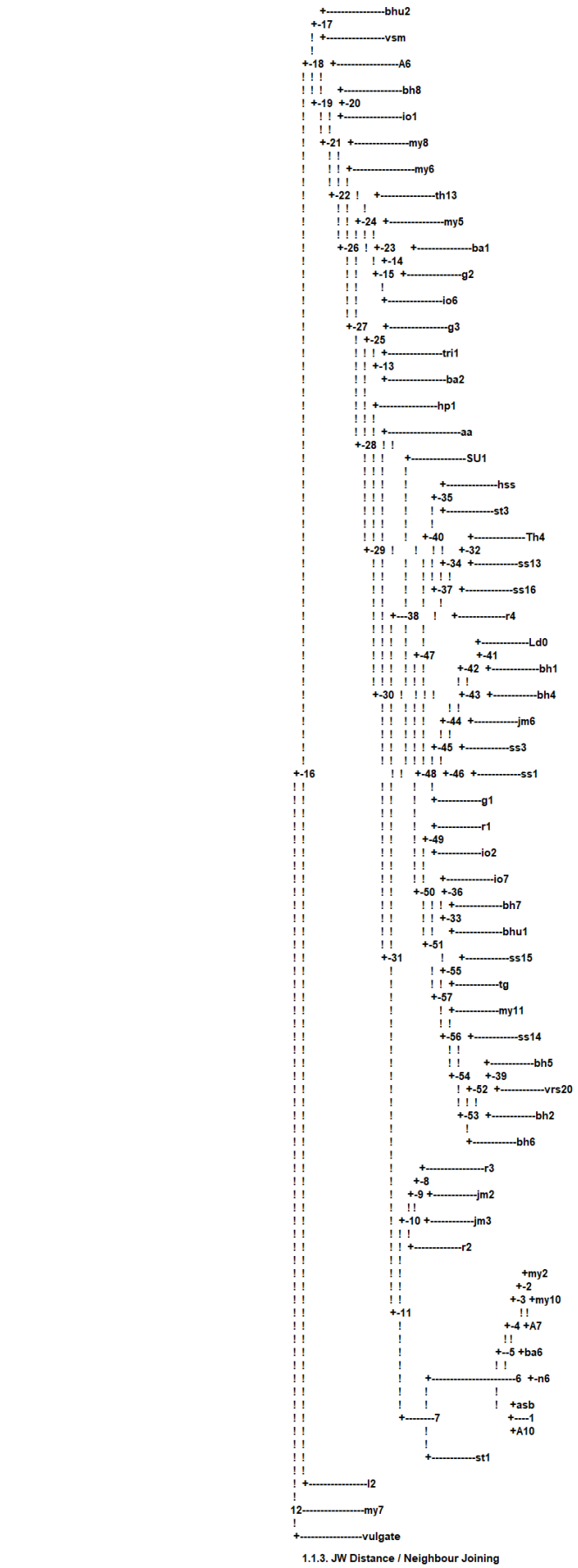
2.2.6. JW Distance / NJ



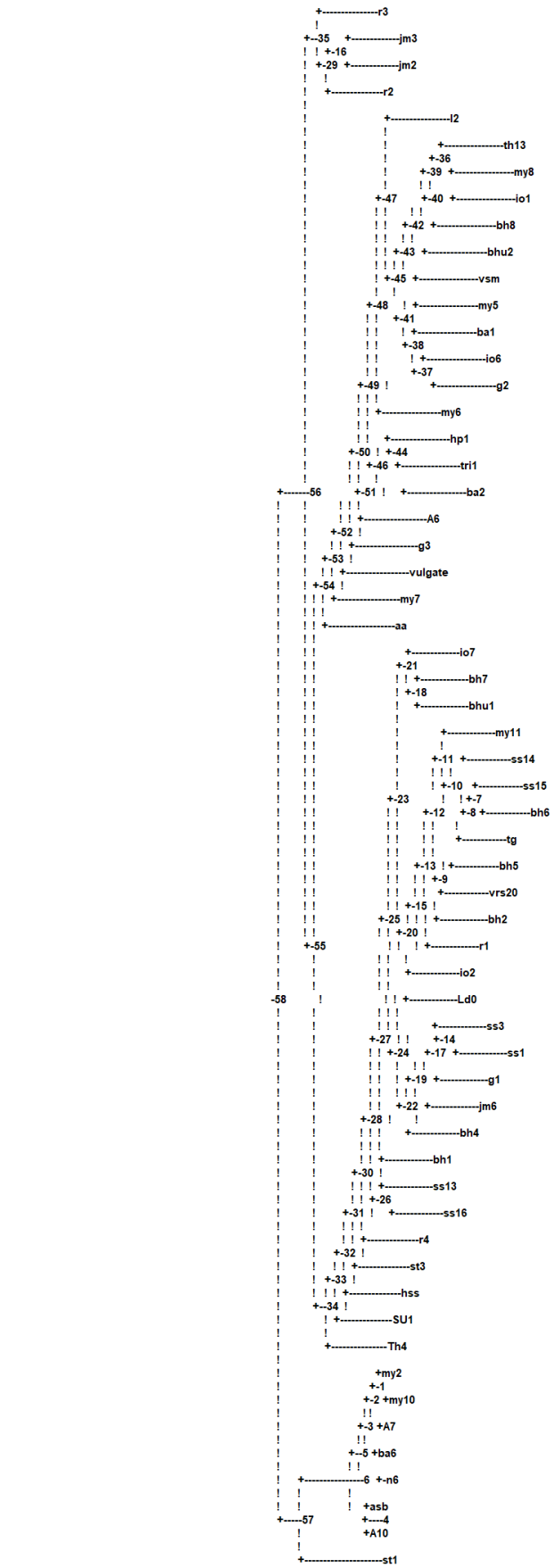
2.2.6. JW Distance / UPGMA

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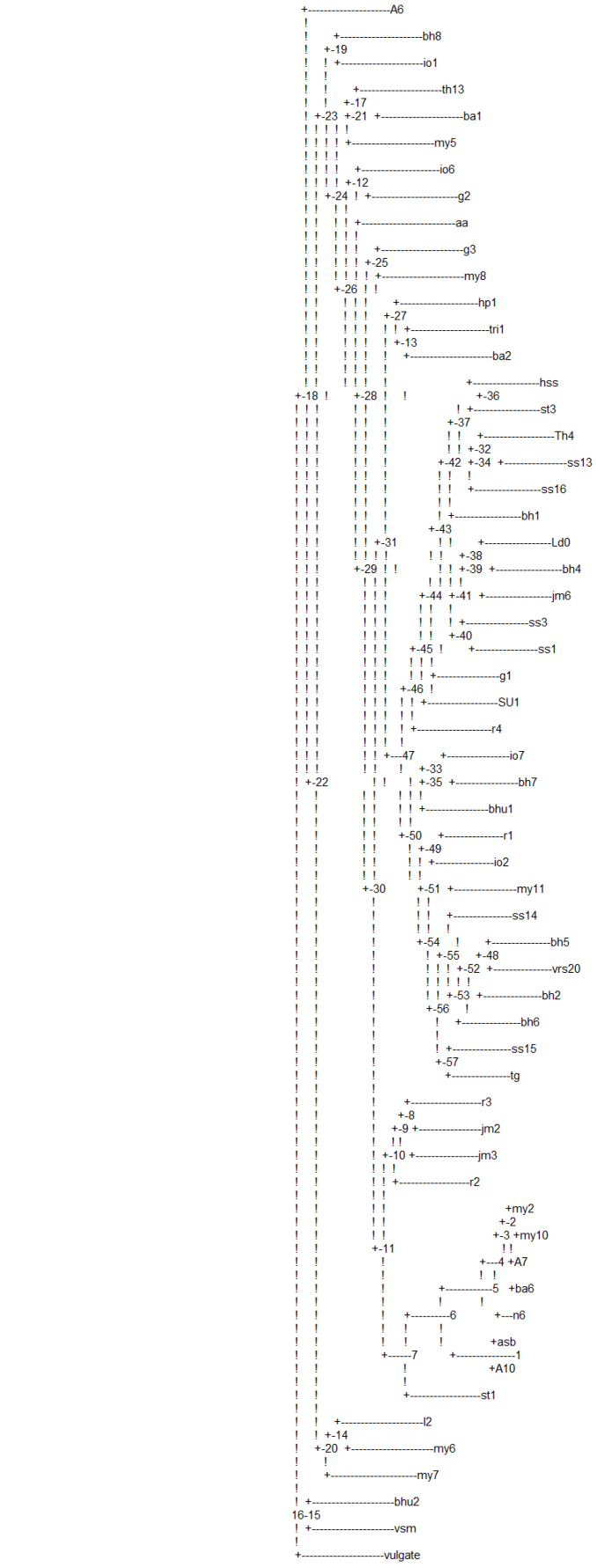
Each tree is shown on a single page for reading clarity. Please see the pages below.



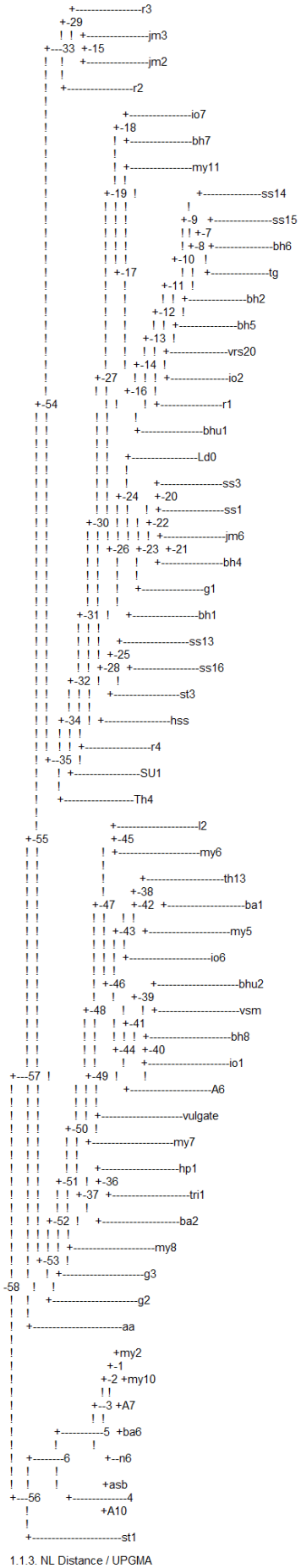
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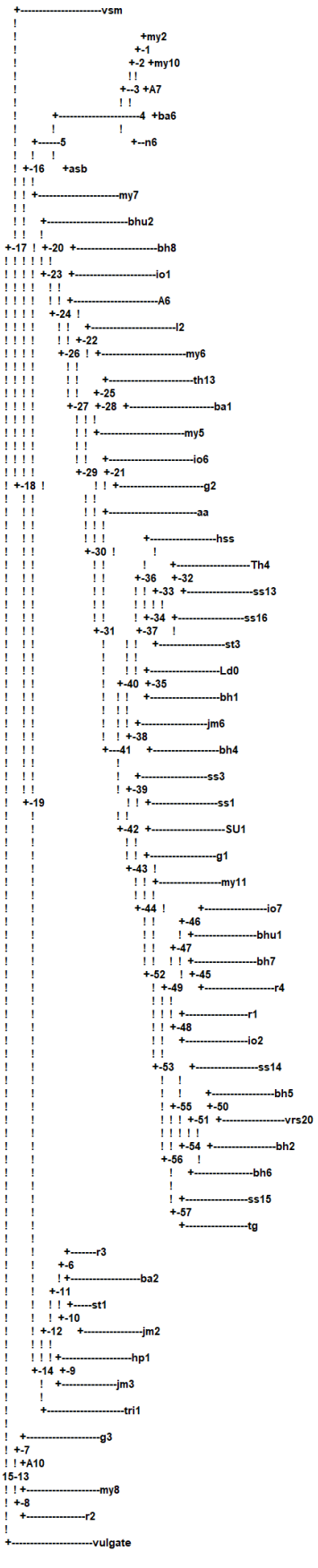
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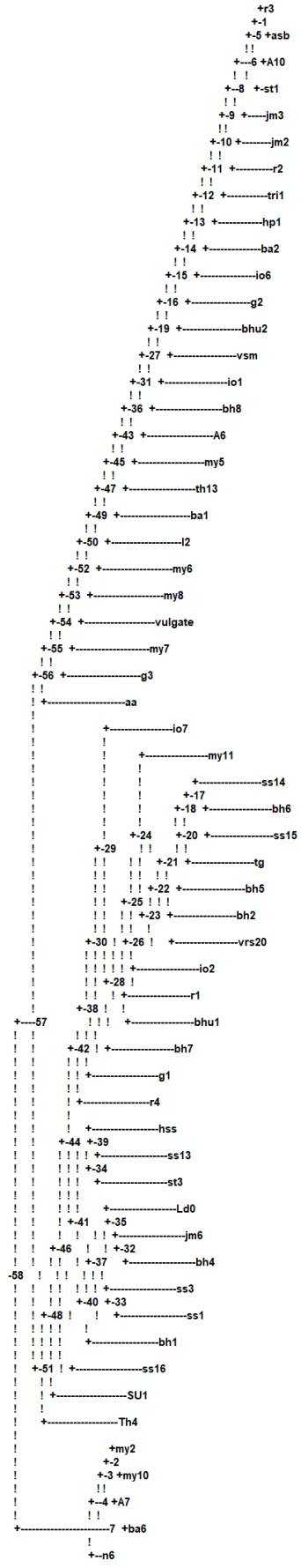
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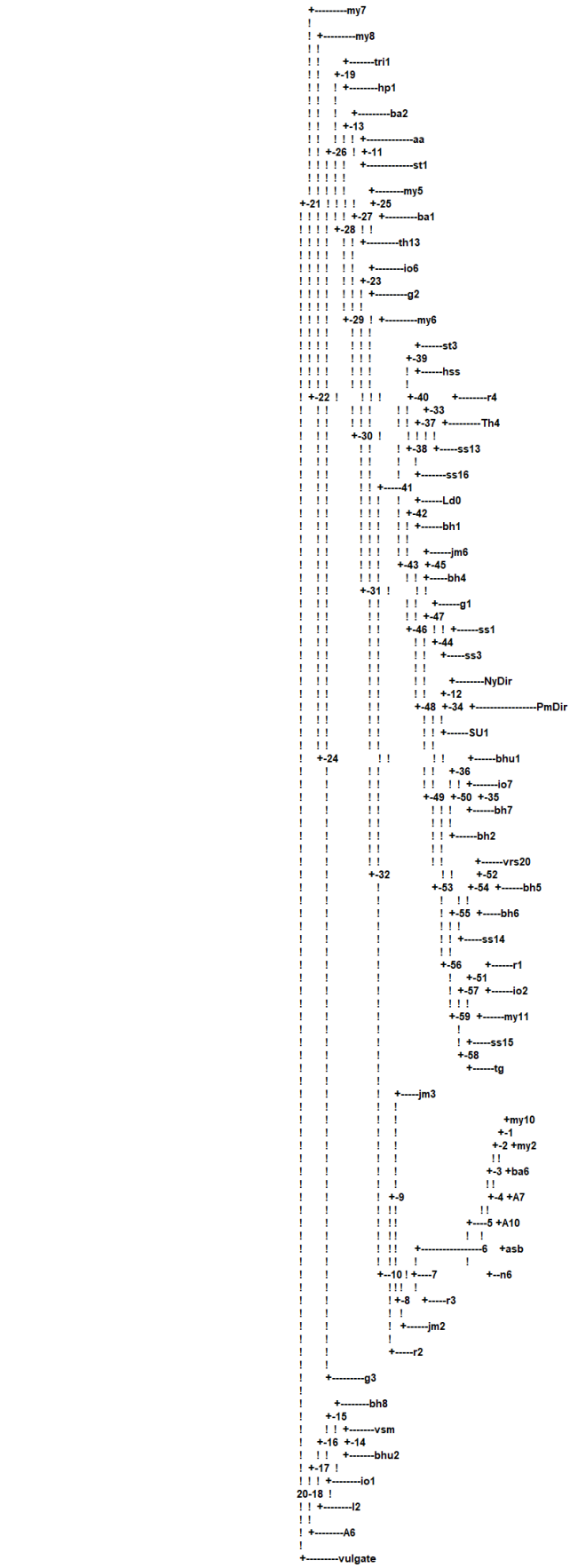
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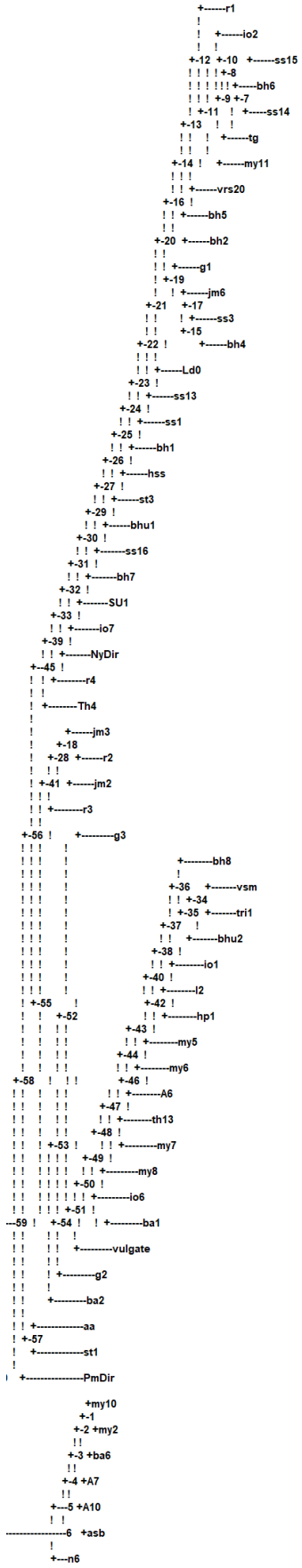
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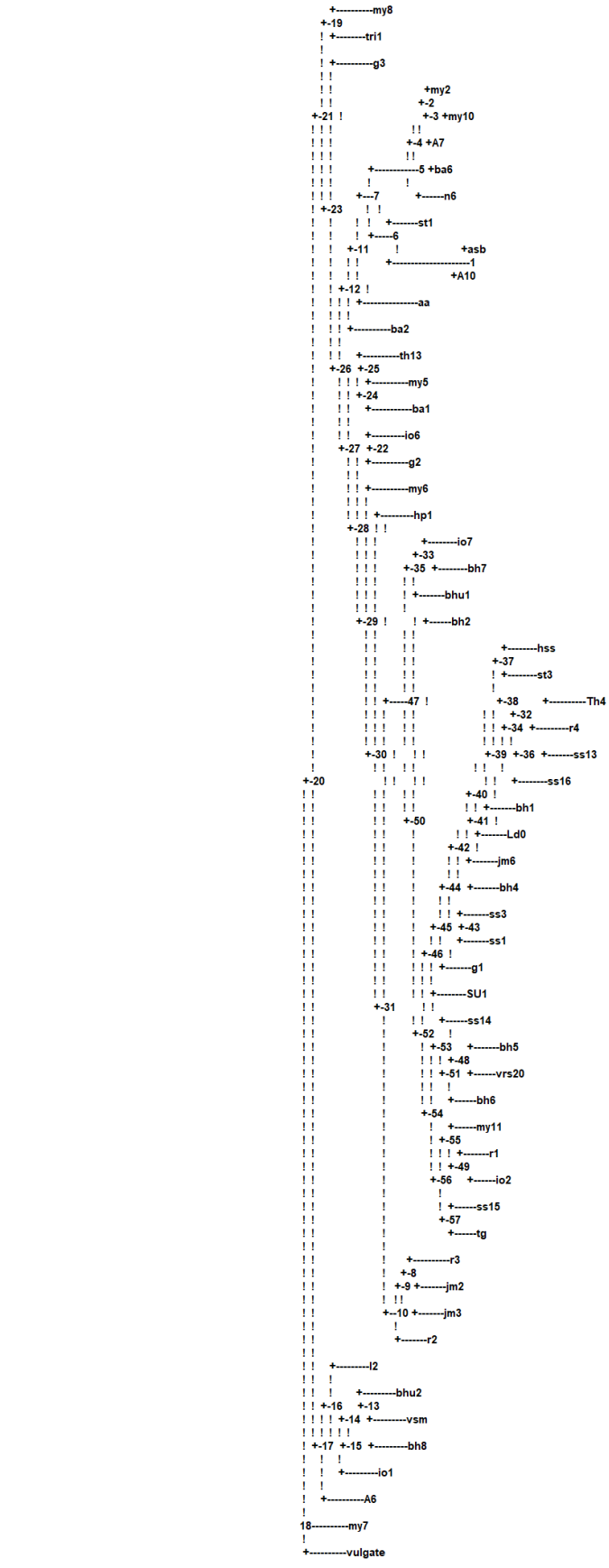
1.1.3 Cosine / UPGMA



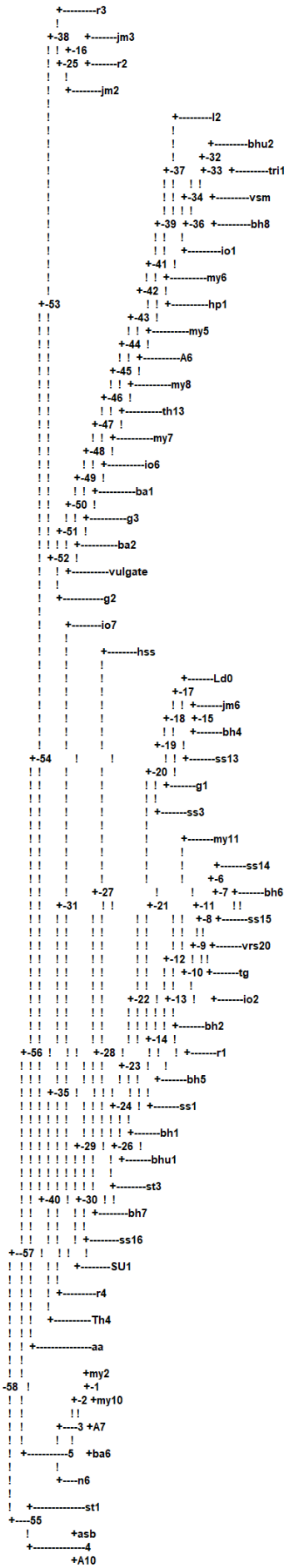
1.1.3 Cosine 50 Dimensions / Neighbour Joining



1.1.3. Cosine 50 Dimensions / UPGMA



1.1.3 Cosine 100 Dimensions / Neighbour Joining



1.1.3. Cosine 100 Dimensions / UPGMA